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Mexico

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OECD ECONOMIC SURVEYS

2002-2003

Mexico

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BASIC STATISTICS OF MEXICO

THE LAND

Area (sq. km)	1 964 375
Agricultural area (sq. km) (1990)	394 000
Forests (thousand sq. km)	65

Inhabitants in major metropolitan areas (million, 2000)	
Mexico City	18.1
Guadalajara	3.7
Monterrey	3.3

THE PEOPLE

Population (thousands, Census 2000)	97 483
Inhabitants per sq. km, 2000	49.6
Annual population growth, 1990-2000	1.85

Employment[1] (thousands, 2002)	41 086

PRODUCTION

Structure of production, 2002 (per cent of total, 1993 prices)	
Agriculture	5.4
Industry	26.7
of which: Manufacturing	19.8
Services	67.9

GDP in 2002 (US$ billion, current prices and current PPPs)	935.3
GDP per capita in 2002 (US$, current prices and current PPPs)	9 224
Gross fixed capital formation in 2002 (per cent of GDP, 1993 prices)	19.3

THE GOVERNMENT

Public sector indicators (per cent of GDP, 2002)	
Public sector total expenditure	23.7
Federal government total expenditure	18.4
of which: Capital expenditure	2.2
Federal government revenue	16.1
Net debt of public sector (December 2002)	25.2

Composition of Parliament (Sept. 2003)	Senate	Chamber of Deputies
PRI	20	222
PAN	46	151
PRD	16	95
Other	6	28

FOREIGN TRADE

Exports of merchandise (per cent of GDP, 2002)	25.2
Main exports (per cent of total, 2002):	
Manufactures	88.3
Petroleum products	9.0
Agriculture	2.4

Imports of merchandise (per cent of GDP, 2002)	26.5
Main imports (per cent of total, 2002):	
Intermediate goods	75.0
Capital goods	12.4
Consumer goods	12.6

THE CURRENCY

Monetary unit: Peso

Currency unit per US$, average of daily figures:	
Year 2002	9.6605
September 2003	10.9231

1. People economically active according to results of the Quarterly National Employment Survey.

This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of member countries.

•

The economic situation and policies of Mexico were reviewed by the Committee on 20 October 2003. The draft report was then revised in the light of the discussions and given final approval as the agreed report of the whole Committee on 31 October 2003.

•

The Secretariat's draft report was prepared for the Committee by Bénédicte Larre, Stéphanie Guichard and Ann Vourc'h under the supervision of Nicholas Vanston.

•

The previous Survey of Mexico was issued in April 2002.

Assessment and recommendations

Performance improved over the 1990s, but not by enough

In most respects, Mexico's economic performance improved over the 1990s. GDP growth was vigorous, inflation fell steadily, and the current account deficit was of only moderate size. Even the downturn which followed on the heels of the United States slowdown in 2001 was mild by Mexico's standards. Since the 1995 peso crisis, the financial system has been strengthened and, relative to its own past history and experience in many other Latin America countries, the Mexican economy has been stable. This improvement in economic performance owes much to sound macroeconomic policies. Yet, the wide-ranging structural reforms of the past fifteen years, including the entry into NAFTA, have not yet led to an unambiguous rise in labour or total factor productivity growth. Even abstracting from the recent cyclical weakness, Mexico's growth performance since the restoration of macroeconomic stability has been unsatisfactory. Potential GDP growth estimates have been revised down to below 4 per cent, too slow for a country with low levels of income and productivity and high rates of population growth, and too slow to narrow the gap in living standards relative to other OECD countries. Furthermore, the delayed impact of previous reforms is uncertain and is unlikely to raise potential growth sufficiently in the near term.

More reforms are needed to boost income growth

Potential growth is constrained by low levels of human capital, while additional fiscal resources are needed to improve and expand inadequate physical infrastructure, and combat widespread acute poverty that lead to poor health and social marginalization. Furthermore, there are poor incentives for labour to work in the formal sector, where productivity is higher, but where there are blunted incentives for the private sector to invest and innovate.

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There is thus a need for further reforms that will raise the rates of growth of labour and total factor productivity. The objective should be to increase the growth of potential GDP by at least two percentage points, to above 6 per cent annually. The main challenges are to:

- remain committed to macroeconomic stability;
- put public revenue and expenditure on a more solid and predictable footing;
- ensure that resources for education and training are used more effectively;
- raise and improve the stock of infrastructure capital;
- pursue labour market reforms;
- ease regulatory measures and other impediments, including failings of the judicial system and high perceived levels of corruption, that weigh on entrepreneurial activity and business investment.

Growth-oriented policies must also be flanked by measures specifically addressed at alleviating extreme poverty and ensuring that benefits of stronger growth are shared more broadly across population groups.

Economic activity turned up in 2002, but the recovery has been slow

Mexico did not escape the economic slowdown that hit OECD countries in 2001, and the recovery has yet to become firmly established. GDP growth is likely to be below 2 per cent in 2003 (not much above population growth of 1½ per cent). Economic activity is expected by the OECD to gather momentum in the latter part of the year and expand by around 3½ per cent in 2004, as exports quicken in line with foreign markets. And business investment, which has been particularly sensitive to global economic developments and uncertainties surrounding them, should gradually pick up. The main risks to the outlook concern world oil prices, financial markets and, above all, the speed of the United States recovery and how this translates into rising demand for Mexico's exports. On the domestic front, advances in the structural agenda would help create an environment more supportive of private investment, including foreign direct investment, and could lead to stronger GDP growth as early as in 2004.

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The firm stance of the 2003 Budget is appropriate but there is no room for slippage

The public sector financial deficit would have come in at 0.7 per cent of GDP, just below target in 2002, but the cost of liquidating a small development bank towards the end of the year and creating a new institution for the rural sector pushed it up to 1¼ per cent of GDP. Meanwhile, the broader public sector borrowing requirement (PSBR) was slightly above 3 per cent of GDP, excluding non-recurrent revenues. The 2003 budget deficit is targeted at 0.5 per cent of GDP, with the objective of achieving balance by 2005. By then, the PSBR is to be brought down to around 2 per cent of GDP. Thereafter, a reasonable medium-term objective would be to put the PSBR on a track consistent with a steadily falling ratio of broad debt to GDP.

Further action is needed on the tax front, while…

Although the public sector deficit has been reduced over the years, the allocation of public spending has been improved and tax revenue has been increased, further improvement in public finances is achievable and necessary. General government revenue is still low relative to GDP and highly dependent on volatile oil-related revenue. The scope of the 2002 tax reform fell short of initial proposals, especially as regards indirect taxes, and the actual impact on revenue of the incomplete reform also fell short of official estimates. Spending in areas conducive to economic development (poverty relief, human and infrastructure capital) needs to be financed at a higher level and on a more predictable basis, while explicit and implicit liabilities will continue to affect the budget over the medium term. Hence further action on the tax front is indicated. A reasonable benchmark would be to raise tax revenue in the medium term by 2 percentage points of GDP. Even then and considering that Mexico uses tax concessions where other countries use social transfer payments, the ratio of tax revenue to GDP (excluding oil-related revenues) would still be one of the lowest among OECD countries. A widening of the VAT base, with the elimination of exemptions and zero rating would facilitate tax administration. Changes in that direction could be revenue-enhancing even if rates are lowered. A number of issues will have to be addressed, such as how to limit the impact of the VAT change on low-income groups. Although there is wide agreement in Mexico that more tax revenue is needed, what has been lacking is an agreement on how to achieve this. Political parties

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are currently discussing with the government various options for another set of tax measures.

… the on-going efforts to put public finances on a sounder footing have to continue

The medium-term national programme for financing development (PRONAFIDE) should be broadened. The government has a commendable record of meeting annual budget targets even in adverse conditions, but the large volatility in revenue has often led to *ad hoc* adjustments to spending, especially on public investment. Another side-effect is that fiscal policy has typically had a pro-cyclical bias. On the tax side, the existence of numerous special regimes and exemptions has meant that the government has had to make recurrent changes to the tax law, complicating tax collection and adding to the cost of compliance. A medium-term goal for Mexico should be to set public expenditure programmes in a multi-year framework while maintaining fiscal prudence. This would encourage both policy-makers and Congress to focus more attention on the longer-term development needs of Mexican society, how best to finance them, and when.

Monetary policy has succeeded in reducing inflation pressures in 2003

The inflation objective of the monetary authorities for 2002, of 4.5 per cent year-on-year by December, was missed, mainly because of increases in food and government-regulated prices. However, core inflation declined steadily to below 4 per cent by end-year. The inflation objective is to bring the CPI inflation rate down to 3 per cent, with a variability margin of plus or minus one percentage point, by December 2003, and keep it in that range over the medium term. At the beginning of 2003, to drive inflation expectations down, the Bank of Mexico tightened the policy stance. By July 2003, headline inflation was down to 4.1 per cent and core inflation to 3.6 per cent, with inflation expectations also down. In that context, the Central Bank did not try to oppose the strong decline in market interest rates induced by lower country risk perception in financial markets. The Bank's readiness to respond quickly to changes to the inflation outlook is appropriate.

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The inflation target is appropriate, and moving towards an interest-rate targeting framework could be considered

Looking ahead, continued caution on the part of the monetary authorities is required to meet their objective and stabilise inflation at around 3 per cent. Although the targeted inflation rate is higher than the average in major trading partners, it is adequate over the next few years. A moderate, predictable and stable inflation environment is necessary to further reduce risk premia required by domestic and foreign lenders, and to encourage long-term investment by the private sector. As inflation recedes and the pass-through from exchange rate fluctuations to domestic prices diminishes, thus obviating the need for large changes in interest rates, the Bank of Mexico could consider changing its operating procedure to set interest rates directly, instead of bringing about changes in market conditions through its announcement about the overall position of banks (*corto/largo* mechanism). This would enhance transparency of monetary decisions.

Recent financial reforms should help revive banking credit to the private sector

Actions taken in the aftermath of the 1995 peso crisis have strengthened the Mexican banking system: it is now as solid and profitable as in other OECD countries and the supervision and regulatory framework are close to best practices. Although there has been a resumption of credit growth to consumers especially after the credit crunch of the mid 1990s, lending to private firms remains somewhat hesitant. The test will come as demand picks up. To address some remaining weaknesses in the legal and regulatory framework that increased credit costs and might have deterred banks from taking risks, the Congress in April 2003 approved several welcome amendments to strengthen the legal framework for secured credit transactions. To bring about the revival of bank credit in practice, especially to small and medium-sized enterprises (SMEs), these amendments need to be translated into action. Other channels for SMEs' financing might be explored, as long as they do not create distortions. The state-owned development bank segment is currently being rationalised, and this is welcome. Credit unions and Savings and Loans institutions have increased their share in the SME and housing credit markets, calling for a strengthening of the sector's supervision. Measures to address this last concern have been decided, and they should be implemented without delay.

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To raise productivity growth and narrow the gap in living standards, more reforms are needed

Previous *Surveys* have underlined the solid progress made over the last decade on the structural reform front. The size of direct government intervention and participation in the productive economy has been substantially diminished, and the business environment has become more market-oriented. There nevertheless remain several areas where efforts could be stepped up to correct inadequacies in the implementation of earlier reforms and overcome delays in others, in order to raise potential growth. The focus of reform should be on the basic determinants of growth: investment in human capital and the fostering of conditions conducive to investment and innovation.

Efforts to reach a consensus on key reforms are welcome

The authorities are committed to a comprehensive strategy that aims at raising investment rates and improving the country's international competitiveness, as underlined in the PRONAFIDE and the Development Plan 2001-2006. But on many fronts, it has been difficult to reach consensus and reforms have stalled, even in areas where there is general agreement on their desirability. Progress in coordination and dialogue between the government and the newly-elected Congress will be key to moving ahead decisively in areas which need urgent action, including electricity and labour market reforms, in addition to the tax reform. At the same time, it is essential to step-up progress in areas where legislative changes are not necessarily required, such as the education sector.

Reforms should focus on further enhancing human capital…

Improving the effectiveness of the education system, and thereby strengthening human capital, is indeed a necessary condition for high sustained growth in the long term. Despite major improvements over the past decades, especially in increasing basic school enrolment for a rapidly growing school-age population, Mexico's human capital lags far behind most other OECD countries. The quality of education services and training is well below OECD best practice, many school-leavers have poor literacy and numeric skills, and the cost effectiveness of education programmes needs to be improved. Additional resources could be used to address some of the shortcomings, but more importantly, what is required is further modernisation of curricula, providing better training to teachers, enhancing the accountability

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of schools and rewarding individual effort by teachers. Resources to be allocated to pre-school education should not be at the expense of greater efforts to ensure that more children remain in secondary school and for longer periods. New options should also be investigated to improve access of the less well off to tertiary education. To avoid massive redistribution in favour of higher income families, tuition fees should be raised while increasing the availability of grants for poor students and making student loans more readily available. Adult training, including in basic skills, plays a decisive role in increasing labour force mobility and is the key to removing unskilled adults from poverty traps. On-the-job and formal training programmes should therefore be continued, and publicly funded programmes should be made more cost efficient.

… meeting basic needs of the population…

The scale and acuteness of poverty in Mexico is an important factor influencing policy making. While in the longer run, poverty can be reduced as more people get more and better education and work opportunities, in the short-term targeted programmes, such as those currently in place, are necessary to ensure that the basic needs of the population are met. Since the mid 1990s, public spending on poverty alleviation has been increased, and targeted action for the most disadvantaged groups has widened in scope. The integral programme for education, health and nutrition (PROGRESA/O*portunidades*), which was extended to cover more than 4 million families (21 million people) has had a significant positive impact on the welfare of supported families. It should be continued. Progress made in specific actions for the poorest, to provide food assistance and improve access to basic education and health services is welcome. Attention should now focus on improving the quality of services provided. Beyond such targeted measures, the development strategy that is required to lift the population out of poverty on a lasting basis must aim at facilitating access to goods and financial markets, and fostering the expansion of productive jobs in the formal sector.

… and improving incentives to work in the formal labour market…

The main labour market problem that policy has to tackle is to increase the relative size of the formal sector while maintaining a high degree of real wage flexibility in a low inflation environment. Since there are few effective

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sanctions against working in the very large informal sector, it is the attractiveness of the formal sector that needs to be enhanced. Social contributions and benefits need to be reviewed. For employers, social security contributions, relatively high for the lower salary scales, create a possible obstacle to hiring low skilled-workers with a formal contract. Moreover, workers in the formal sector view the social benefits for which they qualify as being of low quality in relation to their contributions. A comprehensive reform of the housing fund (INFONAVIT), which gets a significant share of social contributions, should be considered to address this issue, as well as improvements in the quality of health services financed by the social security institute (IMSS).

… including implementing the recent labour market reform proposal, and going beyond it

If approved, the labour market reform proposal presented to Congress in December 2002 would represent a decisive step in making it more attractive for employers to hire in the formal sector. In particular, it would allow probationary periods and ease some of the regulations on permanent contracts. The proposal needs to be adopted and it should be followed by additional elements to improve incentives on both sides to raise formal employment. For example, overly strict regulation of atypical forms of employment and dismissals should be relaxed, while introducing an unemployment compensation scheme, carefully designed to avoid both creating unemployment traps and giving incentives to "surf" between the formal and informal sectors.

Much remains to be done to make the business environment friendlier…

To take full advantage of other growth-enhancing reforms, actions are desirable to improve the business environment. Measures have been rightly taken in recent years to reduce bureaucracy and accelerate firm creation and exit procedures, but their effects are not yet fully visible. Moreover, a serious and persistent problem is the poor rule of law, as illustrated by long judicial processes, poor enforcement of judicial decisions especially at the local level, weaknesses in the protection of intellectual property rights and corruption. Actions to address these issues are required at the Federal and the State levels.

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… and to increase competition in key sectors

Cross-country evidence shows that competition is an important driver of innovation and productivity levels and growth. Product market competition has been sharpened in recent years and trade liberalisation has made the Mexican economy more responsive to market forces. However, there are legal and regulatory obstacles to investment in certain key sectors, electricity in particular. In telecommunications, maintaining a level playing field between the incumbent and new entrants remains a big challenge. Given the importance of these sectors for future growth, reforms should not be delayed.

- In the *electricity sector*, a coherent reform strategy will have to be implemented (whether based on the current proposal or a new blueprint) so as to ensure reliable and growing generation and distribution capacity at low prices for Mexican industry and households over the medium to long term. Since the State does not have the resources required to finance investment on a sufficient scale, it is essential that the private sector be allowed a bigger role. Political obstacles to this need to be resolved quickly.
- In the *telecommunication sector*, despite the privatisation and deregulation of the 1990s, and the growth of the industry and the drop in tariffs in real terms, telecommunication density remains lower and tariffs higher than in most other OECD countries. To enhance competition in the market, the authority of COFETEL, the regulatory body, has to be strengthened.

More use of economic instruments is needed to ensure a sustainable, unpolluted water supply…

Water use in Mexico is on an unsustainable path. A large number of underground aquifers are being depleted. In many cases this is the result of over-extraction by the agricultural sector for irrigation use. Farmers are now paying a larger part of the operating costs of irrigation systems but the remaining direct subsidies need to be ended and the agricultural sector should be obliged to pay the market price for electricity used for pumping. As well, the irrigators should pay for extraction rights, rather than being exempted. In urban areas, the challenge is to convince local authorities, who control the utilities, to place water distribution on an economic footing. Raising water charges would also help

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finance the major investment programme that is needed both to improve the treatment of wastewater, more than three-quarters of which is discharged without treatment into rivers, and to expand the provision of potable water in rural areas and its quality in both rural and urban areas. Ensuring that both local authorities and industry pay the legislated penalties is also essential in order to lower pollution.

… and greater efforts to reduce the high level of air pollution are required

The need for stronger policies to reduce air pollution is clearly evident in Mexico. Urban areas in the country suffer from some of the highest pollution levels in the OECD area, bringing significant social costs. New emphasis is being given to the reduction of air pollution through national and local programmes, but further initiatives may be necessary. Diesel fuel quality needs to improve at a quicker pace using economic instruments to ensure adequate investment by the state-owned petroleum company (PEMEX). Vehicle inspection programmes need to be tightened. Registration taxes for diesel vehicles should be made dependent on emission characteristics so speeding the scrapping of old trucks and buses. There is also scope for reducing emissions from large stationary sources. A trading system might be difficult to introduce as there would be few participants in the market, given that the nearly all electricity plants and petrochemical plants are state-owned. However, a tax-based system could be considered. Finally, enhanced cross-border co-operation is essential, given the rapid economic development in areas that form common air sheds with the United States.

Emigration is an important economic and social phenomenon for Mexico

Emigration plays an important role in the increasing integration of the Mexican and US economies. The equivalent of one tenth of the Mexican-born population lives and, for the most part, works in the United States, and gross flows of temporary migrants across the US/Mexico border amount to several hundred thousand annually, well above the legal quota limits. Mexican emigrants to the United States are overwhelmingly low-skilled young adults looking for work and (usually) intending one day to return to Mexico. Traditionally, such migrants were seasonal agricultural workers, mostly males, coming disproportionately from a limited number of rural areas, and working in Mexican States just across the border. Although this pattern remains important, migrants increas-

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ingly eventually settle in the United States, more of them are women, they more often work in low-paid service jobs in the US, they come increasingly from urban areas, and large numbers of them now work in the north of the United States. High-skilled emigrants remain comparatively rare.

There are economic benefits, which could increase

Migration confers economic benefits on both Mexico and the United States, though the social costs in terms of disruption of family and social life are borne mainly by Mexicans. The US economy benefits from the presence of a sizeable and prime-age labour force willing to work for low wages. Although Mexican migrants work in low-paid jobs, often informally, their wages are typically six times higher in the United States than in Mexico, and migrants typically remit a quarter of their earnings to their families, as well as accumulating savings during their stay in the United States. Thanks to enhanced competition and better technology, the cost of remitting funds to Mexico has fallen in recent years, and the financial authorities of both countries should encourage this trend by requiring the publication of comparable information on service costs. The remittances themselves are mostly used immediately to finance consumption. Many recipient households are thereby lifted out of acute poverty, with its associated ill-health and social exclusion, and children can more easily attend school. The high proportion of low-skilled emigrants from specific regions and sectors partly reflects low levels of economic development there. The project to develop local financial networks under the aegis of BANSEFI should raise financial intermediation in rural and low-income populations and at the same time spread the benefits of remittances more widely across the community and over time.

Efforts to reach a migration agreement with the United States should be pursued

Mexican migration policy was traditionally one of benign neglect, emigration being seen as a safety valve to offset the chronic inability of the Mexican economy to create jobs fast enough for the expanding working-age population, and as an important source of dollar income. During the 1990s, the Mexican authorities started to take a more active stance, trying in particular to reach an agreement on several aspects of migration with the United States. After the 11 September 2001 events, negotiations were interrupted. Instead, border security has been tightened further, making

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illegal migration even more risky and costly. But the economic incentives remain huge and once across the border and into a job, migrant workers can live comparatively openly in many areas. Current stringency of border security may also encourage Mexican immigrants to remain in the United States for longer spells than they would otherwise have chosen. Efforts to reach a migration agreement should be pursued in the interests of both countries. It would strengthen the position of Mexican workers in the United States, thus possibly increasing remittances and reducing the deadweight costs of illegal border crossings.

In summary

Fiscal rectitude, progress towards macroeconomic stabilisation, and past structural reforms have been necessary and desirable, but have not yet been sufficient to raise potential growth to rates that would allow closing the gap in living standards with other OECD countries. Prolonged cyclical weakness, with no unambiguous signs yet of a vigorous upturn, has depressed private investment, which is also hampered by legal and regulatory obstacles in key sectors, electricity in particular. Mexico's catching-up is further hindered by low human capital accumulation. The administration has insufficiently solid and stable revenue to finance necessary social spending and public infrastructure investment on the required scale. Policies should therefore give priority to broadening the tax base and creating conditions – economic, financial and legal – in which a competitive private sector has the ability and incentives to invest more. It is also important to spend more productively in areas such as education; efforts there should concentrate on making the existing school system, and the teaching body, more effective, and on allocating more resources to the training of adults. Although the large informal sector provides a kind of safety valve for many of the low-skilled, the formal sector must become a more attractive place in the longer term in which to work and to employ. Emigration also provides a safety valve, and remittances lift many households out of acute poverty. A migration agreement between the United States and Mexico would bring benefits to both. Levels of water and air pollution are unacceptably high in Mexican urban areas, and though policies are addressing this, the (implicit or explicit) pricing of natural resources and of polluting activities is far from optimal.

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Overall, Mexico needs to move ahead with comprehensive structural reforms, including most immediately approval of the tax, electricity and labour reforms, so as to fully release the country's growth potential and provide resources to deal with important issues of human capital and poverty relief.

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I. Fostering faster growth

Introduction

Even abstracting from recent cyclical weakness, Mexico's underlying growth performance since the restoration of macroeconomic stability after the 1994-95 crisis has been insufficient to narrow the gap in living standards with other OECD countries.[1] Structural reforms in product and financial markets have been implemented since the late 1980s and NAFTA has opened the rest of the North American market to Mexican exports over the past decade and introduced keener competition in Mexico itself. The twin policies of inflation targeting and rigorous adherence to strict fiscal targets have been rewarded by falling interest rates and insulation from financial crises in other Latin American countries. Yet, at about 4½ per cent, underlying growth remained sluggish over 1996-2000, and an extended period of cyclical slack thereafter with concomitant low investment may have lowered potential growth further. This is all the more serious in the case of a country like Mexico, where a significant proportion of households live in acute poverty, and the provision of basic health and education services is not yet assured for many others. This chapter focuses on impediments to faster growth, and identifies policies to overcome them.

The first part of the Chapter reviews Mexico's growth performance in 1990-2000. The second part considers more recent macroeconomic developments and forces acting on short-term prospects. A third part reviews medium-term prospects to highlight potential gains from reform.

Why has Mexico's growth not been stronger over the past decade?

Mexico's growth performance

A series of external and domestic shocks in the 1980s brought to an end a prolonged period of rapid expansion in the two previous decades. Real output stagnated until nearly the end of the 1980s. Net job creation in the formal sector was not strong enough to absorb a fast growing labour supply (2½ per cent per year on average), so that workers either remained in unskilled jobs in agriculture or drifted into the informal urban sector. The recovery phase that started in the

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late 1980s entailed major external financing, created large imbalances, and led to the 1995 financial crisis and substantial currency depreciation.[2] Over the more recent expansion of 1996-2000, Mexico's economic performance had clearly improved, auguring well for the future (Figure 1). GDP growth averaged almost 5 per cent per year while macroeconomic conditions were far more stable: inflation came down, and the current account and the public sector deficits fell to relatively low levels as a percentage of GDP. This expansionary phase, however, was cut short by a downturn largely originating in the United States, Mexico's main trading partner, and the recovery has yet to become firmly established.

Mexico's GDP per capita, the second lowest in the OECD, is equivalent to about one quarter of that of the United States and 36 per cent of the OECD average (measured at purchasing power parity exchange rate PPPs). Over the 1990s, it increased by only 1½ per cent per year, preventing catching up with more advanced OECD countries. In terms of GNP, the growth performance of Mexico is not very different (1.6 per cent).[3] Over the same period, GDP per capita rose by 3 per cent or more in Greece, Portugal and Korea, three other OECD countries with comparatively low income levels.[4] A better measure of economic performance is the ability of the country to support private consumption of its residents. In a country like Mexico, where economic growth has been concentrated in exports with a significant part of the associated value added going to foreign capital, and domestic producers, mostly small- and medium-sized enterprises (SMEs), generally less productive than their foreign counterparts, the growth performance of private consumption would be expected to be significantly weaker than GDP (as in Ireland, for instance). But in Mexico, trends are very similar, perhaps as a result of the increasing scale of migrants remittances. In terms of levels, per capita private consumption measured at PPPs is in fact slightly higher relative to the OECD average than is its per capita GDP (Figure 2).

Potential output growth is estimated at a little over 3 per cent over the past decade (which has seen a major crisis). Over the 1996 to 2000 period, "trend" growth of GDP accelerated to an average 4.6 per cent, but the recent slowdown brought trend growth back down below 4 per cent in 2001-02. Gross fixed investment has not yet recovered to its 2001 level, further reducing potential growth.

The positive contribution from relatively high employment growth…

Growth of output depends on the growth of the quantity and quality of inputs, and changes in the efficiency with which they work together. Quantitatively, growth in labour input and labour productivity account for most of GDP growth in OECD countries (Table 1). Some OECD countries with high income per capita growth have recorded stable or rising employment and large productivity gains, for instance the United States, Canada, Australia and Ireland. Other countries, many of them European, also recorded high productivity gains but these were

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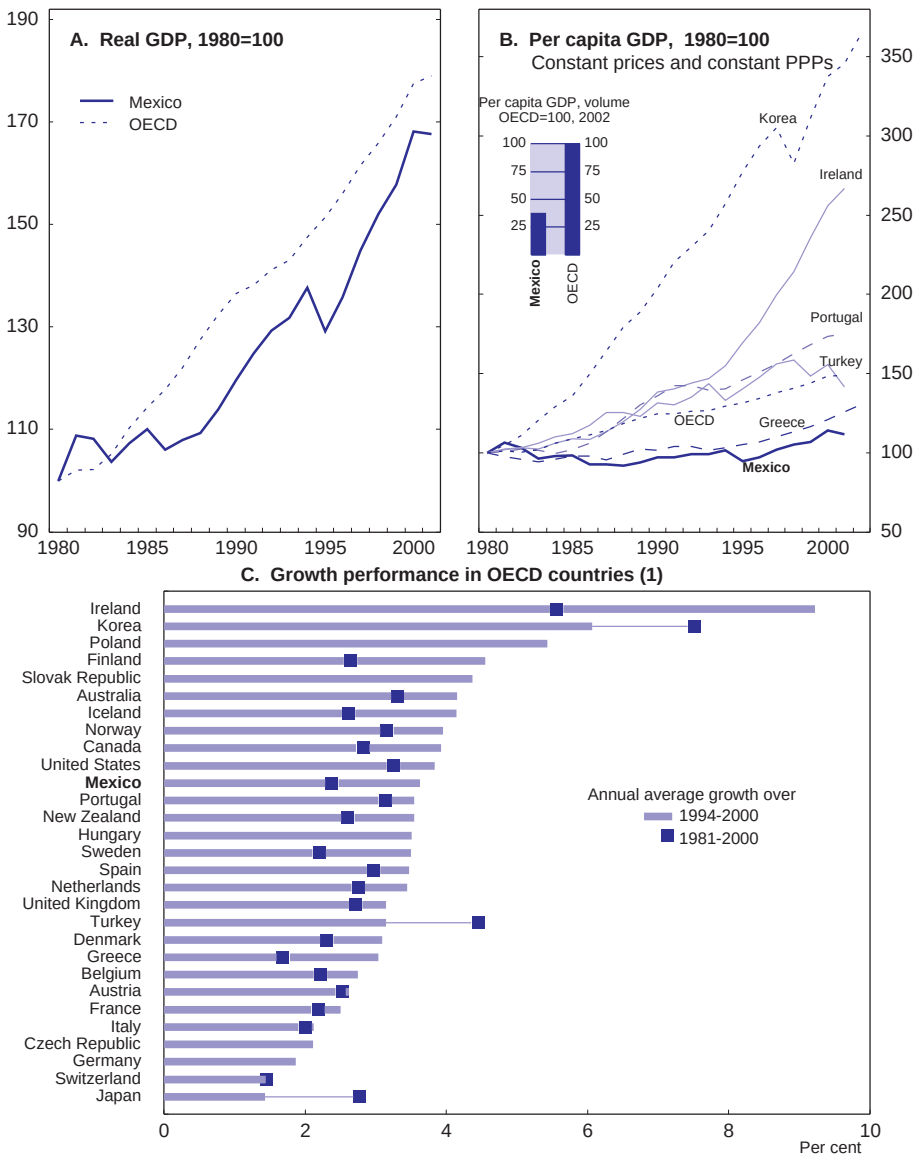
Figure 1. **Mexico's output and growth performance in comparison**
Measured by real GDP



1. The series for the Czech Republic, Germany (due to unification), Hungary and Poland are incomplete before 1991.
Source: OECD, *Main Economic Indicators*; OECD, *National Accounts*.

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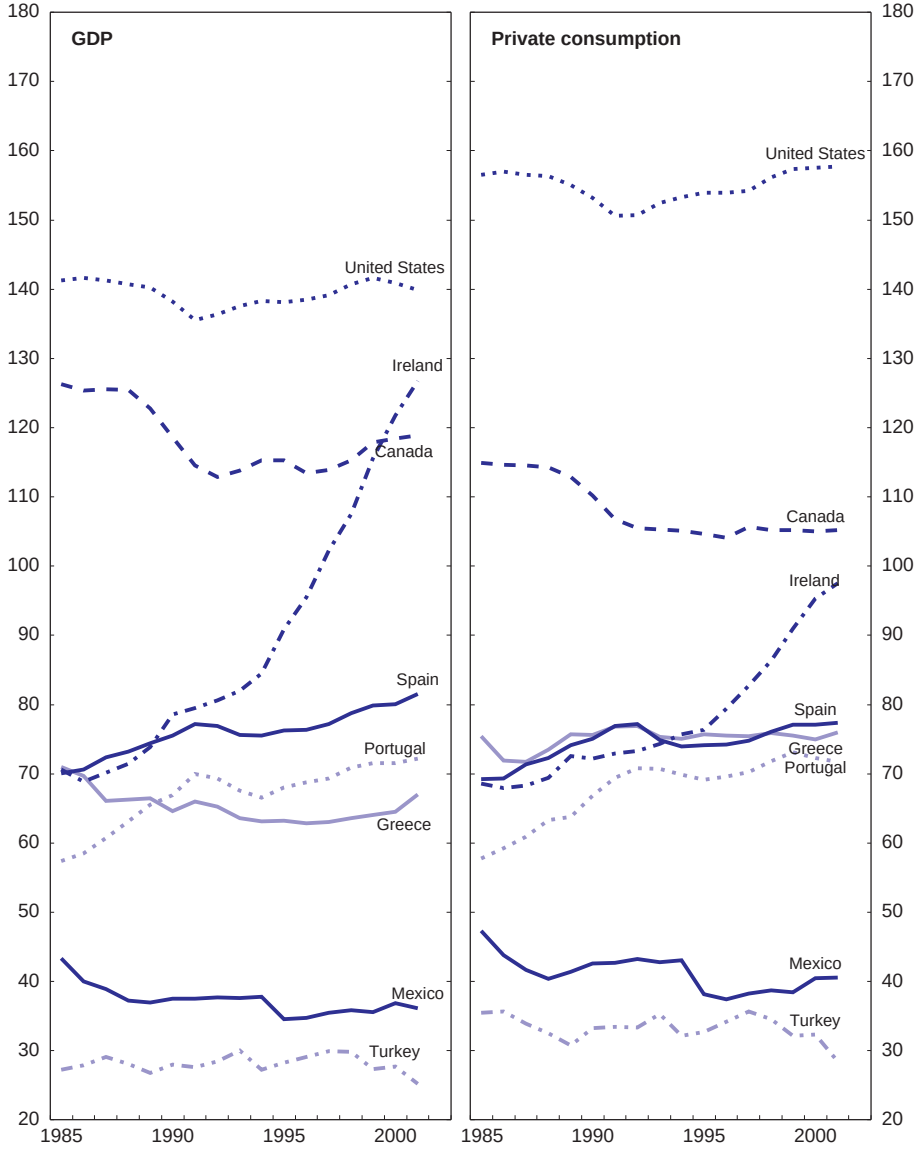
Figure 2. **GDP and private consumption levels in selected OECD countries**
Per capita measured in purchasing power parity exchange rates
OECD = 100



Source: OECD, *National Accounts*.

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Table 1. **Sources of growth in real GDP per capita: selected OECD countries**[1]

Average of annual changes in per cent

	Mexico[2]		Canada		United States		Greece		Ireland		Portugal		Spain		Korea	
	1994/1987	2000/1994	1994/1987	2000/1994	1994/1987	2000/1994	1994/1987	2000/1994	1994/1987	2000/1994	1994/1987	2000/1994	1994/1987	2000/1994	1994/1987	2000/1994
Real GDP	3.5	3.5	2.0	3.8	2.7	3.8	1.8	3.2	4.7	9.8	3.2	4.0	2.6	3.7	7.7	5.7
Population	2.0	1.6	1.3	1.0	1.0	0.9	0.6	0.3	0.2	0.9	–0.2	0.3	0.2	0.3	1.0	0.9
Real GDP per capita	1.5	1.8	0.6	2.8	1.6	2.9	1.1	2.9	4.6	8.8	3.4	3.7	2.4	3.3	6.6	4.8
Labour input	1.0	1.7	–0.4	1.2	0.3	0.9	0.1	0.4	1.2	4.6	1.5	1.0	0.4	3.6	1.8	0.1
Contribution from:																
Working-age population (share of total population)	0.0	0.9	–0.2	0.2	0.0	0.4	0.3	0.0	0.7	0.9	0.6	0.2	0.5	0.2	0.7	0.3
Labour force participation (share of working-age population)	1.0	0.5	–0.0	0.3	0.2	0.1	0.2	0.6	0.1	1.7	0.8	0.3	0.3	1.8	1.1	0.1
Employment (share of labour force)	0.0	0.3	–0.2	0.6	0.0	0.4	–0.3	–0.3	0.3	1.9	0.1	0.5	–0.4	1.5	0.1	–0.3
Labour productivity	0.5	0.1	1.0	1.6	1.3	2.0	1.0	2.6	3.4	4.0	1.9	2.6	2.1	–0.2	4.7	4.6

1. Growth in real GDP per capita is decomposed into growth in labour input variables and growth in labour productivity. Growth in labour input is derived from following the identity ET/POP = POPT/POP*LF/POPT (participation rate) * ET/LF (employment rate); while labour productivity is defined as: GDPV/ET; where ET = total employment; POP = total population; POPT = population of working age; LF = labour force; GDPV = real GDP.
2. For Mexico, employment data refer to broader definition (National Employment Survey).

Source: OECD, Analytical Database.

Table 2. **Determinants of labour force growth**
Percentage changes, annual rates

	Labour force		Working-age population	
	1980s	1990s	1980s	1990s
Aged 12 years and over				
Mexico[1]	**4.1**	**2.8**	**2.7**	**2.4**
Aged 15-64 years				
Mexico[1]	**. .**	**2.9**	**. .**	**2.5**
United States	1.5	1.4	0.9	1.0
Canada	1.7	1.2	1.2	1.1
Korea	2.6	1.6	2.3	1.2
Greece	0.9	1.3	0.9	0.5
Ireland	0.4	2.9	0.6	1.7
Portugal	1.5	1.0	0.5	0.6
Spain	1.4	1.4	0.9	0.5
Turkey	1.8	0.7	3.1	2.5

1. The 1990s data are calculated from 1991 to 2000.
Source: OECD, *Labour Force Statistics*; Mexican Ministry of Labour.

associated with declining employment (or total hours worked), as low-skilled labour was shed. Mexico stands in sharp contrast to both groups. Employment generation was strong in the 1990s, reflecting increases in: *i*) working-age population, because of high birthrates earlier; *ii*) labour force participation, because of rising female participation; and *iii*) employment as a share of labour force. The labour force increased by close to 3 per cent per year over the past decade (equivalent to 1.5 million new entrants per year)(Table 2). In addition, the quality of labour supply has also improved [measured by educational attainment or by returns on schooling (Bosworth, 1998)].

... is largely offset by weak gains in productivity

In most OECD countries, labour productivity gains account for at least half of per capita GDP growth over the long term. In the case of Mexico, given the reduction in import barriers and the increase in trade exposure over the past 15 years and the stronger competitive pressures that should have resulted, substantial gains in productivity would be expected. Furthermore, the share of the low-productivity agricultural sector in total employment was reduced from more than a quarter to 18 per cent over a decade. This shift of labour to the rest of the economy would also be expected to yield overall productivity gains, as in other countries. Reliable estimates of multi-factor productivity growth (MFP) are not available for Mexico, and a number of problems plague the measure of labour productivity. Several indicators need to be used to evaluate developments.[5] Based

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on the broad employment aggregate (national employment survey), measures of economy-wide productivity suggest quasi stagnation in 1990-2000 (following a decline over the previous decade).[6]

The agricultural sector, including livestock, forestry, fisheries and hunting, accounts for nearly one fifth of employment; and productivity gains have remained weak overall, ten years after NAFTA. Some parts of the sector have been exposed to increasing competitive pressures from trade liberalization, and they have in fact seen substantial increases in productivity (those producing exportable goods mostly). However, the sector's overall adjustment to the trade liberalisation shock did not take place on the originally expected scale: a large segment made of small production units in remote rural areas, face high transportation and other transaction costs, and lack access to credit. They have continued to produce for own consumption, using traditional technologies, and they have remained isolated from market forces.[7] As a consequence, the sector as a whole has continued to record weak productivity gains.

In the non-farm sector, some gains in labour productivity were recorded during the second half of the 1990s (based on national employment survey data), but not large enough to offset the decline in earlier years (Figure 3, Panel A). The discouraging picture of productivity over the 1990s reflects an abundant and flexible supply of unskilled labour, much of it coming from the shrinking agricultural sector, which has led to buoyant employment growth in low-productivity sectors, such as construction or retail trade, without generating wage pressures. Indeed, real wages have declined almost continuously since the early 1980s, which has been conducive to high-labour intensity of production in most sectors of activity.[8] Until the late 1980s, there was strong job creation in the public sector, but this was no longer the case during the 1990s. Emigration has been an important outlet (see Chapter IV). But excess labour was mostly absorbed by the informal sector in urban areas, largely made up of self-employed occupied in very small units (of family size), in often unstable jobs, but also in undeclared jobs in registered establishments (Box 1). The quasi stagnation of productivity in the non-farm economy thus reflects the growing concentration of employment in such activities, exacerbated by the still-low average skill level of the employed population and very low levels of training, making difficult the diffusion of production techniques requiring high degrees of literacy and/or numeracy.

Based on a narrower measure of employment in the formal sector (that includes essentially wage earners with a contract and self-employed or employers in registered establishments, as measured by the national accounts), productivity shows a slightly positive trend, with less than 1 per cent annual growth since 1993 (Figure 3, Panel B).[9] Manufacturing has recorded substantial productivity gains, especially in the machinery and equipment branch, which includes capital- and technology-intensive activities (automobiles for instance), where the strongest

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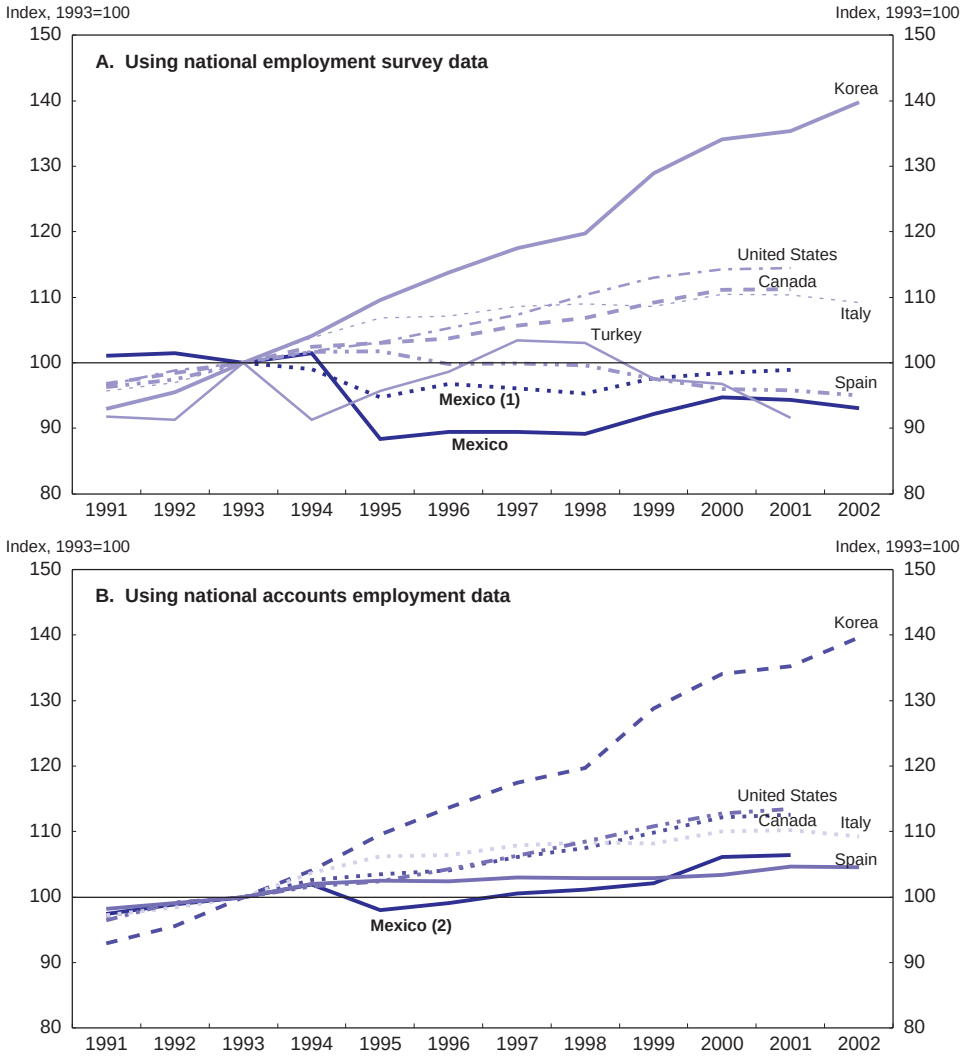
Figure 3. **Productivity in the non-farm sector**

Value added divided by employment



1. Productivity in the formal sector, where employment in the formal sector is based on survey data (broad definition) less INEGI's definition of the "employment in the non-structured sector". GDP in the formal sector is total GDP less informal sector GDP in the INEGI National Accounts ("satellite accounts").
2. Employment refers to remunerated workers with contracts (jobs occupied).

Source: OECD; INEGI, *Cuenta Satélite del Subsector Informal de los Hogares*; STPS-INEGI, Quarterly National Employment Survey (ENET).

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Box 1. The informal labour market

The informal labour market is very heterogeneous. One of the reasons for its development has been the absence of income support for those who cannot find a job in the formal sector. This has meant that the gap between labour supply and demand has been reflected in informal activities, rather than open unemployment. Domestic services, retail trade, cleaning services and to an extent construction have been typical activities where informality has developed. The statistical office now produces a strictly-defined measure of the informal sector, which only considers employment and output of "non-structured sectors", such as street-vendors or micro-enterprises (repair shops…). These informal units rarely comply with the various registration obligations. They very often rely on family ties for labour contracting. There are other areas where informal labour is found. For instance, there are people working in registered establishments and drawing steady income, but having no contracts and no social security. There are also unpaid workers employed by families. There was already a large informal labour market at the beginning of the 1990s. The 1990s have witnessed an explosion of the labour supply, while adjustments in the productive sector and the liberalisation induced sizeable labour shifts across sectors, and real wages declined. Over that period, net job creation in wage-earning positions increased less than the number of self employed, and real wages fell during most of the decade.[1]

Between 1991 and 2002, total employment rose by roughly 10 million, an annual rate of 2½ per cent, considerably faster than the OECD average. With the agricultural sector losing about 1 million jobs over the 11-year period, a total of 11 million jobs were created in the non-farm economy. Six million jobs have been created in that sector since the 1995 peso crisis. Of those, about 4 million were created in the formal sector, the majority of jobs being wage earners who are registered with a social security institution, the rest being employers in formal establishments and some self employed. Over the same sub-period, employment in the informal sector ("non-structured" sectors) expanded by just below 2 million people, reaching 10½ million people in 2002.[2] The share of the (non-farm) informal sector thus defined has remained broadly stable, at about a quarter of total employment since 1995. Even when employment and output of the "non structured" sectors are excluded from the measure of productivity, gains are weak.

1. According to the 1996 Survey on micro-enterprises, characterized by a high incidence of informal work, 25 per cent of the workers starting to work on own account had left their previous wage-earning position because wages were too low, 27 per cent because of dismissal or closing down of enterprise (computed by Norma Samaniego, 2000).
2. Estimates by INEGI, based on the quarterly national employed survey, in accordance with ILO methodology *c.f.* "Informal employment, a conceptual framework developed by ILO", Bureau of Statistics, ILO, 2002. Measuring the informal sector is clearly difficult, since these activities are generally performed without being officially registered. Although informal retail trade might include trade of stolen goods, the concept of informal activities as understood in Mexico, does not include illegal activities, such as drug dealing, etc.

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Box 1. **The informal labour market** (*cont.*)

However, other measures are also relevant. For instance, if informal labour is considered in a wider sense, to include also people employed by enterprises or households, but having no work contract and no payment, about 7½ million more persons would need to be added into informal employment (data for 2000), for a total of over 17 million, almost half of total employment (and 54 per cent of non-farm employment). By 2002, informal labour is estimated to have reached 18.5 million persons, its share in employment having also slightly risen.

The large size and increasing trend of the informal labour market has a high social cost: it is associated with deterioration of human capital. It calls for policies which improve the functioning of labour markets, so as to encourage job creation in the formal sector, as well as other policies that are more directly linked with the strategy for poverty alleviation (see Annex I.C).

productivity gains have been recorded since the onset of NAFTA.[10] At the same time, output and exports both grew at spectacular rates from 1993 to 2000.

Productive plants in the sector achieved gains in efficiency thanks to capital deepening, better quality of labour, better technology, and better management. However, in the late 1990s, even before the economic slowdown, productivity gains were weakening (Figure 4).

The above analysis suggests that the two main forces driving productivity growth in other OECD countries over the medium-term have operated only moderately in Mexico over the past 10 to 15 years. First the reallocation of resources away from low productivity sectors (mostly agriculture) to high productivity sectors (mostly manufacturing and high-skilled services) did not occur on a large scale. Instead, most of the movement has been between low-productivity sectors. Second, productivity growth within sectors has been relatively slow, except for a few specific cases. This "stickiness" could reflect a number of factors. Insufficient firm dynamism, due to low exit rates of low-productive firms and low entry rates of innovative and potentially high-productive firms, might have contributed to weak productivity growth. Heavy regulations and strict and costly employment protection (dismissal rules in particular) can discourage firms from taking the risk of expanding, while poorly designed bankruptcy rules (until the recent reforms) probably delayed the demise of poorly performing enterprises.[11] Recurrent problems in the banking sector until the very late 1990s have meant that there was essentially no market-based financing available for a large part of the productive sector, for start-ups in particular and for small and medium-size enterprises.[12] The credit crunch is likely to have impeded investment by SMEs

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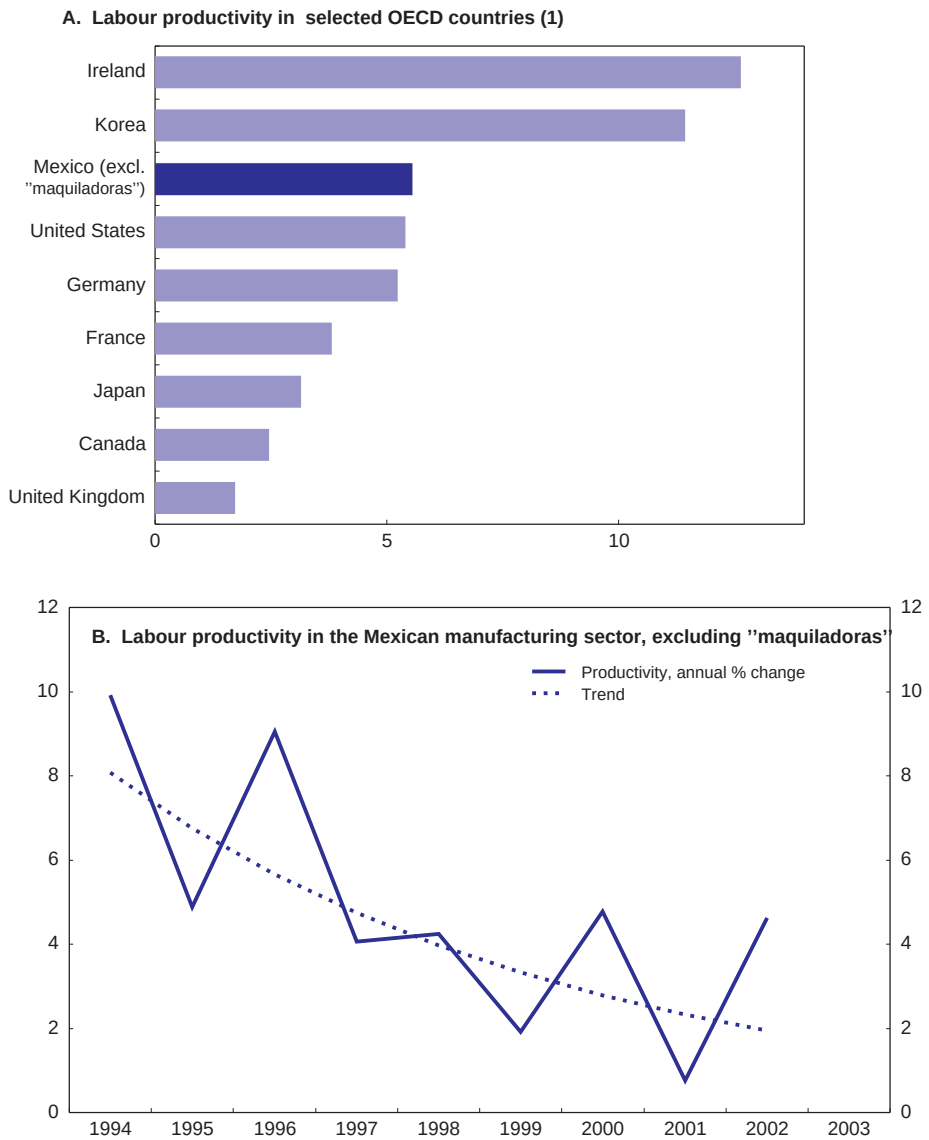
Figure 4. **Labour productivity in the manufacturing sector**



1. Over 1993-2002. Output per employees, except for Canada, Japan and the United States where output per hour worked is shown.

Source: OECD, *Main Economic Indicators*; INEGI.

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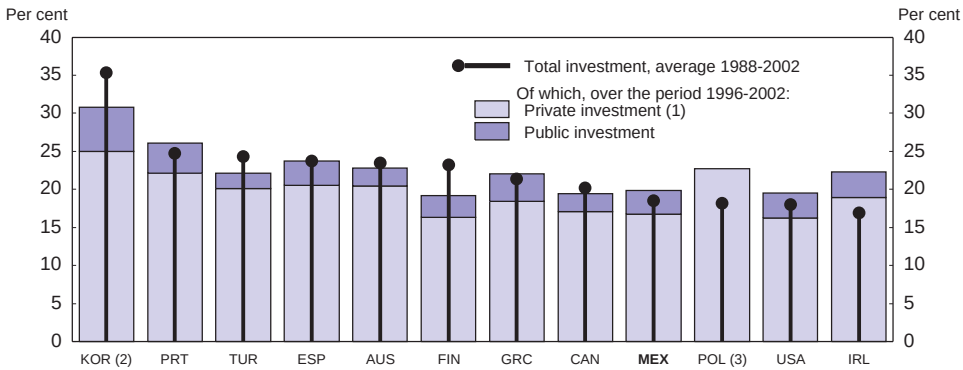
Figure 5. **Fixed investment ratios and their main components**
Per cent of GDP, in nominal terms



1. Includes state-owned enterprises (PEMEX, CFE, in particular, for Mexico).
2. Data available till 2000 only.
3. Data before 1990 and breakdown by component are not available.

Source: OECD, Analytical Data Base.

to enhance product quality, modernize and update technology. For many sectors, lack of capital, including a low use of investment in information and communication technologies (ICT), is probably one of the main factors impeding productivity gains.[13]

There is some evidence of inadequate levels and quality of investment, although data limitations constrain the scope for research. The investment ratio averaged 20 per cent in the latest expansion phase, 1996-2001, including residential construction and investment by large state-owned companies, PEMEX and CFE. This ratio is not high by OECD standards (Figure 5). Following the extensive privatisation operations of the early 1990s, the public sector share of investment declined, part of it offset by investment of the privatised enterprises. Infrastructure spending was cut for budget consolidation purposes following the 1995 crisis. Furthermore, the "quality" of some public investment projects has been adversely affected by the fact that their implementation was disrupted by changes in oil revenue. The low level and lack of predictability of public spending have impeded medium-term planning[14] and might have created shortages in various areas – communication, transportation, electricity, sanitation, water.[15] Due to regulatory and legal restrictions, it was not always possible for private investment to fill the gap in some of these key sectors, even

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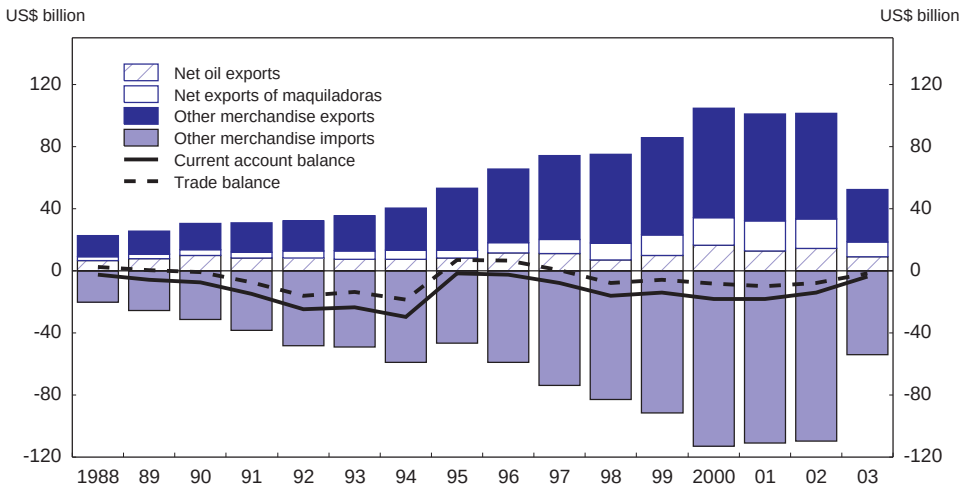
though, the use of PIDIREGAS and other investment projects involving private financing has cushioned shortages of public investment in the energy sector (Chapter II, Box 6). Finally, the investment boom of the second half of the 1990s was largely focused on consumption-related facilities (shopping malls, fast-food outlets), rather than industrial capacity.[16]

On the demand side, growth has been export-led

In Mexico, exports now amount to 30 per cent of GDP, up from 17 per cent in the early 1990s, and the country has been relatively successful in raising its market share in world total exports to 2.6 per cent in 2000-01 from 1.4 per cent 15 years earlier. The growing predominance of manufacturing goods in exports has lowered the vulnerability of export revenue to changes in oil prices. The increased openness of the economy has been accompanied by the development of intra-industry and intra-firm trade.[17] Although there is a close synchronisation of the industrial sector with the US, the current account balance is less vulnerable to the cycle in the partner country than in the past. When exports slow down, so do imports (Figure 6). The *maquiladoras* exporting sector (in-bond industries) illustrates the growing interdependence between Mexico and the United States (Box 2). Differences in the recent performance of these industries exemplify the adjustment process of the manufacturing sector and the importance of competitiveness.

Figure 6. **Foreign trade and the current account**



Note: For 2003, data refer to the first half-year.
Source: Bank of Mexico.

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Box 2. *Maquiladoras* exporting industries

Maquiladoras (in-bond industries) are assembly plants, working in virtually tax free zones for re-exporting. First located close to the northern border, they have also spread to central and southern regions. The sector has been the most dynamic sector in Mexico since the early 1980s. The fact that it benefited from a highly deregulated legal framework in comparison to the rest of the Mexican economy has been a key factor in its development.[1]

In the post NAFTA period, in particular, its expansion has been spectacular, with exports growing at an annual 20 per cent and job creation amounting to nearly 110 000 jobs per year. By 2000, the maquiladoras sector employed 1.3 million workers, about 30 per cent of manufacturing employment (based on permanent insured employees). And it accounted for almost half of total manufacturing exports. But activity slowed in 2001 and prospects for growth over the near term may be weaker than in recent years. In addition to the impact from the US slowdown with which they are closely synchronized, there is evidence that some maquiladoras suffer from a loss of cost-price competitiveness, as productivity gains have fallen short of real wage increases. This affects some sectors more than others: in labour intensive activities (*e.g.* clothing, electrical and electronic material) the decline in activity is likely to be durable, with several multinational companies moving their projects to China or other competitive locations in Asia and Latin America. But the recovery is already manifest in capital-intensive sectors. These have received large FDI flows, which would be expected to be accompanied by technological transfers and better organisational processes.[2]

1. See J. Mattar *et al.* (2002). The authors focus their analysis on gross fixed capital formation in manufacturing, trends in FDI and the activities of maquiladoras.
2. The Ministry of Economy reports that in 2001 and early 2002, more than 20 firms in total (mostly in electronics and textiles, clothing) have moved from Mexico to more competitive locations (July 2002 information).

The manufacturing-for-exports phenomenon is important in other respects. Because the factories there are essentially part of the US manufacturing sector, the large increases in "foreign trade" in manufactured goods do not in practice impinge as much on the Mexican economy as the trade-growth figures suggest. The lowering of tariff and non-tariff barriers has undoubtedly had an impact on the strength of competition on the domestic economy, but much of the economy remains comparatively closed, because of over-regulation, and high transport and communication costs. The failure of productivity to pick up in a sustained fashion, 10 years after entry into NAFTA, suggests deep-seated

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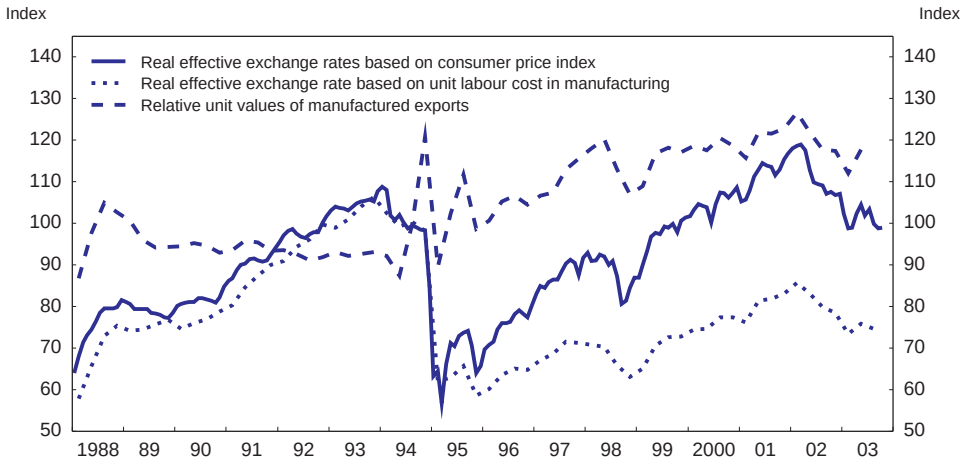
Figure 7. **Real effective exchange rates**
Index 1994 = 100[1]



1. Indices in US$ terms.
Source: OECD.

problems of adaptation and lack of competitiveness outside the narrowly-based manufacturing export sector, which are only slowly being resolved.

In the longer term, loss of competitiveness can lead to a deterioration in export performance and reduced FDI. The appreciation of the real effective exchange rate, measured by relative export prices and consumer prices, which have now reached levels above those of the early 1990s, points to a steady erosion of Mexico's price competitiveness (Figure 7). However, the slower rise in relative unit labour costs suggests that Mexican exporters have been able to preserve their unit profits.[18] Other factors besides price-cost competitiveness have been at work. On the one hand, the more diversified manufacturing export base, with a trade specialization evolving towards medium and higher technology manufactures has helped Mexico's export performance (Table 3). On the other hand, the competitiveness of a country, hence its attractiveness for investors, is assessed not only in terms of direct labour costs, but also in terms of tax incentives, flexibility in working time and other indirect cost of labour, user cost of telecommunications and energy, efficiency of distribution including transport infrastructure, degree of legal certainty, access to financing and restrictions on foreign investment. On this account, Mexico does not fare well (see the aggregate competitiveness indicator in Chapter III).

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Table 3. **Change in the trade specialisation of Mexico**

SITC-3 code	Main comparative advantages	RCA[1]	Share of exports[2]	RCA[1]	Share of exports[2]
		2001		1990	
78	Road vehicles	7.37	17.69	8.58	11.51
33	Petroleum, petroleum products and related materials	7.24	7.38	33.01	35.87
76	Telecommunication and sound recording apparatus	6.54	12.15	–4.50	0.19
75	Office machines and automatic data processing machines	3.50	8.39	–0.42	2.09
84	Articles of apparel and clothing accessories	2.97	5.09	–0.88	0.33
05	Vegetables and fruits	1.55	2.16	4.11	5.44
82	Furniture and parts thereof	1.37	2.09	–0.27	0.17
11	Beverages	0.88	1.05	0.64	1.01
81	Prefabricated buildings, sanitary, heating and lighting	0.50	0.69	0.04	0.18
03	Fish, crustaceans, mollusks and preparations thereof	0.31	0.40	1.13	1.26
79	Other transport equipment	0.24	0.57	–1.21	0.64
66	Non-metallic mineral manufactures, n.e.s.	0.17	1.13	0.71	1.66
00	Live animals other than animals of division 03	0.13	0.26	0.97	1.30
07	Coffee, tea, cocoa, spices and manuf. thereof	0.08	0.25	1.23	1.50
06	Sugar, sugar preparations and honey	0.07	0.20	–1.68	0.23
85	Footwear	0.06	0.23	0.00	0.31

1. RCA: Revealed comparative advantage indicator (Xi/X-Mi/M)*100, where Xi(Mi) and X(M) represent respectively Mexican exports (imports) of products and total Mexican exports (imports).
2. As a percentage of total exports in 1990 and 2001, respectively.

n.e.s. = not elsewhere specified.

Source: OECD.

Recent economic developments and prospects

Recent trends in perspective

Recent economic developments underscore some of the weaknesses that have hindered Mexico's growth performance in the expansion phase of the cycle. The global slowdown starting at the end of 2000 has had a significant impact on activity in Mexico. Mexican exports and investment declined in 2001 and GDP contracted for several consecutive quarters, in close synchronization with activity in the United States (Figure 8). By contrast, Canada at the time, whose trade is even more closely linked with the United States, recorded positive, albeit small, GDP growth rates. The lack of domestic sources of dynamism in Mexico can be explained at least partly by the underlying drivers of the recent cycle, described above: first, the insufficient level and quality of investment in the second half of the 1990s; second, the credit squeeze in the aftermath of the peso crisis, constraining more severely smaller-size enterprises not linked to the export sector; third, high real interest rates until recently, weighing on construction and other

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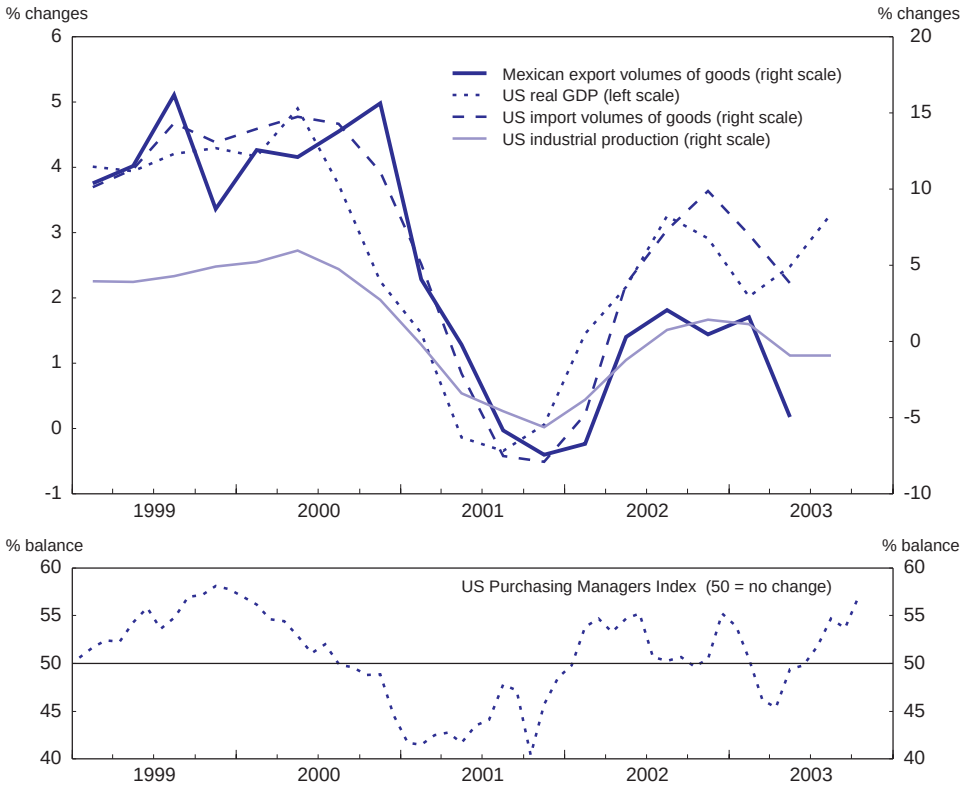
Figure 8. **Activity in the United States and Mexican exports**
Year-on-year change



Source: OECD.

private sector investment; finally, the fragile employment situation, with more than one third of the jobs created in informal activities, where income is typically low and unstable.

Despite its lack of domestic resilience, however, the Mexican economy has fared better in the current cyclical downturn than in previous ones. The downturn has been relatively mild and it is the first recession that was not accompanied by, or precipitated by, a currency or financial crisis. Financial crises in the rest of Latin America did not spread to Mexico. This is in large part due to sound macroeconomic policies, so that the economy has been able to ride through such a slowdown without the added burden of having to deal with underlying macroeconomic

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imbalances or confidence crisis. The improved macroeconomic fundamentals have also been reflected in Mexico's external balance. In the context of economic weakness, the current account deficit came down to 2.2 per cent of GDP in 2002, with estimates for 2003 below that level. Capital inflows have continued to be dominated by FDI, well in excess of the current account deficit.

Short-term prospects and risks

Mexico's real GDP turned up in the middle of 2002, in pace with the United States; but the Mexican recovery has been hesitant. Evidence from recent developments suggests that export growth will continue to be the main driver of faster growth over the short-term. Because of the high degree of intra-industry and intra-firm trade between Mexico and the United States, Mexican exports are dependent on US industrial production, which has remained sluggish during the upturn. The leading indicator of US industrial production became clearly positive only in the middle of 2003, so that the prospect of a firm pick-up in Mexican output growth has been postponed to the latter part of the year (see Annex I.C).[19] GDP growth is expected to gain momentum during 2004. But, with no stimulus expected from macroeconomic policies and given the standstill on the reform agenda, the recovery in investment is likely to be slow, so that both in 2004

Table 4. **Short-term outlook**

	2000	2001	2002	2003	2004	2005
	Current prices billion pesos	Current prices Percentage changes, volume (1993 prices)				
Private consumption	3 683.7	2.7	1.2	2.8	3.6	4.4
Government consumption	609.7	–1.2	–1.3	2.1	2.3	2.2
Gross fixed capital formation	1 174.1	–5.8	–1.3	–1.2	4.8	6.3
Final domestic demand	5 467.6	0.6	0.5	2.0	3.7	4.6
Stockbuilding[1]	136.4	–0.2	0.5	–1.0	0.3	0.0
Total domestic demand	5 603.9	0.4	1.0	1.0	3.9	4.5
Exports of goods and services	1 704.1	–3.6	1.4	–0.3	6.5	7.8
Imports of goods and services	1 810.6	–1.5	1.6	–1.7	7.2	8.4
Net exports	–106.5	–0.7	–0.1	0.5	–0.4	–0.5
GDP at market prices	5 497.4	–0.3	0.9	1.5	3.6	4.2
GDP deflator	_	6.4	4.6	5.2	3.3	3.3
Memorandum items						
Private consumption deflator	_	7.2	4.8	4.6	3.4	3.1
Current account balance[2]	_	–2.9	–2.2	–1.9	–2.5	–3.0

1. Contributions to changes in real GDP (percentage of real GDP in previous year), actual amount in the first column.
2. As a percentage of GDP.

Source: OECD. Estimates and projections as prepared for *Economic Outlook* No. 74. Based on data available on 3 November 2003.

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and 2005 GDP growth could be more subdued than in the late 1990s. The current account deficit is expected to widen from its 2003 record low, reaching 3 per cent of GDP by 2005, but continuing to be mostly financed by foreign direct investment.

The main risks to the outlook concern external developments, including world oil prices, financial markets and, above all, the speed of economic activity in the United States and its impact on Mexico's exports. However, the main uncertainty on Mexico's growth potential is a domestic one: it relates to the country's structural agenda, where discussions have not always led to proposals, nor proposals to approval, nor approval to implementation, despite a political consensus that reforms are necessary. There is a wide consensus in the private sector as well that the lack of structural reform is one of the main factors holding up economic activity in the coming months. Approval of the labour market, tax and electricity reforms in particular would boost business confidence. It would create an environment more supportive of private investment, including foreign direct investment, and the cyclical recovery could turn into a sustained brisk growth era.

The challenge is to boost medium-term growth

Medium-term growth prospects are unsatisfactory…

As noted earlier, raising the potential growth rate by a significant amount (if possible by 2-3 points) and keeping actual growth close to this higher potential rate over a prolonged period (decades) is the greatest policy challenge facing Mexico. Although very ambitious, it is by no means impossible, as the experience of countries as varied as Ireland and Korea proves. The particularity of Mexico is both the conspicuous deficits in all types of capital together with management expertise and technological know-how, and the overriding social need to raise not just living standards, but also nutritional and basic health standards of a substantial proportion of the population. Despite the policy efforts made to increase spending on social development and widen access to basic health and education services, the problem of poverty remains widespread and especially acute in remote rural areas. In 2002, about half of the Mexican population lived in poverty and one out of every five Mexicans was in extreme poverty concentrated in rural areas, which has important bearing on policy priorities (Box 3). The growth strategy for Mexico (outlined in Chapter III) has to find ways for benefits to be shared more broadly across the population and deal with the poverty problem.

The government has outlined a medium-term policy strategy (National Programme for Financing Development, PRONAFIDE for 2001-06). This can contribute to foster domestic and foreign investors' confidence. The government's macroeconomic policy objective is to secure a stable environment, with sound public finances, low inflation and hence low real and nominal interest rates. Furthermore, the financial system should be able to fulfill its role of financing the private sector.

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Box 3. Poverty alleviation

Poverty alleviation is one of the greatest policy challenges for Mexico. In the aftermath of the Mexican crisis of 1994-95, real per capita GDP fell sharply; employment in the formal sector declined as well as real wages. This led to a drastic increase in poverty at a national level, reversing most achievements of poverty alleviation policies of the late 1980s and early 1990s. However, since 1996, there has been a notable turnaround, and the share of the population in poverty has declined. Mexico has made a solid progress in combating poverty, through more sustained income growth and macroeconomic stability, leaving behind some of the most devastating effects of the 1994-95 crisis. Despite this progress, the poverty levels in Mexico are still high. In 2002, one of every five Mexicans was living in acute poverty, defined as having income insufficient to cover basic food needs. A wide range of public programmes have been implemented over the years to reduce poverty and improve the well being of the population. The current administration organises the strategy for poverty alleviation along several lines of action, including bothprogrammes targeted to the poorest categories (the PROGRESA/O*portunidades* programme essentially) and actions seeking to ensure wider access to basic education and health services. An increase in public housing is also part of these actions. Spending on poverty alleviation programmes has increased from 0.7 per cent of GDP to 1.3 per cent in 2002. Spending in 2002 amounted to 77.6 billion pesos (over US$7 billion) of which: 18.4 billion pesos for PROGRESA/O*portunidades*, 3.9 billion pesos for the Temporary Employment Programme, 8.6 billion pesos for the education effort, 5.7 billion pesos for the health component and 5.3 billion pesos for nutritional programmes. Through the years, there has been a greater decentralization of programmes, giving states and municipalities increased control over the allocation of resources. There has also been a gradual transition away from general subsidies towards targeted programmes. (See Annex I.D.)

The various programmes for poverty alleviation by themselves will not be sufficient to eliminate poverty. Mexico needs economic growth and expansion of employment opportunities in the formal sector to ensure that families who see their welfare rise as a result of some of these actions do come out of the programmes, avoiding poverty traps, and once out do not fall back into poverty.

This strategy is discussed in more detail in Chapter II. The government's baseline scenario, which assumes that none of the important structural reform is approved, foresees low inflation over the medium term, with the current account deficit at a level which can be easily financed through stable capital inflows (FDI); potential GDP growth is projected at around 4 per cent, close to the average of the past decade. This outlook is comparable to the OECD medium-term reference scenario, in which projections are essentially driven by the supply side of economics, also assuming no new major structural reforms.[20] For Mexico, GDP growth is

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projected to hover between 4 and 4½ per cent, at rates that are too low to close the gap with the OECD countries in the foreseeable future, while job creation in the formal sector is insufficient to absorb new entrants in the labour force and make significant reductions in the informal sector. The conclusion is that although macroeconomic stability is a *sine qua non* for domestic and foreign business to invest in the Mexico economy on a long-term basis, such stability by itself will not generate fast enough growth.

... they would be boosted by forceful action in key areas

A number of bottlenecks would have to be removed in order for GDP growth to reach a rate of 6 to 7 per cent per year, the target announced in the current administration programme, and the growth rate required to absorb the large number of people entering the labour force every year (and reduce informal labour). The authorities have drawn up an alternative scenario in their medium-term programme, based on a different set of policy assumptions, whereby a comprehensive package of reforms – in structural and fiscal areas as well as in education – are implemented; they estimate that through their impact on productivity and private investment, these reforms would eventually boost potential output growth by up to 3 percentage points. Reaching this ambitious objective would require sustained reform efforts, that would lead to much higher human and physical investment rates than recorded lately.

The low average level of human capital is perhaps one of the main factors hindering growth in Mexico. It needs to be raised, through more and better education and training. The importance of additional schooling for economic growth has been put in evidence in an OECD study based on panel data for 21 member countries: over a period during which low education cohorts were being replaced by workers with higher levels of education, one extra year of schooling raised output per capita by as much as 4 to 7 per cent in the long run.[21] A labour market reform that facilitates mobility and flexibility of labour is another key ingredient of the comprehensive growth strategy. In particular, steps that make formal sector employment more attractive to employers and workers, would facilitate the re-allocation of labour to more productive activities, away from precarious jobs in the informal sector.

Furthermore, productive investment clearly needs to be stepped up to ease the supply constraint on the Mexican economy. Based on various methods, Shiau *et al.* (2002) estimate that in order to bring the growth of potential output up to 7 per cent, investment would have to grow at an annual rate of 8½ to 12½ per cent (depending on the productivity assumptions). Such growth rates of fixed capital formation are necessary to bring the investment ratio up to levels more comparable to those of fast catching-up economies and are comparable with rates already reached in Mexico in the 1970s. A significant proportion of the investment effort should go in ICT

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investment. In the best performing OECD countries, including the United States, Canada, Australia, Ireland and the Nordic European countries, there is strong evidence that higher ICT in the 1990s enhanced multifactor productivity growth.[22] In Mexico, more skilled and adaptable labour would contribute to enhance the use of ICT in productive processes.

FDI can play a role. Besides helping capital accumulation, it can have an impact on economic performance through reallocation and technological effects.[23] The evidence of technological spillovers from FDI to other firms is sparse. Early studies showed that the rate of convergence of labour productivity in Mexican industry to the US growth rate was higher in industries with a higher share of multinationals; but it was not possible to distinguish between the rise in within-firm productivity and the "creative destruction" effect.[24] The large scale of FDI in financial services after 1999, when the banking sector was fully opened has contributed to strengthening the banking sector, which should be a key factor in supporting stronger economic growth over future years (Chapter II).

Changes to the regulatory and legal framework are needed to create conditions conducive to higher domestic and foreign investment. Two lines of action are particularly promising: first, removing direct obstacles to private investment in key sectors, such as the electricity sector; second, enhancing competition by lowering barriers to entry, strengthening the institutional framework for competition policy and ensuring enforcement of decisions. This would also contribute to raise multifactor productivity by increasing incentives to adopt new technology and to meet best practices.[25]

In creating a friendlier climate for business, a key challenge facing Mexico is providing a more transparent and effective legal system in which business can be carried out more efficiently and at lower cost. The "rule of law" is easier to recognize by its presence (or absence) than to define, but necessary elements are well-defined property rights, that are upheld in the courts, a *sine qua non* for contracting parties to be confident that terms will be honoured. The Fox administration has taken actions to counteract corruption at all levels, but it is far from having been eradicated. It is important, too, that economic reforms, democratically agreed upon, are not successfully resisted at lower levels of government, or by the private sector, in order to curry votes or extract rents.

In sum, Mexico has great potential for catching up with more advanced OECD countries. Forceful action to release this potential is required, identified as follows (and discussed in details in Chapter III):

– *To boost human capital development and facilitate the adjustment of the labour force.* A direct impact is expected from additional education, via improvements of the quality of labour, and a more permanent effect, by facilitating the adoption of new technologies. The issue here is not so much to spend considerably more on basic education, but to ensure that such

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spending is more effective. Training also contributes importantly, because skilled labour is more adaptable. And reducing obstacles to formal hiring facilitates re-allocation of labour to higher productivity activities.

– *To create an environment supportive of business investment, including* ICT. Through capital deepening, higher investment impacts directly on labour productivity, and by introducing more quickly embodied technological change it has a indirect and sustained effect. Real interest rates that are low and nominal interest rates that are not volatile are required, as well as necessary legal changes that ensure that investors will enjoy the fruits of their efforts if successful, and withdraw and reallocate capital quickly and cheaply, if necessary.

– *To improve access to financing* (*including* FDI). Domestic conditions and foreign capital play a critical role in investment, including venture capital for higher-risk innovative activities.

– *To ease regulatory measures which put barriers to entrepreneurial activity*. Lifting legal restrictions on market access, alleviating administrative burdens on new firm creation, having a bankruptcy regime that allows firms to exit, all foster technological progress, thereby facilitating the catching up process.

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II. Challenges ahead for macroeconomic policy and the financial sector

This chapter reviews recent macroeconomic policy developments and outcomes. It also explores ways in which fiscal and monetary policies can contribute to Mexico's core priority of raising actual and potential growth rates of per capita GDP. The fiscal policy setting can support growth, via higher and more predictable spending on infrastructure, basic health programmes and targeted poverty relief, provided the overall size of the government is not excessive. It also requires a tax system that provides adequate financing, without creating distortions and inefficiencies. To this end a tax reform is needed to generate higher, more stable and broader-based revenue than is currently the case. Monetary policy can help by achieving and maintaining a low-inflation environment in which allocation of investible resources becomes more efficient,[26] provided that the financial sector is fulfilling its intermediation role. In this context, it is noteworthy that Mexico has already made significant progress towards low inflation and avoiding excessive macroeconomic fluctuations. Fiscal discipline was maintained in 2002, with the public sector borrowing requirement being brought down to close to 3 per cent of GDP, while monetary conditions have eased, with interest rates reaching a historical low by mid-2003. The chapter first reports on fiscal policy and public debt management, including a discussion of medium-term issues in tax policy and budget planning. The second section reports on monetary policy. The final section highlights recent improvements in the financial sector, addressing the key question of whether it is now in shape to resume broad-based lending to the private sector.

Fiscal policy

Fiscal policy in the short run

Budget outcomes are typically close to projections

Over the period from the early 1990s, the public sector financial deficit (including the federal government and the main public enterprises) has fluctuated

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between ½ and 1¼ per cent of GDP.[27] Credibility has been gained following the 1995 peso crisis by the fact that outcomes have usually been closely in line with budget targets, notwithstanding unexpected cyclical developments and volatility in oil-related revenue (these account for almost a third of total budget revenue). Thus, the severe revenue shortfall from the oil price fall in 1998 was immediately followed by spending cuts; since then there have been revenue windfalls, in 1999 and 2000. The largest part of the revenue windfalls was used to reduce the public debt and augment the oil stabilisation fund. A smaller share has been used to finance spending, a counterpart to spending cuts on other occasions. Although highly effective as a means of achieving deficit targets and thus calming financial markets, the "stop and go" behaviour of public spending has delayed financing of core programmes, and made the fiscal impact procyclical. In more recent years, fiscal policy has been oriented towards avoiding large fluctuations in public sector balances, and reducing deficits (both broadly and narrowly defined) as a percentage of GDP (Figure 9). In this, the authorities have been broadly successful.

The more comprehensive public sector borrowing requirement (PSBR) provides a more accurate view of public finances. It excludes non-recurrent privatisation proceeds, and includes the net financial requirement of development banks and the net costs of deferred investment projects (PIDIREGAS) (see Box 6 in section below). It also takes fuller account of debt servicing related to financial support programmes.[28] The PSBR widened in the aftermath of the peso crisis, reaching a peak of 6.3 per cent of GDP in 1998, under the combined impact of the financial sector rescue programmes and the development of PIDIREGAS. Between 1998 and 2002, there has been a 3 percentage points reduction in the PSBR, marking clear progress in fiscal consolidation, even when economic activity was flagging (Figure 10).

The broad public sector debt-to-GDP ratio is sizeable

The net debt of the broad public sector, augmented to include liabilities created by the financial sector rescue and long-term investment schemes (PIDIREGAS), amounted to 41.7 per cent of GDP at the end of 2002, up 1½ percentage points on 2001 (Box 4). This ratio is not unduly large when compared with other OECD countries. But as financial savings are particularly low in Mexico, the domestic component of the public debt, at 25 per cent of GDP, is large relative to the stock of private domestic financial assets. This implies that private lending has been to some extent crowded out by the public sector.

Meeting budget targets was not easy in 2002

The budget for 2002 was prepared in a context of great uncertainty regarding the external environment and the strength of the recovery in Mexico. Despite

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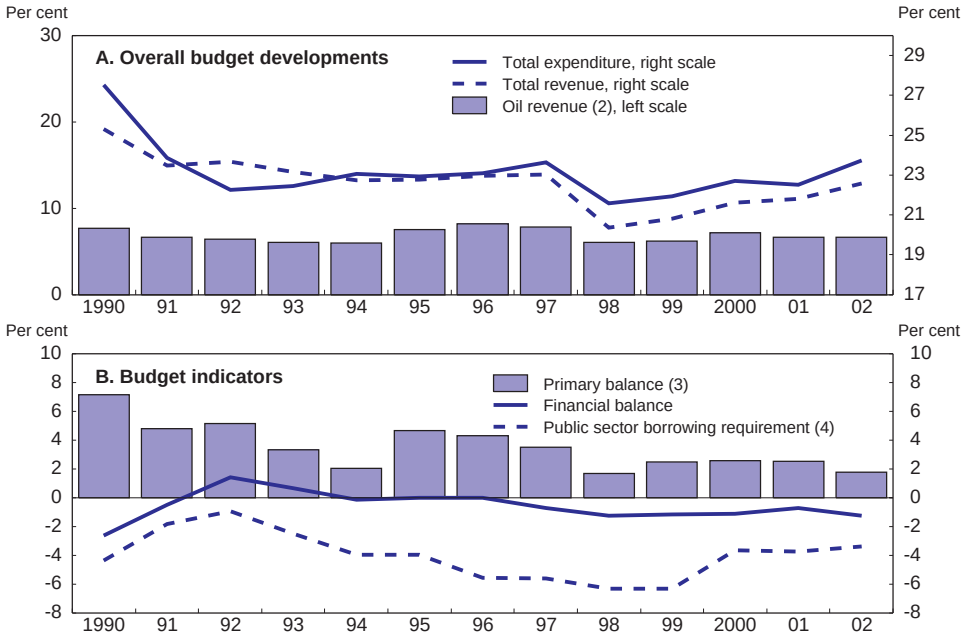
Figure 9. **Public sector budget aggregates**[1]
Per cent of GDP



1. The public sector comprises federal government and public enterprises under budgetary control (such as PEMEX). Financial intermediation by development banks is not included.
2. Includes oil extraction royalties, VAT and excise taxes on oil products.
3. The primary balance is the financial balance less net interest payments.
4. Public sector borrowing requirement (PSBR) includes net costs of "PIDIREGAS", inflation adjustment of indexed bonds, imputed interest on bank-restructuring and debtor-support programmes and financial requirements of development banks. Non-recurrent revenues (privatisation) are not included. Further adjustment to include the net non-recurrent capital costs of the financial sector support programmes would increase the PSBR.

Source: Ministry of Finance; Bank of Mexico.

lower-than-anticipated GDP growth, the public sector deficit would have come out close to target at 0.7 per cent of GDP. However, it was affected late in the year by the cost of the liquidation of B*anrural* (the development bank for agriculture), which raised the deficit to 1.2 per cent of GDP that year (Table 5). The PSBR, which was not affected by the B*anrural* operation,[29] stood at 3.3 per cent of GDP in 2002 against 3.7 per cent a year earlier (excluding non-recurrent revenue). Some tax measures were approved with the 2002 budget, though many of them come into force only in 2003 (see below). Although tax revenue in real terms increased markedly from the previous year, it fell short of budget projections, partly because of weak activity, but also because the impact of the changes to the tax system had been overestimated.[30]

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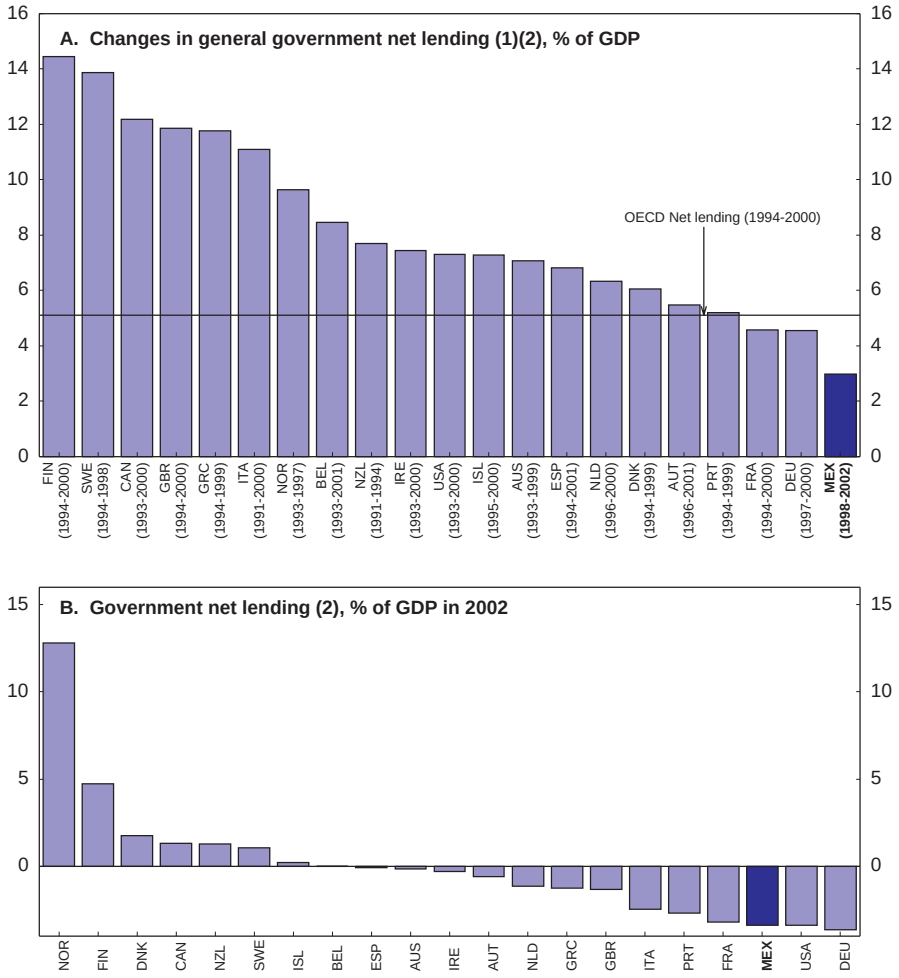
Figure 10. **Fiscal consolidation in selected OECD countries**[1]

Stocks outstanding, end of period

Per cent of GDP



Note: Fiscal consolidations are defined between 1990 and 2002 as periods of protracted (more than three years) improvements in the annual general government's budget balance in per cent of GDP, as compared with the previous year, where such periods are allowed to be interrupted if the worsening of that balance does not exceed 0.5 per cent of GDP and does not last for more than one year.

1. Value in the last year of the consolidation minus the value in the year before the consolidation.
2. For Mexico, public sector borrowing requirements as a % of GDP.

Source: OECD, *Economic Outlook* No. 73, June 2003; Ministry of Finance.

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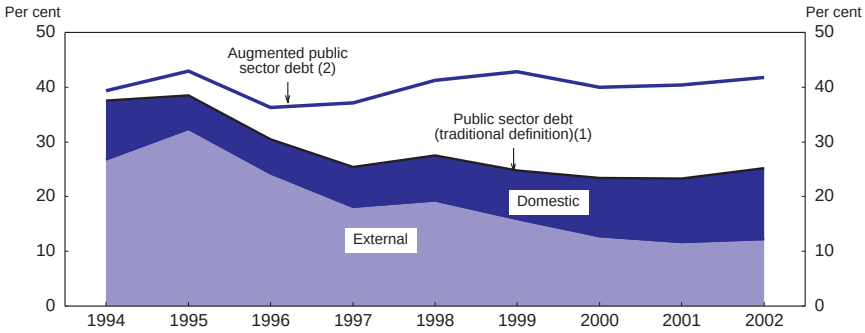
Box 4. The public sector debt and its management

At the end of 2002, net public debt, in the traditional definition, reached 25 per cent of GDP, slightly higher than a year earlier, because of the impact of the nominal depreciation of the peso on the external component, the liquidation of B*anrural* and the low rate of output growth. (Figure 11). The "augmented" public sector debt also increased slightly in 2002. Beside the financing requirement of the federal government, the increase in the "augmented" debt reflects mostly higher liabilities related to PIDIREGAS investment. On the other hand, as noted above, the liquidation of B*anrural* in 2002 had no impact on the broad public sector debt, since the increase in the federal government debt was fully compensated by a reduction in net indebtedness of development banks. In the first half of 2003, the net public debt/GDP ratio came down slightly.

Figure 11. **Total net debt of the public sector**

Stocks outstanding, end of period
Per cent of GDP



1. Gross debt less deposits and liquid financial assets of the Federal government, plus foreign debt of the non-financial public enterprises and development banks. Official trust funds are excluded.
2. Gross debt less deposits and liquid assets of the Federal government, non financial public enterprises, development banks and trust funds, plus liabilities related to the banking sector rescue programmes and PIDIREGAS.

Source: Ministry of Finance.

In addition to reducing the outstanding stock of public debt, the Mexican authorities have sought to lower reliance on external debt and lengthen the average maturity of debt instruments. As part of this strategy, they continued to redeem Brady Bonds in 2002 and 2003 (Brady Bonds were to mature in 2019); with a last operation in July 2003, they fully cancelled the Brady Bond debt. The transactions served to reduce the external public debt, extend the payment horizon, release collateral and generate important savings on debt service, because instruments issued in substitution offer more favourable conditions

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Box 4. **The public sector debt and its management** (*cont.*)

than Brady Bonds. At the same time, debt arising on account of PIDIREGAS schemes has been financed on external markets. The debt related to PIDIREGAS investment, mostly undertaken on account of PEMEX, is issued through an independent debt issuer (under the supervision and explicit authorisation of the Federal government).

On the domestic public debt market, the authorities have continued to issue longer-term instruments at fixed nominal rate, to replace floating-rate and indexed instruments, which until 2000 made up the bulk of government issues. This strategy has served to reduce the vulnerability of public finances to changes in domestic interest rates; it has contributed to consolidating the yield curve for government instruments on the domestic market and to set a benchmark for private sector debt instruments. The shift has brought the average maturity of the domestic federal debt to 2½ years in June 2003, against 1½ in December 2000. Given the decline in (nominal and real) domestic interest rates over the recent period, the government strategy is appropriate in that it is helping to reduce interest rate risk. Regarding management of the broader public sector domestic debt, at present, the Ministry of Finance and IPAB (the agency in charge of administration of the financial support programmes) act separately, each operating on a different segment of the market. An integrated approach to debt management, with a single government agency in charge of a consolidated portfolio, including IPAB liabilities, would probably produce a more efficient strategy.

Interest payments on the public sector debt fell below 3 per cent, the sharp drop from 2001 reflecting lower levels of interest rates. Nevertheless, over the year as a whole, public spending was higher than budgeted. The closing out of *Banrural* and creation of *Financiera Rural* contributed to the unexpected increase in spending, part of which was paid for by non-recurrent resources.[31] This operation resulted in a strengthening of the development bank sector, and will reduce fiscal risks for the future. Excluding the *Banrural* operation, primary spending increased by 4.5 per cent in real terms, significantly more than GDP growth that year. Current expenditure was affected by large increases in salaries, pension payments and budget transfers to local governments.[32] Regarding the latter, the federal government provided to the states the resources that had been budgeted, as an advance on future income from revenue sharing in 2003, so as to limit the impact of the lower tax collection on transfers to states through revenue sharing.[33]

The budget for 2003 foresees further fiscal consolidation

The approved budget for 2003 foresees a public sector financial deficit of 0.5 per cent of GDP, slightly lower than the underlying deficit outturn for 2002,

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Table 5. **Public sector financial accounts**
Per cent of GDP

	1998	1999	2000	2001		2002		2003
	Outturn	Outturn	Outturn	Budget[1]	Outturn	Budget[1]	Outturn[2]	Budget[1]
Revenue	20.4	20.8	21.7	21.2	21.8	22.7	22.6	22.4
Expenditure	21.6	21.9	22.8	21.9	22.5	23.4	23.7	22.9
Financial balance	**–1.2**	**–1.1**	**–1.1**	**–0.7**	**–0.7**	**–0.7**	**–1.2**	**–0.5**
Primary balance	**1.6**	**2.6**	**2.6**	**2.8**	**2.6**	**2.7**	**1.8**	**2.3**
Memorandum items:								
Interest payments, total	2.8	3.6	3.7	3.4	3.2	3.3	2.9	2.8
of which:								
Provisions for support programmes[3]	0.3	0.5	1.1	0.6	0.7	0.7	0.7	0.5
Cost of IMSS reform[4]	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Revenue sharing	3.0	3.1	3.2	3.2	3.4	3.5	3.5	3.4
Public sector borrowing requirement (PSBR)[5]	**–6.3**	**–6.3**	**–3.7**	. .	**–3.7**	. .	**–3.3**	. .

1. The budget projections for 2001, approved budgets for 2002 and 2003.
2. The financial balance is affected by the liquidation of BANRURAL (increasing the deficit by about 0.5 per cent of GDP). The PSBR is not affected.
3. Part of the cost of the support programmes for banks and debtors.
4. IMSS is the social security institute for private sector workers; the total cost of the reform includes both reduced revenues and higher expenditures.
5. Excluding non-recurrent revenue. The PSBR includes borrowing requirements related to direct long term investment on infrastructure projects (PIDIREGAS), to the agency in charge of administration of the banking sector rescue programmes (IPAB), the toll-road fund (FARAC), debtor support programmes and development banks.

Source: Ministry of Finance.

based on a GDP growth assumption of 3 per cent. The PSBR is expected to stand at 3.1 per cent of GDP in 2003. The oil price assumption underlying the budget projection has been set at US$18.35 per barrel for the Mexican mix, which is equivalent to world prices around US$25 per barrel, much lower than the average level in January-August 2003.[34] Very few changes were made in the tax law for 2003, but a number of the measures approved at the end of 2001 came into force in 2003 (see Box 5 below). Indirect taxation remained practically unchanged, while the changes made to direct taxes (including widening the tax base, simplifying the system and lowering tax rates) together are expected to be revenue enhancing. In addition, sales of government assets planned for 2003 (including real estate) are budgeted to bring in around 11 billion pesos (0.2 per cent of GDP).

"Programmable" expenditure of the federal government, *i.e.* excluding interest payments and revenue sharing, is projected to remain broadly unchanged

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Box 5. The oil stabilisation fund

The oil fund was created in 2000 with the intention of smoothing the impact on the permanent sources of revenue and on public finances from abrupt movements in world oil prices or other factors. The large revenue windfall recorded in 2000 provided the resources for its creation (40 per cent of the windfall went into the fund, 60 per cent served to amortize public debt, as established in the budget).

The oil fund mechanisms are in place. Contributions are specified every year in the budget, typically less than half of revenue windfalls go into the fund, where revenue is not related only to oil.* In the 2003 budget, for instance, half of extra resources is to be allocated to increase infrastructure investment by the states, another 25 per cent to reduce the public sector debt and the balance for the oil-stabilisation fund.

If oil export revenues are lower than projected in the revenue law in a given year, public spending could be financed with disbursements from the Fund, these being limited to not more than half the fund balance at the end of the previous year. If the revenue shortfall exceeds that amount, expenditure cuts must be made to complement resources withdrawn from the fund. Furthermore, resources can be released only if the price of oil is lower than the assumption by a specified amount. If the price of oil falls by less than that (or if the revenue shortfall is related to lower-than-expected sales, foreign or domestic), then an adjustment in spending is required to match the revenue shortfall.

Every quarter, the government reports to Congress on the evolution of the stabilisation fund.

* Revenue windfalls refer to both tax and non-tax revenue, including rights (oil fees; extraordinary oil fees and additional oil fees); fees (recovery on oil surplus yields; the Central Bank's operation surplus and fees not related to privatizations and indemnities); royalties and improvement contributions.

at 11½ per cent of GDP in the 2003 budget as in the 2002 budget. An increasing share of government spending goes into current expenditure, which is now above 10 per cent of GDP for the federal budget and shows a lot of inertia from one year to the next (Annex II, Tables). The share of capital spending by the Federal government has been shrinking almost continuously over recent years.[35] A programme of voluntary redundancies in the Federal government was introduced in the 2003 budget to promote a rationalisation of the public administration, with indemnities paid from an increase in the public deficit. The savings expected over future years from the suppression of jobs should pay back its cost.[36]

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As in the past few years, specific safeguard rules have been included in the budget for automatic adjustments in spending in case of public revenue surprises.[37] If government revenues turn out to be lower than budgeted, adjustments will be made automatically to compensate the shortfall: if the shortfall comes from oil-related revenues, then the rules of the oil stabilization fund apply (Box 5); but if non-oil tax revenues are lower than projected, expenditure cuts will be implemented.[38] Rules have been set on how to use revenue windfalls: half of any extra resources are to finance infrastructure projects, and the rest is to be split equally between the oil stabilization fund and debt reduction.[39] These rules are an attempt to smooth budget execution. The role that the stabilisation fund can play is highly dependent on the oil price assumption retained in the budget. A problem here is that this price is negotiated each year with the draft budget, rather than being based on a transparent medium-term approach.

There is no room for manoeuvre on fiscal policy

Although economic activity is likely to be weaker in 2003 than assumed in the budget projection, it should be possible to achieve the budget target without major difficulty. Low growth implies lower-than-projected tax revenue; but improved tax collection might mitigate the results and oil-related revenue is likely to be higher than expected. However, it is clear that there is no margin to use fiscal policy to boost the economy. As in past years, public finances remain highly vulnerable to oil price changes. During the first eight months of 2003, world oil prices dropped from a peak of US$35 per barrel to below US$25 in early May, creeping up subsequently, and the export price for the Mexican mix averaged US$25 (Figure 12). In this context, oil-related revenue in the first half of 2003 was 10 per cent higher than programmed. This extra revenue, combined with unexpected resources from non-recurrent operations, allowed both strong increases in public spending and a sizeable increase in the primary surplus. If revenue for the year 2003 as a whole remains higher than projected, part of the extra revenue will be spent on investment. Although this move would provide some stimulus to activity, there remains some uncertainty on how stable the financing of investment projects can be. Extra spending this year could well be reversed in the near future, because of the unresolved dependence of the budget on volatile funds. The existence of an instrument, such as the oil fund, designed to reduce the impact derived from volatility in public revenues does not replace and should not delay the implementation of a tax reform (see below).

Strengthening the fiscal position should not be delayed

While there have been successful efforts to reduce the public sector deficit, improve the allocation of resources and strengthen tax revenue, further progress is required. The target is to balance the public sector financial deficit by 2005. By then

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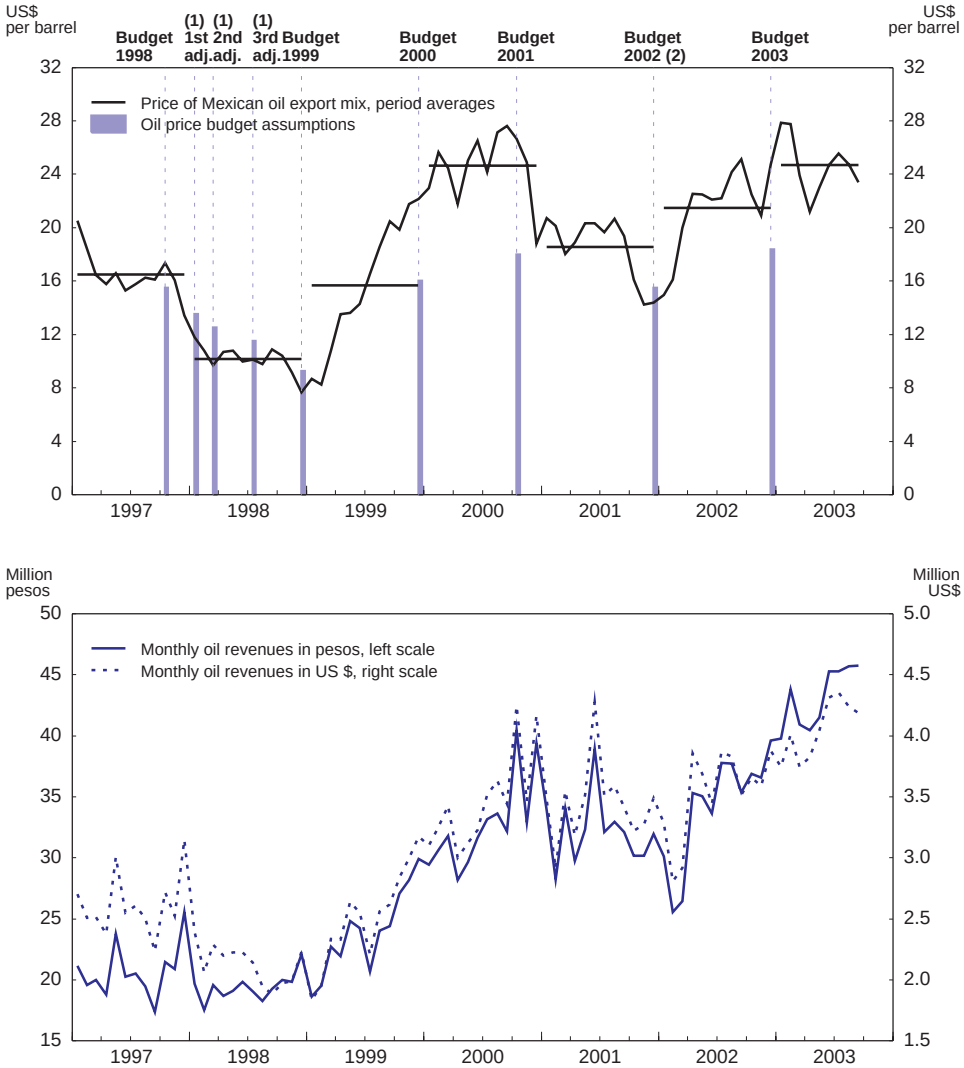
Figure 12. **Oil prices and budget assumptions**



1. Adjustments made to the 1998 budget assumptions.
2. According to the approved budget for 2002.

Source: Ministry of Finance; PEMEX.

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the PSBR is to be brought down close to 2 per cent of GDP. Achieving the objectives will not be easy. Once there, a reasonable medium-term objective would be to put the PSBR on a track consistent with a reduction in the broad public sector debt-to-GDP ratio. This would help create room for manoeuvre in changing cyclical conditions. The low ratio of non-oil tax revenue to GDP and volatility of the oil-related component constrain public spending, while core programmes need to be financed on a stable basis (notably, human and physical capital development) to secure stronger and better balanced-growth. There are also spending pressures over the medium-term to finance the cost of support to banks and debtors and the cost of pension reform – some of which, but not all, have been broadly quantified. Deficit financing is not an attractive option, because it would add to government liabilities, which are already sizeable, when defined in the broad sense. Furthermore it would have an unfavourable impact on investors' confidence, likely to outweigh the immediate benefits of the additional spending financed. The best option is thus to move ahead with a revenue-enhancing tax reform, while reducing tax distortions. A reasonable benchmark would be to raise the tax/GDP ratio by about 2 percentage points in the medium term so as to finance additional expenditure with high marginal social and economic benefits. At the same time, the government will need to continue its efforts to improve the transparency of public finances and enhance the cost-effectiveness of spending.

Strengthening revenue

The tax/GDP ratio in Mexico is the lowest in the OECD and it is also one of the lowest in Latin America. Including social security contributions and all payments by PEMEX to the government, tax revenue amounts to 18.5 per cent of GDP, half the OECD average (Figure 13). Excluding oil-related revenue, the ratio drops to 15 per cent. It should be noted, however, that such ratios are not fully comparable across countries, as countries use the tax and transfer system differently: for instance Mexico uses tax credits instead of social transfers to compensate low-income wage earners in the formal sector, implying (all else being equal) a somewhat lower level of taxes or spending than in countries where transfers are used to compensate individuals. If tax expenditures are taken into account (6.3 per cent of GDP in 2003), including tax incentives for promoting investment and R&D-type activities, Mexico still ranks at the low end of OECD countries. Given the low burden of both direct and indirect taxes at present, there is some margin to strengthen the tax base through a revenue-enhancing tax reform, with a view to reduce the vulnerability of the budget to oil price changes. Only part of the government tax reform proposals were approved at end 2001. They were expected to strengthen revenue and reduce tax distortions. While not much was done with VAT, measures approved on direct taxation have gone some way in widening the tax base by getting rid of various preferential regimes both for corporate and personal income taxpayers, and reducing direct tax rates (Annex III). Nonetheless, as noted above, the outturns for 2002 fell short of original projections. And it is

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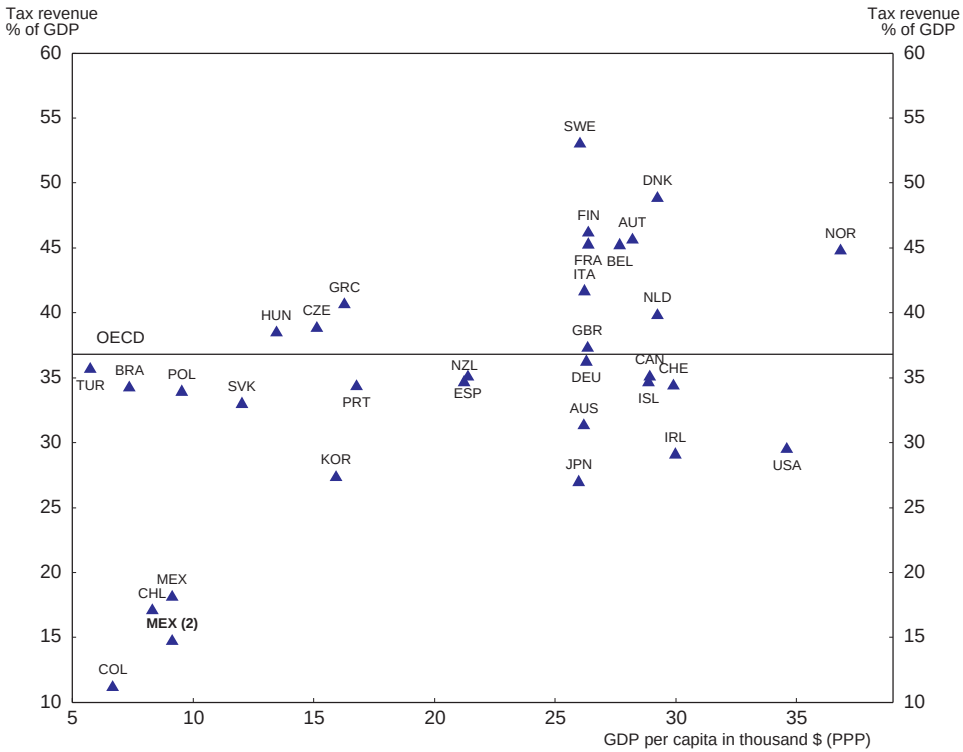
Figure 13. **Tax revenue and the level of income in comparison**[1]



1. 2001 or nearest year available. Including social security contributions. Inweighted average for OECD.
2. Excluding PEMEX contributions to the Federal Government.

Source: OECD (2002), *Revenue Statistics 1965-2001*; OECD, *National Accounts*; Inter-American Centre of Tax Administrations; World Bank.

not clear yet that the measures on direct taxation coming into force in 2003 will have a large revenue-enhancing impact.[40] There is a wide agreement that the government requires more revenue, but what is lacking is an agreement on how to achieve this.

A recurrent problem lies in the numerous changes that are made to the tax system from one year to the next. They tend to complicate the system and impose a high cost on both the taxpayer and the administration. These changes, which result from compromises made during the tax law discussions, often reflect short-term financial considerations in a particular year. They have sometimes introduced distortions, discouraging production or employment (the payroll tax,

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for instance). When tax reform is brought back on the agenda, a more comprehensive approach to the reform would be appropriate. Any reform proposal should focus on VAT, where a widening of the base and elimination of special regimes would facilitate tax administration. Several options might need to be considered to make broadening of the VAT base politically viable. One such option is to directly compensate low income groups by transfer mechanisms.[41] Another, less preferable, option would be to retain zero-rates on a few basic staples (milk, bread, tortillas, eggs, for instance). VAT is normally easy to administer and control. The application of a uniform VAT rate across all goods and services would bring substantial budgetary revenue, by suppressing the existing possibilities of evasion and fraud, even with a lower rate than the current standard rate and even if some compensation to low-income groups is applied, provided the mechanism chosen is strictly defined so as not to dilute the revenue impact of the base broadening.

The reform of the tax administration (SAT) has been approved. It defines strictly the role of the administration, which is responsible for tax collection and for providing information and exercising statistical control over all taxes and customs duties. The responsibility for tax policy lies with the Ministry of Finance. The SAT is required to reduce the costs of tax collection and of tax compliance.[42] Among the recent measures to improve tax administration, some advances have been made towards income globalisation; and tax collection and payment are now done electronically for 3 million taxpayers (individuals with large income, including those with business activity, and small taxpayers under the special regime). It is important to step up efforts to build up and use information systems, so as to facilitate cross-checking of sub-systems. Continued progress in modernising tax administration and improving its efficiency can help build a consensus for revenue-enhancing tax measures, by increasing perceived fairness. Attention should also be given to the way revenue is spent. Changes that raise tax revenue, even when well designed, usually face strong political resistance. They may be more acceptable if the marginal benefits from higher taxes are clear. And economic agents are more likely to comply with tax rules when they perceive that their taxes are well used.

Budgeting and public administration

The 2000 OECD *Economic Survey* of Mexico identified a number of ways in which the budget process and public administration could be improved to enhance the effectiveness of public spending. Since then, various measures have been taken.

The approval of the "transparency law" in April 2002 facilitates access to information. However, it remains difficult to assess the real extent of government activities in the economy. A large volume of information has typically been provided through the expenditure budget decree (*Presupuesto de Egresos de la*

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Federación), the federal revenue law (*Ley de Ingresos de la Federación*) and the "Criteria of economic policy". But it was not always clear nor strictly consistent with international practices. Important innovations in this respect were made with the 2003 budget, with a view to improve the quality of information supplied and facilitate discussions in the Chamber of Deputies.[43] The coverage of the budget data does not provide a good picture of government activities. Some major state-owned enterprises are included in the budget (PEMEX, CFE…), while development banks are not, for instance.[44] The more comprehensive public sector borrowing requirement, published since the end of 2001, has been recording more operations, and the information provided on the liabilities stemming from PIDIREGAS and IPAB is now more detailed. Another important step has just been accomplished with the requirement for the government to publish tax expenditure projections.[45]

The government has made great strides in modernizing the budget process over recent years, in particular in 2003, in the context of the preparation of the 2004 draft budget. The various steps – setting objectives, making the links with indicators from institutions, coordination with spending units – are now spread over several months ending in September, and starting earlier in the year than was previously the case. The use of a new information system is streamlining the whole process. The Budget Committee at the Chamber of Deputies is also now participating in the budget drafting process, which is useful because it allows a closer scrutiny of the budget at the preparatory stage.[46] These good practices should be embedded in legislation to ensure that they become permanent. There have been no changes yet regarding the deadline for submitting the budget, although this does not seem to be a key issue since, in practice, the budget has regularly been submitted in October rather than at the end of the deadline (mid-November). Likewise, there is still no provision in case the budget is not approved by the end of the legislation, but in practice, with a divided Congress over the past six years, the budget has always been approved in time.[47]

Guidelines for a new investment system have been set. The system seeks to address problems inherent in public investment spending: many projects executed without a cost-benefit analysis or after insufficient evaluation; lack of planning for investment spending in some sectors and poor definition of the scope of the projects; finally, at the execution stage, differences between the original amounts planned and actual spending. Under the new investment system, the use of cost-benefit analysis is being generalized and each government agency will prepare a planning document with a horizon of 5 to 6 years (training will be provided to officials involved). Systematic registration of the investment project will be required for it to be included in the budget proposal. Following the execution, a review of the project will be undertaken. The recently-created audit body in Congress (*Auditoria Superior de la Federacion*) will have a larger role in *ex post* control of spending, including quality evaluations using cost-benefit analysis.

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Setting fiscal policy in a medium-term framework

There is a general consensus about the need to move in the direction of medium-term programming, including a comprehensive budget projection set in a macroeconomic framework, with a rolling time horizon. Some elements are already in place, such as the medium-term projections for investment projects. The PRONAFIDE 2001-2006 also is a partial attempt in that it provided broad projections under different policy assumptions. A number of pre-requisites are needed for a more complete exercise: substantial progress in fiscal consolidation, based on fiscal accounts which include details about contingent liabilities (such as PIDIREGAS) and getting public enterprises out of the public sector budget deficit. Medium-term projections should spell out clearly what proportion of contingencies is taken into account. Over the medium-term, a change in the Constitution would be required to introduce a "fiscal responsibility" principle. Once the public accounts are balanced or close to balance and credibility is built up further, with a stronger revenue base in place, budget discussions could become focused on a concept of an expenditure "norm", or a cyclically-adjusted budget balance, rather than on the actual budget balance. A medium-term framework would give more flexibility for operating fiscal policy from one year to the next, and temporary deviations, as with automatic stabilizers in the traditional sense, could then be envisaged.

Public spending requirements to strengthen growth in the longer term

In Mexico, sustained higher levels of public spending are desirable in several areas to create conditions for higher sustainable growth of per capita income. Such growth will mainly be fuelled by private sector activity over any time span, but productivity in the private sector will rise faster if there are higher levels of basic infrastructure and a rising share of healthy, well-educated people in the labour force. Targeted poverty-relief programmes such as PROGRESA/*Oportunidades* have been successful in raising levels of nutrition and building up human capital for the poorest groups, and these should be continued. In some areas, such as education, measures should be taken to improve the quality of services provided without necessarily spending significantly larger sums (Chapter III). However, there are areas where additional spending is needed. In particular, increasing the supply of medicines and nursing staff in public hospitals is an immediate priority. In the longer run, an ageing population will entail higher spending in the health care system and pensions, even if Mexico's demographic structure make the problem less urgent that in many other OECD countries. The private sector pension system has been transformed from a pay-as-you-go to a fully-funded system and the transition cost is currently being paid. But a large share of workers in the private sector who will not have accumulated enough savings on their individual accounts will be entitled to the minimum state pension. Furthermore, there are other spending requirements, related to the government pension system (ISSSTE), still unreformed.[48]

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Box 6. PIDIREGAS and other investment projects involving private investors

Long-term projects for infrastructure (PIDIREGAS). Since 1997, the government has made use of private investment to carry out direct investment for infrastructure projects in the energy sector (electricity and oil) through a hybrid scheme of direct infrastructure investment projects with postponed impact on expenditure (generally called PIDIREGAS). These are being used in the construction of infrastructure in the electricity and oil sectors, where there are legal barriers to private investment. Under the direct investment PIDIREGAS, the private sector company finances and builds the plant, with no payment due by the public sector until completion. When the project is concluded it is delivered to the public sector, which pays for the contracted obligations,* theoretically using the revenue flow the projects will generate. While recording in the financial balance takes place when payments actually start, the PSBR records the resources involved – *i.e.* an estimation of the financial requirement during the year at the time the project is initiated, hence when the public sector obligation is created and when public policy is having an impact on aggregate demand. In theory, the revenue flow allows to reimburse the cost of construction, so that it should be budget neutral. But the private sector does not share the financial risk of the project. Funding for the entire project is legally approved by the Chamber of Deputies at the time the project is launched, even if the impact on the financial accounts is postponed, so that it has a *de jure* preferential status concerning future disbursements.

Another kind of public investment projects financed by the private sector (for CFE notably), is one under which the government acquires the plant only if some previously established eventuality occurs ("conditioned investment"). Under these projects, the government guarantees the purchase of electric power during the plant's life; it assumes the risk of the investment and must record the investment as a contingent liability, but it is not recorded in the financial balance nor in the PSBR. It becomes a PIDIREGAS, only if the project is taken over by the public sector. In recent years, most of CFE-financed investment projects have been "conditioned investments", while PEMEX typically uses "direct" investment PIDIREGAS.

The "augmented" public debt, which records the traditional public sector financial balance plus government liabilities, only takes into account those related to the PIDIREGAS "direct" investment projects, since the "conditioned" investment (equivalent to about one-tenth of "direct" investment) is not public investment.

Independent Private Producers of Energy (IPP). Under this scheme used for electricity generation, a long-term contract is established between the private sector and the Federal Electricity Commission (CFE) for the sale of electricity, the government agreeing to purchase all the electricity generated by the plant through its life (usually 25 years). In that case, the plant remains privately owned and privately operated.

* After the projects are received by the public sector, the payment obligations for the first two years are considered as direct liabilities, and recorded as investment spending in the public sector financial accounts; the remainder is registered as contingent liabilities – to be covered by the flow of income that will be generated.

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Box 6. **PIDIREGAS and other investment projects involving private investors** (*cont.*)

Projects for the Provision of Services (PPS). Their scale is very small relative to PIDIREGAS or IPPs, and it will take some time before they are in place. Budgetary restrictions are the most important raison d'être for these, but they also seek to provide better quality service.

There is also a clear need to address the backlog in physical infrastructures (for example roads, drinking-water supply and water treatment).[49] A better and more widely-spread transportation system, in particular, would help raise productivity levels in the medium to long term. Spending on infrastructure has not been neglected: indeed, according to budget rules, part of windfall oil revenues must be spent in this sector. But there are two major problems. First, fiscal revenues under the existing tax system are insufficient to finance infrastructure spending by federal, state and local governments at an adequate level. Second, the overarching priority accorded to meeting pre-announced budget targets, combined with the incompressible nature of much current spending, means that investment spending is typically subject to unpredictable surges and wasteful cuts. These considerations underline the need for reforms that raise core revenues, and enable fiscal balances to move to levels that would justify taking a more relaxed view of cyclical influences. Involving the private sector in the provision of some public services can contribute to developing infrastructure, and further privatization cannot be excluded (Chapter III). In the absence of legislative reforms that open up the energy sector to private initiatives, the government has chosen to develop a hybrid scheme for financing infrastructure building, through investment projects with a deferred impact on the budget (PIDIREGAS). Under the scheme, the private sector finances and builds infrastructure, with no payment due by the public sector until completion (Box 6). This has proven useful in allowing investment in the electricity and oil sectors for capacity enlargement and modernization sooner rather than later, but it can create fiscal risks, in that the State would be obliged to take over the project from a failing private company.

Monetary policy

The monetary stance remains cautious

After the 1995 crisis, the Mexican central bank successfully reduced inflation from more than 50 per cent to below 4½ per cent by end 2001 and rebuilt a solid

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credibility. Nonetheless, the 2002 target for headline inflation was missed (Figure 14).[50] As a result, and to bring down inflation expectations in a context of a depreciating peso and of inflationary pressures from administered and farm prices that risked contaminating core inflation, the Bank of Mexico started to tighten monetary policy in September 2002. The *corto* was increased on several occasions since then, including three times in the first quarter of 2003 (see Figure 15, and Box 7 below for a description of the *corto* mechanism). These actions resulted in a rise of short-term interest rates through early 2003. Conditions changed rapidly in April, and there has been no increase in the *corto* since March 2003, as monetary conditions, given signs of inflation easing, satisfied the central bank.

From April, following the reduction of geopolitical uncertainties and the return of capital flows to emerging markets, the exchange rate appreciated while interest rates dropped sharply.[51] Overall monetary conditions seem to have eased (Figure 16). The peso appreciation was subsequently reversed around mid-year, as low interest rates and uncertainties regarding the recovery reduced the relative attractiveness of the peso. However, the pass-through to domestic prices has remained subdued. Inflation expectations have declined in response to positive inflation outcomes and lower contractual wage increases and by October they were broadly consistent with end-2003 inflation target of 3 per cent, with a variability interval of plus or minus 1 per cent.

The monetary framework has been strengthened

Several changes to consolidate the inflation-targeting framework and to make monetary policy more transparent and effective have been implemented since end-2002. First, the communication policy of the central bank changed at end-2002, with monetary policy decisions since then being announced on a predetermined schedule (twice a month except in December) and accompanied by press releases (once a month). More recently, in April 2003, the Bank of Mexico modified the way it operates the *corto/largo* mechanism from accumulated balances to daily balances (Box 7). This change was introduced in order to simplify the operating mechanism for the *corto* and make it easier to understand. It also contributes to eliminate some distortions arising at the end of the accumulation period for the *corto*. The central bank also increased banks' compulsory deposits at the central bank to maintain the effectiveness of the *corto* mechanism.

Monetary policy in the medium-term

The monetary authorities' disinflation strategy has worked well since 1995, and the 2002 slippage does not change this overall diagnosis. The target is to bring inflation down to 3 per cent with a variability interval of plus or minus 1 per cent at end 2003 and maintain inflation in that range thereafter. The 3 per cent target is considered to correspond to the steady state level of inflation in Mexico.

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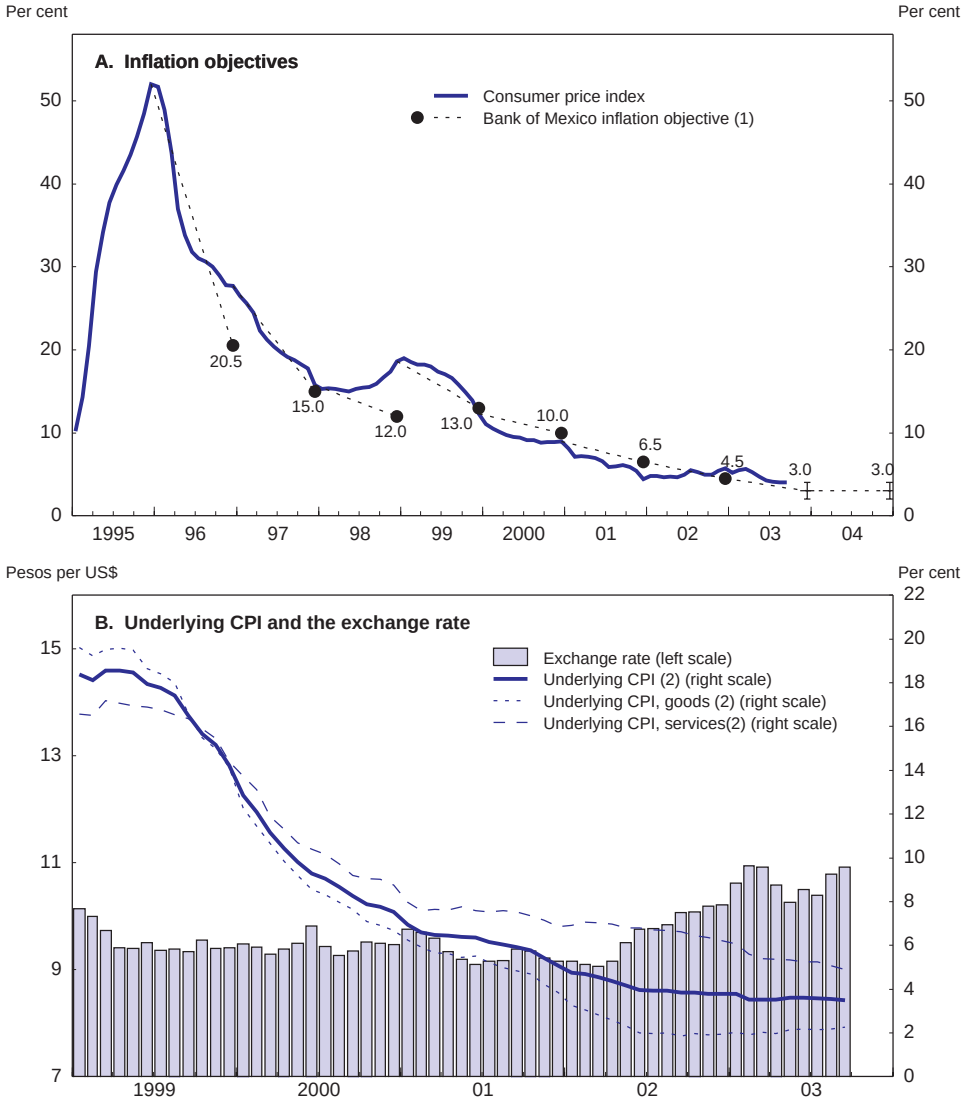
Figure 14. **Inflation objectives**
Percentage changes over 12 months



1. From 2003 onwards a variability interval of plus/minus one per cent is established over the 3% target.
Source: Bank of Mexico.

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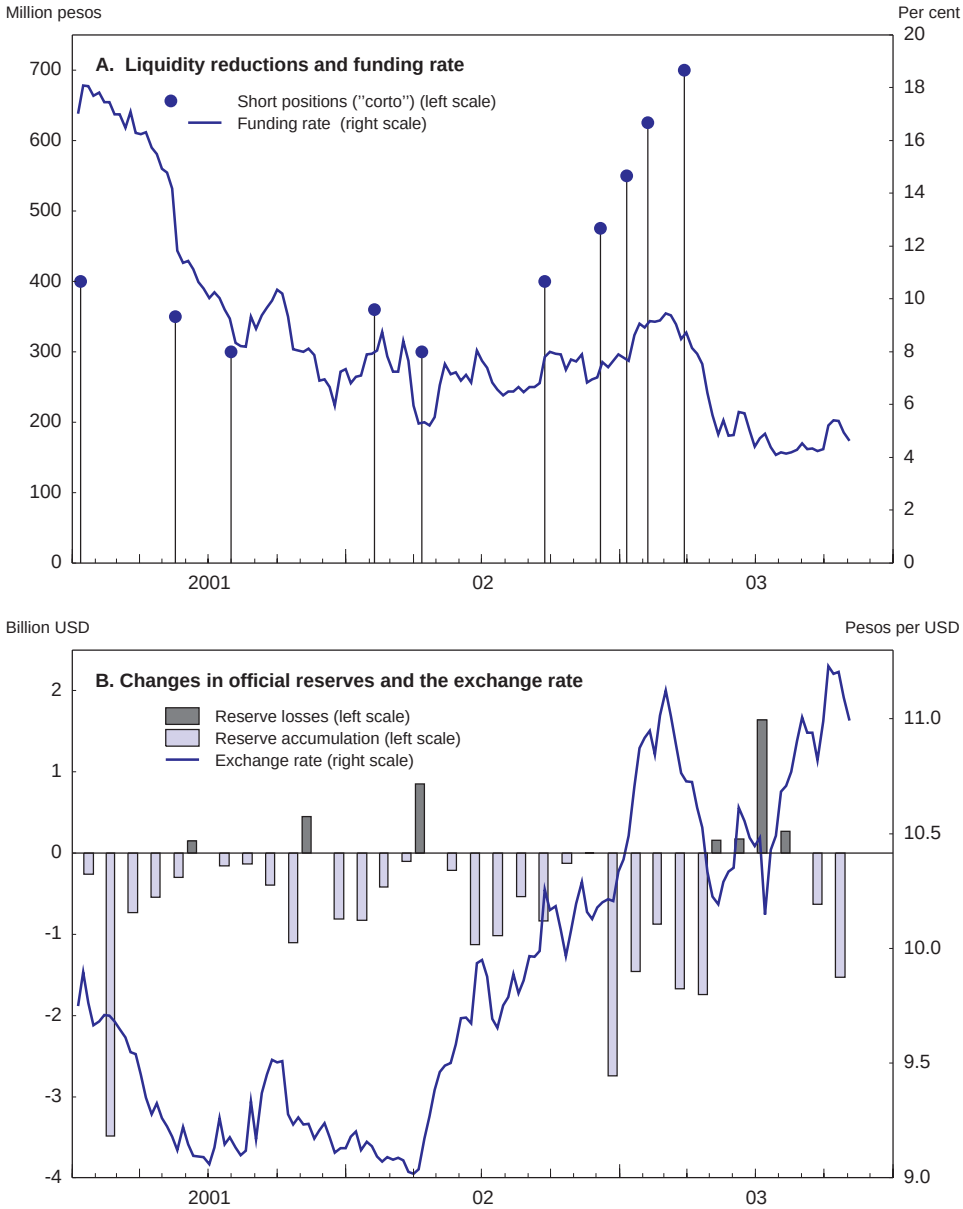
Figure 15. **Monetary policy actions**



Source: Bank of Mexico.

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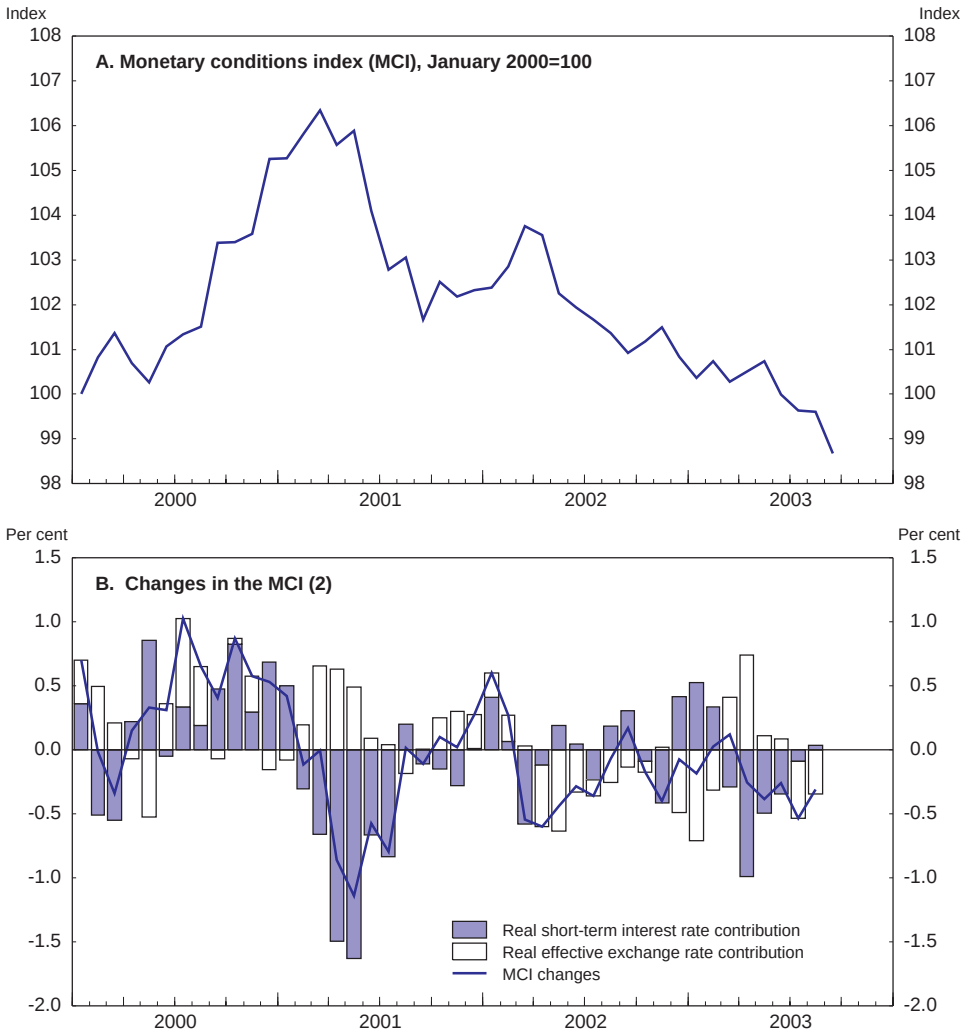
Figure 16. **Mexican monetary conditions index**[1]



1. The monetary condition index at time t (MCI[t]) is defined as:
MCI[t]=MCI[t – 1]* (1+((1 – XGS[t]/GDP[t])*(r[t]-r[t – 1])+(XGS[t]/GDP[t])*(exchr[t]/exchr[t – 1] – 1)), where:
r is the real short-term interest rate (one-month CETES rate minus year-on-year inflation in current month);
exchr is the real effective exchange rate based on consumer price inflation, expressed as foreign currency per unit of domestic currency;
XGS[t]/GDP[t] is the share of exports in total GDP at time t.
An increase in the index reflects a tightening of monetary conditions.
2. Three-month centred moving averages.
Source: Bank of Mexico; OECD.

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Box 7. The *corto/largo* mechanism and recent changes

The main policy instrument of the Bank of Mexico is the *corto/largo* mechanism: it induces changes in interest rates through the announcement of a target for the sum of the balances of all current accounts held by credit institutions at the Bank of Mexico. These balances are required to be null, a penalty rate (double the Cetes rate) is applied to overdrafts and positive balances are not remunerated. When the Bank targets an overdraft or negative balance (*corto*), it signals its intention not to provide credit institutions with sufficient resources at market rates. Although the *corto* represents only a small amount of the total liquidity, increasing it induces market participants to bid up interest rates to avoid paying overdraft charges and puts upward pressure on market interest rates. The reverse mechanism applies when the central bank targets a positive balance (*largo*).

The main advantage of this mechanism in the Central Bank's view has been its flexibility, very adapted to a country submitted to frequent shocks as is Mexico. In particular, the Central Bank does not fix the interest rates, but lets them be determined by market forces so that they can adjust daily to market conditions. In practice, short-term changes in market-determined interest rates can be very large, far more so than the sort of changes in official rates usually instigated by the monetary authorities in other countries. The main drawbacks, stressed by analysts, are the difficulty to understand fully this mechanism, and a lack of transparency due for instance to the fact that changes in the target give information only on the direction of the interest rates change desired by the Central Bank and not the level. Studies have also shown that the impact of changes in the balance target on interest rates is usually temporary (see for instance Bancomer, 2003 and Sanchez, 2002).

To maintain the effectiveness of the *corto* mechanism, in mid-1998, the Bank of Mexico introduced a remunerated compulsory deposit for commercial banks. Liquidity is then reintroduced through the money market, putting the Bank of Mexico in a position of net creditor to the banking system and reinforcing its control over short-term interest rates. The compulsory deposit was increased in April 2003 from 25 billion pesos to 30 billion pesos. This action was presented by the Central Bank and seen by market participants as purely technical, to increase the efficiency of future *corto/largo* changes. In April 2003, the Bank of Mexico also changed the procedure by which it measures the balances, switching from a 28-day period accumulated basis to a daily basis. This was done with a view to simplify the operating mechanism for the *corto*, make it easier to understand, and has eliminated the distortions caused by some market participants closing out large positions as the end of the accounting period drew near.

This medium-term objective is somewhat higher than inflation in main trading partners.[52] This is appropriate over the next few years given the relative price volatility inherent in Mexico's still-developing economy. Achieving the target will nevertheless require continued caution and readiness of the central bank to respond quickly

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to any risk of inflation pressures. Furthermore, monetary policy will have to take into account the reactivation of the credit channel: as banks resume lending, the timing and magnitude of the impact of monetary decisions are well likely to change.

In the medium term, once inflation stabilises around the target of 3 per cent and the credibility of the central bank becomes more solidly established, the required changes in interest rates are likely to become smaller, and a change in the monetary framework could be considered. The Bank of Mexico could consider acting directly on leading interest rates rather than relying on the indirect impact of announcements on the overall liquidity position of banks. While the *corto* mechanism has worked well in terms of communication with markets, such a move would result in increased transparency of monetary decisions for the general public. The counterpart might be more volatility of the exchange rate, which implies that this change could only occur when the pass-through of exchange rates movements to inflation has been structurally reduced. In the recent episode of peso depreciation, there has been no evidence of a pass-through of exchange rates to inflation. This results from a mix of temporary factors, such as the weakness in domestic demand and falling international prices, but also of structural factors, such as the increased credibility of inflation targets, changes in firms' pricing practices in a lower inflation environment and changes in the behaviour of economic agents who are getting more used to a floating exchange rate system. However, some impact of the exchange rate depreciation on prices later on cannot be discarded. Therefore, efforts to consolidate the credibility of central bank targets must be continued. Moreover, structural problems have to be solved (especially in public finances) to reduce further the Mexican risk premium and make it less volatile.

Is the financial system in shape to resume broad-based and sound lending to the private sector?

The health of the banking sector and the regulatory framework have continued to improve

A well-developed financial system, which efficiently channels resources towards the most rewarding activities and stimulates investment, is important to support stronger growth in Mexico.[53] The deregulation of banks and privatisation operations date from the late 1980s and early 1990s.[54] After the peso crisis of 1995, the sector had to be rescued by massive government support. Following the introduction of various schemes to support banks and debtors, the reforms of 1999 marked the starting point for a more solid recovery. In addition, banks have improved substantially their credit assessment processes. In a detailed review of these developments, the 2002 OECD *Economic Survey of Mexico* concluded that the Mexican banking system is increasingly solid and as profitable as in other OECD

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countries, and that the supervision and regulatory frameworks are now close to best practices.

In 2002-2003, the health of the banking sector has continued to improve.[55] Additional measures, in line with past OECD recommendations, have been prepared to enhance its supervision. The financial authorities are completing a new law on bank resolution that will include a prompt corrective action system, to be sent to Congress for approval in 2003 or early 2004. The creation of a single autonomous agency covering the various entities in charge of the supervision of financial institutions is still under discussion. Measures are now being taken to increase transparency on bank services and fees, in order to boost competition.

However, while some components of commercial bank credit started to pick up in late 2000 (especially when adjusted for the decline in non-performing loans), the revival of credit to the private sector is still fragile (Figure 17). Consumer credit has been the only dynamic component since 2000. At early 2003, credit for housing was still contracting. Lending to enterprises, which is very low by international comparison, has only recently started to show a positive trend.[56] According to the Bank of Mexico surveys, by mid-2003, only 30 per cent of those companies interviewed were using commercial bank financing.[57] While this reflects partly lower demand for credit in a context of depressed activity and international and domestic uncertainty, several factors have limited credit supply. Surveys show that low credit has been partly due to too high costs of credit and a reluctance of banks to lend to riskier firms, especially to small- and medium-sized enterprises.[58] This in turn has reflected among other factors weaknesses in the legal and regulatory framework. Furthermore, the expansion of credit to the private sector has been crowded out by a strong demand from the public sector. In the current context of a reduced PSBR, and low interest rates, the incentives for banks to lend to the private sector have increased, helped by legal changes described below.

A new bankruptcy law was enacted in 2000, but its effectiveness in raising bank lending was limited. Experiences in Eastern Europe and the former Soviet Union suggest that when such changes are made to the legal framework, time is needed to prepare the legal staff and the potential users for the changed practices and institutions must be modified for the implementation of the new law. In the case of Mexico, the new law itself had some shortcomings. Measures were taken recently to address them and increase incentives for banks to finance private sector investment. In April 2003, Congress approved a reform of the legal framework for secured credit transaction (*Miscelanea de garantías*) including amendments to seven related laws.[59] These amendments include faster seizure of collateral and a broadening of the asset range that can secure credit to include commercial units. The aim is to grant greater judicial certainty to creditors and borrowers and reduce risks associated with credit transactions. This should in return reduce the cost of credit. However, further strengthening of the judiciary system

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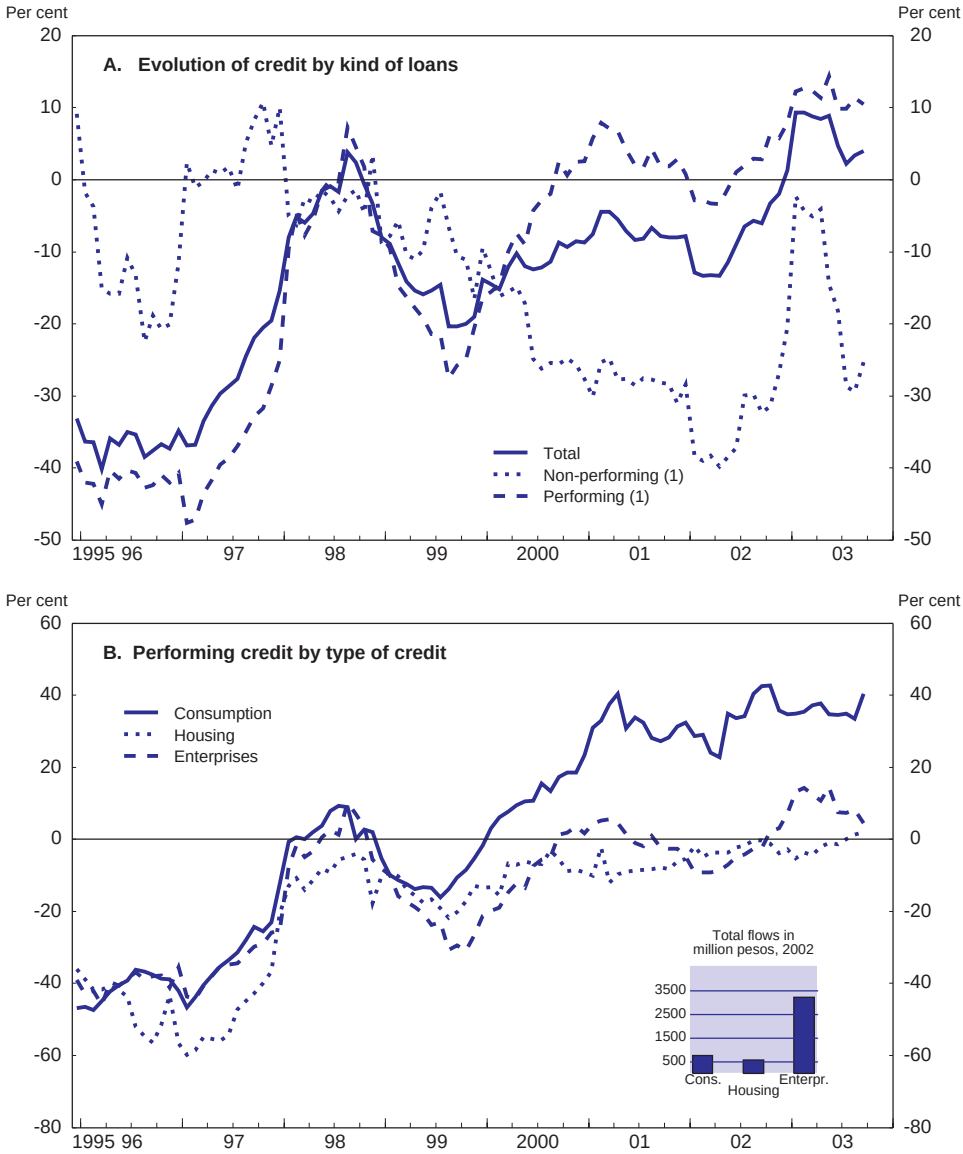
Figure 17. **Recent trends in domestic credit to the private sector**[1]
Year-on-year percentage change in real terms



1. Direct credit by commercial banks.
Source: Bank of Mexico.

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seems necessary to guarantee fully effective implementation of these new amendments. This issue of the quality of the rule of law in Mexico, which is especially weak at the State and local levels, is discussed below in Chapter III.

Overall, conditions seem to have been met for banks to channel savings to agents seeking funds for investment when demand increases.[60] This process will be facilitated if, as mentioned earlier, government bond remuneration remains low, making it relatively more profitable for banks to lend to the private sector. Furthermore, fiscal consolidation can also help, by reducing the outstanding stock of public debt relative to GDP. However, the credit recovery is likely to profit first the larger/export-oriented companies that present lower credit risks (and already have access to foreign credit). Small- and medium-sized enterprises (SMEs), which have been very much affected by the lack of credit and rely mainly on supplier-credit for external financing, will also benefit from the credit recovery. Given the size of their credit requirements and the fact that, as in many OECD countries, the inherent higher risk of SMEs will continue to make them relatively less attractive to the banking sector than larger ones, it is also necessary to promote other sources of financing.

Strengthening and developing alternative sources of financial intermediation

Other financial institutions have in fact gained market share over banks on the credit market, contributing to lowering the share of bank lending. While foreign banks have increased lending to large/exporting companies, credit unions and savings and loans institutions have increased their market share on the SME and housing credit markets. This trend raises the need to continue developing a level playing field of regulation and supervision between different types of intermediaries.

In this regard, Congress approved the Popular Savings and Credit Law (*Ley del Ahorro y Crédito Popular*) in June 2001, which regulates all institutions that accept savings and deposits and give credit to people with low incomes. Savings and loans institutions and credit unions operating under the former legal framework have to change their status and transform into Popular Financial Societies (*Sociedades Financieras Populares*) by 2004. This law establishes a delegate supervision system, with the planned creation of Confederations and Federations that will supervise the Popular Financial Societies together with the National Banking and Security Commission (*Comisión Nacional Bancária y de Valores*, CNBV). A working group has been created to propose a new framework for the savings and loans sector, and the newly created *Banco del Ahorro Nacional y Servicios Financieros* (*Bansefi*) is an important actor in the strengthening of this sector.[61] In addition, a new department in charge of supervision and surveillance of credit unions was created by the CNBV in March 2002 and unhealthy institutions are being closed. Moreover, the CNBV is currently adopting a risk-based supervision for credit unions (including

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features such as capitalization based on risk analysis, portfolio classification based on risk, liquidity indexes, etc.).[62]

The development bank segment accounts for slightly more than one fifth of the financial system assets. This segment is necessary where there are market failures. In Mexico, for instance, the state-owned development sector represents in some cases the only source of financing in the agricultural sector and for many SMEs. However, as highlighted in the 2002 *Economic Survey*, the sector needs to be rationalised. Several measures have been taken in that direction in 2002 and 2003:

- In the rural sector, as noted above, the former insolvent institution *Banrural* was closed and a new one was created on sounder basis (*Financiera Rural*). See Annex I.B on Agriculture.
- Congress approved amendments to the credit institutions law and the organic laws of other development banks (*Nafin*, *Bancomext*, *Banobras*, *Bansefi* and *Sociedad Hypotecaria Federal*) aiming at improving their efficiency and governance and strengthening the prudential criteria in their management.

Developing the domestic financial market

A recent development has been increased issuance of corporate debt, on increasingly long term and denominated in pesos. While this is a positive development, a stronger capital market is needed to allow financing of medium-size companies, and not only the large and export-oriented ones as has typically been the case in Mexico. Venture capital – seed money for ICT business in particular, especially directed to the early stage of enterprise formation and to the higher technology sector as in the US or Canada – has been shown to have a significant impact on growth. So has equity financing, in particular a secondary share market with easier rules of access. These sources of financing play a negligible role in Mexico. Among the obstacles to the development of the capital market, an important one is the small size of the investor base. A decisive step has been recently taken with the easing of the regulatory guidelines on investment by private pension funds designed to allow diversification of investments away from the public sector. The former limit of 35 per cent of funds invested in the private sector has been now eliminated and replaced by limits set in terms of credit rating of the issuers and concentration limits have been tightened.[63] These moves should help accelerate the development of the capital market in Mexico.

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III. Structural reforms for sustaining high growth

Mexico's development aspirations have been frustrated by low underlying growth rates over several decades. Chapter I has identified several major impediments to sustained high growth: low human and physical capital; an inefficient formal labour market; weaknesses in sectoral regulatory frameworks; and an unfriendly business environment. Further enhancing skills of the present labour force and future generations of workers is essential to increase potential growth and raise income. The magnitude of the benefits to be drawn from increased human capital depends however on progress in other structural areas. A well-functioning labour market is needed to translate human capital improvement into higher output, productivity and wages. Moreover, business costs arising from labour and product market regulations, infrastructure bottlenecks, and the functioning of the legal system need to be addressed. Both human capital and improved regulation could underpin the expansion of investment (both domestic and foreign) and the diffusion of new technologies. This chapter reviews the policy measures that are needed in these areas. A synthesis of the main recommendations for structural reform is presented in Box at the end. A final section looks at three specific issues of sustainable development: air pollution, use of water and water pollution.

Enhancing human capital

Human capital plays a crucial role in growth. Accordingly, improving access to education of a rapidly increasing population has been a priority of government policies in past decades, and these have succeeded in raising enrolment rates, especially in basic education. However, despite substantial improvements, Mexico's human capital lags *vis-à-vis* the rest of the OECD (see Figures 18 and 19).

Reducing this gap is especially important at a time when competition from less developed countries is putting pressure on labour intensive sectors and Mexico needs to enhance diffusion of new technologies and move towards more value-added and competitive products. It implies dealing with three main challenges:

– increasing further the coverage of the education system, especially for secondary education;

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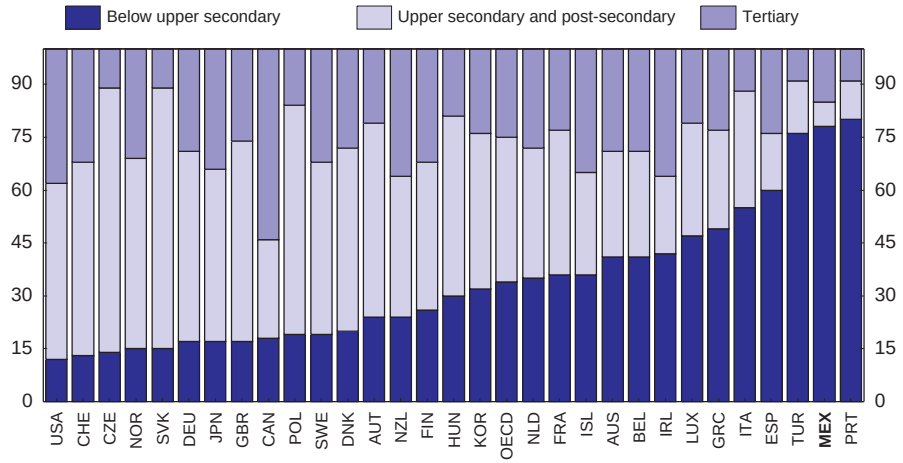
Figure 18. **Educational attainment of the population in OECD countries**
Distribution of the population of 25 to 64-year-olds
by highest level of education attained in 2001



Source: OECD, *Education at a Glance*, 2002.

– improving the quality of the education system, which has become a major concern in recent years;

– and providing more and better training to the labour force, including basic education programmes for adults.

The National Education Programme 2001-2006 draws up a framework to address these challenges, and both the Ministries of Education (*Secretaría de Educación Pública*, SEP) and of Labour and Social Welfare (*Secretaría de Trabajo y Previsión Social*, STPS) coordinate these efforts.

The general education strategy may not in itself suffice to deal with the widespread problem of poverty, and acute poverty in particular. A large share of the population still has only very limited access to basic education and health services; and poor nutrition and health hinder students' cognitive skills as well as workers' productivity. Moreover, an important factor behind the poverty and low educational achievement of the young generations is the poverty and low education of their parents. While, in the medium-term, the best way to alleviate poverty is to enable people to benefit from education and work opportunities, in the shorter run, specific policies are called for to ensure that the basic needs of the most vulnerable groups be met and that the negative dynamics that keep people in poverty be reversed (Annex I.D).

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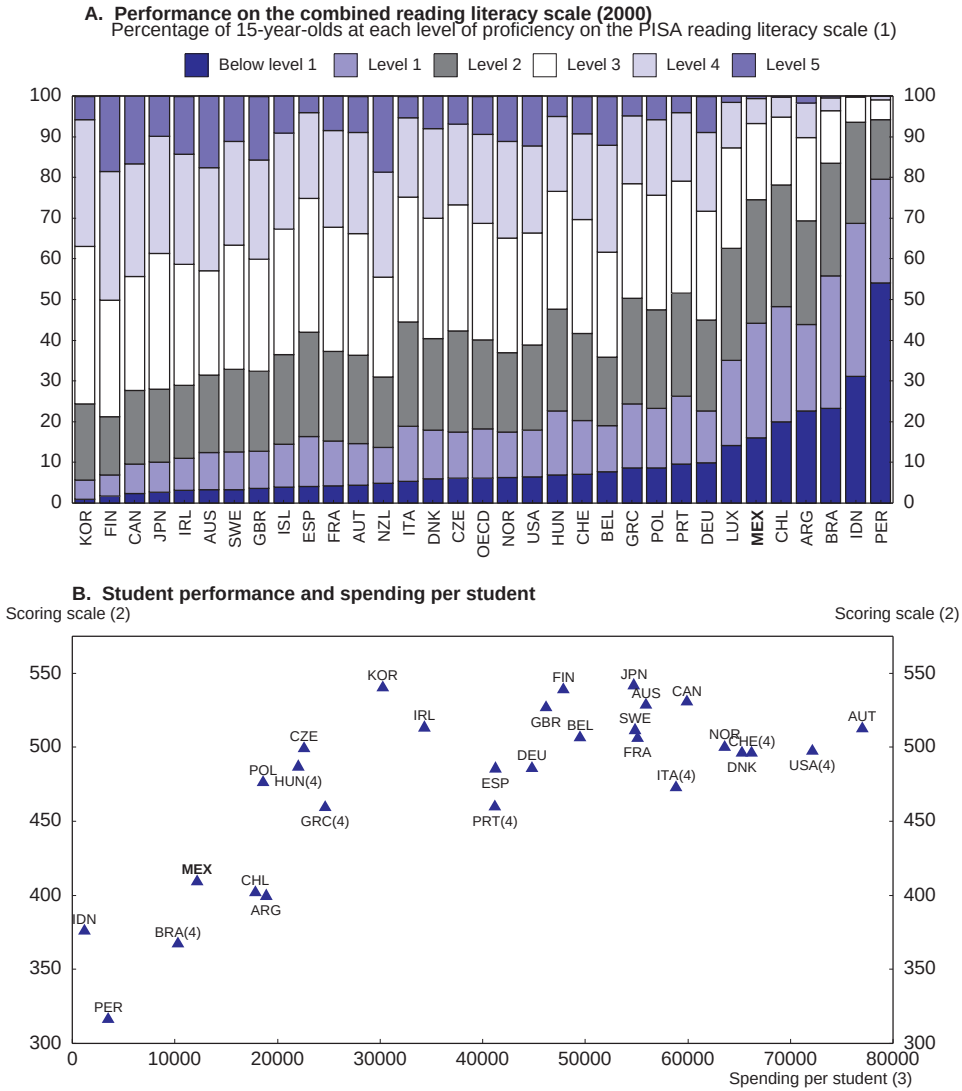
Figure 19. **Student performance in selected countries**



1. For the definitions of levels see OECD, *Literacy Skills for the World of Tomorrow: further results from Pisa 2000* (2003).
2. Average performance across the combined reading, mathematical and scientific literacy scales in 2000.
3. Cumulative expenditure on educational institutions per students up to age 15 in USD converted using PPPs in 1999. (1998 for Brazil, 2000 for Indonesia.)
4. Public institutions only. For the United States public and independent private institutions only.

Source: OECD, *Literacy Skills for the World of Tomorrow: further results from Pisa* 2000, 2003.

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The new challenges ahead for the education system

There have been dramatic achievements in education…

Education has been the priority of Mexican public policies for a long time, and there has been major progress in increasing school enrolment, with a special focus on lagging regions. Programmes such as *telesecundaria* and PROGRESA/*Oportunidades*, as well as the compensatory programmes launched in the 1990s have played a decisive role in this regard (see Annexes I.D and III.A). The enrolment rates of a rapidly growing school-age population went up for all education levels from pre-school programmes to superior education (Table 6).[64] As a result, the average number of years of schooling attainment of the working-age population has increased.[65] The quality of education services has also improved, especially in the poorer states, as shown by the reduction in the dropout and repetition rates.

… but the coverage and quality of education services are still insufficient

Nevertheless, school enrolment remains too low, even in basic education. Compulsory education was extended from 6th to 9th grade starting in 1993, but enrolment in lower-secondary education is still only 85 per cent a decade later. There are also marked inequities in education, with some groups having particularly low access: indigenous population; marginalised rural communities; temporary agricultural migrants; "street children" in urban areas. Moreover, given demographic trends, as primary education cohorts start shrinking and the lower-secondary ones stabilise, the system will have to face larger cohorts at the upper-secondary school. This will require a change of focus towards 10th to 12th grades in general education, as well as the development of technical education.[66]

The quality of education services has not increased at the same pace as coverage, and it remains low. Dropout and repetition rates are still high for secondary education, resulting in poor terminal efficiency.[67] High dropout rates reflect partly financial problems and the high opportunity costs of enrolling children in schools for low-income families; but various international and national surveys suggest that they might also result from inadequate curricula and poor quality of education services.[68] According to the OECD 2000 Programme for International Student Assessment (PISA), Mexico ranks last in the OECD on the combined reading literacy scale for 15-year-old students, with 44 per cent of this age category at the lowest benchmark,[69] more than double the average OECD ratio. National surveys of achievements in mathematics and Spanish language in primary schools show that approximately half the students have not reached the objectives of the programmes they attended. The entrance examinations to upper-secondary and tertiary programmes also reveal weak mathematics and reasoning skills. In addition, the surveys show that the quality of education services is uneven both from one state to the other and within states.[70] While improving the

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Table 6. **Performance of education programmes in 2001-2002**[1]

Basic Education	Preschool	Primary			Lower secondary			
	Enrolment	Dropout	Repetition	Terminal efficiency	Absorption	Dropout	Repetition	Terminal efficiency
Total	67.9	1.6	5.7	87.7	93.4	7.3	19.7	77.7
Change from 1990-91	12.3	–3	–4.4	17.6	11.1	–1.5	–6.8	3.8
Two richest States								
Federal District	77.7	0.7	2.1	94.6	104.9	6.7	19	82.9
Change from 1990-91	5.1	–0.8	–3.8	5.8	–2.1	–2.3	–18.3	9.5
Nuevo Leon	71.6	0.8	2.6	94.2	98.5	2.1	16.8	92.2
Change from 1990-91	12.3	–2.1	–2.8	10.7	6.6	–4.3	3.4	9.9
Two poorest States								
Oaxaca	70.5	2.7	10.9	82.0	91.9	8.1	15.1	77
Change from 1990-91	13.3	–3.7	–6.7	25.8	22.4	0.3	–6.8	5.1
Chiapas	71.2	2.7	10.8	74.7	88.6	5.6	12.1	81.1
Change from 1990-91	29.5	–11.2	–4.3	36.7	14.1	–2.8	–10.1	6.5

	Technical				Upper secondary[2]				Superior
	Absorp-tion	Dropout	Repeti-tion	Terminal efficiency	Absorp-tion	Dropout	Repeti-tion	Terminal efficiency	Absorp-tion
Total	11.8	25.4	25.4	46.6	84.6	15.8	39.2	58.8	87.2
Change from 1990-91	–2.6	0.3	–2.9	8.8	23.6	–1.6	–8.4	–1.3	17.5
Two richest States									
Federal District	19.9	30.6	40	37.6	115.7	19.0	53.6	44.8	127.3
Change from 1990-91	2.9	1.5	6.0	1.4	34.2	3.1	–5.0	–18.9	49.2
Nuevo Leon	30.4	30.7	18.5	42.3	74.8	21.0	48.9	56.1	106.4
Change from 1990-91	7	9.4	2.5	3.9	11.4	–1.2	–9.6	–1	16.9
Two poorest States									
Oaxaca	5.0	21.6	34.7	49.0	81.9	17.7	42.5	57.9	70.0
Change from 1990-91	–5.2	–2.7	–1.0	–9.6	26.3	–4.0	3.2	2.0	5.3
Chiapas	6.9	33.4	32.1	45.2	86.1	11.3	29.1	60.9	53.0
Change from 1990-91	–5.4	24.5	0.4	10.7	31.3	3.7	–6.6	5.8	3.5

1. 2002-2003 data were not used here since they are still estimates.
2. Includes general and technical programmes.
Source: Ministry of Education.

quality of lower- and upper-secondary education is a critical step for developing access to tertiary education, the quality of tertiary education itself and its matching with employers' needs are also sources of concern. At that level, dropout rates are also high and time for completion is relatively long. Combined with a heavy administration, this might explain why the ratio of expenditure per student on tertiary education to educational expenditure per student in primary education is twice the OECD average. Moreover, it seems that a large number of students are not able to find a job in line with their specialisation.[71]

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Low quality of education services results from the combination of: *i*) outdated curricula with little relation to future employers' needs and little emphasis on active learning; *ii*) poor school infrastructure and lack of investment in new equipment, especially for upper secondary education, including laboratories, libraries and other equipment that could improve access to information technologies (among 15-year-olds, 23 students share one computer *vs.* 13 on average in the OECD); and *iii*) weaknesses in school administration and teacher careers reducing incentives to improve performance.[72] Several initiatives were launched over the past 10 years to deal with these issues, including the teachers' career programme (*Carrera Magisterial*) launched in 1992 for basic education and more recently the programme of performance incentives for teachers at higher level.[73] However, the system still does not provide the right incentives for teachers to perform better, train, or work in difficult areas.

The tight budget constraint has also been an important limitation on education outcomes. Although spending on education accounts for more than one fifth of total public spending (compared with 13 per cent on average in the OECD), in proportion to GDP it is the lowest in the OECD, and is especially low when corrected for the relatively large share of the youth (aged 5-29) in total population (Figure 20). Moreover, according to OECD estimates, expenditure on primary, secondary and upper secondary education even fell as a percentage of GDP between 1995 and 2000. To reach enrolment rates similar to the OECD average, Mexico would need to increase resources by about 2 percentage points of GDP and this does not include the required improvement in quality and equipment detailed above. More cost-efficient education spending can however contribute to easing the financial constraint, as argued below.

Current reforms and further options to improve human capital

Strengthening education services

Following previous administrations' efforts, several important actions have been undertaken since 2001 to address the issues mentioned above and further increase the coverage and quality of education services (see Annex III.A). Keys actions include:

- Scholarships for low-income families have been increased via PROGRESA/*Oportunidades* which now concerns also upper-secondary students.
- *Escuelas de calidad* provides government grants for schools that are committed to, and succeed in, increasing quality (see Annex III.A).
- Modernised curricula, better integrated from one level to the other and more in line with students' and future employers' needs, and programmes to retrain teachers, especially in upper-secondary, are under preparation, and pilot implementation is projected for 2004-2005.

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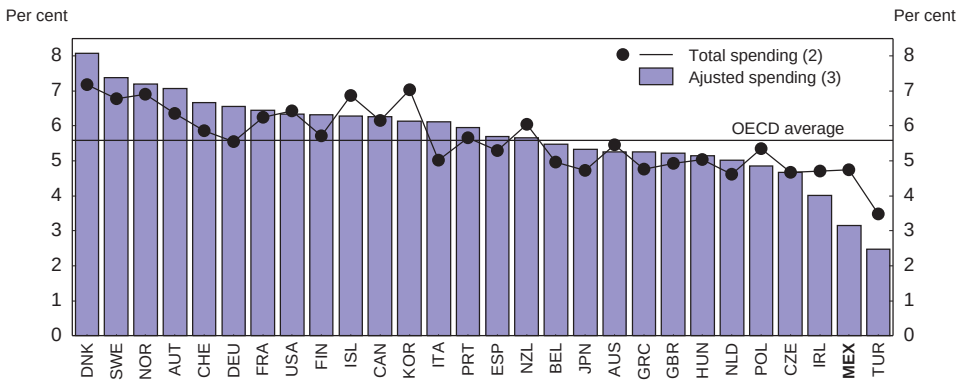
Figure 20. **Expenditure on education in OECD countries**[1]

Per cent of GDP



1. In 1999 or nearest year available. Countries are ranked according to adjusted spending in that year.
2. Public and private spending.
3. Adjustment to take into account the difference in the proportion of population aged 5 to 29 of each country relative to the OECD average.

Source: OECD, *Education at a Glance*, 2001.

– In August 2002, an institute to evaluate all institutions (except in superior education) was created (INEE, *Instituto Nacional de Evaluación de la Educación*). The recent law on transparency will ensure that these evaluations are made accessible to the general public.

– The education administration is under review to identify areas for improvement and make concrete proposals for its restructuring (regarding especially further decentralisation) by 2005.

– Several initiatives have been also launched to promote use of ICT in education programmes within the e-Mexico initiative (see Annex III.C).

While the government strategy to address quality and coverage problems in education services is going in the right direction, some recent decisions are questionable and several other actions should be considered to complement this strategy (these recommendations are summarized in Box 10.

A law has been passed targeting public education spending at 8 per cent of GDP by 2006 (compared with below 5 per cent in recent years),[74] a ratio that is broadly in line with the National Education Programme. However, the availability of these additional resources will clearly depend on progress on the fiscal reform (see Chapter II), and it is hard to see how the volume of resources could be raised so rapidly and be allocated efficiently. Whether these additional resources for education are made available or not, given the importance of the sector in public spending, it

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is necessary that the authorities clearly define spending priorities and make education programmes more cost effective. Most resources allocated to education go to staff compensation (Mexico showing the highest share in the OECD, at above 90 per cent together with Turkey and Portugal) while resources for investment (infrastructure and teaching equipment), although also relevant for the quality of teaching, are scarce.[75] It is therefore important to review education expenditure and examine possible reallocation of human and financial resources to priority areas.

The recent Congress initiative to make pre-school education compulsory starting in 2004-2005 seems unwise. As a result, by 2008, compulsory basic education will be raised to 12 grades, longer than in most other OECD countries. P*er se* developing pre-school programmes is known to have some positive effects as it usually results in higher literacy scores later at school and could allow mothers to work more and increase family income. However, in Mexico, this initiative is likely to divert scarce resources from necessary improvements of the quality and coverage of the already compulsory 9 grades.

Improving efficiency requires a better evaluation of institutions, programmes, and teachers and a close monitoring of the implementation of reforms. Further efforts should be made in some specific areas. In particular, it is important to accelerate work of the INEE and the publication of evaluation results. To make schools accountable, there is a need to go beyond the *escuelas de calidad* programme and, in particular, sanction poorly performing schools.

The low efficiency of vocational programmes, due to high dropout and repetition rates, needs to be dealt with and the introduction of apprenticeship contracts or alternation programmes could be considered to facilitate job market entry of school leavers, while providing better incentives to stay in the programmes. New financing options should be investigated to improve access to tertiary education. In Mexico, even more than in other OECD countries, children from poor and less-educated families have limited access to tertiary education. In this context, low tuition fees result in a massive redistribution in favour of middle and upper-middle class families. They tend therefore to be a regressive form of public spending and money could be used elsewhere. It would be reasonable to expect students to pay a portion of fees, while increasing spending on grants for students from poor backgrounds and implementing a student loan system, possibly backed by the government, for low-income families. For now, there is no student loans market as such, although the high returns on tertiary education suggest that such a system would be profitable.[76] Such reforms should be considered, even if it requires amending the constitutional requirement to provide free education up to the tertiary level.

E*xpanding labour force training*

Enhancing human capital requires also specific programmes to train the labour force. Lifelong learning provides an opportunity for individuals to complement

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previously received education or training. It is a major challenge for Mexico, where 8 per cent of the working-age population did not receive any education, 40 per cent left school at the primary education level, and where the young currently entering the labour force are still characterised by low school achievement. In many cases, teaching, including literacy and basic education, and training could help bring the working-age population out of poverty traps by providing the basic skills needed to move into better positions in the formal market. Training could also play a decisive role in increasing labour force mobility across sectors and facilitating the adoption of new technologies and the restructuring of the Mexican economy under the pressure from competition. However, only one fourth of the labour force has ever received some form of training from their employers and, among these, qualified workers are the main beneficiaries, while unskilled people received only 10 per cent of total training provided by firms.[77] In this context, public intervention is needed to improve access of the under-educated workers to training.

The National Institute for Adult Education (*Instituto Nacional para la Educación de los Adultos*, INEA), and the newly created National Council of Education for Life and Work (*Consejo Nacional de Educación para la Vida y el Trabajo*, CONEV*y*T) are devoted to articulating and improving adult learning coverage. The Education Ministry also provides basic education to adults. Moreover, several public programmes have been launched in the last decade in Mexico such as PROBECAT (renamed SICAT), a training programme for workers who lose their jobs, and CIMO (renamed PAC), that subsidises on-the-job training in small firms; both are managed by the Labour Ministry (Table 7).[78] These programmes form a substantial part of active labour market policies. Nevertheless, total spending on active labour market policy remains by far the lowest in the OECD. Adult training and validation of qualifications are an integral part of the National Education Programme, which insists on the need to promote both training for work and on the job, as well as skills development and certification, equivalencies and revalidation of studies and labour qualifications. In these areas too, there is a strong need to insure cost efficiency and a good matching between training and future employers need. The willingness to link the education system with industry and individuals was already at the origin of the Mexican National System of Competency Standards, Skills Testing and Certification (CONOCER)[79] created in 1995 under the joint supervision of the Ministries of Education and of Labour, and such efforts should continue. The new labour law proposal emphasises the need to promote private training by firms, which is also welcome. It considers reducing bureaucracy associated with training procedures and making training compulsory for both employees and employers instead of only employers, as in the current law (see Box 9 below).

Towards a more efficient labour market

A well functioning labour market is needed for achieving high growth, translating on-going human capital improvement into effective poverty reduction

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Table 7. **Active labour market policies**

Thousands

	1990	1995	2000	2001	2002*
Public employment services					
Job applications	241.2	533.5	431.7	521.8	570.7
Vacancies	240.6	326.4	476.2	452.7	457.2
People placed in a job	n.d.	123.7	169.4	173.4	183.2
Total expenditure (millions of pesos)	n.d.	16.8	84.3	91.7	114.7
Training support for SMEs (CIMO)[1]					
Supported workers	11.4	104.6	733.9	333.5	233.2
Enterprises supported	33.1	368.1	329.5	98.2	51.8
Enterprises receiving other services					
Total expenditure (millions of pesos)	1.8	74.4	201.5	172.2	134.9
Training scholarships (PROBECAT)[2]					
Scholarships for the unemployed	64.1	412.3	593.2	397.0	230.2
Trainee courses	2.1	23.0	27.1	18.8	12.2
Total expenditure (millions of pesos)	50.6	476.6	1 249.9	1 184.8	740.9
Temporary farm workers to Canada					
Job applications	. .	. .	. .	13.0	12.6
People placed in a job	5.1	4.9	9.2	10.5	10.7
Total expenditure (millions of pesos)	0.18	0.24	1.1	1.3	1.4
Support to formal employment[3]					
"Apoyo a buscadores de empleo"					
Job applications	–	–	–	–	40.5
People placed in a job	–	–	–	–	14.1
Total expenditure (millions of pesos)	–	–	–	–	64.0
Support to agricultural workers[3]					
"Apoyo a jornalero"					
Job applications	–	–	–	–	35.1
People placed in a job	–	–	–	–	8.8
Total expenditure (millions of pesos)	–	–	–	–	40.0
Productive Investment Projects[3]					
Job applications	–	–	–	–	7.1
People placed in a job	–	–	–	–	5.4
Total expenditure (millions of pesos)	–	–	–	–	76.7
Temporary employment programme, PET					
Workers supported	–	495.2	495.4	533.2	..
Total expenditure (millions of pesos)	–	710.2	1 262.1	1 307.0	..

* Preliminary data.
1. In 2002, the CIMO programme became PAC.
2. In 2002, PROBECAT became SICAT.
3. Programmes started in 2002.

Source: Ministry of Labour and Social Welfare (STPS); OECD, *Employment Outlook*.

and, more generally, ensuring that benefits of growth are shared more widely across the population. The adaptability and mobility of labour is also key in minimising the potential hardships created by structural changes, while ensuring an efficient re-allocation of labour resources across sectors and firms.

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Labour market performance and issues

On many measures, the performance of the Mexican labour market appears relatively satisfactory, when compared with other OECD countries. Open unemployment is low, real wages have been quite flexible in the last decade and the labour force relatively mobile.[80] Overall, the market has demonstrated its ability to cope with shocks. However, this performance has been achieved *de facto* through the expansion of the informal labour market (see Chapter I). Moreover, a need for more adaptability is emerging. Real wage flexibility becomes more difficult to achieve in a low inflation environment and the labour market has to cope with increasing competition pressures within NAFTA and from low-income countries.

The informal labour market represents an unavoidable buffer in a country where there is no unemployment insurance. However, its persistent high share of labour raises several concerns:

- It reveals the inability of the Mexican economy to generate enough formal jobs to absorb the growing labour force.
- As mentioned in Chapter I, most informal jobs are of low productivity.
- The existence of an informal sector contributes to the reduction of the tax base, implying higher tax rates (for the same revenue) for formal workers; and more importantly, it increases taxpayers' perception of unfairness.
- Informal workers do not contribute to – nor benefit from – the social security system. This has two consequences: first it results in limited coverage of informal workers for health care provided by the social security institute IMSS; second, as workers appear to move frequently in and out of the formal sector, their accumulated pension rights are limited.[81]
- The population in the informal labour market does not have access to on-the-job training opportunities, which limits human capital development opportunities.

Therefore, the main problem that labour market policy has to tackle is to reduce the relative size of informal activities. While a better enforcement of legal requirements through an increase in the number of tax and labour inspectorates could help reduce certain types of informal activity, the priority is to address the main factors behind the expansion of the informal labour market. These include relatively high non-wage costs of formal activities for firms, and a perception by workers that the rewards for remaining in the formal labour market (access to health, pension and rudimentary social programmes) are marginal relative to their costs.

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Labour costs are too high in the formal sector

As in many other OECD countries, a major reason for the expansion of the informal activities has been to avoid the tax wedge (*i.e.* the difference between labour costs paid by employers and take-home pay of employees). According to evidence from OECD countries, lower labour taxes, including social security contributions, are generally associated with stronger employment performance. Whether or not the impact on employment translates into higher (or lower) labour supply depends on the extent to which benefits (or cost) accrue to employees take-home pay. If take-home pay is too low, Mexican workers would choose to move into informal activities; in most OECD countries, they would move into unemployment or out of the labour force, depending on the characteristics of the safety net.[82] If firms are not able to pass high non-wage labour costs on net wages to employees, they would reduce labour demand, at least in the formal sector.

Although social contributions in Mexico have decreased following the 1997 reform, they remain a relatively high proportion of total wage costs for many workers.[83] Compared with other OECD countries, average social contributions per employee are relatively low (Figure 21). But this picture is misleading. A regressive pattern of employers' contributions results from much higher contribution rates for workers earning less than the "Average Production Worker" used as a benchmark for international comparison (Figure 22).[84] Furthermore, there are additional contributions weighing on labour costs (Table 8). An important one is the housing fund INFONAVIT, financed by a compulsory 5 per cent payroll contribution, with the purpose of providing subsidised loans for housing. In addition, a federal payroll tax was introduced in 2002; it was then suppressed in 2003.[85] Experience of other OECD countries suggests that payroll taxes tend to have negative effects on employment, affecting especially low-productivity workers, who are pushed out of the workforce. Fringe benefits further increase labour costs for workers with permanent contracts; and firms are required to share 10 per cent of their profits with their employees.[86]

Strict employment legislation contributes to increasing labour costs and creates incentives to opt for informality, especially for SMEs. Mexico is one of the few OECD and Latin American countries where there has been no major change in labour legislation for decades. The absence of probation period combined with relatively high severance payments[87] increases the cost of hiring people on permanent contracts. At the same time, fixed-term and temporary contracts are strictly regulated, limiting the possibility of firms to circumvent hiring costs in the formal market.[88] Overtime hours are paid double the normal time rates and therefore much more than in most other OECD countries. In addition, red tape and administrative burdens reinforce incentives to hire workers informally. Overall, even if in practice its application is patchy, formal labour market legislation

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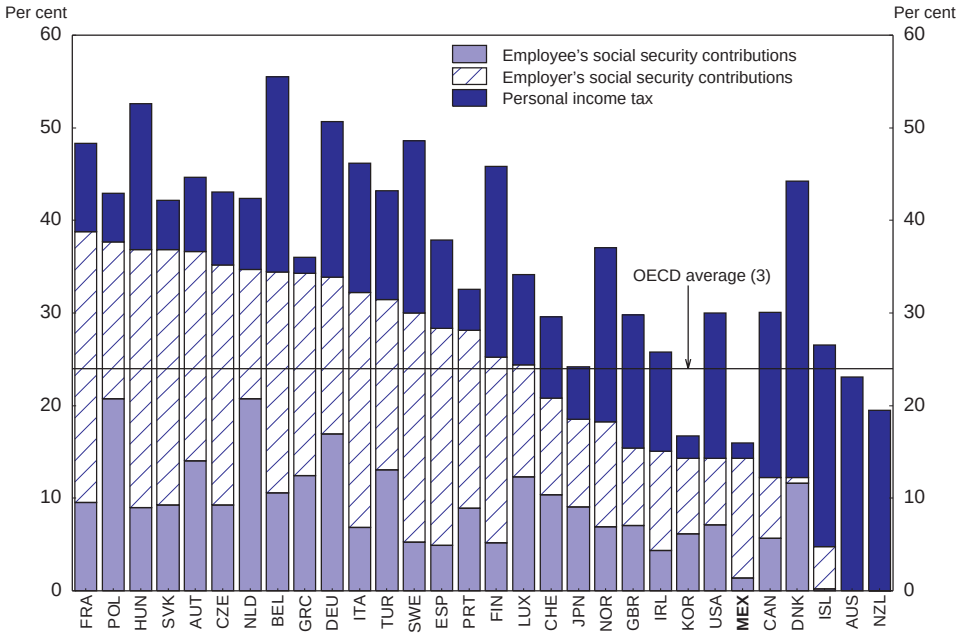
Figure 21. **Tax wedge on labour in OECD countries**[1]
Per cent of gross labour costs[2]



Note: Countries are ranked according to the total social security contributions, from the highest to the lowest. For Mexico, contributions for discharge and old age insurance, to INFONAVIT and SAR are not included (see Figure 22 and Table 8)

1. In 2001. For a single individual at the income level of the average production worker. Data are based on estimated wage levels of the average production worker.
2. Gross wage plus employer's contributions.
3. Unweighted average of total social security contributions.

Source: OECD.

regarding entry, exit and permissible contracts appears stricter than in many other OECD and Latin American countries.

Work in the formal market has limited advantages

Besides the cost of social contributions itself, there seems also to be a problem with the way social benefits (access to health and housing, pension and rudimentary social programmes) provided in the formal market are perceived by workers, raising the question of the quality relative to costs. Although informal employment is largely involuntary, resulting from the lack of formal job opportunities, anecdotal evidence and some surveys have shown that it can also be the

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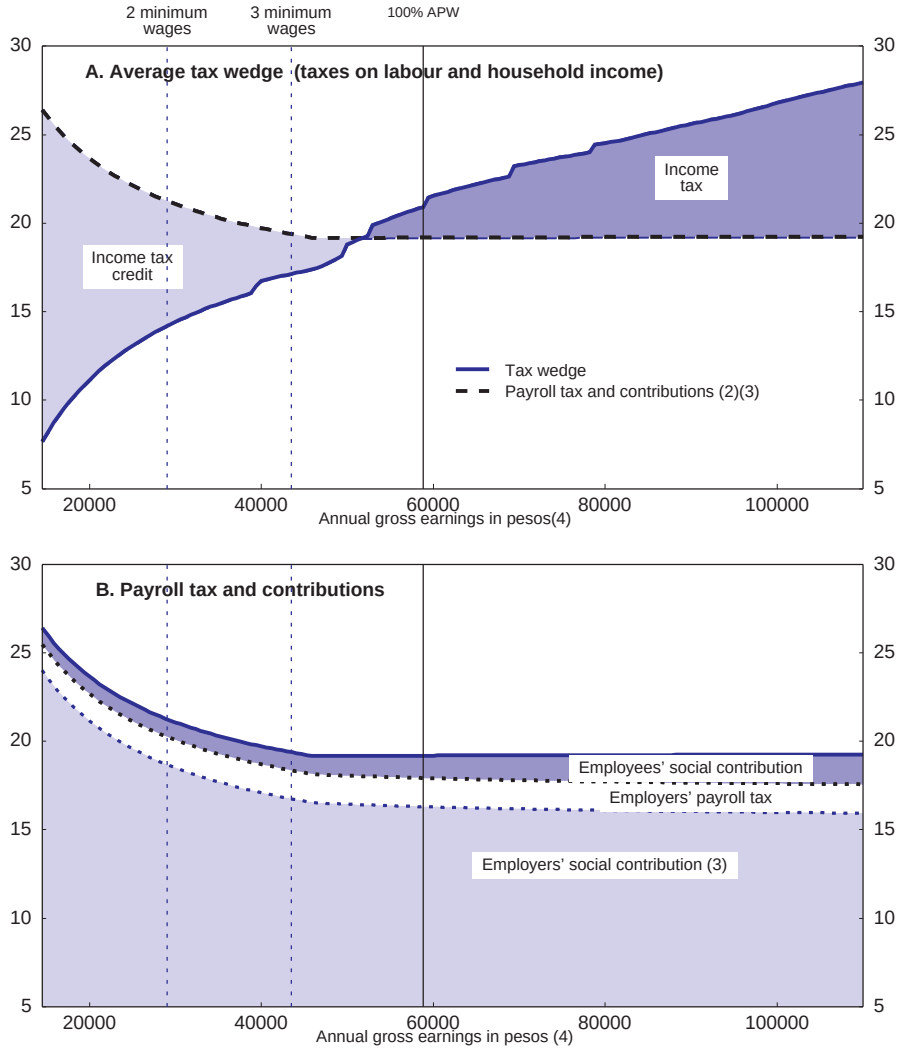
Figure 22. **Average tax wedge by level of income, 2002**

Per cent of gross labour costs[1]



1. Gross wage plus employers' contributions, for a single worker.
2. Employers' and employee's contributions plus employers' payroll tax.
3. Includes contributions made by employers to the retirement Fund (SAR) and the housing Fund (INFONAVIT) as well as for discharge and old age insurance.
4. The first left point on the scale refers to the minimum wage. The APW line is the income received by the "Average Production Worker".

Source: OECD.

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Table 8. **Social contributions and fringe benefits**

	Employer	Worker	Government
A. Payroll contributions: (as a percentage of gross earnings)			
IMSS			
Health and maternity care	Fixed amount[1] + 4.04[2]	1.36[2]	Fixed amount[1]
Salary replacement	0.7	0.25	0.05
Inabilities, life insurance, old-age severance[3]	4.9	1.75	0.35 + Fixed amount
Retirement (SAR)	2	–	–
Daycare centres	1	–	–
Housing (INFONAVIT)	5	–	–
Occupational risks	2.6[4]	–	–
	Annual bonus (days)		**Wage equivalent (per cent)**
B. Fringe benefits			
Rest days paid	52		14.3
Obligatory paid holidays	7		1.9
Year-end bonus (minimum)	15		4.1
Vacation bonus	1.5		0.5
Paid vacation	6		1.6
Total	**81.5**		**22.4**

1. For employers, the fixed amount is egual to 16.5 per cent of minimum wage in the DF (Federal District) in 2002. According to the 1997 Law on Social Security, this percentage is increased every year to reach 20.40 per cent by 2007. For the government, the fixed amount was 6.17 pesos daily in 2002 (calculated as an initial 13.9 per cent of the general daily minimum wage in the DF indexed on the CPI).
2. Percentage (for 2002) of the difference between gross earnings and 3 minimum wages in the DF. According to the 1997 law on Social Security, both rates are decreased every year to reach 1.10 per cent for employers and 0.4 per cent for workers by 2007.
3. With a ceiling of 20 minimum wages in the DF for the contribution base. Fixed amount for the government was 2.44 pesos daily in 2002.
4. IMSS estimate of average rate considering Class III as the average risk premium.

Source: IMSS; Ministry of Labour and Social Welfare (STPS).

result of a choice by workers.[89] The 1997 IMSS reform, combined with buoyant economic activity until 2000, might have contributed to the stabilisation of the informal labour share. However, there has been no marked reduction of the share. Further efforts to improve the quality and cost efficiency of social services are needed to make the formal jobs more attractive. Although in all countries, many workers pay more in individual contributions than they are likely to get back in individual benefits, the issue seems more acute in Mexico, where the services provided are perceived as of poor quality, even for the ones receiving relatively more benefits. Services are still insufficient, not only regarding IMSS social services such as health or day-care, which workers who can afford private services do not use, but also pensions and INFONAVIT benefits. On pensions, one problem has been the high commissions of private managers (AFOREs), and more generally the low rate of return on savings accounts. As a consequence, the pension

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benefits to which workers are entitled are in many cases no higher than the minimum pension.[90] The minimum guarantee being provided after 24 years of work, the pension scheme provides no incentives for low-income workers to remain longer than 24 years in the formal sector. Measures have recently been taken to lower fees charged by AFOREs, and returns should increase following the changes in investment limits presented in Chapter II. Concerning INFONAVIT, it is estimated that while the majority of workers bear the cost of its financing, only a small share has benefited from the loans. Since 2001, the situation has improved somewhat as all workers have been entitled to use the money they put into INFONAVIT as a collateral for loan credit.

In the current legal framework for the labour market, workers have little incentive to seek formal employment. The flexibility of working hours and the possibility to work part-time are for instance mentioned in surveys as factors for deciding to work outside the formal labour market; self-employment or informal employment is sometimes a response to the lack of career perspectives in the formal sector (where promotion is based on seniority).[91] The organisation of industrial relations, which in particular includes an exclusion clause (*i.e.* requirement of official union membership to new hires), might also have affected decisions to work in the formal market.

Making the formal market more attractive

Reviewing the social tax and benefits

In Mexico, a lower tax wedge would be expected to translate into a more favourable split between formal and informal activities. Several options could be considered to achieve this result. First, as mentioned in Chapter II, an in-depth tax reform is necessary. It should focus on indirect taxes, continue the planned reduction in income tax rates and avoid payroll taxes, so as to prevent negative effects on employment. Second, as advocated in previous OECD *Economic Surveys of Mexico*, there is a need to review the financing and operation of INFONAVIT. Third, employers' contribution rates to IMSS according to the salary scale should be reviewed, as they are relatively high for workers in the lowest wage categories. This pattern puts Mexico's social security contributions above most Latin American countries for low-paid jobs, which account for an important share of the labour force (Figure 23) and are the more likely to move to the informal sector.[92] The effect of relatively high contributions on the tax wedge for low-income workers is compensated by the existence of a progressive "salary tax credit" and a progressive income tax. About 70 per cent of wage earners paid no tax at all in recent years, but rather received payments through this means. There are various options to reduce labour costs at the low end of the salary scale such as wage subsidies or making employers' contributions proportional to income up to a certain

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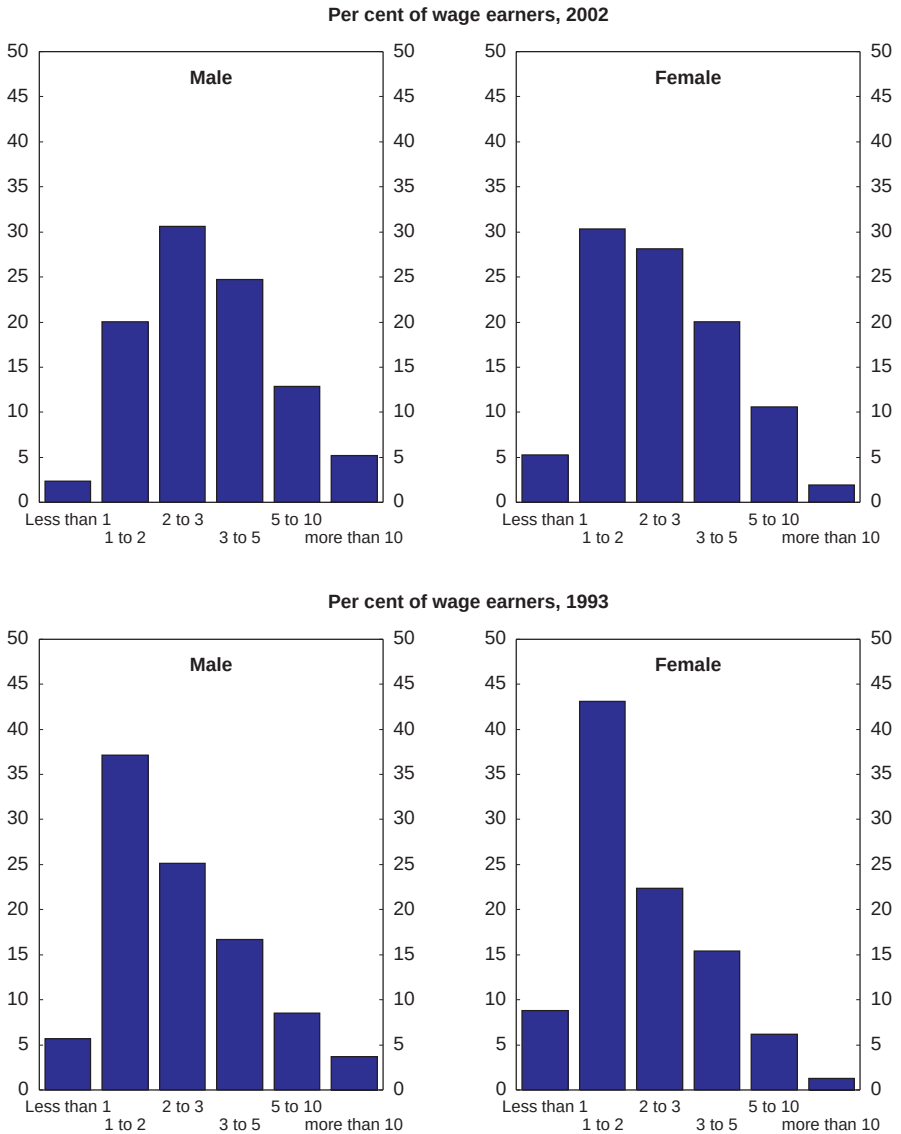
Figure 23. **Wage distribution in Mexico**

Multiple of the minimum wage in urban areas[1]



1. In the formal and informal sectors in areas of more than 100 000 inhabitants.

Source: INEGI, *Encuesta Nacional de Empleo*, 1993 and 2002.

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low ceiling. However, in the current context, there is little room to finance such measures, except by improving the efficiency of the IMSS, which would also help improve the effective and perceived quality of the health care system.[93]

Modernising labour market legislation

With almost half of the labour force working with near impunity in the informal labour market and evidence of regular breaches in the application of the law, the labour law as it stands does not serve its aim to protect workers.[94] Indeed, it probably contributes to the development of informal activities. A reform is therefore indispensable. According to the Bank of Mexico survey on business opinion, labour market reform ranks in the top three reforms deemed to be necessary for domestic and foreign investment in Mexico to increase. It would be preferable to have less rigid rules, but ones that are applied, than the current legislative framework. Easing employment protection legislation, hiring procedures, and restrictions on atypical forms of employment, and modernising industrial relations could *de facto* increase the average protection of workers by widening the scope of the formal sector. Such moves would also increase firms' ability to adjust better to shocks in the formal sector and foster efficient resource allocation and productivity growth. In particular, part-time contract regulations should be eased not only to increase firms' ability to respond to changes in demand, but also to make this market more attractive for women.[95] The remuneration of overtime hours should also be reduced. Lastly, given currently low compliance, the requirement for firms to share 10 per cent of their profits with employees should be abolished.

A difficult issue is the one of the reduction of dismissal costs, which can be seen as substitutes for almost non-existent unemployment benefits. Mexico is the only OECD country that does not have a system of unemployment insurance. Only a rudimentary system of unemployment support exists (Annex III.B). Putting in place a comprehensive safety net on the scale which exists in many European countries is not an option, because experience has shown that it can create disincentives to work and lead to unemployment traps, as well as being potentially costly. However, combining a more systematic form of job loss support with an easing of firing constraints could make the Mexican formal labour market more fluid and efficient and better able to adapt to new technologies. The existence of limited benefits would facilitate formal employment search in line with workers' skills, without giving incentives to remain unemployed. Easing firing procedures and lowering severance payments could reduce labour costs (if designed to more than compensate for the existence of an insurance scheme) and would in any case allow firms to adjust better to shocks. This would in turn increase formal employment demand. It would also facilitate voluntary changes in jobs.

Several options could be considered to extend job loss support while relaxing employment protection. In particular, Mexico could consider creating

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Box 8. Main characteristic of unemployment insurance savings accounts in Chile

Contributions:

Employees: 0.6 per cent of wages to individual accounts.

Employers: 1.6 per cent of wages to individual accounts, and 0.8 per cent of wages to a solidarity fund.

Withdrawal conditions:

Withdrawal reasons include job loss due to dismissal, end of a fixed-term contract, or voluntary separation. Workers have to have been working for at least 12 months to be entitled to access their savings. Workers can draw 30 to 50 per cent of their previous wages depending on the number of years they have been employed for up to five months.

Insufficient balances are covered by the join solidarity fund financed by employer contributions, in case of dismissals only.

Balances upon retirement are rolled into the workers' pension accounts.

unemployment insurance savings accounts. Such schemes have been adopted by several Latin American countries including Chile (Box 8).[96] Formal workers and employers would be required to save a small percentage of wages on these accounts. Workers could draw on the account, in case of dismissal, including for economic reasons.[97] In some countries, workers can also withdraw money from their account when they resign or to finance education or for health reasons. But in Mexico, where the government would not be able to check if beneficiaries of the scheme resign to take on an informal activity, money withdrawal should not be allowed on resignation. When workers do retire, the unemployment savings accounts would be rolled into pension accounts, thereby creating an incentive to minimise length of unemployment periods so as to increase future savings. To avoid creating disincentives to seek a new formal job, the specific features of the scheme would have to be thought through carefully, in particular the contribution rates of workers and employers, the length of previous formal employment to be entitled to draw money (which has to be long enough to limit moral hazard) and the amount and timing of withdrawals, which in any case will have to be limited.[98] When the system is introduced, care should be taken to link the new scheme with public employment services and with training programmes offered to unemployed workers. In the meantime, the cost efficiency of these training programmes

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should be improved further. Moreover, the PET (*Programa de empleo temporal*) has to be reviewed. It seems quite expensive and, although the Programme was meant essentially as an income support instrument in times of emergency, the public work projects may have not always been as productive as could be wished.[99]

The reform under discussion

The debate on the modernisation of labour legislation that started in 1997 (with the signing of "the agreement for a New Labour Culture") resulted in 2002 in a formal law proposal that is still under discussion in Congress. The reform proposal could represent an important step in making the formal market more attractive. Its main aspects are described in Box 9. Several items address the issues mentioned above, and are in line with OECD recommendations re-iterated since 1996 to make the labour market work better. These include: the introduction of probation periods; first steps in easing regulations concerning permanent contracts (with the introduction of a system of cumulative work hours); efforts to modernise industrial relations; emphasis on merit-based promotions; some simplification of bureaucratic requirements for SMEs; measures to promote labour force training; and measures to reduce corruption and legal uncertainty. Some of the proposed measures, however, might have unwelcome effects, for instance as regards obligations in subcontracting.[100] Moreover, the new law does not address many of the employment protection issues mentioned above, in particular the introduction of elements of flexibility in labour contracting.

Beyond changes to labour market legislation, other initiatives should be taken in the short to medium-term to improve the balance between the formal and informal sectors. They include enhancing the quality of services provided to formal workers and reducing the tax wedge, in particular employers' social security contributions for low-skilled workers. Even though the recommendations, summarised in Box , could in principle be implemented on an "*à la carte*" basis, such measures would be more balanced if implemented as a whole and would probably prove more effective than individual measures taken separately.

Strengthening competition policy and moving forward on the regulatory reform agenda

Ensuring a pro-competitive environment is essential to increase investment and productivity in Mexico. OECD cross-country analysis suggests that pro-competitive regulations generally improve resource allocation and result in higher output and investment, more important innovation efforts and a better diffusion of innovations.[101] Moreover, domestic product-market regulations that impose unnecessary costs on businesses and create barriers to entry tend to affect negatively FDI.[102] A competitive environment facilitates performance comparisons among firms; the associated risk of losing market share encourages the elimination of slack; and the need for

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Box 9. Key features of the labour law reform proposal

Probation period

- Introduction of a maximum 30 day trial period as a general rule and a 180 day trial period for managers and directors, as well as for professional specialized technical positions.

Modernisation of industrial relations

- Several measures aim at ending "white unions" and "protection contracts" practices.[1] Secret ballots are now the rule for elections to win collective bargaining authority, in strikes and to choose union leaders.[2]

Working hours

- Creation of a system of "bank of hours" enabling workers to benefit from longer, continuous vacation periods, while meeting production demands through a system of a cumulative work hours, provided that at no time is the worker forced to work excessively long hours.
- Flexibility to make use of official holidays any other day of the week such as the Monday or Friday closest to the specific holiday, provided there is a previous agreement between employers and workers.

Training

- Introduction of new types of contracts for initial training of workers. Workers hired under these contracts receive an income even if they do not have previous work experience. The use of these contracts is regulated and the period worked under these types of contracts may be considered as seniority if the labour relation continues.
- Reduction of bureaucracy for training procedures.
- Modernisation of Article 153 related to training and skill development in order to *i*) enable workers who request it to begin, continue or finish elementary school, high school or college; *ii*) allow training and skill development to be provided by the regular educational system, instead of trainers recognized by STPS.
- Under the Bill, not only do employers have to provide training and skill development, but workers are also required to receive training in order to improve their skills and income. (Article 153 A).

Others

- Skills should become the main criteria for promotion, before seniority.
- Guarantee of same right and obligations as those of regular workers for seasonal workers.

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Box 9. **Key features of the labour law reform proposal** (*cont.*)

– The law promote non-discriminatory practices consistent with ILO Guidelines.
– Measures to improve labour law enforcement and secure the legal framework.

1. "White unions" are phantom non-combative unions, the formation of which is traditionally supported by the employers. A "protection contract" is an arrangement reached between the official union and the employer to protect the employer from claims of the workers and independent unions.
2. According to the draft law in the case of union leader elections, the secret ballot is only a possibility, but discussions in Congress suggest that the law might be amended to make it compulsory.

firms to be cost efficient to survive pushes them to adjusting technology to best practices. While trade liberalisation and privatisation have already considerably increased competitive pressures on Mexican producers, these pressures have been felt on some sectors but not on others, and key industrial infrastructure development is still hampered by inappropriate regulations.

Reinforcing the institutional design

An institutional framework to protect market competition was set up in 1992 with the Federal Law of Economic Competition and the creation in 1993 of the Federal Competition Commission (CFC) in charge of enforcing it. Although the competition law is formally close to best practice in OECD countries, its implementation has faced resistance, and a competition culture has yet to become solidly established in Mexico. In this context, the powers and independence of the CFC, which has demonstrated its determination to defend competition and has already become a credible enforcement body, have to be strengthened. First, the CFC is significantly under-resourced given its responsibilities. It has fewer staff today than in 2000 despite a much heavier workload, and there is a pressing need for more, and more senior, lawyers. The CFC also needs more powers to make surprise searches for business records and to block anticompetitive decisions in the application of trade laws. It should also have the opportunity to comment publicly on regulatory decisions (with the regulator having a duty to respond publicly) and should be more involved in public debates on competition issues. It is necessary to propose a new law that would increase the autonomy and authority of CFC. In particular such a proposal should establish the CFC as a decentralized agency with budgetary independence from the Executive Branch.

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Some problems faced by the CFC are linked to broader issues concerning the legal and judicial system (see below). For instance, the constitutionally guaranteed *amparos* (a legal procedure for challenging the government's actions) have been increasingly used to delay administrative enforcement. The number of *amparo* cases in which CFC was involved increased from 122 in 1993-1997, to 513 in 1998-2002. Nevertheless, the Commission has been active and quite successful in the Courts defending its decisions and the constitutionality of the Competition Law.[103] The antecedents provided by these favourable outcomes should make it more difficult in the future to use the Courts to unduly postpone the application of the Law through the abuse of the legal system. The *amparo* process needs nevertheless to be streamlined and a single, expert appellate court for all competition matters, replacing the current rather complex system, is needed. CFC faces also difficulties in collecting the fines it imposes. In the 10 years to 2002, it has collected 9.7 per cent of imposed fines and revoked another 17.1 per cent, leaving uncollected the remaining 73.2 per cent (equivalent to US$23 million).

Progress in sectoral reforms

Because of their spillover effects on all activities, the development of infrastructures in telecommunications, transport, and energy (gas, oil, and electricity) is essential for growth and competitiveness. In all these sectors, investment should be stepped up to address the lack in infrastructure illustrated by Mexico's poor ranking compared with other countries, in particular for transportation and telecommunications (Figure 24). Better infrastructure would help the catching-up of lagging regions.[104] In some of these key sectors, even though liberalisation was implemented during the 1990s, important regulatory and competition issues remain to be solved. For instance, in both telecommunication and energy, sector regulators are still not in practice independent from ministries. Moreover, in energy, current legislation, including the Constitution, puts strict limits on private investment.

The reform of the electricity sector must move forward

Without an in-depth reform of the electricity sector, power shortages are likely to occur in the coming years and hamper economic growth. According to the Bank of Mexico survey on business opinion, in 2003, this reform area ranked first in requirements to see an increase in domestic and foreign investment (with 24 per cent of the private sector votes). Under the current framework set by the Constitution, the generation, transmission and distribution of electricity are the exclusive domain of the Mexican State, which is unable to finance investment on a sufficient scale to meet expected demand.[105] According to official estimates, in order to satisfy projected demand over the next 9 years, net generating capacity would have to be raised by over 50 per cent by 2011, and the transmission and distribution systems

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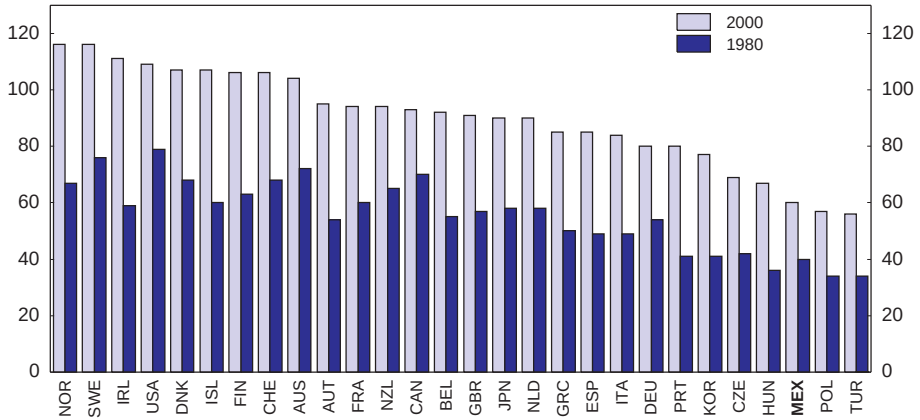
Figure 24. **Aggregate infrastructure indicators in OECD countries**
USA 1995 = 100



Source: Nicoletti *et al.* (2003), "Policies and international integration: Influences on trade and foreign direct investment", OECD *Economics Department Working Paper,* No. 359.

modernized. Under optimistic assumptions concerning the depreciation rate, this implies that annual investment should more than double by 2011, with 38 per cent of the new investment going to generation, 25 per cent to transmission, 22 per cent to distribution, the rest for maintenance and other. The Ministry of Energy considers that CFE (*Comision Federal de Electricidad*) could at best afford 47 per cent of total investment requirements.[106] The remaining 53 per cent would hence have to come from private investors. Private investment is allowed in electrical generation capacity only for self-supply and small-scale cogeneration, in which cases all excess production must be sold to the national power companies, and for the independent power producers (IPP) supplying power to CFE (see Chapter II). These schemes will not be able to provide the necessary additional capacities. In particular, the amount the IPPs are allowed to sell is limited and, in 2002, the Supreme Court declared unconstitutional a presidential decree to increase it.

The reform proposed by the previous administration in 1999, which considered splitting in small entities and privatising the two state-owned generation companies CFE and LFC (*Luz y Fuerza del Centro*), was not approved by Congress. In 2002, the current administration submitted a new proposal to Congress, which is less ambitious than the previous one. It does not consider privatisation, but instead relies on a new regulation of the two companies to create market conditions. As in the previous proposal, it is envisaged to separate the generation,

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transmission and distribution, with transmission and distribution remaining monopolies of the CFE and LFC under an independent system operator (ISO) control. The regulatory agency (*Comisión Reguladora de Energía* (CRE)) would be strengthened to ensure proper regulation of those entities. The generation market would be fully open to private producers, and users with consumption higher than 2 500 MWH would be allowed to choose their supplier. This proposal, while falling short of what is needed and in comparison with OECD best practices, is still politically difficult. It is also quite a sensitive issue for international capital markets, which tend to see the approval of this reform as a test of the Mexican authorities' ability to go ahead on the reform agenda. If the reform cannot be approved, the Ministry of Energy is planning, as a third best policy, to implement a new, fairer, tariff system for selling excess capacity in order to boost private generation.

Whether the reform is approved or not, there is a strong need to bring prices further into line with costs. The reduction in subsidies, which started in early 2002, should continue, and the subsequent increase in electricity prices will have to be offset by direct financial support to the poorest households. However, pressure groups are using *amparos* to block the 2002 price increase and, in April 2003, the Senate approved an increase in subsidies for residential electricity prices in some regions in the North and South-East ("with difficult climatic conditions"). This decision is clearly a step backward, and it can only further reduce the ability of CFE and FLC to invest in new capacity.

Increasing competition in the telecommunication sector

Mexico's telecommunications sector was liberalised in the 1990s, starting with the privatisation of TELMEX more than 10 years ago, and new private operators have penetrated the market over the following years. However, as in several other OECD countries, the incumbent has retained a dominant position. It still represents close to 70 per cent of the long-distance market,[107] 90 per cent of commercial fixed lines and 96 per cent of residential ones. In the mobile market, TELCEL, a subsidiary of America Movil has a market share of 77 per cent. The sector's performance is lagging. Mexico still has the lowest telecommunication density in the OECD (Figure 25) and one of the lowest in comparison to emerging markets in Latin America. The mobile lines to fixed lines ratio is among the highest in the OECD (1.6 versus 1 on average in the OECD). Tariff reductions have been slow in the 1990s. Since 2000, successive agreements on interconnection charges between TELMEX and other market participants have brought the charges more in line with international standards. Overall, however, telecommunication tariffs in Mexico, while on a declining trend, are still high compared with most OECD countries, increasing business costs (Figure 26). TELMEX, as the carrier with the largest market share of out-going calls, is also in a situation in which it still has the ability to establish the terms and competition on termination of international calls.[108]

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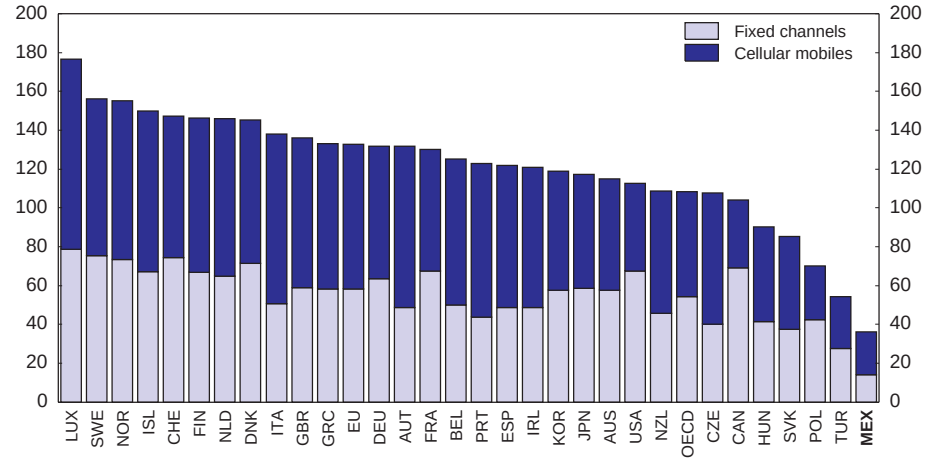
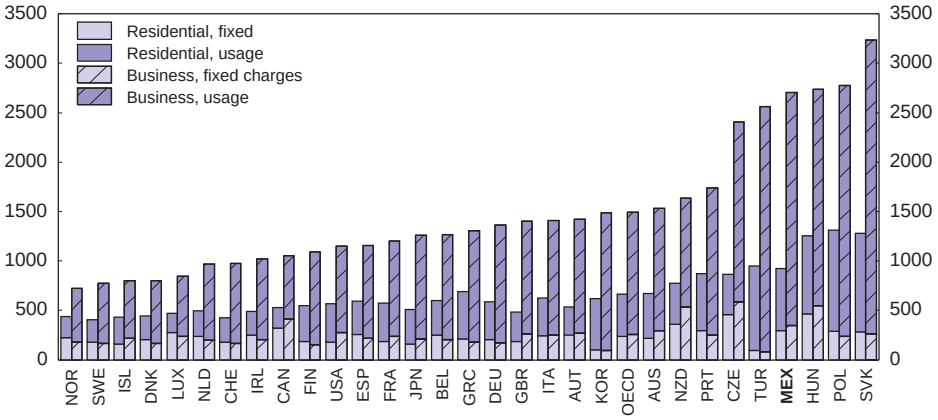
Figure 25. **Telecommunication networks in OECD countries**
Access paths per 100 inhabitants, 2001



Source: OECD, *Communications Outlook,* 2003.

Figure 26. **Telecommunication prices in OECD countries**[1]
US$ PPP, February 2003



1. OECD composite basket of residential and business telephone charges. Calls to mobile networks and international calls are included. VAT included for residential basket, excluded for business basket.

Source: OECD, Telecommunication database.

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The regulatory agency COFETEL (Federal Telecommunications Commission), set up in 1996, has faced difficulties regulating TELMEX. Despite the efforts made by CFC to declare TELMEX as dominant operator, its resolutions have been contested in Court. Nonetheless, COFETEL continues to control TELMEX's tariffs, information and quality of service. COFETEL has not been able to enforce fully certain decisions in this regard as a consequence of legal challenges in Court.[109] Concerning the powers of COFETEL and its ability to resist political pressure, the Mexican authorities are moving in the right direction. A proposal to modify the legal charter of the Commission and its relationship with the Ministry of Communications and Transportation is currently under discussion to be enacted in the coming months. Specifically, the powers of COFETEL will be increased by strengthening its functions of enforcement. As recommended by the OECD, penalties will now be imposed by COFETEL. Independence will also be increased. In line with OECD recommendations, the Members of the Board of COFETEL will have fixed and staggered terms which will disconnect their tenure from the political cycles. Such measures should increase the stability of the regulator and favour the establishment of regulatory benchmarks for the development of the telecommunications sector. These proposed changes to COFETEL powers and status should be approved and implemented without delay. Modifications to the Federal Law on Telecommunications were submitted to Congress in May 2002 for discussion. The modification proposal seeks to reduce TELMEX's control over communication networks by allowing an extensive development of new service providers, giving a non-discriminatory treatment, promoting fair competition and using resources more efficiently.[110] The draft law maintains a 49 per cent limit on the share of foreign direct investment in the fixed line sector.

To achieve the objectives of universal service and a greater coverage, a social fund to finance investment in infrastructure in poor areas was created in 2002. This initiative is in line with earlier OECD recommendations to set up a universal service fund that could subsidise network expansion. In early 2003, the Ministry of Communications and Transport initiated studies to allocate US$75 million as an incentive for companies to provide approximately 750 000 new phone lines in low-income areas. The government is to select the areas and take bids. For now, the fund is financed on public funds. Its medium term, financing is not yet clear. One option under consideration is to create fiscal incentives to promote private service providers' participation. Another option would be a compulsory financial participation of all operators.[111]

Other infrastructure sectors

Apart from electricity, in the rest of the energy sector, where investment needs are sizeable, reforms are also needed. PEMEX remains the national monopoly in oil. There has been a lack of investment in exploration, which PEMEX blames

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on budgetary control. In refinery activities, private capital is not allowed. In petrochemicals, which have been opened to private initiative (up to a 49 per cent participation), the private sector has shown no interest in partnership with PEMEX, perhaps because of the lack of certainty that PEMEX will supply crude oil at favourable prices. In the gas sector, where PEMEX still has a monopoly in extraction (but not transportation and distribution) a reform is all the more needed in that it could help lower electricity prices. A first step forward was taken in 2003, with PEMEX calling for its first exploration and production bids using multiple service contracts, which will allow private companies to invest in an ambitious plan to increase the country's sagging natural gas production.

In the transportation sector, privatisation has led to a mixed performance. The State had to take over a number of private airlines a few years after the privatisation as well as bail out (renegotiating the concessions on) toll roads after the financial crisis. The privatisation of Mexican airlines as separate entities was put off in 2002, the market conditions being unfavourable to the operation.[112] This operation should take place as soon as market conditions permit. For merchandise freight, ports and railroads seem to function well, but there is a need to move forward with inter-modal systems. In particular, some ports are not well served by road and/or rail connections. Railroads for passenger transportation are plagued with problems. Although there are plans to expand the network, regulatory and competition issues remain to be solved.

In urban water and sanitation services, which are run by local authorities, there are large and pressing needs for more infrastructure; investment is both insufficient and has been declining in recent years. This raises the question of the low cost charged to users, the low recovery of these charges, the limited independence of public distribution companies from local authorities and the inadequate regulations that limit private investment.

Other competition and regulatory issues

Mexico is, with Austria and Turkey, one of the most restrictive countries in the OECD concerning the regulation of professional services, such as lawyers, accountants, engineers and architects. These regulations are estimated to result in prices which are 10-15 per cent higher, thereby imposing costs on business operations which could be eliminated by appropriate measures. In addition, restrictions still persist on FDI.[113] Mexico ranks poorly in comparison with other OECD countries, its position reflecting in part restrictions in the electricity sector, but also in business, distribution and tourism activities.

Fostering entrepreneurship, investment and new technology diffusion

Strategies to enhance productivity growth through human and physical capital development need to be complemented by measures facilitating the creation

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and exit of companies and reduce the cost of their operations. This implies removing restrictions that seem to hinder the entry of new innovative and potentially highly-productive firms or slow the exit of low-productivity ones, and thereby impede productivity gains. It is also important to improve the business environment, since a number of weaknesses continue to weigh on Mexican enterprises, especially SMEs, and reduce Mexico's attractiveness to foreign investors, as is clearly revealed in international surveys. Institutional infrastructures (in particular bankruptcy procedures and the judiciary system) have to be strengthened, and bureaucratic processes simplified.

Facilitating exit and entry of firms

Well-designed bankruptcy rules contribute to increasing aggregate productivity by helping poorly performing firms to exit the market. It was only in 2000 that a new bankruptcy law in line with international best practices was enacted, as well as a law on guarantees. However, the new law was hardly applied subsequently, suggesting that there were still shortcomings and that further reform was needed. The law on guaranties has already been reformed in 2003 (see Chapter II). In March 2002, a new system to start a business within one day (SARE – *Sistema de Apertura Rápida de Empresas*) was created for most activities.[114] The Federal Regulatory Improvement Commission (*Comisión Federal de Mejora Regulatoria*, COFEMER) has signed SARE agreements with several State and municipal governments, to ensure full cooperation and participation of these local levels, where an important part of the formalities related to the start-up of a business is concentrated. In particular, COFEMER agrees with local authorities to assist them in implementing the SARE with the counterpart that local authorities are committed to report monthly on the results obtained. By easing firm creation, the new system is likely to make formal activities more attractive.

Improving the business environment

While there is cross-country evidence that good governance and a "friendly" business environment contribute to growth, international surveys show that Mexico ranks poorly in the views of the business community, compared with other countries in the OECD or in Latin America. For instance, according to the Global Competitiveness Report 2002-03, Mexico ranks 60th out of 75 countries, well below other OECD countries and Brazil, regarding the quality of the business environment (Figure 27).

This poor rating reflects persistently important bureaucratic burdens on firms despite clear progress in the past 10 years to "cut red tape" and improve the quality of the institutional framework. In the second part of the 1990s a systematic review of all business licenses was undertaken, resulting in the elimination or simplification of almost 80 per cent of the pre-existing formalities and a substantial reduction in the excessive levels of discretion being exercised by the lower levels of

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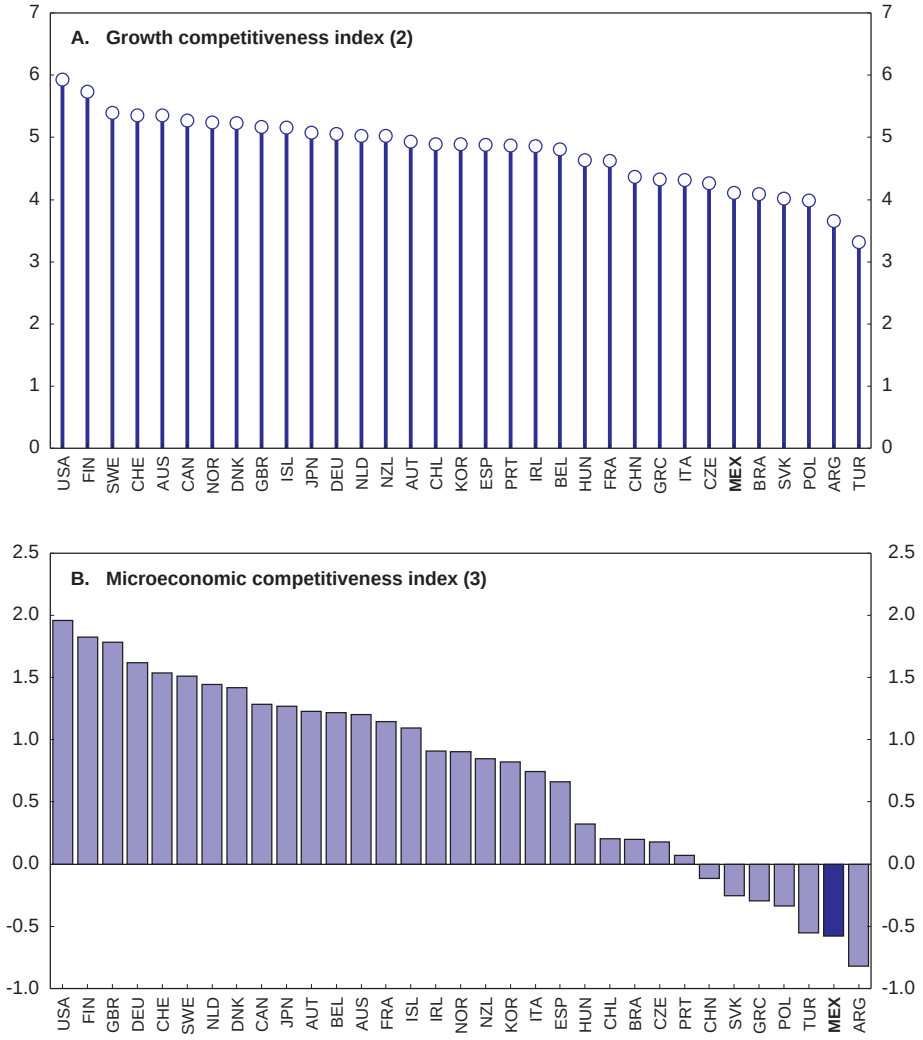
Figure 27. **World competitiveness indicators in selected countries**[1]



1. The OECD area, some selected Latin American countries and China are covered.
2. Based on technology, public institutions and macroeconomic environment variables driving economic growth in the medium and long term.
3. Computed from indexes measuring the quality of the microeconomic business environment and the sophistication of company operations and strategy.

Source: World Economic Forum, *Global Competitiveness Report, 2002-2003.*

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the bureaucracy, thereby reducing opportunities for corruption.[115] In March 2000, the Federal Administrative Procedures Law was amended to improve the quality of regulations through analysis and review. It aimed at increasing the degree of public participation and transparency in the drafting of federal regulations and enhancing legal certainty of existing formalities and obligations imposed by government. A Federal Registry of Formalities was also created to identify all federal formalities (available on the web in 2003) and then eliminate or simplify them. More recently, a new approach to administrative simplification has emerged in the framework of the 2001-2006 "Good Government Initiative", which integrates administrative simplification into a management-for-quality programme.[116] The electronic system of government procurement introduced several years ago, called Compranet, has been important to improve transparency and efficiency. The new and broader e-Mexico network should contribute substantially to these improvements (Annex III.C).

Another major problem affecting the business environment in Mexico is the weakness in the rule of law. Despite some improvements in the late 1990s, Mexico was still at the bottom of the league in 2002 on the rule of law and perception of corruption when compared with other OECD or Latin American countries (see Figure 28).[117] The results of these international surveys are in line with perceptions of investors at the national level as illustrated in the 2003 Bank of Mexico survey, as investors see improvements in the judicial framework, certainty, and a stronger "rule of law" (*Estado de Derecho*) as among the top prerequisites for increasing investment. Key concerns are:

- The excessive length of judicial processes, often delayed by lack of financial means but also by abuse of the right to file *amparos*.
- Poor enforcement of judicial decisions, especially at the local levels, as illustrated by the difficulties of CFC to collect fines or for banks to recover collateral associated with unrecoverable loans. This also negatively affects the perception of property rights in Mexico.
- Deficiencies in the protection of intellectual property rights, despite the modernisation of the related Law in 1995, that result in low numbers of patents registered in Mexico compared with other OECD countries (see OECD, 2002, *Main Science and Technology Indicators*).
- Widespread corruption. For instance, according to the *Consejo Coordinador Empresarial* survey, 15.3 per cent of firms were required to pay extra-official sums (averaging 11 348 pesos) to start a business (the percentage of firms in some states being as high as 30 per cent and the amount required 40 000 pesos, about US$4 000).

It is therefore important to further increase transparency, better protect property rights, improve the functioning of the judiciary system and ensure credibility and enforceability of laws. At the federal level, changes to the *amparo* law should be considered to limit the number of *amparos* per case and consolidate complaints in

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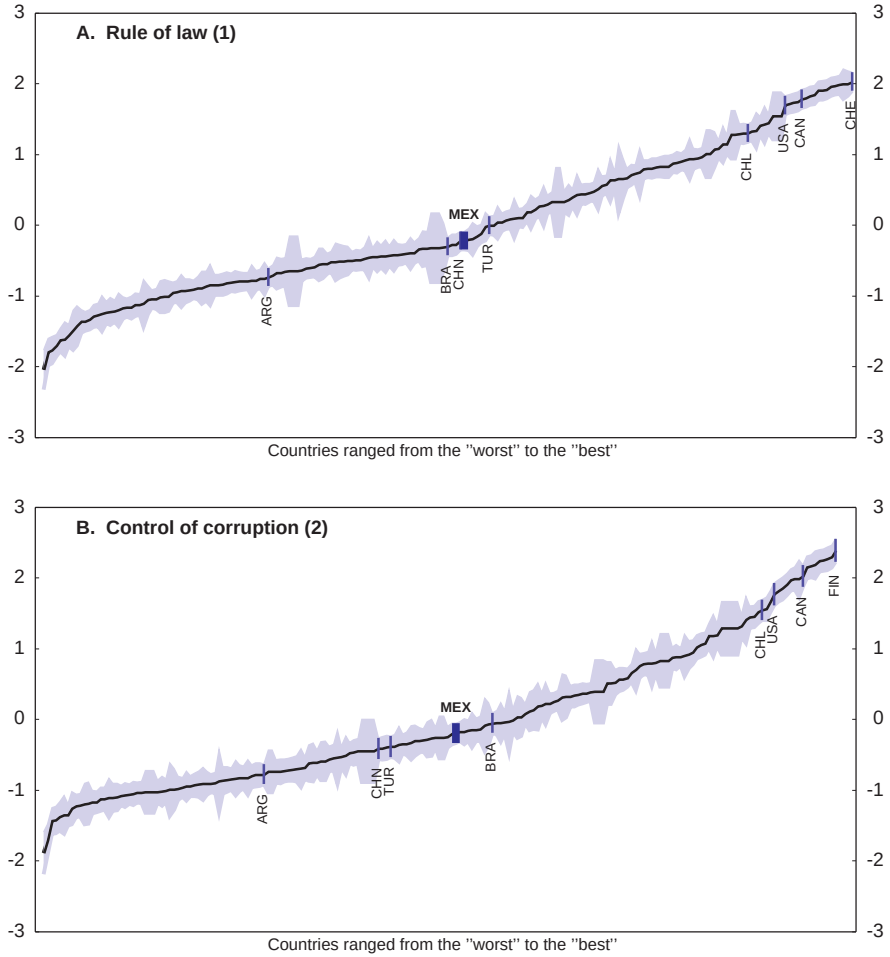
Figure 28. **Governance indicators**
2002



1. The rule of law represents the extent to which agents have confidence in and abide by the rules of society. This indicator includes perceptions of the incidence of both violent and non-violent crimes, the effectiveness and predictability of the judiciary, and the enforceability of contracts.
2. Control of corruption measures perceptions of corruption conventionally defined as the exercise of public power for private gain.

Note: In this figure the grey area shows the statistically-likely range of the indicator. The midpoint on the line corresponds to the best single estimate. In addition to NAFTA countries and some Latin America countries, the graph indicates the last and first OECD countries besides Mexico in the ranking, as well as China.

Source: World Bank, 2003 Governance Indicators.

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one file. The Federal government could provide financial support to reward progress in judicial modernisation and reform at the local level.[118] At the same time, ongoing rating of States by international agencies as well as surveys ranking states according to the quality of local rule of law can be effective in putting pressure to improve business environment at the local level.

Supporting the diffusion of new technologies

Mexico lags well behind other OECD countries for use of new technologies, in particular ICT, which have been shown to play a role in boosting growth.[119] In particular, SMEs still have limited access to information on technologies and markets, despite the active policy implemented by the Mexican authorities in the last 10 years to improve their awareness and access, including training and financial support *via* loans by the development bank NAFIN.[120] New measures were introduced in 2002. The e-Mexico programme already mentioned is an important initiative that can improve ICT diffusion via several subprogrammes designed to develop Mexico's ICT industry and market, improve access of the population to internet and support the development of the digital economy in the business sector (in particular SMEs).

To sum up, in order to increase its growth potential and reduce the gap with other OECD countries, Mexico needs to implement a comprehensive strategy aimed at enhancing its human capital, improving the functioning of its labour market, easing regulations that currently limit investment in key infrastructure areas, reducing business costs arising from regulations and from bureaucratic burdens, actively enforcing competition rules in the private sector, and promoting the use of new technologies. Detailed recommendations are presented in Box 10. Overall, the required reforms have been well identified by the authorities but implementation has been slow, except in the financial sector (see Chapter II), and has even stalled in some areas. It is therefore necessary to accelerate the pace of reform. Enhancing human capital is the key challenge for Mexico's long-term growth, but closing the human capital gap will be a slow process. It is therefore necessary to make progress on all fronts. Key reforms, such as the ones pertaining to the labour market and the energy sector, could have an immediate effect on growth potential, by eliminating important bottlenecks. This comprehensive strategy also requires progress on fiscal reform in line with the recommendations presented in Chapter II.

Sustainable development in Mexico

There is growing concern that long-run sustainable development may be compromised unless measures are taken to achieve balance between economic, environmental and social outcomes. This section looks at three specific issues of sustainable development that are of particular importance for Mexico: air pollution; sustainable use of natural resources, especially water; and water pollution. In each

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Box 10. Summary of assessment and recommendations[1]

Proposals[2]	Current actions	Assessment and Recommendations
1. Enhancing human capital		
A. Increase coverage of post compulsory education		
Facilitate access of poorer students to upper secondary education	Extension of PROGRESA/ *Oportunidades* to upper secondary	Monitor and assess results
Increase fees for tertiary education and develop an unsubsidised student loan market	No action	Consider
increase scope of scholarships for low income students	Creation of PRONABES grants (2001) for low income students	Continue
B. Increase quality of education services		
Modernise curricula and better integrate them between levels	Ongoing	Continue
Better train teachers and administrative staff	Ongoing	Continue
Make schools more accountable; make government transfers conditional to student characteristics and to outcomes.	*Escuelas de calidad*	Go further and consider a broader mechanism, including sanctions for poorly performing schools
Review incentives for teachers	No recent action	Consider
Evaluate schools and publish results	In progress	Accelerate implementation
C. Facilitate the transition from school to job		
Consider the introduction of apprenticeship contracts or alternation programmes.	No action	Consider
Further strengthen vocational education	Ongoing	Continue
D. Adult training		
Continue to develop skill standardisation and certification	Ongoing	Continue
Strengthen public training programmes	Ongoing	Continue and monitor cost-efficiency of programmes
Facilitate development of on-the-job training	Measures included in the Labour Law reform proposal	Continue
2. Labour market issues		
A. Increase the flexibility of employment regulations		
Ease permanent contracts regulation	Partly addressed by the Labour Law reform proposal	If law is approved, implement and go further
Allow probationary periods so as to facilitate job creation through long-term contracts	In the Labour Law reform proposal	Proceed with reform
Broaden scope for using short-term contracts	No action	Make progress
Broaden scope for using part-time contracts	No action	Make progress
Review overtime remuneration	No action	Make progress
Ease employment protection provisions while putting in place minimal revenue support in case of job loss.	No action	Consider options

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Box 10. Summary of assessment and recommendations[1] (*cont.*)

Proposals[2]	Current actions	Assessment and Recommendations
B. Reform the tax and transfer system		
Increase IMSS efficiency	Ongoing	Continue
Reform public housing fund (INFONAVIT)	Money can now be used as collateral	Review financing and operation of the fund
Avoid use of payroll tax	Payroll tax introduced but eliminated	Consider other options to raise tax revenue
Review social contribution rates, rather high at the low end of the salary scale	No action	Consider
C. Modernise the collective bargaining framework	Partly addressed in the Labour Law proposal	Proceed with reform and go further.
3. Energy		
Electricity: reform the sector to open it to private investors and establish a transparent, effective, non-discriminatory and competition enhancing regulatory framework.	Reform proposal in Congress	If the law is approved, proceed with implementation
Continue liberalisation and opening up of the gas sector	Tenders for multiple services in July 2003	Continue
Advance liberalisation in petrochemicals: Step up efforts to restructure PEMEX petrochemicals; ensure that the process leads to a more competitive environment, while preserving significant economies of scale.	No action	Make progress
4. Telecommunication		
Use market-based mechanism to ensure universal service	In progress	Continue implementation
Improve the regulatory framework to enhance competition	Reform proposal in Congress	If the law is approved, proceed with implementation
5. Transportation		
Proceed with privatisation of airlines as separate entities.	Postponed because of unfavourable economic conditions	Proceed with privatisation when possible
6. Promote a business friendly environment		
Facilitate entry and exit of firms	Ongoing	Continue implementation
Reduce business costs	Ongoing	Continue implementation
Improve the functioning of the judiciary system and ensure enforceability of laws	Measures under consideration	Step up efforts at federal, state and local levels of government
7. Promote use of new technology		
Emphasise the importation and diffusion of technologies, for now concentrated in large exporting companies	Ongoing	Continue implementation

1. Assessment of the 2002-2003 tax measures and proposals for an additional tax reform are discussed in the previous chapter.
2. Proposals are based on analysis in this and previous OECD *Economic Surveys on Mexico*.

Source: OECD.

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Box 11. The integration of policies across sustainable development areas*

The major administrative instrument for ensuring the integration of economic, social and environmental policy in Mexico is the National Development Plan that the Constitution requires the federal government to produce every six years, at the start of a new administration. This plan reflects that the Constitution also requires that problems relating to the environment and the exploitation of natural resources should be considered a national priority. Against this background, the 1995-000 plan promoted the integration of policies in different areas. The most recent plan (2001-2006) makes explicit the goal of reaching a sustainable development path. This plan has three main objectives: improving social and human development; achieving economic growth; and improving the rule of law. The machinery of government has been changed with a view to meeting these three goals. Three inter-ministerial committees have been established to deal with each of the goals. The Environment Ministry is represented in all three committees, to ensure that adequate attention is paid to the integration of environmental concerns into all areas of policy. In addition, each ministry is required to produce sustainable development plans. The Environment Ministry is monitoring progress in developing these plans.

The new structure of government should improve the extent of policy integration that has been lacking in the past. For example, policies in the area of agriculture, livestock and agrarian reform have favoured deforestation, excessive use of water and degradation of soil quality. The government intends that, in evaluating future policies, full attention should be paid to the depletion of natural resources. The use of cost-benefit analysis in the environmental area should draw on the experience of the Federal Regulatory Improvement Commission (COFEMER). This body has to review regulations and verify that new proposals provide an overall benefit to society (Macias, 2001).

On the basis of past experience, though, there must be some doubt about the extent to which these plans will be implemented. The analysis of past reforms in the environmental area illustrates that often laws are not enforced (as with pollution charges); bills are not collected for the use of natural resources such as water; or measures announced in plans are carried over from one plan to the following plan. A major change in the effectiveness with which policies are carried out is required.

Further integration of policies is also required between the different levels of government. At the federal level, the transfer of the production-related activities of the Environment Ministry (fisheries, forestry and water) into agencies may help the Ministry to focus on its regulatory role, giving greater weight to wider goals. Environmental control is delegated to the states by the Constitution, unless specifically reserved for the federal government. Municipal authorities, moreover, are given responsibility for all areas of public sanitation. Local environmental policy has

* The sections in this report dealing with air pollution, sustainable use of natural resources and water pollution are inputs into the OECD's follow up on Sustainable Development, as mandated by the Ministerial Council decision in May 2001.

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Box 11. **The integration of policies across sustainable development areas*** (*cont.*)

not tended to take into account inter-relationships, being typified by a sectoral approach and being administered by staff without adequate training in their area. A variation in standards has emerged, especially as there is a lack of an appropriate mechanism for supervising the implementation of local laws and sanctioning non-compliance with federal standards.

case, indicators are presented to measure progress and the evolution of potential problems, and an assessment is made of government policies that affect the area. The section also considers whether institutional arrangements are in place to integrate policy-making across the different elements of sustainable development (see Box 11).

Air pollution

Main issues

Air pollution can damage health, quality of life, ecosystems and structures. The economic costs, which are often indirect and can appear with a long lag, are substantial in highly polluted environments. Mexican air quality has improved from a very serious state since the late 1980s, but not for all regions nor for all pollutants. With the authorities targeting further increases in ambient air quality, notably for particulate matter and ozone, where problems remain severe, the issue is to use effective instruments that impose the least cost on the economic pillar of sustainable development.

Performance

A significant reduction in air pollution has occurred in Mexico since 1990. By 2001, national air quality standards for sulphur dioxide and nitrogen dioxide were nearly always met in major Mexican cities (Table 9), though the position was less favourable for ozone and for particles. However, in an international context, the levels of pollution are still high. In the mid-1990s, the average concentrations of sulphur dioxide and total suspended particles in Mexico City were 160 and 77 per cent above the average for OECD metropolitan areas (Table 10). Since then, sulphur dioxide concentrations in Mexico City have fallen by 7 per cent, as result of switch to low sulphur fuels, while concentrations of particles fell by 11 per cent respectively, leaving concentrations of both pollutants in Mexico City still above

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Table 9. **Percentage of days above national air quality standards 2001 for selected Mexican cities**

Area	03	SO_2	NO_2	Particles
Mexico City	81	5	0.3	13
Guadalajara	10	1	1	22
Monterrey	4	0	0	25

Source: National Institute of Ecology.

the OECD average. Moreover, particle concentrations in expanding areas, such as Guadalajara and Monterrey, were, by 2002, above those in Mexico City. Nitrogen dioxide concentrations were below the OECD average, reflecting relatively low car ownership. Nonetheless, ozone pollution remains a problem in Mexico City and is becoming a problem in other metropolitan areas (Figure 29). As to other pollutants, concentrations of lead and carbon monoxide fell significantly as unleaded gasoline gained ground and three-way catalysers were fitted to cars.

A comprehensive picture of the sources of this pollution will become available when the National Emission Inventory is completed in 2004. The first stage of this work was finalised in July 2003 when an inventory for the border area was completed. Prior to this, an inventory for the Mexico City area was completed for 1998 emissions. This study showed that for particles (other than those related to soil), two-thirds of emissions came from diesel trucks, buses and private cars. Industry was a minor source. For emissions of NO_x, private cars and taxis were a significant source but once again heavier diesel vehicles were the dominant source. For unburnt hydrocarbons, private cars dominated but the distribution of liquefied petroleum gas (LPG), used in most households for heating and cooking, was also an important source. Outside the Mexico City area, it is noticeable that electricity generation is estimated to produce three times more sulphur dioxide per kilowatt hour than in the United States (Miller, 2003), mainly due to the use of high sulphur content of the fuel oil used in some power stations. However between 2000 and 2003, sulphur dioxide emission intensities in the electricity sector have fallen 37 per cent as more natural gas was used, bringing an overall fall of 15 per cent in emissions from the electricity sector. Oil refineries are another significant source of sulphur dioxide. In 1997, sulphur dioxide recovery per barrel of crude oil processed in Mexico was only one third of that processed in the United States, because the higher sulphur content of the crude raised the cost of purification (Gilbreath, 2002). Differences elsewhere in the economy are not as great, with sulphur emissions per unit of GDP being some three-quarters higher in Mexico than in the United States. However, with the United States having a much higher income level, per capita emissions were twice as high in the United States as in Mexico.

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Table 10. **Ambient air quality: an international comparison**

OECD countries and the ten most polluted non-OECD countries, around 1995

	Particles	Sulphur dioxide	Nitrogen dioxide
	Micrograms per cubic metre		
Australia	45	13	51
Austria	47	14	42
Belgium	78	20	48
Bulgaria	195	39	122
Canada	34	14	41
China	325	106	73
Colombia	120	n.a.	n.a.
Czech Republic	59	14	33
Denmark	61	7	54
Ecuador	148	18	n.a.
Finland	40	4	35
France	14	14	57
Germany	43	13	40
Ghana	137	n.a.	n.a.
Greece	178	34	64
Hungary	63	39	51
Iceland	24	5	42
India	280	25	27
Indonesia	271	n.a.	n.a.
Iran	248	209	n.a.
Ireland	0	20	0
Italy	87	31	248
Japan	44	34	54
Korea	84	50	59
Mexico	**201**	**47**	**56**
Netherlands	40	10	58
New Zealand	26	3	20
Norway	15	8	43
Philippines	200	33	n.a.
Poland	61	18	35
Portugal	61	8	52
Slovak Republic	62	21	27
Spain	57	14	42
Sweden	9	3	20
Switzerland	31	11	39
Thailand	223	11	23
Turkey	57	55	46
United Kingdom	n.a.	22	66
United States	n.a.	18	73

Note: Data are annual means (measured in micrograms per cubic metres) in selected metropolitan areas (see 2003 *World Development Indicators*, table 3.13, page 168, paragraphs 4 and 5).
Data for Mexico City correspond to weighted annual means registered by the Air Quality Monitoring Network in the year 2000.

Source: World Bank, *World Development Indicators*; Federal Government of Mexico.

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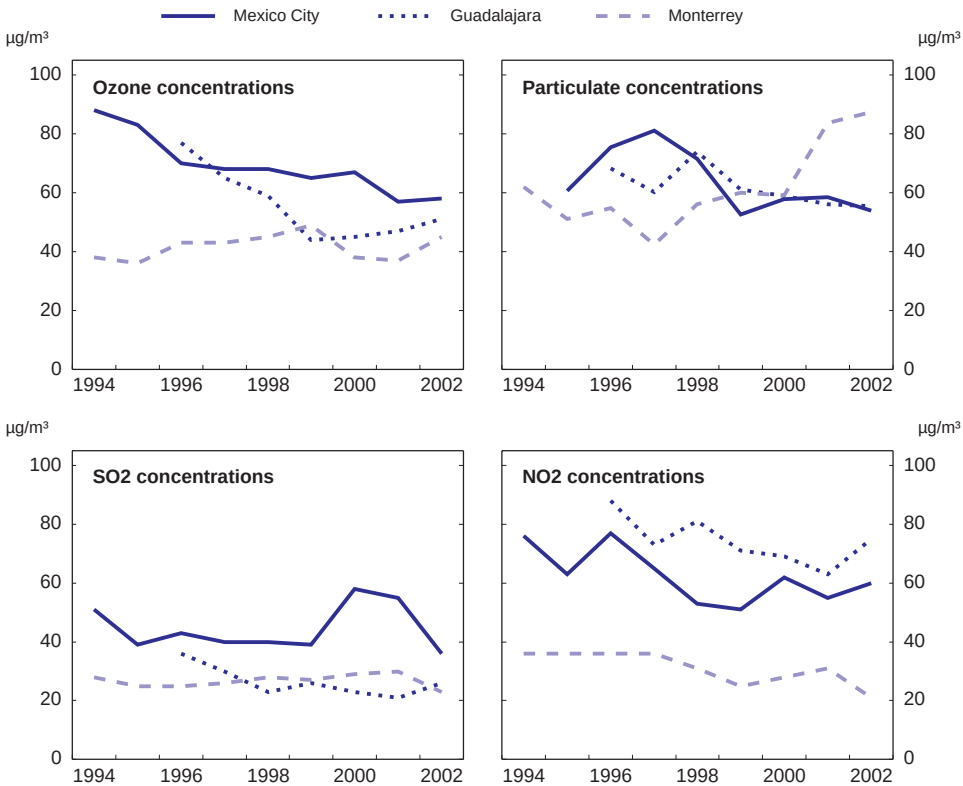
Figure 29. **Ambient air quality for three metropolitan areas**
Annual average of hourly concentrations



Source: National Institute of Ecology (INE).

Policy

Significant welfare gains are possible from lowering air pollution. For example, if pollution were cut by just over half in the Mexico City metropolitan area by 2010, then annual welfare gains are estimated to be equivalent to 1.5 per cent of GDP in the area (Vergara, 2002). The primary gain from reducing air pollution is estimated to come from reduced particulate concentrations (85 per cent of the total) rather than reduced ozone concentrations. There are four key factors that could help control particulate pollution from road vehicles: fuel quality, improved vehicle technology, inspection programmes and fleet renewal. Significant investments have been made to improve the quality of gasoline as high fuel quality is essential to

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ensure that pollution control technology works well. At present, the average sulphur contents of gasoline and diesel are considerably higher than in a number of other countries (Favela, 2001). For regular grade gasoline ("Magna" gasoline), the average sulphur content was 408 ppm in 1999, against an average of 160 ppm in United States metro areas, elsewhere the average was 714 ppm against 280 ppm in the remainder of the United States (except California where the average was below 50 ppm). Pemex plans to lower the average sulphur content of Magna gasoline to 300 ppm in 2005 and 30 ppm by 2008 at a cost of around 6 cents per gallon. Premium grade gasoline, with a content of 30 ppm similar to that planned in the United States and Europe, will be available by 2006. The incremental costs of producing high quality fuel are limited, suggesting that a tax-based system could speed the planned introduction of low-sulphur gasoline and diesel, whose maximum sulphur content is planned to reach 300 ppm by 2006 and 15 ppm by 2008. A new legal standard for diesel and gasoline will be published in 2003.

Mexico has introduced increasingly stringent emission standards for new vehicles in the last decade. Standards equivalent to United States Tier One limits were introduced in 1999, five years after the United States. However, Tier Two standards will be adopted gradually starting 2006, just two years after the United States, leaving domestic manufacturers (40 per cent of the market) time to adapt to these limits. As importantly, the proposed Mexican standard for Tier 2 vehicles expected to be published in 2004, and already agreed with the industry, envisages that emission standards must be met after a vehicle life of between 80 000 and 194 000 kilometres, depending on the availability of low sulphur gasoline (at present, limits apply only to vehicles when new). As a consequence, it will be particularly important to speed up the introduction of low sulphur gasoline. If not, consumers will find that the pollution gains from the introduction of Tier 2 emission limits in the period 2006 to 2009 will be small. More importantly, the limit for sulphur in diesel needs to be reduced in order to ensure that health benefits actually materialise from the particulate reduction technologies foreseen for new imported trucks. Such a policy would appear to be welfare enhancing, based on studies for the United States (United States Environmental Protection Agency, 2000).

The vehicle inspection programme is important in limiting pollution from older vehicles, which are more prevalent in Mexico than in other OECD countries. The inspection and maintenance (I&M) programme in the Mexico City area has been markedly improved since it was first introduced, with better monitoring of combined test and repair sites and also monitoring of wider range of pollutants that enables fraudulent results to be better detected. Although trucks and buses produce most transport related particles, effective testing for diesel vehicles does not exist. This is particularly serious, because the average age of trucks in Mexico is much higher than in other OECD countries and these vehicles are likely to account for a disproportionate amount of emissions (Howitt *et al.*, 2000).

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Fiscal instruments have been used to try to reduce the average age of the vehicle fleet and so lower its pollution intensity. Subsidies have been offered to the owners of older taxis and similar schemes have been in place for replacing buses, but the later programme would take 30 years to replace existing micro-buses. At one point the Mexico City government encouraged the use of compressed natural gas (CNG) fuel for buses, but the scale of the programme was small. In any case, experience in California suggests that equipping diesel buses with particle filters is a better option than subsidising the use of CNG. Annual registration taxes in the current system decline with the age of a vehicle, in contrast to the profile of emissions per vehicle. Vehicle registration taxes therefore work against quick renewal of the vehicle fleet.

Physical limitation of the use of vehicles and industrial premises has also been used in an attempt to reduce pollution. Regulations that restrict the use of cars in Mexico City to 4 days from Monday to Friday have been in force since 1989. The plan has been modified over the years. At one point it was only in force during the winter months when ozone pollution was high. At present, it operates around the year, but more recent vintage cars can circulate on all days, and cars equipped with pollution control equipment are also exempt provided they have passed an inspection test. Some studies suggest that the programme has been unsuccessful in reducing pollution in that it led to the purchase of extra cars and to extra car use at the weekends (Eskeland *et al.*, 1995). Regulations that directly restrain economic activity on days of extremely high pollution are also in force but there has been a decreasing need to use them as the number of days of extreme pollution has fallen.

Permissible pollution norms for fixed point sources are high by international standards. For electricity plants, permissible emissions of new plants are three to five times higher than in the United States. Moreover, old plants are allowed to use high sulphur fuel oil without the installation of pollution control equipment. Norms are lower in areas that are subject to high pollution but the difference is much less than in the United States. As a result, the difference in norms between California and Baja California is tenfold. Such gaps have led to cross-border investment projects in which the US regulator allows domestic companies to claim credit for pollution reductions that occur in Mexico. However, international emission trading of criteria pollutants requires the establishment of an operational framework, based on legislation in both countries. Mexico and the United States are working towards the definition of such a framework in the border area. There has been only limited investment in retrofitting generating plants, even in areas with the highest pollution. Rather, the emphasis of government policy has been to encourage the state-owned electricity generator to use natural gas. The government has considered a trading system for sulphur dioxide reduction; but with only two major players in the market (both state-owned), a competitive market might be difficult to establish. Instead emphasis is being given to introducing an internal market between the individual plants of the two state-owned companies.

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Conclusions

The reduction of air pollution in Mexico has stalled in recent years but new emphasis is being given to the reduction of pollution through the national environmental programmes and the PROAIRE programme in the Mexico City Metropolitan Area, designed to reduce emissions by 40 to 50 per cent between 2000 and 2006.[121] Similar plans have been introduced before but many elements of the plans were not implemented and the fall in pollution stalled in the mid-1990s. The reduction of particulate pollution should be a priority and would require speeding foreseen action in a number of areas within the national and local plans to reduce pollution. The sulphur content of both gasoline and diesel fuel needs to be reduced at a quicker pace. Achieving this goal would be helped through introducing higher taxes on lower quality fuel, but will also require investment by the state-owned petroleum company that would need to be authorised by the government. Inspection and maintenance programmes need to be tightened for diesel vehicles, with strict limits on acceptable emissions limits backed by roadside inspections. In view of the slow turnover of the diesel fleet, registration taxes should be made to depend on emission characteristics. Elsewhere, scope exists for reducing emissions from large stationary sources. A trading system might be difficult with few participants in the market but a tax-based system could be considered. Finally, enhanced cross-border co-operation is essential, given the rapid economic development in areas that form common air sheds with the United States.

Sustainable management of natural resources: water

Main issues

The depletion of water resources may eventually make certain locations in Mexico unsuitable for economic development. The challenge for water policy makers is to ensure that extraction is placed on a sustainable footing and to make more use of the price mechanism to ensure that a scarce resource is used efficiently.

Performance

Nation-wide, water consumption amounts to only 15 per cent of the renewable water supply, and water supply in Mexico would be more than adequate if it were evenly distributed across the country. But in the areas of high population and strong economic activity, *i.e.* in the northern and central areas of the country, water use, especially by the agricultural sector, exceeds the renewable supply and underground water stocks are being depleted (Table 11). The consequences of over-exploitation are most noticeable near the border with the United States and in the Valley of Mexico, where the capital is located. In the former area, surface water extraction (mainly for agricultural use) has led to the border river (the Rio Bravo,

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Table 11. **Underground water use by region**

	Underground water				Population	GDP
	Usage			Recharge as per cent of usage		
	Industry	Public supply	Agriculture			
	Percentage of total				Per cent	Per cent of total
North Central	3.1	9.8	87.1	61.7	3.8	3.4
Lower California	9.3	10.1	80.6	61.9	3.1	4.3
Valley of Mexico	10.6	68.2	21.2	79.7	20.1	31.8
North West	1.3	14.5	84.1	103.4	2.4	2.6
Rio Bravo	4.4	9.9	85.6	103.6	9.7	15.0
Total of above regions	5.3	19.3	75.4	84.5	39.1	57.1
Santiago Pacific	4.3	23.4	72.3	121.0	3.8	16.0
Northern Gulf	4.9	16.3	78.8	132.6	4.8	3.4
Pacific North	2.2	34.4	63.4	266.1	3.9	2.8
Balsas	11.5	37.9	50.6	321.5	10.3	6.7
Central Gulf	14.7	41.9	43.4	589.7	9.3	5.1
Southern Pacific	3.8	62.4	33.8	782.6	4.0	2.1
Yucatan peninsula	8.4	25.1	66.5	1 391.7	3.4	4.0
Southern Border	15.2	27.6	57.2	3 796.9	6.0	2.8
Total of above regions	6.5	27.2	66.3	512.8	45.5	42.9

Source: Mexican government submission to the OECD.

called Rio Grande in the United States) running dry. Moreover while official data shows that there is no aggregate depletion of aquifers in this area, serious depletion of groundwater is occurring in a number of underground reservoirs that straddle the border region.[122] In the latter area, over-exploitation of the aquifer results in annual land subsidence of over 5 cm, causing extensive damage to property and the network of pipes and canals used for wastewater disposal. In addition, subsidence has led to a major flooding risk in the capital area.

Water use can be expected to rise in the future not the least because substantial sections of the population in Mexico are still without access to potable water. In urban areas nearly all families (95 per cent) had access to supplies in 2000. However, in the Mexico City Metropolitan Area, according to the 1990 census, one-third of households had only a stand pipe on their property, and in thirteen administrative areas within the metropolitan area that accounted for one quarter of the national population, 18 per cent of households had no access to water on their property, against 3 per cent elsewhere in the region. Instead, these families had to rely either on water trucks or illegally drilled wells. In rural areas, access to potable water is worse than in urban areas, with 30 per cent not linked to piped water.

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This proportion has decreased only slightly in the past decade, although the rural population has been stable.

Policy

The government has the possibility to charge fees for extraction rights both for surface and underground water but major users pay little for these rights. The constitution specifies that water is the property of the Mexican State, so giving the National Water Commission (CNA), which is an agency of the Ministry of Environment, the power to charge for water extraction. The pricing structure adopted by the CNA varies according to the extent of water scarcity and the sector of the user. Agricultural users are exempt from charges, while industry pays between 30 and 40 times the price charged to drinking water distribution companies. By 2001, industry provided over 80 per cent of the water extraction revenue, amounting to US$550 million raised by the CNA. An attempt was made to introduce charges for the agricultural sector in 2003 but, while the principle has been accepted, the charge applicable to farmers is only 0.1 cent of a dollar per cubic metre.

While the government does not charge for scarce water rights, it has put in place the framework for the trading of extraction rights. The rights are granted to irrigation districts and are for a period of time that can vary between five and 50 years. In 1998, the commonest concession length was for 10 years for final users, but users who replace extracted water, such as power plants where water is used for cooling, have been granted concessions of up to 50 years. The registered rights are not completely secure as the 1992 law mentions the possibility of forfeiture for reasons of public interest if the water has not been used "efficiently" or if it has not been exploited for three years. Although the rights are specified in volumetric terms, in practice the rights are proportional since the water user associations are to allocate deficits or surpluses proportionately across all existing rights in times of drought (Holden and Thobani, 1996). Nonetheless, some trading has occurred, mostly within the same sector (Moreno, 1998). Uncertainty about the value of the rights also arises because their total may exceed optimal extraction rates. At present, water basin authorities have not established hydrological balances that would determine sustainable extraction rates, nor have decisions been taken on how to permanently reduce rights in the case of over-extraction.

Within any irrigation district, the users of water still do not have adequate incentives to take into account the scarcity of water. In 1992, the government introduced a major reform of the irrigation systems. Most of these had been created and funded by the federal government, often in conjunction with resettlement of peasants and land reform. By 2001, these systems had almost been entirely transferred to user associations that were covering 80 per cent of operating costs, up from 20 per cent at the beginning of the 1990s. However, the government is still investing heavily in the sector to rehabilitate and modernise existing systems, expand the

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area of irrigated land and cover the remaining running costs of the associations. Total spending is running at the rate of US$400-500 million per year. In addition to this expenditure, farmers using underground water are able to buy electricity from the state-owned distribution company at a preferential rate. In the mid 1990s, this subsidy amounted to US$250 million per year but fell to US$10 million five years later. However, under the Agricultural Sectoral Programme, the subsidy may rise to US$150 million per year from 2003 onwards.

The inefficiencies in the urban water distribution sector are typified by the problems in the Valley of Mexico area, where the overall supply-demand balance is worst. The federal government has recognised that the problem of water over-exploitation in the Mexico City area required a more market-related approach and has moved some way in this direction. In 1993, it decided to transfer authority over water in the Federal District to a water commission. The commission then contracted-out the provision of water service to four companies, a means of overcoming the difficulty of reforming the public sector water distribution body.[123] These were responsible for the metering, billing and distribution service but did not collect revenue. The policy has been successful to a certain extent. The proportion of households with meters has risen and the extent to which bills are issued has increased to 83 per cent. However, collection of the bills was left with the government treasury, which is legally unable to cut the water supply for non-payment, and the real value of collections barely rose between 1994 and 1998 (Haggarty *et al.*, 2001), with collections from large users rising while those from small users fell in real terms. Revenues would need to rise 50 per to balance operating costs. As revenues are low, there is little incentive to reduce leakages from the water distribution network that, at 37 per cent, are much higher than in many other Latin American countries.

The reform did have some success in reducing the over-exploitation of the aquifer. Pumping rates fell by nearly 20 per cent between 1988 and 1998, mainly as a result of a reduction in consumption by large consumers who faced a major rise in prices in this period. On the hand, private consumption use continued to rise in the 1990s, with this type of consumer experiencing falling real prices. Even before real prices fell, apparent consumption per person per year in the Federal District (379 litres per person per day) was over one-third higher than in a number of major South American cities. Moving prices back into line with costs, both private and social, would help reduce demand. Infrastructure investment is also planned to bring more water to the Valley area but it seems unlikely, given past pricing policies, that the National Water Commission will charge full economic prices for the provision of supply to water utilities in the Valley of Mexico.

The decentralisation of water distribution has meant that it has been difficult to co-ordinate policy in the Federal District and the surrounding areas in the State of Mexico. The water sources in the Federal District and State are interconnected and so co-ordination is required but has been ineffective in the past due to

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the devolution of water policy to local authorities in areas surrounding the capital. An institution capable of improving this situation in the areas around the capital, a unified water management commission, was put in place in 1999. However, the institution has been slow to act. It was only in 2002 that analysis of the possibility of creating a regulatory authority and involving the private sector in water distribution was started (Espinosa, 2002).

Public authorities have designed their tariffs inefficiently. Prices for potable water differ between industrial and domestic users, with the former almost twenty times the price paid by the modal domestic user. For domestic users, prices rise sharply with the volume consumed, almost tripling as annual consumption rises from a typical 120 m^3 per year to 360 m^3 per year (National Research Council, 1995). Such a price tariff has been justified on social grounds, as a redistributive policy. However, the very poorest, who do not have access to piped water, pay a higher price than nearly all other consumers of domestic water as they purchase from water trucks. Over the country as whole, revenues amount to only 35 per cent of total costs, requiring an overall implicit subsidy of nearly US$4 billion per year (OECD, 2003).

Conclusions

Water use in Mexico is on an unsustainable path. A large number of underground aquifers are being depleted. In most cases this is the result of over-extraction by the agricultural sector for irrigation use. Farmers are now paying a large part of the operating costs of irrigation system, but the remaining direct subsidies need to be ended and the agricultural sector should be obliged to pay the market price for electricity used for pumping. As well, the irrigation sector should pay for extraction rights, rather than being exempted. In urban areas, one challenge is to place water distribution on an economic footing but this is unlikely when utilities are controlled by local authorities. The newly created river basin councils could serve as water distribution authorities as well as producing sustainable water extraction plans.

Water pollution

Main issues

Water pollution can pose a threat both to public health and the environment. In the case of Mexico, the challenge to policy makers differs from most OECD countries in that there is still considerable room to improve the basic infrastructure for wastewater treatment. Policy makers will need to find efficient ways to free resources for a major investment programme.

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Performance

The extent of ground and surface water pollution is high in areas where economic activity is situated. Indeed, in the area to the northwest of the Tuxoan (on the Gulf coast) and Papagayo (on the Pacific coast) river basins, nearly every river is polluted according to national definitions (Table 12). This area receives 80 per cent of the overall discharge of pollutants. Only 5 per cent of water bodies are classed as in excellent condition and could be used as drinking water without further treatment. No information is available of the trends in water pollution at a national level. However, in the basin of the Lerma river (a source of drinking water for Mexico City and Guadalajara), pollution rose markedly in the 1990s (Saade Hazin, 2002). There is some evidence of nitrate pollution of water, but most eutrophic pollution stems from excess phosphate content in water, usually associated with urban pollution.

The main cause of water pollution is the discharge of raw sewage into rivers. At a national level, over three-quarters of the population is connected to a sewage system, with a much lower connection rate in rural areas. Some parts of the network need improvement, as sewage can leak from unlined canals or broken pipes, opening the possibility of infiltration to aquifers and the potable water network. Moreover, there is only the capacity to treat one-quarter of wastewater flow from the sewage system and one-third of this capacity does not function correctly. Consequently, more than four-fifths of wastewater is directly discharged into rivers. As there is no river in the Mexico City area, a portion of the discharge is used for agricultural irrigation without further processing. The treatment of industrial wastewater is limited to 13 per cent of total discharges, with the sugar industry being the principal polluter. Nearly all drinking water is now disinfected before distribution, though only water from surface sources is subject to further treatment. However, in the mid-1990s, studies indicated that in the areas around Mexico City, the level of residual chlorine in household water was low. When coupled with the lack of wastewater treatment

Table 12. **Pollution status of river basins**

	Per cent of total water flow in area	Per cent of national water flow
Polluted river basins		
Northern Gulf Coast	100	8.8
Northern Pacific coast	80	15.5
Interior rivers	63	0.1
Southern areas	0	0.0
All polluted rivers		24.4
Unpolluted basins		
Northern Pacific	20	3.9
Southern areas	100	71.7
All unpolluted rivers basins		75.6

Source: Ministry of the Environment.

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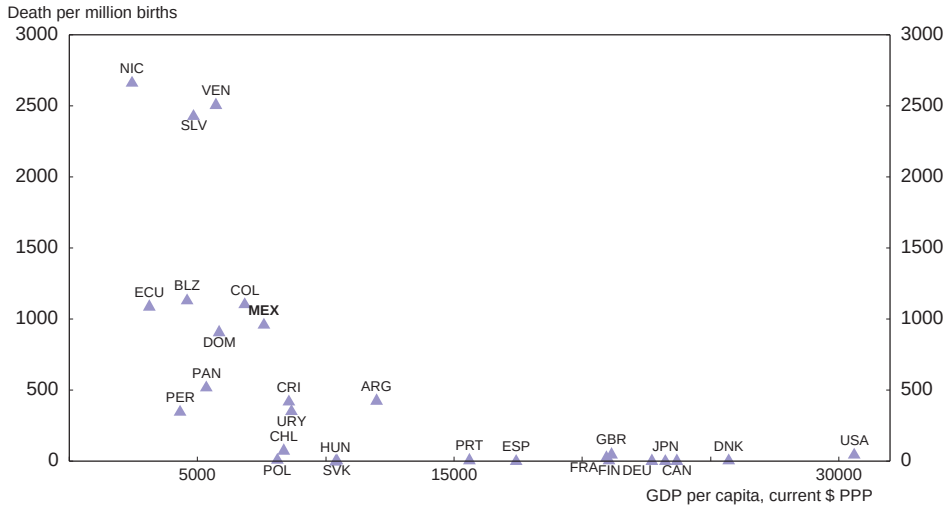
Figure 30. **Infant mortality from water infectious intestinal diseases**



Source: World Health Organisation, UN, *World Development Indicators*.

and inadequate maintenance of the water distribution network, low levels of chlorine constitute a health hazard in some parts of the country. In 1991, infectious intestinal diseases were the second cause of infant mortality. The infant death rate from this cause was higher than nearly all other OECD countries and is even higher than in many other Latin American countries (Figure 30).

Policy

The government put in place more than a decade ago a system designed to give polluters an economic incentive to reduce discharges and has coupled this with a regulatory system. Polluters are required to pay a charge based on the extent of the discharge and the degree of pollution of the river into which the discharge is made. Both water utilities and industry are liable to pay the tax. In addition to the tax, the government introduced a simplified set of norms for pollution discharge in 1996. These norms are to be phased in over the period 2000 to 2010.

These regulations have not been effective in lowering pollution from water utilities that are almost totally run by municipalities or states. In the case of the pollution tax, the local authorities did not pay the tax, accumulating a debt of US$6 billion by the end of 2001. At that point, the debt was written off by the federal government, provided the authorities undertook to make progress in wastewater

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treatment. In the future, revenue from the tax will be paid to the CNA and used to finance infrastructure investment. Compliance with industrial norms has also been poor. The sugar industry does not comply with the norms (Saade, 2002) and the CNA estimates that only 35 per cent of industrial wastewater treatment plants comply with regulatory norms. In response to poor compliance, the government has strengthened the Office of the Environmental Prosecutor, while the National Water Commission has created a new Inspection and Monitoring department with 160 inspectors out of a total of 20 000 employees.

The government has responded to the poor condition of water resources by introducing a five-year plan to improve all aspects. It establishes the river basin as the basic unit for decision making. A significant increase is planned in the supply of potable water to rural areas and higher-quality water in urban areas, while the coverage of waste-water treatment will be expanded nationwide. The government has estimated that investment of US$2 billion in the period to 2006 will be required to meet these goals, seven times the existing level of spending. Expanding the treatment of waste water would also allow a degree of recycling of water in areas, such as Mexico City, where water resources are over-exploited. Private participation in water distribution and wastewater treatment has been limited. Most companies have not renewed wastewater treatment contracts and only 40 per cent of all water distribution contracts signed in the 1990s were still in force at the end of the decade.

Conclusions

Mexico is faced with the need for a major investment programme both to improve the treatment of wastewater, and to expand the provision of potable water in rural areas and improve its quality in both rural and urban areas. The scale of spending required is large and an adequate financing plan is needed. Significant government resources could be freed for such investment if the pricing of potable water were to be placed on an economic footing, as the current subsidy for water provision amounts to US$4 billion per year. However, such a policy may be difficult to implement when control over water is vested with municipalities whose elected officials are limited to one term of three years. The 25 river basin councils now in place might serve as the base for a system of water distribution but their powers would need to be strengthened. At the same time, the orientation of the federal water commission needs to be progressively re-oriented, away from irrigation projects and large water projects towards expanding the water to smaller, principally rural communities.

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IV. Migration: the economic context and implications

Introduction

At least 10 million Mexicans live in the United States today and they are joined by as many as 400 000 every year. They affect the Mexican economy both through the labour market – the balance of emigration and return migration influences the level of the labour supply and its composition – and through remittances, which are important in household incomes and the balance of payments. Migration also plays an important role in the increasing integration of the Mexican and US economies, a process also spurred by NAFTA.

This chapter reviews the main characteristics of migration in Mexico and then discusses migration policies in Mexico and in the United States insofar as they affect Mexico. The last section turns to economic aspects of migration, including those associated with illegal entry into the United States, labour market incentives and impacts, human capital issues and remittances.

Characteristics of migration in Mexico

Emigration has been growing

Mexican migration flows are dominated by emigration to the United States. Immigration is rather limited, with 500 000 foreigners residing in Mexico in 2000, representing about 0.6 per cent of the population. Foreigners mainly come from North America, and a number of them are in fact children of migrants returned from the United States.[124] They are often professionals working in multinational corporations located in the northern border area. There is also a tradition of seasonal migration of agricultural workers from Guatemala in the south of the country, involving between 50 000 and 75 000 legal entries a year as well as an unknown number of illegal entries.[125] But those immigration flows which have been growing fastest over the last two decades are transit flows of Central Americans emigrating to the United States. About 200 000 persons transit legally through Mexico every year, and although there is no available estimate of illegal transit flows, the number of apprehensions led by the Mexican authorities reached 150 000 in 2001 and

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138 000 in 2002 (against 10 000 in 1980).[126] (The related policies are reviewed in some detail at the end of the next section.)

Mexicans themselves have a long history of migration to the United States. Large-scale flows began around 1900 with the extension of railroads in the south of the United States and the connection with the existing Mexican railroads. They grew after the First World War, and again during the Second World War, when the first bilateral B*racero* (day labourer) programme was set-up by the two governments, providing a six-month renewable visa for seasonal agricultural work in the southwest of the United States. Large-scale repatriation occurred in periods when economic conditions were poor in the United States, sometimes organised by the US government. The phasing-out of the B*racero* programme in the early 1960s, under pressure from US religious and labour organisations, did not halt the upward trend in migration flows, and net outflows of permanent migration from Mexico grew steadily over the last four decades, to reach about 360 000 per year in the period 1995 to 2000 (Figure 31, Panel A). In fact, in the second half of the 1990s, Mexico has been the country with the highest emigration rate in the OECD (by far) and one of the highest in the world (Figure 32). As a consequence, in 2000, 8.5 million Mexican-born were residing in the United States, amounting to about 30 per cent of the foreign population in the United States, and almost 9 per cent of the Mexican population. According to a demographic survey, in 1997, 18 per cent of households in Mexico had direct family relatives with some migration experience to the United States.

Migration flows are diverse in nature, in particular in terms of permanence of residence, and associated return flows. Emigrants may stay in the United States for temporary periods or they may settle permanently. Temporary migrants may remain from days to months but consider Mexico to be their place of principal residence; they are often "circular" migrants who regularly work short periods in the United States, a traditional example being the agricultural worker migrating for the harvest season in the south of the United States. Other emigrants consider the United States to be their permanent residence, even if they return regularly to Mexico for visits.[127] Often though, the distinction between these two migration patterns is more a matter of degree than of type, since an emigrant may start with a number of temporary jobs in the United States but may stay longer as time passes and may decide to establish residence there. A decrease in net flows was expected in the second half of the 1990s as a result of a reduction in the demographic pressure, according to the Binational study on Migration between Mexico and the United States (1997). Yet it did not materialise, and flows remained about as high as in the first part of the decade, highlighting the importance of inertia in migratory flows associated with the building of networks and economic hardship in Mexico after the 1995 peso crisis. A survey of migration conducted at the northern border (EMIF, see Annex IV) provides information on temporary flows of people residing in Mexico (which came in addition to permanent migration flows). On this

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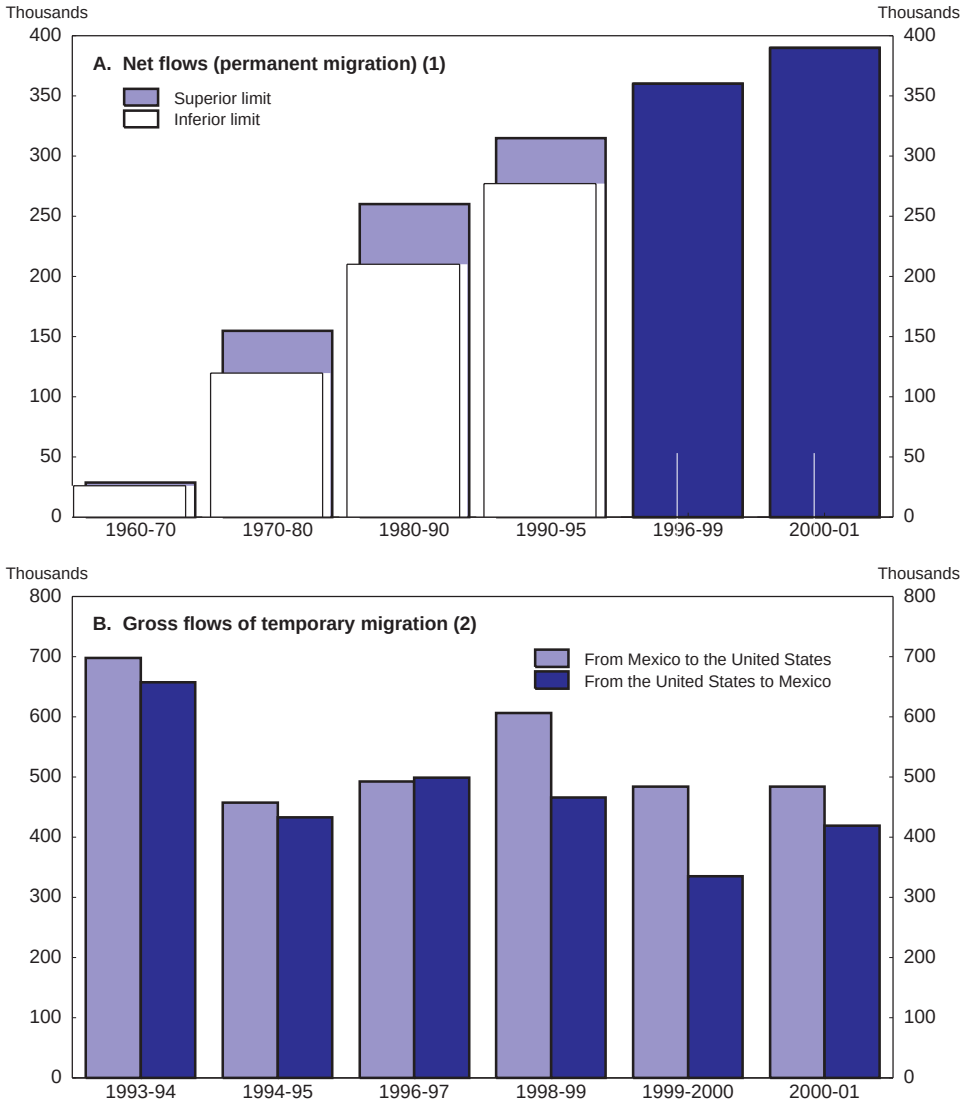
Figure 31. **Migration flows between Mexico and the United States**

Yearly averages



1. For years 1960 to 1995 interval estimates provided by the Ministry of Foreign Relations; for years 1996 to 2000 estimates from CONAPO.
2. Refers to all temporary flows of people residing in Mexico, most of which cross the border for work motives.

Source: CONAPO, *Surveys of Migration to the Northern Border* (EMIF); Ministry of Foreign Relations, Binational Study on Migration between Mexico and the United States, 1997.

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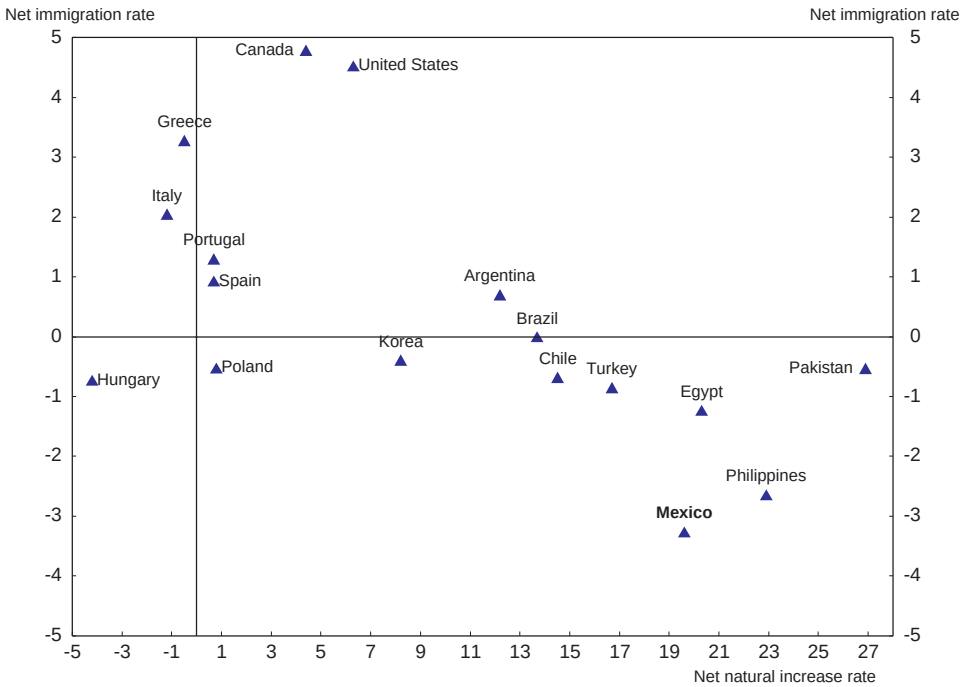
Figure 32. **Net immigration and natural increase of population in selected OECD and non-OECD countries**[1]

Rates per 1 000 inhabitants 1995-2000



1. The net immigration rate is the number of immigrants less the number of emigrants relative to the average population of the country. The natural increase rate is the annual number of births less the annual number of deaths relative to the average population.

Source: OECD, SOPEMI; United Nations Population Division.

basis, temporary emigration is estimated at about 540 000 per year, on average since 1993, with return flows of 470 000, with no clear trend (Figure 31, Panel B). Temporary outflows for labour motives, on the other hand, have tended to decrease and in 2000-01 they were lower than return flows.[128]

Migrants differ also in terms of their legal status. Although by nature difficult to estimate precisely, the numbers of unauthorised Mexican migrants – who have either entered in the United States without permission or have overstayed the term of their visa – are very large. According to CONAPO (2001), they were 3.5 out of the 8.5 million Mexican-born residing in the United States in 2000, while the US Immigration and Naturalization Services (INS) estimated that 4.8 out of

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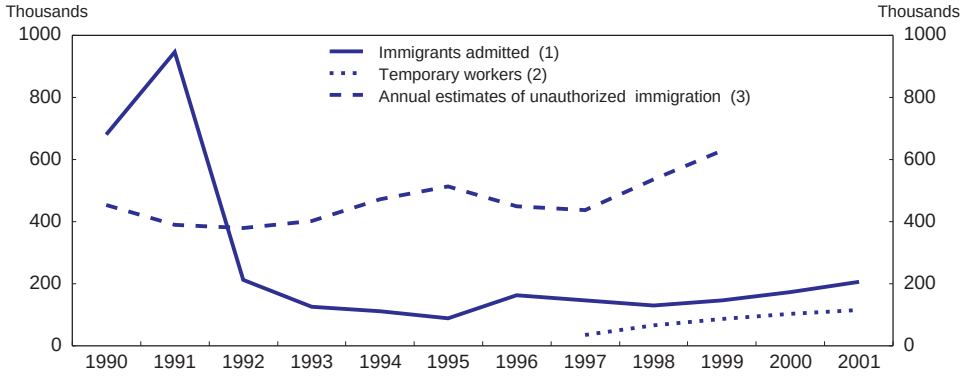
Figure 33. **Gross immigration flows of Mexicans in the United States by legal status**



1. Fiscal years. Persons lawfully admitted for permanent residence in the United States. Include both new entries and adjustments of previous status (temporary workers, undocumented, students).
2. Fiscal years. Includes visitors and intracompany transferees.
3. Estimation of gross entries for calendar years.

Source: United States Department of Justice, Immigration and Naturalization Service (INS).

9.1 million Mexicans were staying without authorisation. Legal permanent immigration into the United States increased substantially at the start of the 1990s, mainly due to the massive legalisation programme started in 1986, but unauthorised entries seem to be largely dominating the flows since 1992, exhibiting a rising trend (Figure 33). As to the temporary/circular flows, two-thirds of the people were crossing the border without papers at the end of the 1990s.[129] Overall, the US border patrol apprehended 3.9 million deportable Mexicans between 1999 and 2001.

A number of Mexican-born people have also acquired US citizenship. Naturalisation, however, is less frequent for Mexican-born people than for other foreign-born, with 20 per cent of the Mexican-born residing in the United States legally being naturalized in 2000, against an average of 37 per cent for the foreign-born population – a feature probably related to the geographic proximity of Mexico and the corresponding lasting cultural ties with the country of origin.

The emigrant profile is changing

The bi-national migration study conducted jointly by the Mexico and US governments in 1997 identified two main migration patterns: the sojourner migrants, who tend to be young men, with little schooling, working in agriculture and with low earnings; and the settlers or permanent residents, who tend to resemble more the US population as a whole, being more balanced between genders and better educated than the sojourners (Table 13). Over time however,

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Table 13. **Characteristics of Mexico-born migrants compared with Mexico's and US' populations**

Characteristics	Mexico resident population[1]	Mexico ENADID 97 Population having worked or working in the US at the date of the survey	EMIF 1998/2001[2]		US Resident population Census 2000	
			Migrants going to the United States	Mexican migrants coming back from the United States	Mexican-born	Total Residents
A. Demographics						
Age (median)	22.0	. .	. .	. .	32.6	35.3
Age (average)	26.0	. .	34.0	33.0	. .	. .
Male proportion	48.8	83.6	79.5	87.3	54.2	49.1
B. Schooling[3]						
Years (average)						
Primary incomplete	35.0	40.7	25.1	28.1	16.5	1.6
Primary complete[4] and lower secondary	38.7	41.6	48.5	54.0	31.9	7.5
Upper secondary complete[4]	12.7	10.4	}26.4	}17.9	38.2	40.0
Tertiary	12.5	6.1			13.5	51.0
Not specified	1.0	1.2	. .	. .	. .	. .
C. Labour force[5]						
Total participation	54.5	82.4	. .	. .	69.2	67.2
Male	75.1	. .	. .	. .	87.4	74.7
Female	35.9	. .	. .	. .	47.7	60.2
Unemployment rate	1.9	. .	. .	. .	7.3	4.0
Employment Sector						
Agriculture	17.9	. .	35.7	33.2	13.3[5]	2.4
Construction/ manufacturing	24.8	. .	22.6	29.6	36.2[5]	22.1
Services	57.3	. .	41.3	36.9	50.5[5]	75.4

1. For parts A and B data source is the 2000 Mexican Census. For part C data source is the 2002 National Employment Survey (ENE).
2. Corresponds to phases spreading from 11 July 1998 to 10 July 2001. For parts A, B all emigrants. For part C only worker emigrants.
3. Educational attainment of population aged 25 years and older, except for EMIF 12 years and older.
4. In the case of EMIF includes tertiary education.
5. Current Population Survey (CPS) March 2000 data for Mexican-born.

Source: INEGI; CONAPO; US Census Bureau.

although there is no single comprehensive and reliable source to analyse the profile of Mexican migrants, data from the various available sources (see Annex IV) suggest that the characteristics of migrants are becoming increasingly diverse.

The origin of migrants remains quite concentrated in a few traditional sending states as well as in rural areas, although increasingly, migrants tend to

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come from other states in Mexico and from urban areas. Between 1998 and 2001, 42 per cent of the temporary migrants crossing the northern border still came from the traditional sending states located in the western part of the country[130] – Zacatecas, Michoacán, Guanajuato, Nayarit and Durango being the most important –, which concentrated a much smaller share of the total Mexican population. There has been a steady increase of emigration from some central and southern states, such as Puebla and Oaxaca, which made only a minor contribution until the 1980s, and from the state of Mexico. With the progressive urbanisation of the Mexican population, urban areas have become the main source of migration flows since the 1980s, even in the sojourners group, where the share of migrants coming from urban areas now matches the composition of the total population.[131]

Along with this diversification in the place of origin, the occupational structure of migrants is also evolving. Temporary migrants are still more often employed in agricultural activities than the average Mexican before and after crossing the northern border, but less and less so (one out of three in 2000/2001 against one out of two in the early 1990s). Settled migrants, on the other hand, are much more likely to be employed in construction and manufacturing or in services than the temporary migrants (Table 13).[132] California remains the main destination, absorbing about half of the flows, along with Texas and Illinois; but due to the particularly strong reinforcement of border controls and other repressive policy measures *vis-à-vis* undocumented immigrants in California, the other states are gaining importance.

Increasingly too, migrants tend to stay longer in the United States. First, the share of settlers is increasing, as evidenced from the above statistics on flows. Demographic surveys also show that the return rate of migrants has decreased in the more recent cohorts.[133] In Guanajuato for example, 8 out of 10 international migrants have established their residence in the United States over the 1990s.[134] Circular migration also seems to have become less frequent, since it is more and more common to cross the border without a first migration experience.[135] Second, the average duration of temporary stays is lengthening. Traditionally, it used to be of about 7 months for seasonal agricultural workers, but migrants now frequently combine seasonal jobs with temporary jobs in other sectors. Urban migrants also tend to stay longer: a survey conducted in Mexico City found an average duration of stay of 22 months (Lozano, 2000). A number of factors explain that trend, including: the massive legalisation started by the United States in 1986, the family reunification process, the reinforcement of controls at the border which makes crossing much more difficult and expensive, and the shift in occupations. Although men still largely dominate temporary flows, with four men out of five migrants at the end of the 1990s, more and more women are coming to the United States, very often to settle.

Migrants do not generally come from the lowest socio-economic groups in Mexico nor from the most marginalised areas. According to CONAPO (2002), only

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26 of the 386 most marginalised municipalities had a high or very high intensity of international migration.[136] Thus, indigenous municipalities, often among the most marginalised, make a small contribution to international migration. Settlers are clearly more educated than Mexican residents (Table 13). As to temporary migrants, those who had returned from the United States by 1997 were less educated than the average Mexican, in part reflecting the relatively low education level in the traditional sending regions. Temporary migrants crossing the northern border have become significantly more educated over the 1990s, reaching an education level comparable on average to that of the Mexican-born residing in the United States of the same age category.[137] Temporary migrants now also include educated Mexicans from urban areas going to the United States to earn extra revenue – although they occupy the same low-skilled jobs as the other Mexicans in the United States (Lozano Ascencio, 2000).

Leaving Mexico requires the accumulation of a certain amount of resources for the journey, as well as for supporting the family until the migrant having a job can send back money for the first time. This is all the more the case if the migrant does not have access to a network of relatives already in the United States. Those who lack "connections" rather migrate inside the country, to the northern border area, for instance, where they can find jobs in *maquiladoras*, or, as many indigenous people, to the northern agricultural regions to work as *jornaleros* (day workers). Some of them may then decide to cross the border, when they have accumulated enough money and information. Available evidence shows that having already made an internal migration increases the probability of emigrating.[138]

Migration policies[139]

A large number of policies affect migration flows, including economic policies in both origin and destination countries. This section concentrates on specific migration or migration-related policies, while economic aspects will be addressed in the following sections.

Emigration from Mexico to the United States

US immigration policies are having an impact on migration of Mexicans to the United States. The extent to which such policies can determine the level of flows is debatable, but they are certainly affecting the nature of flows. In the recent past, the two most significant US policies enacted have aimed at reducing illegal immigration. The Immigration Reform and Control Act (IRCA), passed in 1986, allocated more funds to the Immigration National Services and to border enforcement, and established sanctions on employers who knowingly hired undocumented workers. At the same time, IRCA offered legal permanent status to almost three million persons out of a total undocumented population estimated at between five and six million at the time, about two thirds of which were of Mexican

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origin. Ten years after IRCA, though, as illegal immigration continued, the 1996 Illegal Immigration Reform and Immigrant Responsibility Act, which was passed in combination with laws dealing with welfare reform[140] and terrorism, followed up on some IRCA provisions: penalties on smugglers and immigrants entering illegally were further increased, deportation was facilitated, and a doubling of the Border Patrol by 2001 was mandated (see section below). The changes were only partly successful, and illegal entry has remained at very high levels, building up the stock of undocumented Mexicans in the United States.

The limited size of authorized immigration in the United States – relative to US employers' demand for low-skilled labour and migrant workers supply – together with the fact that sanctions on employers are rarely enforced – largely explains the failure to halt illegal immigration. Only a small share of permanent immigrants comes into the United States specifically for work reasons (a maximum of 7 300 Mexicans, mostly highly qualified). The vast majority comes under the family reunification motive (196 000 in 2001), and they are allowed to work. The number of Mexicans authorised to enter the United States through temporary work visas was also relatively small until 1997. It has been growing since then, reaching approximately 115 000 in 2001, but is still far below the estimate of unauthorised inflows (Figure 33). Two-thirds of the temporary workers enter the United States to work in low-skilled jobs, in agriculture (with H2A visas) and increasingly in other sectors (with H2B visas). Highly-skilled Mexicans, who represented only about one tenth of temporary workers in 2001, come into the United States with a different visa (H1B visa). Since 1994, highly-skilled migrants can enter through the provisions of chapter 16 of NAFTA; unlike Canadians, Mexicans have been subjected to an annual cap (set at 5 500 persons). However, the cap has not been binding and it is to be removed in January 2004.

Regarding the outlook for migration flows, the US Office of Labour Statistics estimates that the United States will face a shortage of workers over the coming years, and that a little more than half of the jobs being created will require personnel with modest training only and formal education below high-school level. Thus, with the number of people of working age in Mexico growing by 1.3 million per year,[141] and wage differentials likely to remain large (see below), migratory flows from Mexico to the US are expected to continue to grow.

The position of the Mexican authorities *vis-à-vis* Mexico-US migration flows was for long one of deliberate non-engagement. However, since the 1990s, in particular after NAFTA came into force, the Mexican authorities have started to take a more active stance on migration matters. Although the country has no "emigration policy" as such, it is trying to pursue certain objectives regarding the situation of Mexicans residing in the United States on the one hand, and regarding migration flows on the other.

There has always been an interest in protecting the rights of Mexicans living in the United States, but since the mid-1990s, the Mexican government has

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been strongly involved in reaching out to the Mexican-born community, with a view to improving its situation. At end 1996, the Mexican Constitution was changed to allow Mexicans to keep their nationality even if they opt for another. This means that those who get naturalised have a political voice in the United States, while keeping their Mexican citizenship. The network of Mexican consular offices in the United States has been extended over the years, to provide general assistance to Mexicans travelling or residing there and to ensure that they have access to the justice system. These consulates also provide consular IDs (*matricula consular*) to Mexican migrants, including unauthorised migrants, in the United States. The new matricula consular is a sophisticated identity card with enhanced security features that encodes a great deal of information. IDs for Mexicans abroad have existed for more than a century. They were relaunched by the Mexican consulates in the United States after the setting of the bilateral agenda in 2002. By August 2003, the matricula consular was recognised by more than 14 states, 100 cities as a valid identity document by the police in particular or to obtain a driving license, by 100 financial institutions when opening an account.[142] Apart from this "legal" support, in the early 1990s, the Mexican authorities established a Programme for the Mexican Communities Living Abroad – which has now become the Institute for Mexicans Abroad – to institutionalise relations with the emigrant community and its organisations, through activities in community organisation, education and culture, and with the creation of a network of cultural institutes, healthcare, and information. The programme also aims at promoting business ties between Mexicans and Mexican-Americans, in particular through the Council for Business Promotion. Given the role that they play in their communities of origin and the amount of remittances that they send, some groups of Mexican residents in the United States are now also militating to keep their voting rights in Mexico, but no agreement has been reached on this issue.

Regarding migration flows, the position of the Mexican authorities has moved over time towards increasing dialogue with the US authorities and accepting some responsibilities. The Binational Study on Migration between Mexico and the United States (1997), which started in 1994 at the initiative of both governments to analyse the nature, causes and consequences of migration for both countries, provides an illustration of this new attitude. After the Mexican elections in 2000, migration issues became high on the agenda of the new administration, and in February 2001 the presidents of both countries set a bilateral agenda on migration, comprising five items:

- regularising about 3.5 million undocumented Mexicans living in the US;
- establishing a jointly managed guest-worker programme, along the lines of that existing between Mexico and Canada for agricultural workers;[143]
- increasing the number of immigration visas provided to Mexicans;

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– strengthening border security through co-ordinated actions, in particular to prevent the death of migrants and fight the trafficking of persons;

– promoting regional development in Mexican regions of high migratory intensity.

In the field of border security, the Mexican authorities have taken a number of steps over recent years. Some progress has been made regarding immigration into Mexico (section below). On other items, however, progress has been slow, after the negotiations with the United States to reach a bilateral migration agreement were interrupted in the aftermath of 11 September 2001. In the United States, a guest-worker programme is no longer envisaged. On the Mexican side, the Labour Ministry is now working on a plan to better organise the recruitment of temporary workers going to the United States with H2A and H2B visas, which currently suffers from deficiencies. US employers have been using intermediaries to recruit migrant workers in Mexico, and irregularities have been commonly reported in the way these operate, charging workers large fees for the provision of a visa.[144] The plan would be to allow workers willing to work in the United States to put advertisements on a website or with a telephone employment service (Chambanet and Chambatel, both already operating for domestic employment). While reducing the risks of fraud and the cost of recruitment intermediaries for US employers, this would also allow workers to be informed of their rights once in the United States. The Mexican government has also undertaken a number of initiatives to try to reduce the costs for migrants sending back remittances and promote the investment of remittances in the regions of origin (see below).

Given that there is little possibility of negotiating a migration agreement with the US government in the near future, the Mexican authorities now seem to be shifting to a more bottom-up strategy, trying to build support from various sectors in the US society. Increasingly, they are also working at the state (or even municipal) level rather than at the federal level. Migration matters are under federal jurisdiction in both countries, but in practice the impacts of migration are predominantly felt at the state (and local) levels, and responsibilities in some affected areas are increasingly devolved to state and local authorities, as is the case for welfare in the United States or education in Mexico. Traditional sending states in Mexico are trying to increase their knowledge regarding migration affairs and are seeking more active intervention through the creation of specialised institutions.[145] The objectives of the state offices are: *i*) encouraging potential migrants to remain in Mexico through the development of their regions of origin; *ii*) establishing cultural and commercial links with source localities and migrants clubs in the United States; *iii*) improving the health status of migrants communities in both countries; *iv*) promoting programmes of temporary work visas with Canada and the United States; and *v*) favouring cost reduction for transfers of remittances. There has not been a clear statement of the objectives of Mexican policy towards

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emigration flows, and both federal and state governments seem to want both to reduce emigration flows and to keep "the door open" to the United States.

Immigration

The legal basis for immigration policy in Mexico is provided by the General Population Law. Family connections and employment skills are emphasized as a basis for admission, and the lack of details provided in the law on objectives and rules confers significant discretion to the Mexican authorities. Highly-skilled migrants coming into Mexico from the other NAFTA countries benefit from a special regime and in 2000, 170 000 businesspersons and professionals from the United States and Canada entered Mexico.[146] More generally, immigration in practice is relatively easy for highly-skilled people, and much more difficult for the low-skilled. Highly-skilled immigrants can get two types of visas, a "non-immigrant" visa for a temporary period or an "immigrant" visa (initially for one year, and possibly permanent after 5 years of residence). They are provided on a sectoral basis, under the general rule that there should be no more than one foreigner for every 9 Mexicans employed in a firm. Some 22 500 temporary visas were issued to such highly-skilled immigrants in 2002. As to the low-skilled, a visa arrangement (rather than a true guest-worker programme) is in place for seasonal agricultural workers coming from Guatemala to Mexico's southern states; in 2002 about 39 000 persons came as agricultural temporary workers.[147]

Illegal immigration through the Guatemalan border in transit to the United States or to stay in Mexico is important. So has been the effort by Mexican authorities to reduce its scale through expulsion. The General Population Law, which regulates immigration, was changed to increase penalties against those involved in migration-related traffic. The regional conference on migration, or "Puebla Process", was launched in 1996 as an attempt to bring governments from North and Central America together, to address regional migration issues (including migrants' human rights) jointly. It has resulted essentially in Mexico and its southern neighbours (Guatemala and to a lesser extent Nicaragua) tightening their control on undocumented migrants. The controls on transit migrants coming from Central America through Mexico's southern border have been further tightened since July 2001 with the *Plan Sur*, and massive deportations have been conducted.[148] In 2000 and 2001, two regularisation processes have been conducted for immigrants who had been residents in Mexico for more than two years. The number of applications processed is rather low, however, compared with the stock of undocumented Guatemalans estimated to be around 150 000 (OECD, 2003).

Economic aspects of migration

Migration from Mexico to the United States is primarily economically motivated. This section deals both with economic factors and economic consequences

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of migration. A discussion of the cost of smuggling follows labour market and human capital issues associated with emigration. Finally, the size and impact of remittances, as well as the policies to reduce their costs and improve their use, are assessed.

Labour market and human capital

Labour market incentives

Wage differentials between Mexico and the United States, already high in 1980, have continuously increased since then. In 2001, a production worker in the manufacturing sector was earning 5 times more per hour worked in the United States than in Mexico (Figure 34). The difference is even larger when compared with *maquiladoras*, where wages are about one third lower than those in the rest of the manufacturing sector. As to observed earnings, a survey realised in three important sending states found that on average hourly wages of migrants in the United States were about six times higher than those of the employed population in their communities of origin at the end of the 1990s.[149]

As would be expected, given these wage differentials, work is the main motive for emigration of Mexicans to the United States, which is illustrated by the significantly higher participation of Mexicans in the labour force in the United States than in Mexico (see Table 13 above).[150] These labour flows result from a combination of push and pull factors on the Mexican and United States labour markets respectively. The initial motivation for the migrant flow lies largely inside the United States, where US farmers have recruited Mexican workers for decades, creating linkages between jobs in US agriculture and workers of particular Mexican communities (Binational Study, 1997). Since 1990, the strong growth in the demand for low-skilled labour in the tight US labour market has been a new pull factor for Mexican workers. Part of this increase in demand for low-skilled (and low-paid) workers has followed the implementation of NAFTA, when US firms have had to restructure and adapt their production processes and organisation after the trade and investment liberalisation. In the garment industry, for example, after the restructuring prompted by the increased competition from Mexican exports, the share of Mexicans in the sector's employment more than doubled, reaching close to 20 per cent in 2000 (Spener *et al.*, 2002).[151] Spener and Capps (2001) show that there are strong connections between manufacturing trade and migration flows: the sectors where Mexico's exports have been growing most over the 1990s being also those where employment of foreign and Mexican-born are the highest in the United States – and have been increasing most, pointing to some complementarity between trade and investment liberalisation and labour migration.[152] The construction of networks also appears to be key in helping migrants find a job in the United States and ensuring support for newcomers (Munshi, 2002).

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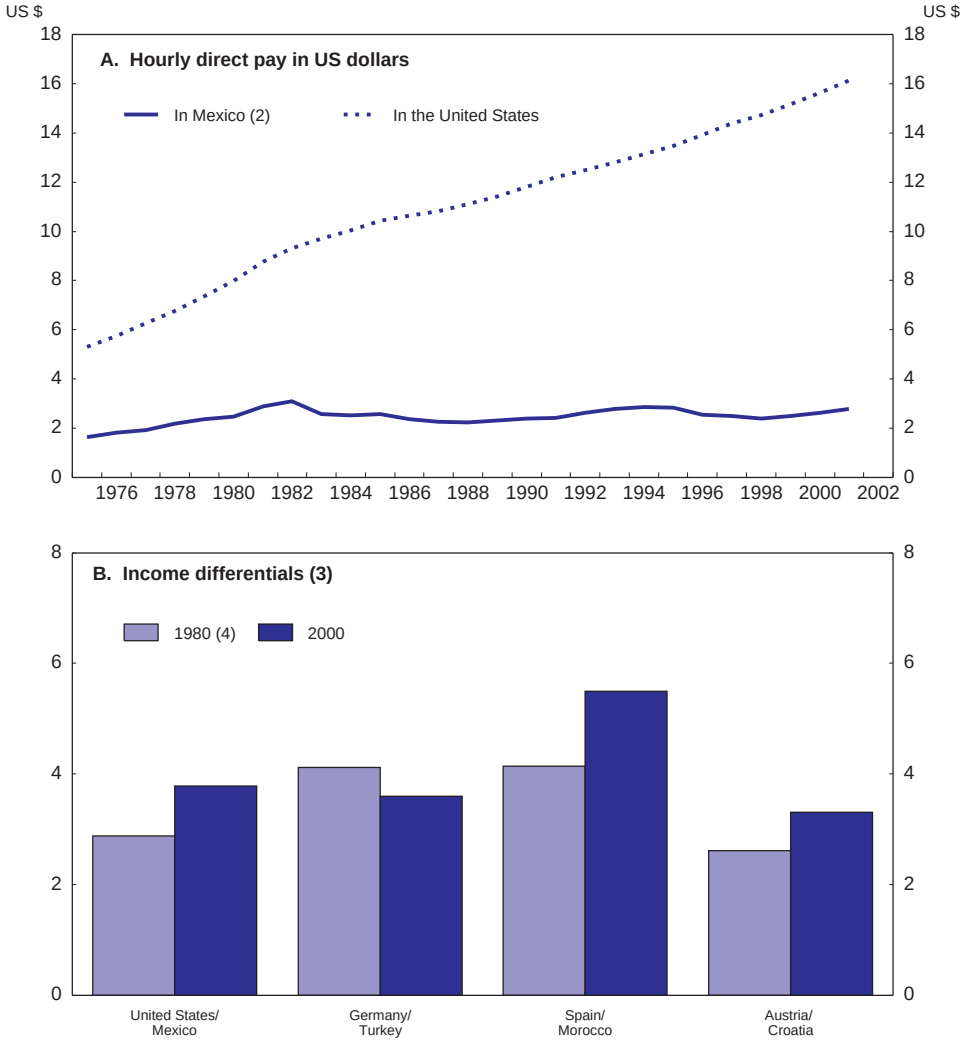
Figure 34. **Hourly pay in manufacturing – Mexico and the United States**[1]
For a production worker



1. Hourly direct pay includes all payments made directly to the workforce before payroll deductions of any kind.
2. Using Purchasing Power Parities (PPPs) for private consumption.
3. Ratio of GDP per capita in current PPPs in wealthy OECD countries to that of source countries.
4. 1990 for Austria/Croatia.

Source: United States Department of Labour, *International comparisons of hourly compensation costs for production workers in manufacturing*; United Nations, 2002 *World Development Indicators*.

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Since the mid-1980s, however, push factors have also become important, as a result of rapid population growth in the 1970s, and recurrent Mexican economic crises, as well as significant labour adjustments, following structural policies, such as trade liberalisation and privatisation, and the restructuring of rural Mexico with low profitability of small-scale farming (Annex I.B). In the absence of unemployment benefits, most people have no choice but to work for subsistence, and the open unemployment rate has remained relatively low in that setting, generally below 3 per cent since 2000, after peaking at 6 per cent following the peso (1995) crisis, but only thanks to an increase in informal employment and a sharp decline in real wages. Over the 1990s, less than half of job creation in Mexico has been in the formal sector (see Chapter I). Formal employment growth recovered after the peso crisis, with net job creation above 500 000 every year for 5 years (based on IMSS registers). Although real wages rose steadily in the late 1990s, the increase did not offset the sharp decline following the 1995 crisis and in 2000, real wages in the formal sector were still substantially below their 1980 level.[153] Wage differentials between the United States and Mexico are commonly reported as being the main driving force for migration, while job availability in Mexico plays less of a role – about 85 per cent of the temporary migrants to the United States over the period 1998-2001 had a job in Mexico before emigrating.

Labour market effects

The large size of migrant outflows could be expected to have significant effects on the Mexican labour market. In 2000, the Mexican-born workers in the US represented the equivalent of 13 per cent of the Mexican labour force (Figure 35). In the few Mexican states with high emigration, the ratio reaches from 25 to more than 50 per cent. Net outflows of permanent migrants represented 14 per cent of the 16-year-old population entering the labour force. Had these migrants stayed in Mexico, the labour force would have grown at a still faster rate than observed between 1990 and 2000 (by 0.7 percentage point per year).

Analysis of the consequences of this reduction of the labour supply is very scant though. Given the low open unemployment level, the reduction in labour supply is more likely to have reduced employment in the informal sector than unemployment *per se* (open unemployment). It has also probably exerted some upward pressure on wages, preventing them from falling more than they did in real terms. This seems to be the case in the traditional emigration regions. Hanson (2003) finds that regional wage differentials have widened in Mexico over the 1990s, in particular, regions with access to foreign trade and investment performing better, and states with high historical migration rates also performing better. Apart from the level of wages, migration may also affect the wage distribution. Relative to the United States, Mexico has high wage dispersion and high returns to education[154] – both having increased with trade liberalisation – implying that the

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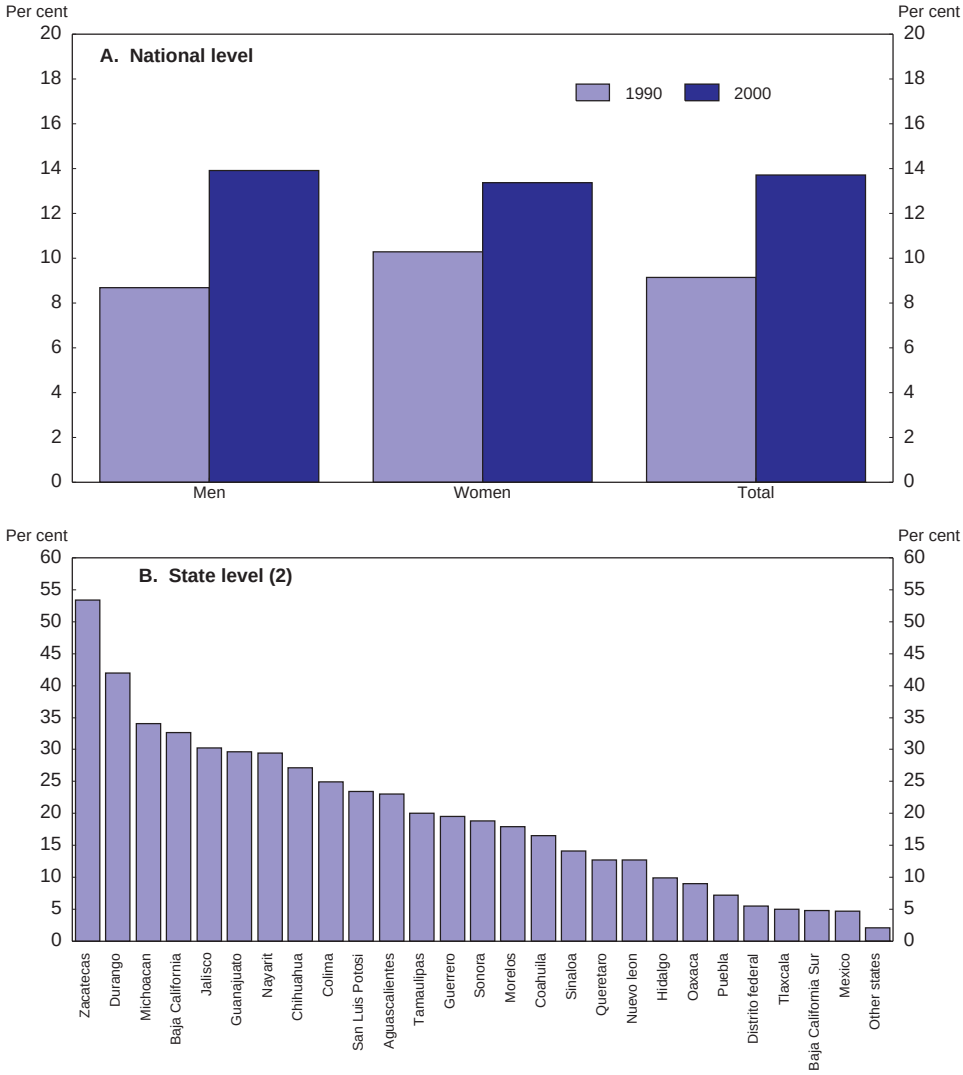
Figure 35. **The Mexican labour force in the United States**

As a percentage of Mexican labour force[1]



1. Persons 15 years and over in Mexico and 16 years and over in the United States.
2. OECD estimates. The Mexican population residing in the US has been allocated across the Mexican states using the data provided in ENADID (1997). The activity rate of immigrants from the different Mexican states in the United States is assumed to be unique and equal to that of all Mexican-born residents in the United States.

Source: INEGI, *XII Censo General de Poblacion y Vivienda 2000 and ENADID97*; US Census Bureau, *Census 2000*.

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United States-Mexico wage differentials decrease with education. However, contrary to what would be expected in that context, Mexican immigrants in the United States, as noted above, are on average more educated than Mexican residents. Chiquiar and Hanson (2002) show that in 1990, the education group of Mexicans most represented in the United States was that with 12-15 years of schooling, corresponding to upper high-school and the level just above, which fall in the middle and upper segments of Mexico's wage distribution. This pattern can be explained by the fact that the cost of migration is high (see section on smuggling cost below) and probably even higher for less educated Mexicans.[155] By removing workers from the middle of Mexico's income distribution, migration is likely to contribute to raising wage dispersion in Mexico.

Is *there a brain drain risk for* Mexico?

With immigration policies in OECD countries becoming increasingly selective and skilled-biased, the concern has been raised that migration of skilled people, while clearly beneficial for the host country, may hamper the innovation capacity in sending countries, especially developing ones, and thus limit their ability to grow and catch up. The possible negative effects for sending countries would arise from the (at least temporary) loss in human capital and productive capacity, but also from the lower return on public investment in tertiary education, *i.e.* a waste of national public resources. On the other hand, sending countries may benefit from highly-skilled migration through a number of channels. The possibility of emigrating may provide an incentive to individuals in the sending country to accumulate skills, both because migrating abroad would increase the return on their human capital investment, and because it may raise the domestic return to skills. Besides, skilled emigrants may return and diffuse the skills they have acquired abroad, as well as favour the exchange of knowledge and the collaboration with foreign institutions and enterprises.[156]

Relative to other less advanced countries, the number of highly-skilled Mexican migrants in the US is not very high. Consistent with the relatively high dispersion in earnings (and in education) in Mexico, graduates and/or post-graduates are relatively under-represented in the Mexican diaspora in the United States In 2000, the highly-skilled represented 4 per cent of Mexican-born residents in the United States aged over 20, against 13 per cent of the population in Mexico. Managers and professionals account for only 6 per cent of the Mexican workers in the United States, against 39 and 26 per cent for the Asian and African worker communities respectively. When it comes to students, who are often a central component of the international migration of highly-skilled persons, the number of Mexicans is also relatively low, despite a significant growth since the entry into force of NAFTA (12 500 in 2001-02, of which 4 400 at the post-graduate level, *i.e.* about 3 per cent of the number of post-graduate students in Mexico).[157] Information on the return

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behaviour of the highly-skilled Mexicans is scarce, but since they are the Mexicans who integrate most easily in the US society, they would also be the most likely to stay. In 2000, 40 per cent of the highly skilled Mexicans in the United States had already acquired United States nationality.

Because the rates of educational attainment in Mexico are relatively low, migration to the US appears to take away a non-negligible share of the population with a tertiary education in Mexico, suggesting that Mexico is experiencing some "brain drain" as a result. Based on official Mexican estimates (CONAPO), Mexican-born population with a bachelor's degree (*licenciatura*) or higher education represented 6 per cent of the population with the same education level in Mexico in 2000. Somewhat higher estimates are provided by Adams (2003). In a study based on data from the US population census, he finds that the Mexican-born population in the US with a tertiary education represents over 16 per cent of the nationals with that level of education in Mexico.[158] As a comparison, for 14 countries out of 20 for which comparable data exist, less than 10 per cent of individuals with a tertiary education have migrated to the United States. In a few smaller Latin American countries, the proportion, however, is higher than 25 per cent.

Apart from wage differentials, the unsatisfactory work environment in Mexico and better conditions for career development in the US are important drivers of the migration decision, rather than job availability (with trade liberalisation and NAFTA, the demand for highly-skilled workers has been growing significantly in Mexico). This seems to be particularly the case in scientific careers, as research and development activities are very little developed in Mexico – in 1999, their share in GDP was 0.4 per cent, against 2.2 per cent on average for the OECD. In 2000, 15 per cent of the Mexican students who were provided with a scholarship to make post-graduate studies abroad at some point in the previous three decades had stayed abroad to work; while less than 2 per cent of those who got a scholarship to study in Mexico were working abroad, two-thirds of them in the fields of mathematics and natural science (CONACYT, 2000). Grantees who do not work for the Mexican government or in research departments after their study are required to repay their scholarship (and usually do so), thus limiting the loss of public resources. The National Council of Science and Technology (CONACYT) has also been managing a repatriation programme since 1991, which supplements the salaries of researchers in the first year after they come back to work in Mexico; its scale is limited though, since it has financed about 200 researchers per year on average.

Other human capital issues

There may be other human capital issues than brain drain at stake in an emigration country such as Mexico. Firstly, even with low- or medium-skills, migrants may benefit from their work experience in the United States, and

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become more productive than they would have been otherwise, if and when they come back to Mexico. Secondly, migration may affect the human capital of the family at large, in particular through education.

In the Unites States, most Mexicans occupy jobs in the low-skill segment of the labour market, as day workers in agriculture (one out of four day workers in the United States is Mexican), or workers in the textile/clothing industry, in construction, domestic and cleaning services or as cooks. Based on the Mexican Migration Project (MMP), a survey conducted since 1982 on both sides of the border, Reyes (1997) finds that migrants with low education and low-wage jobs in the United States, often undocumented, are precisely those with the highest probability of returning to Mexico. These jobs, which are often in the grey/informal sector – 88 per cent of the migrant workers crossing the border to come back to Mexico in 1998-99, had been working in the United States without a contract[159] – often involve repetitive tasks and little possibility of training. According to the EDER survey, only 6 per cent of former migrants had benefited from a professional training in the United States[160], suggesting limited possibilities of human capital enhancement. It cannot be assumed either that acquired skills can be transferred, since this would require that migrants are able to find jobs in Mexico where these can be used, and that they are willing to continue working in the same type of activity as in the United States.

Indirect observation suggests that previous migrants nevertheless generally benefit from their migration in terms of occupation status/level and wage. According to the EREM survey (Papail, 2002), the migratory cycle accelerates the labour force shift away from agriculture (before the migration to the United States) into trade, services and transport. On average, migrants surveyed in the MMP survey tend to occupy jobs slightly more qualified than the ones they had during their last stay in the United States, with for example higher-skilled manufacturing jobs. Using the same MMP data, Zahniser and Greenwood (1997) find that, once controlled for a possible selectivity bias over the 1982-1996 period, the return to United States experience, as measured by earnings, is about eight times the return to Mexican experience.[161] Not surprisingly, the returns are higher for those who have kept the same broad occupation as in the United States, thus allowing more transfer of acquired skills. However, the EREM survey points to migration being above all often a rapid way for migrants to operate a transition from salaried work to self-employment or small business ownership (20 per cent of previous salaried migrants did) – a shift which is taking place along the working life of non-migrants too, but at a slower pace – thanks to savings accumulated during the migration period, rather than a result of enhanced human capital.

Migration to the United States is also likely to have an impact on the family members who remain behind, and in particular on schooling outcomes for children. On the one hand, if migrants send remittances to their families, this may raise the

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family income and allow children in the household to obtain more schooling. On the other hand, having a migrant abroad implies a significant disruption of the family life, which may have negative consequences on a child's schooling outcomes. Providing the only available test of the relationship between household migration behaviour and educational attainment in Mexico, Hanson (2002) finds that the overall effect is positive, *i.e.* that children in migrants households belonging to the 10-15 year cohort complete significantly more years of schooling.[162] Children of migrants who follow their parents and spend part of the year in the United States and part in Mexico may encounter specific problems of repetitive adaptation. To facilitate their entry at school, several state governments and Mexico's federal government have developed a transfer certificate as part of a Binational Programme for Migrants Education, in which some exchanges and training of teachers involved with children of migrants take place.

Fiscal impacts

There is no available estimate of the fiscal costs and benefits associated with emigration. Compared with the hypothetical case in which the migrant is working in Mexico, a number of changes in tax revenues for the Mexican government would be expected in the short term. First, if the migrant is working in the formal sector before leaving, the person stops paying payroll taxes and income taxes. At the aggregate level, this shortfall might be compensated by the entry of a new worker in the formal market, or higher wages if there is labour shortage in the particular sector/qualification of the migrant. Second, the value added tax revenues are reduced by the amount the person was paying on his consumption before migrating, but increased by that paid by relatives on consumption resulting from the remittances the migrant is now sending. Third, the government stops its transfers to the migrant, notably for health services.[163] The net impact on public finances is therefore quite uncertain.

In the longer run, a usual argument made is that Mexico is losing a lot with emigration because it pays for the education costs of migrants as well as health costs during childhood and old age. Assessing such costs would require a generational accounting approach, which is not available. However, even if migration was indeed found to be costly in terms of public revenues/expenditures, it is not obvious what the policy implications should be. For example, Mexico cannot reduce the amount spent on education to equate the benefits associated with education (in the large sense, including economic, equity and cultural) to the costs associated with the fact that some leave the country, since it would imply a sub-optimal education level for those remaining. Besides, this approach would be too narrow, as it would ignore the more global effects of migration on fiscal revenues (for which a dynamic general equilibrium approach would be required).

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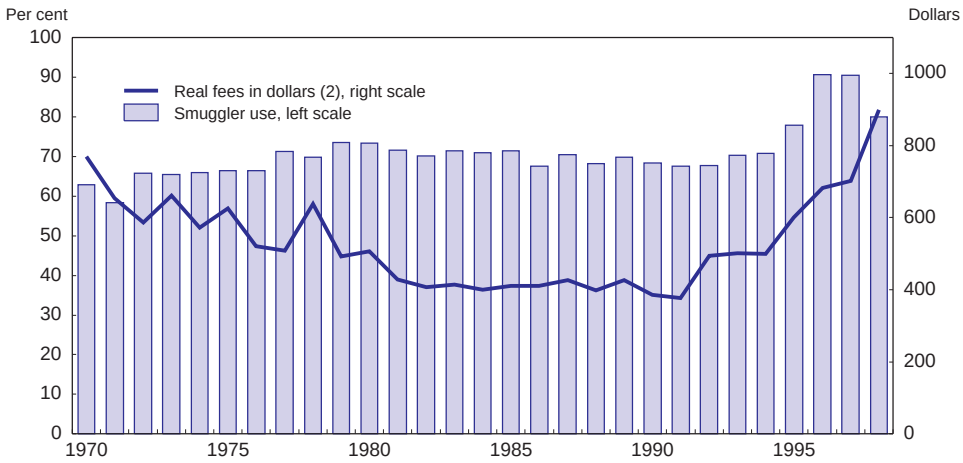
The cost of smuggling

The main response of the US authorities to rising illegal immigration has been to intensify border enforcement effort[164], by increasing the number of border patrol officers, their equipment and the technology used to deter, detect, apprehend and remove unauthorised immigrants. The strategy has been to block entry through traditional routes (*i.e.* the roads going through the main border cities) so as to force migrants to move through remote and sparsely populated areas where they can then be spotted and detained more easily. The strategy has succeeded in significantly raising the cost of illegal crossing. Crossing has become more expensive and more dangerous, migration candidates often risking their life.[165] Increasingly, unauthorised migrants have resorted to smugglers, a survey showing smuggler use rates at about 90 per cent at the end of the 1990s (Figure 36). The smugglers, commonly known as "*coyotes*" or "*polleros*", operate either from the interior of Mexico, recruiting groups of people in their community of origin, or from the border, often through organisations (Spener, 2001).[166] While the fees charged decreased in the 1970s and 1980s, because the supply of smugglers outpaced the demand, the *coyote* prices have trended upwards again since 1994 (Orrenius, 2001). The median *coyote* price reported in the Mexican Migration Project survey was about US$600 in 1998. In any case, with increased border patrol efforts, prices have continued to rise in recent years, and interviews commonly report fees of US$1 000-1 700 at the start of the 2000s.

Figure 36. **Smuggler use rates and fees**[1]



1. From 1996 onwards data points become slightly less reliable due to the smaller size of the sample.
2. Divided by CPI, 1994 = 100.

Source: University of Pennsylvania and University of Guadalajara, *The Mexican Migration Project*.

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This cost obviously reduces the net benefits from migration for the individuals, for their families and for the communities receiving the remittances. A fee of US$1 000 represents the equivalent of about one month of work at the US minimum wage in 2002. The following calculation provides an order of magnitude of the annual cost of smuggling at the national level: assuming an annual gross flow of 500 000 unauthorised migrants (about the average INS estimate over the second half of the 1990s) – of which 90 per cent use a *coyote* – and an average fee of US$1 000 (probably a low estimate), the annual cost of smuggling would reach US$450 million, or almost 5 per cent of the 2002 remittances (and about 0.1 per cent of GDP in that same year). Besides, since migrants most often do not have such an amount in advance and have to contract a debt often at usurious rates, the overall cost is probably higher. However, smugglers (and creditors) being Mexican residents, most of this sum probably remains in the Mexican economy, so that money is mostly redistributed among Mexicans.[167]

Remittances

Size and use

Being able to send money back to families who remain in Mexico is the main motive for emigration to the United States. Migration generates three main types of financial flows to Mexico. Pension benefits received by previous migrants from the US Social Security represent a relatively small amount, US$236 million in 2001 for about 49 300 Mexicans (corresponding to an average yearly benefit of US$4 782). Wages of trans-border workers residing along the northern border of Mexico are more important, as US$1.25 billion were paid to 81 000 workers in 2001 (*i.e.* an average wage of US$15 440 per year). But remittances sent by migrants working in the United States are by far the most important, amounting to close to US$9 billion in 2001 and rising since then, to almost 10 billion in 2002 and as much as 8.3 billion in the first eight months of 2003.[168]

Remittances have been multiplied by 4 between 1990 and 2002 according to central bank estimates – part of the increase being due to improved coverage in data collection and to increased use of formal channels[169] – and they were the second highest in the world in 2002, just after India. They now provide a significant and relatively stable source of foreign exchange in Mexico, equivalent to as much as 90 per cent of oil exports revenues, 145 per cent of tourism receipts, and 72 per cent of the net foreign direct investment flows in 2002. However, compared with other emigration countries, in particular those in Central America or smaller OECD countries such as Portugal and Turkey, the remittances to GDP ratio is relatively low (Figure 37). In fact, the effect of remittances is mostly felt at the regional level. In Michoacán for example, remittances are as high as 8.3 per cent of state GDP and they are equivalent to 60 per cent of the federal public transfers (Figure 38). Their

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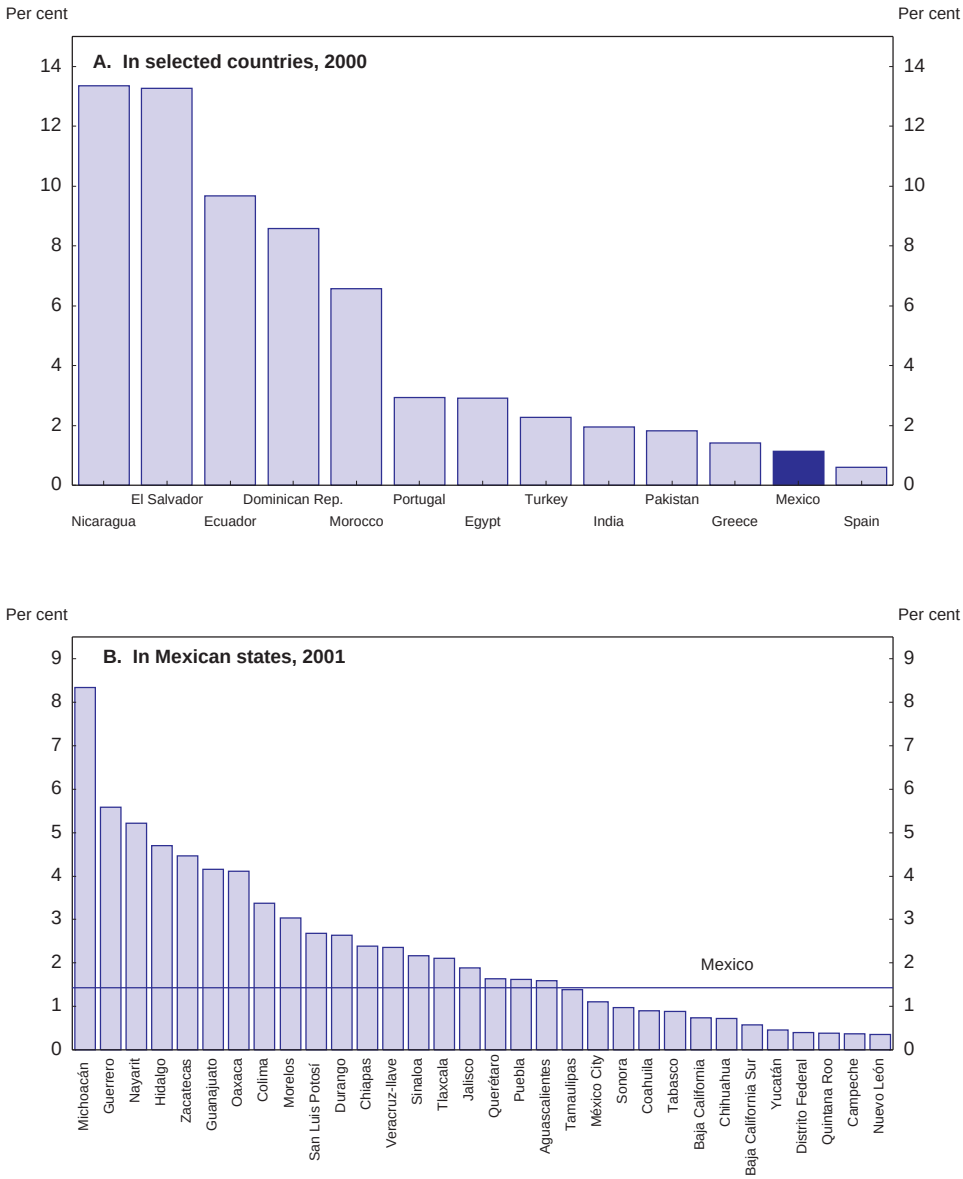
Figure 37. **Workers' remittances**
Per cent of GDP



Source: OECD; Bank of Mexico; IMF, *Balance of Payments Statistics Yearbook*.

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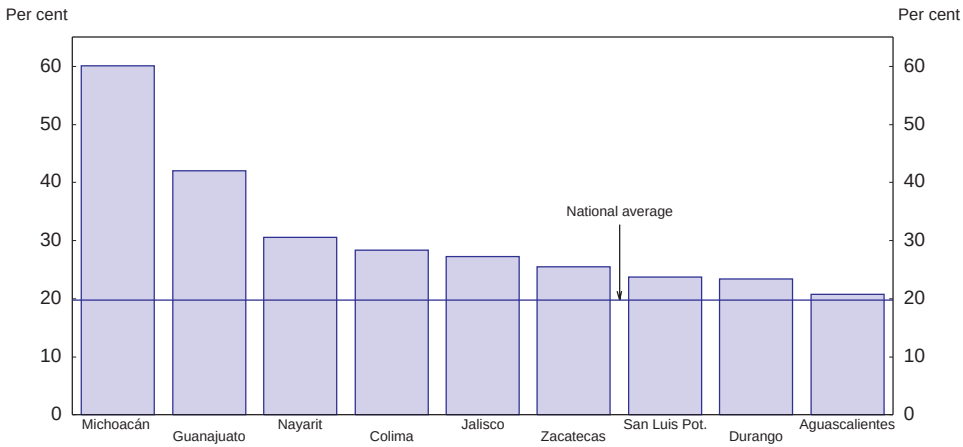
Figure 38. **Remittances and federal public transfers in the traditional states of emigration, 2001**

Per cent of total public transfers received[1]



1. Total federal contributions to the states (appropriation lines 23, 28 and 33 of the federal budget of expenditure).
Source: Bank of Mexico; Ministry of Finance.

relative importance is likely to be even higher in the 109 localities with very high emigration intensities.

According to the national household income and expenditure survey, about 1.2 million Mexican households, or 5 per cent of the total, were receiving remittances in 2000, for an average annual amount of US$3 016 per household. Although the household survey typically underestimates the total amount of remittances, it provides useful insights on the relative importance of these remittances for some of the households receiving them. Households receiving remittances belong to all deciles of the income distribution. For them, remittances represented on average almost 40 per cent of income, being a complement of income in the majority of cases, but the unique source of monetary income for 4 out of 10 households on the receiving end. The highest incidence is among rural households, and although the amounts received are lower than for urban households (Table 14), they are important in relative terms: the average remittance in rural areas is more than three times the average allowance provided through the main poverty programme PROGRESA/O*portunidades* to poor households living in marginalised areas.[170]

On the sending side, migrants from the traditional emigration regions in Mexico were sending back about US$328 dollars per month in the second half of

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Table 14. **Households and remittances**

	1992	1996	2000
Number of households receiving remittances (thousands)	659.7	1 076.2	1 252.5
Proportion of households receiving remittances (per cent)	3.7	5.3	5.3
Localities of less than 2500 inhabitants	6.2	10.0	9.9
Localities of 2500 inhabitants and more	2.9	3.8	4.0
Average remittances (dollars per year)	2 113	1 942	3 016
Localities of less than 2500 inhabitants	1 810	1 582	2 016
Localities of 2500 inhabitants and more	2 323	2 245	3 757
Share of remittances in current income of households receiving remittances (per cent)	28.7	40.9	38.8
Proportion of households receiving remittances where remittances are the only source of monetary income	n.a.	46.5	40.0

Source: INEGI, *Encuesta Nacional de Ingresos y Gastos de los Hogares* (ENIGH).

the 1990s, corresponding to a quarter of their income or one week of work in the United States (Papail, 2002). Proportionately, permanent migrants send money back home less often than temporary migrants, and relatively smaller amounts, but given their number they are responsible for most of the remittances (62 per cent in 1995 according to Lozano (1997)). They tend to send remittances most often to their parents (the majority of households on the receiving end have heads of households aged 50 or more in 2000) or to other relatives, while temporary migrants send money to their close family. When migrants have been staying in the United States for some time, remittances become more occasional and it is not rare that they will stop sending remittances. In fact, the trend towards increasing permanent emigration is raising concerns about a future decline in remittances as family links become looser. This seems to be the case in some localities of the historic emigration regions. On the other hand, transport and communication technologies increasingly allow migrants to keep ties with their community of origin, and networks of Mexicans living abroad continue to develop, allowing them to remain involved in the cultural, economic and political life of their region of origin. And in fact, while individual remittances tend to dry up once migrants get settled, there are still collective remittances being sent through hometown associations to improve infrastructure, public services and recreation facilities in the communities of origin.

According to the research available, households allocate most of the remittances to satisfying basic family needs, buying food, clothes, health, education and transport, and to purchasing some durable goods. Expenditures for the

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maintenance or construction of housing come in second, while the share reserved for savings and productive investment is very limited. Actually, the expenditure profile is quite similar to that of households not receiving remittances, the main differences being a slightly higher saving rate (14 per cent in 1996 against 12.5 in other households) and slightly higher investment in housing (4 per cent instead of 2.6 per cent) (Castro and Tuiran, 2000). The various types of expenditures generally take place at different stages of the migration process (CONAPO, 2002). At first, money is sent to the place of origin to repay debts contracted to pay for the trip to the United States and other debts, and for household basic consumption; subsequently, remittances may be used for buying equipment for the house and improving housing; and later, remittances may serve to buy tools, equipment and inputs for agricultural production (typically cattle in the historic emigration region), or to establish small shops. The use of remittances may also be influenced by the type of migrants. Papail (2002) finds that the share of remittances used for current consumption in a number of municipalities with high historic emigration has gone down from 70-75 per cent in the 1980s to about 64 per cent in the late 1990s, to the benefit of savings and financing of small business, a shift that can be related to the increasing proportion of single persons in migratory flows.

The share of remittances dedicated to productive investment in the traditional emigration regions is estimated to be about 6 per cent in some studies, while others provide somewhat higher figures.[171] This relatively limited share is sometimes seen as a sign that emigration is not improving the development prospects of source regions and of Mexico in general. This view of emigration has to be somewhat qualified, however. First, basic consumption is often vital for migrants' families, so that remittances make in any case a significant contribution to reducing poverty in those families. Second, consumption expenditures also have some multiplier effects. The second round effects of remittances are generally smaller at the local level than at the national level, because part of the demand is leaking to other localities, most often to urban areas.[172] Third, by improving nutrition, health and education (accounted as consumption expenditure), remittances are in reality invested in human capital, which is a key element in the process of development of the region (or the country at large, if migrants' family members change region). This is all the more important in that households receiving remittances are less covered by institutional health services than other households.[173] Overall, there is little doubt that households receiving remittances see improvements in their standards of living, and in the nine major sending states, the Binational Study (1997) found that migratory intensity is generally associated with improved services for housing and a greater use of modern technologies in agriculture.

As to investment, Massey and Parrado (1998) find that, while the act of migration in itself prevents the participation of the migrant in family business affairs, the accumulation of "migra-dollars" over time provides a significant source of capital and, controlling for other determinants, increases the probability of

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business formation. According to the EREM survey, migrants stay longer in the United States when they intend to create a business (6.5 years instead of 5 for those who do not create a business) and the amount invested is generally low – less than US$5 000 in 80 per cent of the cases (Papail, 2002). This underlines the potential tension existing between migration and business creation: accumulating enough savings to invest implies staying for a relatively long period in the United States, which in course may lead migrants to settle and finally stay in the United States and renounce their initial plans. About a quarter of the micro-businesses created in Mexico since 1975 employ other workers (2.7 on average), most of them being unpaid family members though, and the number of paid jobs created is 0.2 per former migrant. Overall, migration and remittances can ease the financial constraint weighing on Mexican households. However, for migrants or their families to invest more, as for most Mexicans (especially in rural areas), measures are needed to overcome a number of constraints: limited access to banking services, lack of basic infrastructure in many localities, marketing problems, and limited entrepreneurial skills (Chapter III).

For a long time, groups of migrants in the United States have also been financing and carrying out some projects in their communities of origin. Hundreds of Mexican home-town clubs exist in the United States, that organise social activities for the migrants in the United States but also work at maintaining links with the community of origin, including through financing projects which provide collective benefit in their home-town. Since the mid-1990s, the Mexican government – initially the states and more recently the federal government – has been trying to promote these "collective" remittances, by adding public finance, and to improve their use. The first initiative was the "Two for one" (D*os por uno*) programme launched in Zacatecas in 1993, in which the state government added two additional dollars for every dollar brought by migrants groups; since 1996, the programme has become "Three for one" (T*res por uno*), with the federal government (the Ministry of Social Development, SEDESOL) also adding a dollar. Similar programme have been in place in Guanajuato and Jalisco since 2000. The federal government is currently trying to extend the T*res por uno* to the national level.

Compared with "family" remittances, collective remittances involve very small amounts. In Zacatecas, for example, migrants brought US$1.2 million in the T*res por uno* programme in 1999, to be compared with US$300-350 million of transfers to families in that state (Papail, *op. cit.*). Overall, between 1993 and 2000, 429 projects were initiated in Zacatecas, for a total value of about US$17 million.[174] The projects financed are mainly basic infrastructure and communications projects (roads, wells, drinkable water, electrification, *etc.*), public service infrastructure related to education, health and social security (schools and clinic building, allowances for needy groups, *etc.*), recreation and other community related projects (like sports fields and church renovation). In times of severe budget constraints, collective remittances contribute to finance projects which would normally fall

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under the state's responsibility. A number of problems seem to be encountered in implementing the projects, notably linked to the mistrust of the migrants *vis-à-vis* local authorities, which are generally in charge of the operation. Most attempts to promote purely entrepreneurial projects through collective remittances seem to have failed, as they provide benefits to a small number of persons rather than the whole community. However, despite their small size and mixed success, these experiences involve political and organisational learning processes at the local level (Goldring, 2003), which, in the longer run, could contribute to "institutional" development.

There are also a number of initiatives taken by state governments and some federal agencies to promote the financing of productive projects by migrants. For instance, since 1996, the programme "My community" promotes the creation of enterprises, co-financed equally by migrants and the state government in poor areas of Guanajuato, with the explicit objective of retaining potential migrants. In that state, 14 *maquiladoras* have been created, of which only four are still operating, mainly because of a lack of real business strategy, and the difficulty of finding able managers in the community of origin. The governments of Jalisco, San Luis Potosi, Michoacán and Zacatecas have created investment funds to promote investment from migrants, but they appear to suffer from excessive red tape and have provided limited results (Garcia Zamora, 2002). A programme called "Invest in Mexico" has also been launched in August 2002 by NAFIN (the development bank for small and medium size enterprises), the InterAmerican Development Bank and three state governments, aiming at directing part of the existing remittances and part of the savings that Mexican workers have accumulated in the United States to the creation of enterprises, by providing loans (or loans guarantees) and technical assistance.

The cost of remittances and related issues

Despite new technology and increased competition in the sector, transferring remittances remains expensive.[175] The remittance industry is composed of two elements, the originator network in the United States and the distribution network in Mexico. On the originator side, transfer agencies – or remittance companies – dominate the market (one of them, Western Union, with a market share exceeding 30 per cent). Over recent years, banks, postal services and non-banking financial institutions such as credit unions have increased their share of the market, as a result both of their willingness to find new markets and clients, and thanks to the Mexican government initiative described above to provide *Matricula consular* to undocumented Mexicans in the United States. In January 2003, 74 US banking institutions accepted the *Matricula consular* as a valid ID for opening an account, and the US Federal Reserve Bank is currently studying the possibility for all banks to recognise it as a valid document.[176] The originator network then channels the money through the distribution network in Mexico, which includes

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commercial banks, post offices and gas stations (often the only possible distributor in remote areas).

There are two components to the overall charge on remittances which are set by the originator: the transfer fee and the fee resulting from the exchange rate applied to the transaction. Originating companies are able to discriminate prices depending on the number of competitors in the various locations (in the United States but also in Mexico). Available research shows that both types of fees have been decreasing since the end of the 1990s as a result of increased competition. In 1999, the charge on a US$300 transfer to Mexico by a wire transfer company could amount to as much as US$60 or 20 per cent, more than half of it through the "exchange rate fee".[177] In the second half of 2003, the total fee for a similar service generally ranged between 10 to 18 dollars (including the charge paid to the distribution network in Mexico which ranges from one to four dollars). Some foreign banks (such as Citibank, Bank of America and BBVA) now provide a "debit card" service in partnership with the two largest Mexican banks (Banamex and Bancomer); this is one of the cheapest way to transfer money. Fees on the more traditional type of transfers also continue to decrease – since April 2003, one bank offers a 5-dollar transfer fee.

While information on the transfer fee is readily available to the migrant, it is not the case for the exchange rate fee, making comparisons difficult – only in California are transfer companies required to disclose both fees to customers. Even in the case of debit cards, which have very low exchange rate fee and advertise themselves as low cost alternatives, there are other types of fees.[178] To improve transparency, the Mexican consumers' federal office (PROFECO) is publishing information on fees charged by some market players on the internet website of the Mexican Consulates in the United States. This information, however, is provided on a voluntary basis by the companies involved, so that information is often incomplete and difficult to verify. Regulations or agreements such as that referred to in the case of California regarding the publication of the fees would be required both in the United States and Mexico to enhance transparency and competition.

The US Federal Reserve Bank and the central Bank of Mexico are also working at a project to connect the automatic clearing houses of the two countries, which would significantly reduce the costs of financial transfers between the United States and Mexico, and could also have an effect on remittances. At the moment, transfers take place between a bank or a wire transfer company in the US and their counterpart or correspondent. Starting from the second semester 2003, an electronic link between the US and Mexican automatic clearing houses should enable all financial institutions in the United States to transfer money to any bank in Mexico at the exchange rate published every day at noon by the Mexican central bank, at low cost (the expected fee is less than one dollar) and in a short delay

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(two days). The effect of this linking facility on the cost of transfers will, however, depend on how banks pass on this decrease in transaction costs in their fees. In practice, many remittances do not originate in banks in the United States, but in other outlets offering money transfer services (including in gas stations), and these normally impose a fee to access the banking system which ultimately processes the transfer.

Another obstacle met in reducing the cost of remittances is that only about one in five Mexicans (37 per cent of the labour force) has a bank account and many large towns in the rural areas of the historical emigration regions have no bank branch. Cashing in remittances transferred through banks would thus often require a very long journey for the recipient, which explains the still high market share of wire transfer companies. For the "debit card" services, banks envisage to develop mobile branches, or to put automatic teller machines into petrol stations in the unbanked areas. On the other hand, non-bank institutions (savings and loans and credit unions) could also play a role in remittance transfers. There are at the moment about 630 such intermediaries in Mexico, which have assets equivalent to less than one per cent of bank assets, but serve about 7 per cent of the labour force, mostly low-income and located in remote and often marginalised areas. Some credit unions have indeed envisaged developing links with counterparts in the US to transfer remittances to the migrants' families.[179]

The project of BANSEFI – the state-owned development bank which will act as a bank for savings and loans and micro-credit institutions – to organise jointly with 17 non-bank institutions a network of small financial institutions called "L@ *Red de la Gente*" may facilitate their participation in remittance services. BANSEFI is already engaged itself in remittances transfer through its 562 agencies (half of which are located in areas where there is no bank in a 20 km periphery). The network would be composed of those small financial intermediaries that would comply with regulations derived from the new Law on Popular Savings and Loans passed in 2001, and BANSEFI would operate as a banker for them, as done in other countries such as, for example, in Germany and Canada (Québec).[180] Regarding remittances, the objectives would be twofold. First, to encourage the intermediation of the sums remitted and hence an increased potential of lending for productive activity. Second, BANSEFI envisages to propose a number of financial products and services that migrants could buy for their family, such as savings products, insurance (including the health insurance package for people not contributing to the IMSS), and initial down payment to get a loan and/or a subsidy for housing,[181] which would possibly reduce the share of remittances exclusively used for consumption.

Clearly, this project goes beyond the remittances aspect as it aims to provide access to financial services not only to migrants but to the low income population without such access at the moment. By providing remittance payment

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services, these small non-bank financial institutions may also be able to retain the remittances recipients as their member/customer. Besides, developing a type of federation of savings and loans and credit unions would permit economies of scale (for liquidity management and accounting services for example) to reduce operating and regulatory costs[182] and, if successful, could go some way towards formalising the sector, making it more efficient, and increasing its reach. The World Bank has committed a US$45 million loan to strengthen the popular banking sector. In the long-run, once regulated under the new international standards, the savings and loans and credit unions may be able to distribute guarantees and/or funds received from international and national development banks (such as NAFIN, F*inanciera* R*ural*, federal mortgage institutions, the World Bank, and the InterAmerican Development Bank). Allowing the small intermediaries to develop is important because they are the only ones dealing with low-income groups in rural areas. Developing such financial services for them could relieve somewhat the credit constraint that they face, thereby helping them to diversify their activity away from agriculture (see Chapter II and Annex I.B on Agriculture).

Conclusions

While it has social costs, emigration seems to be rather beneficial in economic terms for Mexico, both for the migrants and for society, due mainly to the remittances and savings accumulated in the US and brought back when returning. The effects on the labour market and human capital are less clear. In regions with high historical emigration rate, the large reduction in labour supply may have pushed the wages of those remaining upwards somewhat. The emigrant population is heterogeneous, but, by and large, emigration of highly-skilled people is relatively limited, and Mexico does not seem to experience a serious brain drain. Although they most often occupy jobs at the low-end of the US labour market, migrants, if and when coming back, seem to benefit from some return on their US experience. However, since emigration often results from low development prospects in the source regions, policies in Mexico should obviously aim at improving the situation at large, rather than preventing emigration as such.

The benefits from emigration are generated outside Mexico, but policy may nevertheless influence their impact on Mexico. This is especially the case for remittances, and in recent years a number of government initiatives have been taken regarding the cost of these transfers and the uses to which they are put.

Given the importance of remittances both in the balance of payments and in some households' income, the attempts to reduce the still relatively high costs of transferring remittances should be pursued. The establishment of an electronic link between the US and Mexican automatic clearing houses is expected to significantly reduce the exchange rate costs of financial transfers, benefiting migrants and their families provided banks pass on to their customers these cost reductions.

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Some initiatives have also been taken to increase the amount of information available on the fees charged by the various providers of remittance services. More could be done, however, and making obligatory the publication of information on the various costs associated with remittance services by the various providers on both sides of the border would also help increase the competition on that market in general (and thereby favour the pass through of any reduction in exchange rate costs). The Mexican government should therefore work at reaching an agreement with the US authorities on this matter.

Not surprisingly, given the average living standards of migrants families, remittances are almost entirely used by migrants' families for consumption expenditures. This is appropriate, since it tends to reduce poverty levels in those families, and generates some multiplier effect at the local level. Besides, part of this consumption sustains education and health, thus contributing to enhancing human capital. For migrants to invest more of their income, the poor environment for investment (lack of infrastructure in some regions, lack of credit for micro and small enterprises, lack of skills of individuals) would have to be improved. The project to develop the network of existing savings and loans institutions under the aegis of BANSEFI and channelling migrants remittances through it seems promising. If successfully implemented, it may not only improve the access to remittance services of rural and low-income households and reduce their costs, but also ease the credit constraint weighing on these low-income groups somewhat, both through the intermediation of remittances and through the channelling of credits from development banks.

The attempts by some states and the federal government to favour collective remittances from clubs of emigrants to their locality of origin by adding public funding, when the remittances are used for social or infrastructure investment, have met implementation problems, notably due to a mistrust of emigrants *vis-à-vis* local authorities. However, in the longer run, they may contribute to institutional capacity building. Yet, migrants should not be expected to remedy a general problem of lack of public investment.

The benefits of emigration are sufficiently strong for many Mexicans to attempt to cross the border illegally to the United States, often repeatedly. US immigration policy greatly limits legal flows of labour immigration compared with the demand for Mexican labour in the US; hence, illegal immigration has become the most important component of flows. It is often argued that a side effect of US immigration policies is the observed shift from a traditional circular immigration to a more permanent immigration. This cannot be ruled out. Another effect is the increase in smuggling costs, with the strengthening of border enforcement policies, which reduces returns to migration for Mexicans. The forces driving emigration – wage and employment differentials on the one hand, relative demographic developments on the other – are unlikely to diminish in the near future, so these phenomena are likely to persist under unchanged policies.

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Up to the 1990s, the position of the Mexican authorities *vis-à-vis* Mexico-US migration was one of deliberate non-engagement. Since then, Mexican authorities have taken a more active stance, trying to pursue certain objectives for Mexicans residing in the US on the one hand, and for migration flows on the other. The possibility of reaching a bilateral migration agreement with the United States was interrupted after September 11, 2001. The efforts to reach such an agreement should be pursued, however, since higher levels of legal immigration to the United States would be beneficial for Mexico (strengthening of the position of Mexicans in the US and hence their working conditions and salaries, reducing sums lost in smuggling fees, increasing remittances). While the gains to Mexicans in reaching such an agreement are quite clear, it is also the case that there would be economic benefits for the United States. The contribution of Mexicans to certain sectors of the US economy is considerable, even though there may be some costs to certain groups of low-skilled workers competing with Mexicans (see OECD, *Economic Survey of the United States*, 1997). An agreement which would increase the proportion of Mexicans in the United States who work legally would raise their "reservation wage" and hence reduce the downward pressure on low skilled wages; the fiscal position would also benefit from increased number of taxpayers and reduced costs of border enforcement. In the meantime, the plans to set-up a system for postings of Mexicans ready to work in the US on H2A and H2B temporary visas, to the extent that it increases the information available to employers as well as transparency, thereby reducing abusive charging on immigrants by private recruiters, should be encouraged.

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Notes

1. With population growth rates higher in Mexico (between 1½ and 2 per cent per year) than in the rest of the OECD area, but GDP growth close to the average, Mexico is not closing the gap.
2. The peso crisis had many causes and it would be futile to attempt to rank these by order of importance. With the benefit of hindsight, however, it has been possible to understand how the crisis occurred and how it was amplified. This episode is documented in many studies by academics and experts, both in Mexico and abroad. The OECD *Economic Survey of Mexico* September 1995 provides a detailed analysis of the crisis and the adjustment process that followed.
3. The level of GNP is just below GDP, since it does not include profits earned by the large foreign-owned manufacturing and internationally traded service sectors which are not available to domestic residents. On the other hand, it includes net current transfers received from abroad – migrants remittances essentially.
4. See supporting material to Chapter I, in Annex I.A.
5. *Cf.* Annex I.A, Tables and Figures on GDP growth, employment and productivity.
6. The weakness of productivity gains cannot be evaluated with certainty, because complete and reliable labour market statistics are not available. The National Employment Survey reports a broad aggregate, which includes all the self-employed, unpaid dependent employees and uninsured wage earners (*i.e.* formal plus informal sectors). It probably overestimates employment, because it includes "underemployment" according to Mexican terminology. Thus broadly defined as the number of people having an occupation, employment has risen from about 30 million persons in 1991 to 40 million in 2002 (Annex Figure I.A.2). Data reported in the national accounts, on the other hand, include wage earners, but exclude unpaid workers and workers without contracts. Wage-earning jobs reported in the national accounts probably underestimate employment. At the same time both the level and growth of GDP in recent years might have been underestimated. Bearing these caveats in mind, the section reports a few stylized facts, based on several indicators.
7. See Annex I.B. on the adjustment of the agricultural sector.
8. Evidence is only scattered, because of data unavailability. B. Bosworth (1998), based on his estimates of the capital stock finds that in the aftermath of the 1982 debt crisis, physical capital per worker dropped and was lower in 1994 than in 1981. While the cost of labour was falling, the cost of capital rose in the 1980s and most of the 1990s: for imported equipment, it reflected the currency devaluation, but more generally it was due to the unavailability of credit. As a result the capital-labour ratio declined.
9. National accounts employment data exclude the self employed and unpaid workers. They are sometimes used as a proxy for formal labour. These are "estimated" data

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rather than observed data, and they refer to jobs. The same employee can occupy several remunerated jobs. The National Accounts available since 1995 report both series of remunerated jobs and employed persons. Jobs data are used in this analysis because they are available over a longer period. In Annex I.A, employment data are shown according to various definitions.

10. Norma Samaniego (ILO, 2001), on the basis of Employment Survey data.

11. Evidence from firm-level data available for 10 OECD countries provides useful insights on these effects. In the countries under study, which include the United States, France, Italy, the United Kingdom as well as Finland, the Netherlands and Portugal, labour productivity growth in manufacturing appears to be largely explained by evolutions within individual firms, and the exit of lower-productivity firms also makes a substantial contribution (through a "creative destruction" process). Mexico is not included in the OECD study for lack of data. *Cf.* Scarpetta *et al.* (2002), as well as *The sources of economic growth* (OECD 2003), Chapter III and V in particular.

12. Insufficiencies in the credit system and bankruptcy procedures are seen as key impediments to productivity growth in the Mexican economy at large, according to research done by the Central bank of Chile. This work finds that the difference of performance between Chile and Mexico can be explained by the fact that Chile undertook reforms in these sectors roughly 10 years earlier, which generated faster productivity growth, cf. Bergoeing *et al.* (2001). For SMEs in particular, credit has been practically non-existent for a decade. Before the privatization of banks, credit was allocated to privileged sectors which benefited from subsidies. *Cf.* Chapter II below and the more detailed review of the Mexican banking sector in Bonturi (2002).

13. Bosworth in 1998 (op.cit.) concludes that "capital is by far the most binding constraint" on rapid output growth over the medium-term. Evidence of the lack of capital is also found in the more recent article by Shiau *et al.* (2002). Regarding ICT, experience of other OECD countries shows that weak productivity growth might reflect the small size of the ICT sector as well as low use of ICT (Annex I.A. Figure).

14. One-third of budget revenue is still related to oil receipts, hence vulnerable to changes in oil prices.

15. The sole water and waste water sector is estimated to require US$2.2 billion year, which is twice the annual budget of the National Water Commission (CNA), which has full authority in production and regulation for water management (*cf.* OECD *Environmental Performance Review: Mexico* 2003).

16. Net foreign direct investment (FDI), averaging about 2½ per cent of GDP per year between 1994 and 2002, allowed the economy to grow without incurring high foreign indebtedness. About one third of the total went into manufacturing, with the *maquiladoras* exporting sector receiving less than half this amount. The other main destinations of FDI since 1994 have been commerce (retail distribution) which absorbed about 10 per cent of FDI and financial services, especially in the later period.

17. See Table in Annex I.A and OECD, *Economic Outlook*, No. 71, Chapter IV "Intra-industry and intra-firm trade and the internationalisation of production" (2002).

18. Manufacturing labour costs and productivity in Mexico relative to the United States are discussed in Montout (2002) with an application to the automobile industry in NAFTA.

19. Annex I.C presents some details on recent economic developments, including background Tables and Figures.

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20. In June 2003, the OECD foresaw an average 3 per cent GDP growth for the total OECD area (and a little more for the United States) in 2005-08. Mexico's growth was then projected at around 4 per cent over that period. See OECD, "The medium-term reference scenario" in *Economic Outlook* 73, June 2003. These projections will be revised, taking into account the new short-term projections for the OECD.

21. Bassanini, Scarpetta and Hemmings (2001).

22. The importance of ICT is discussed in OECD (2003) ICT *and economic growth: evidence from* OECD *countries, industries and firms*. The intensity of R&D has also been found to play a role on innovation and productivity growth, as put in evidence on the basis of aggregate data (Guellec *et al.*, 2001) and on sectoral data (Scarpetta *et al.* 2002). Lederman and Maloney (*op. cit.*) identify the need for Mexico to catch up in innovation and technological progress. They argue that trade liberalisation and NAFTA, though helpful for this, may not be enough; and that substantial improvements should be made in Mexico's national innovation system to make it more efficient.

23. Higher investment rates, including FDI, are generally recognized to generate positive externalities through the adoption of best practice organisation and technology. Nicoletti *et al.* (2003).

24. *Cf.* Lederman and Maloney (2003).

25. The role played by pro-competition regulations in growth is put in evidence in Nicoletti and Scarpetta (2003). Mexico is not included because data limitations have impeded reliable measures of MFP. Although the estimates made for the 21 countries included in this research are subject to usual caveats associated with econometric analyses, they provide a benchmark, which is of relevance for Mexico.

26. High inflation is negatively associated with the accumulation of physical capital in the private sector, while high government deficits can impair growth. *Cf.* Bassanini, Scarpetta and Hemmings (2001).

27. The "financial deficit" of the public sector is a narrower concept than the public sector borrowing requirement (see below). Nonetheless, it takes into account the cost of the social security reform for the private sector since 1997, a burden on the budget which is equivalent to about 1½ per cent of GDP per year, half of which being the transition cost to a fully-funded system, which means that the government is actually reducing an implicit government liability. It also includes amounts that have been paid into the oil fund, since its creation in 2000, as well as the real component of debt servicing related to banking sector support programmes.

28. Spending related to the deferred public investment projects (PIDIREGAS) are not recorded in the financial balance when the investment is made, but rather on completion of the project, when the financial obligations are serviced (see Annex Box 6). However, the PSBR records the projects from the start, *i.e.* when the investment decision is made. The cost of the financial rescue programmes that is included in the PSBR corresponds to the total liabilities created through: *i*) the bank restructuring operations, which are the largest component, as only the real component of debt servicing is recorded in the budget; *ii*) debtor support programmes; *iii*) the bail-out programme of the toll road concessions. The PSBR also includes the inflation adjustment of the indexed debt and use of resources by development banks to finance private and social sectors.

29. The liquidation of B*anrural* reduces the net indebtedness of the development bank sector, while increasing the federal government debt. The closure of B*anrural* and the creation of a new entity, F*inanciera* R*ural*, aims at solving the structural fiscal problems

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generated by B*anrural*'s operations, while strengthening the rural financial system by establishing a more transparent and efficient financial institution. Resources directed to cover B*anrural*'s liabilities translate into an increase of the Federal's government debt by 37.9 billion pesos. At the same time a capital contribution was made to the F*inanciera* R*ural*.

30. Thanks to the new tax measures as well as improved tax collection, revenue from income tax increased by 6.5 per cent in real terms, but this was significantly less than what had been expected. Revenue from the special taxes on telecommunications and soft drinks, which were created in 2002, also fell short of estimates.

31. Payment of the initial capital of F*inanciera* R*ural* was authorised by Congress and covered by the federal government with 10.9 billion pesos funded from non-recurrent resources related to the IMSS pension reform. The resources were made of unclaimed retirement savings (the SAR/IMSS 92).

32. The main sources of high spending increases have been: spending on staff (+7.6 per cent in real terms), reflecting contractual wage negotiations in the education and health sectors, as well as in state-owned enterprises (PEMEX and CFE) and allocations to strengthen the judicial branch. Social security spending, shown under the classification by function, was boosted by large increases in pension payments and the constitution of pension reserves, while higher investment on highways and the development of e-Mexico were the main factor behind the boom in the item "communications and transports".

33. Resources are transferred to states through revenue sharing (*participaciones*) and through grants (*aportaciones*) ear-marked for specific purposes. Regarding resources transferred through the Programme to develop infrastructure and improve the states' financial position (PAFEF), these were adjusted downwards in the first half of the year, but replenished in the second half, when oil-related revenue increased.

34. The government's initial budget proposal was based on a more prudent assumption, US$17 per barrel. But Congress approved a revised budget, with a higher average oil price. The spread between the average price of the Mexican export mix and the WPI benchmark, which reflects the still large weight of heavy crude oil ("Maya") relative to lighter oils ("Istmo" and "Olmeca") in Mexican exports, fluctuates around 5-6 dollars per barrel, but at times it has reached as much as 8 dollars or as little as 4 dollars. The spread tends to narrow, for instance, when the main suppliers of heavy crude to the US cut back production.

35. It should be noted that at the same time investment projects decided by the public sector and financed by the private sector have been increasing over time (see Annex Box 6).

36. In the 2003 Budget, Congress authorizes additional government indebtedness for implementation of the "Programme for voluntary separation" which is in operation until end August 2003. By July 2003, a little more than 15 500 workers had participated in the programme. The related increase in spending in 2003 is expected to be offset by savings in future years since jobs are being cut. In the next four years, the federal government entities which benefit from the programme in 2003 must have savings of at least the amount they have received.

37. These clauses are called "automatic stabilizers" under the Mexican terminology; their aim is to make fiscal action compulsory to meet budget targets in case of unforeseen events which threaten these targets.

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38. The government can cut "programmable" expenditure without Congress approval (according to a pre-established list of priorities), if tax revenues are lower than budgeted by 5 per cent or less. The so-called "programmable" expenditure items are those on which the federal government has some discretionary power, *i.e.* excluding interest payment on the public debt and revenue sharing with the sates. But for a higher tax revenue shortfall, Congress has to approve the specific spending items to be adjusted.

39. By mid-2003, in particular, states were allocated 7 billion pesos as an advance on likely net revenue windfall. If a windfall of twice that amount does not materialise, the funds will have to be returned to the federal government the following year.

40. The set of tax measures approved for 2002 was estimated to bring in additional resources for an amount of 61 billion pesos that year, according to official estimates made at the time. Most of the increase (24 billion) was to come from the elimination of the special tax rate for retained earnings under the corporate income tax; another 23 billion from the "payroll" tax (substitute tax). The special taxes created in 2002 were to bring in 14 billion pesos (of which 8.8 billion on account of the luxury tax); and changes to the special regimes on income tax were to enhance revenue by 10.5 billion pesos. A number of other changes implied revenue losses. New fiscal measures approved by Congress in 2003 are expected to bring additional tax revenues of 0.3 billion pesos.

41. These and related issues are discussed in Davila and Levy (2000).

42. The composition of the Board of SAT has been reviewed, to include Ministry of Finance representatives, plus three independent members, of which one is nominated by the President, two by the State governments, with a two-year mandate. An effort is being made to increase the scale of tax controls and apply sanctions for non-compliance more effectively.

43. One particular problem in the past has been that budget data are on a cash basis, while the PSBR is on an accrual basis, and the accounting principles have not always been clear. Since 2002, however, new accounting principles are being developed with the objective of providing more transparency.

44. Once the budget is approved, however, the detailed budgets of state-owned enterprises are published, including their programmes, objectives and indicators on their operation. Transfers to public entities that carry out non-commercial activities are recorded in federal government expenditures. And the impact on public finances of financial intermediation by development banks is included in the PSBR.

45. This obligation was introduced with the 2002 budget. The *Presupuesto de Gastos Fiscales* of 2002, published in 2002 provides estimates of revenue shortfall due to exemptions that year. The projected tax exemptions for 2003 were published in September, according to the Law. The projections for 2002 put in evidence the very high share of tax exemptions in proportion of the tax take.

46. For instance, the Committee asked the Ministry of Finance for a breakdown of the IMSS current spending into pensions and other expenditure, information on the fiscal cost of the "Programme for voluntary separation" introduced in 2002 as well as estimates for 2003 and 2004, and a quantified outcome of the "Programme of savings and austerity of the federal public administration". A greater involvement of Congress at an early stage was among the recommendations made in past OECD *Economic Surveys*. Strengths and weaknesses of the Mexican budget process are reviewed in Larre and Bonturi (2001).

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47. Introducing a clause might even have a perverse effect, according to some experts. Under the present situation, in case of non-approval, there would be a provisional appropriation for spending. An explicit clause might be needed for the revenue law, however, because of the risk that the government would be unable to collect taxes).

48. There are no official estimates of the fiscal cost that a reform of ISSTE would entail, but it would be only a fraction of what the IMSS reform is costing.

49. There is a general consensus that underdeveloped infrastructure is a factor limiting Mexico's growth potential and worsening regional imbalances, especially since NAFTA; cf. Davila, Kessel, and Levy (2000).

50. Headline consumer price inflation was 5.7 per cent in December 2002, exceeding the 4.5 per cent central bank target by more than 1 percentage point. According to the Central Bank's estimates if administrated and regulated price had grown at 4.5 per cent, headline inflation would have been almost on target. A low pass-through from import prices and the monetary tightening have helped contain core inflation, which was below 4 per cent.

51. According to market participants, the appreciation of the peso in the second quarter of 2003 reflected to some extent changes in Mexico's policy of managing its foreign reserves. In response to the increasing quasi-fiscal cost of maintaining growing accumulated international reserves, a system to slow their pace of accumulation has been put place starting on 2 May 2003. Part of the foreign currency flows that could potentially increase the level of the country's reserves has been channelled to the market. More precisely, half of the foreign reserves accumulated in the previous quarter, once the amount sold has been subtracted, is sold by auction on a daily basis in pre-established amounts determined at the beginning of the quarter.

52. According to OECD medium-term projections, annual inflation should be around 1.5 per cent by 2005 in the euro area, the United Kingdom and the United States and close to 2 per cent in Canada.

53. A well-developed financial sector *i*) mobilises savings, by channelling the small-denomination savings of individuals into profitable large-scale investments, while offering savers a high degree of liquidity; *ii*) provides insurance to individual savers against idiosyncratic risk through diversification; *iii*) reduces the costs of acquiring and evaluating information on prospective projects, for example through specialised investment services; and *iv*) serves in the monitoring of investments to reduce the risk of resource mismanagement. (see OECD, 2003, T*he sources of growth in* OECD *countries*.)

54. Until then credit allocation was mostly determined by the government (see Bonturi, 2002). Bergoeing, 2001 identifies the insufficiencies in Mexico's financial system, with reforms ocurring much later than in Chile, as a key factor explaining the weaker growth performance of the Mexican economy relative to Chile.

55. In March 2003, the rating agency Moody's qualified as positive the deposit and debt of the main Mexican banks as a result of better management of banks as well as debt profile (in foreign and local currencies).

56. There were signs of a pick-up in credit to enterprises in 2000, before the slowdown, but it did not last.

57. Moreover banks were lending mainly on a short-term basis and therefore bank credit was financing primarily non-investment expenditure (see Bank of Mexico survey "E*ncuestas de* E*valuación Coyuntural del* M*ercado Crediticio*").

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58. For enterprises rated AAA the share of commercial bank financing in total financing was 78.3 per cent in the second quarter of 2003; for small enterprises, it was 24 per cent. See also Martinez Trigueros (2000).

59. *Ley General de Títulos y Operaciones de Crédito*, *Código de Comercio*, *Ley de Instituciones de Crédito*, *Ley del Mercado de Valores*, *Ley General de Instituciones y Sociedades Mutualistas de Seguros*, *Ley Federal de Instituciones de Finanzas* and *Ley de Organizaciones y Actividades Auxiliares del Crédito*.

60. The 1995-2000 recovery took place against the background of a contraction of credit, especially to housing and enterprise. Credit started to pick up only slightly and temporary in 2000. There is also a need to increase the savings base. Only 20 per cent of the Mexican population holds a bank account.

61. Bansefi, formerly PAHNAL, was created in 2001 to promote savings and financial services in remote areas and to low-income families.

62. Given the legal framework in force for saving and loans institutions and popular financial organisations, the CNBV has developed new rules, which can be divided into six groups: organisation and functioning, operations, accounting, prudential regulation, protection savings schemes, and supervision.

63. Since May 2003, the private pension funds (SIEFORES) have been able to invest up to 100 per cent of their total assets in AAA issuers, with a limit of 5 per cent per issuer; for the AA issuers the limits are respectively 35 per cent and 3 per cent respectively, and for A issuers 1 per cent and 5 per cent. Before, a maximum limit for nongovernmental titles was applied equivalent to 35 per cent of the assets of the SIEFORES, accompanied by a maximum limit of 10 per cent in a same issuer. SIEFORES should also be able to invest in foreign security with a share gradually increasing, up to 20 per cent in three years (a law already approved by the lower house and submitted to the Senate).

64. The progression has however been uneven, with enrolment rates declining after the budget cuts that followed the debt crisis and the mid-1990s financial crisis.

65. According to Duryea and Pages (2002), the increase in Mexico's population school attainment was the highest in Latin America between 1980 and 2000.

66. Upper-secondary education in Mexico consists in: *i*) *bachillerato* programmes that can be either general or bivalent (general and technical) and of which the aim is to prepare for superior education); *ii*) technical education *per se* which has a professional aim.

67. The repetition rate is defined as the proportion of students from a cohort enrolled in a given grade at a given school-year who studied in the same grade in the preceding school-year. Terminal efficiency is defined as the percentage of students completing an education level relative to the number of students who started in the same cohort.

68. According to the 2000 population census, out of the 7-29 year old population who left school, 27.5 per cent did it because they did not like or want to study, and 38.5 per cent for economic or familial reasons. The second motive is more important for the older group (aged 20 to 29).

69. At level 1 students are capable of completing only the least complex reading tasks developed for PISA, such as locating a single piece of information, identifying the main theme of a text or making a simple connection with everyday knowledge. Below level 1, while students have usually the technical capacity to read, they have serious difficulties in using reading literacy as an effective tool to advance and extend their knowledge and skills in other areas. Thus, they may be at risk not only of difficulties in their initial transition from education to work but also of failure to benefit from further education and learning opportunities throughout life.

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70. These results might be partly related to the uneven distribution of resource between and within States (see World Bank, 2000 and 2001).

71. See the National Education Programme 2001-2006 and CENEVAL survey on the job situation of the recent graduates (Zubiran, 2003).

72. The National Education Programme recognizes inefficient use of teaching time, insufficient/inappropriate training of teachers and limited accountability of teachers. Absenteeism is estimated to reduce teaching time by 50-75 per cent in rural areas (see World Bank, 2001). Lopez-Acevedo (2001a) provides an overview of these issues in the case of primary education. She notes in particular that the lack of interaction between education professionals even at the school level is an important factor explaining the low motivation of teachers.

73. The teacher career programme *Carrera Magisterial* was created in 1992 to increase quality of basic education by encouraging teachers to continue their professional development. The impact of *Carrera Magisterial* on attainment levels of students has been limited. Actually, the programme does not guarantee continued good performance, since steps cannot be taken back and there are no limits on the number of step advancements by school, encouraging a certain degree of "step inflation" among many managers. Introduced in 1997, the programme of performance incentives for teachers in upper secondary and tertiary education (*Programa de Estímulos al Desempeño del Personal docente de educación media superior y superior*) operates through a point system that is used to calculate cash bonuses given in reward for good performance. Bonuses go from 1 to 14 minimum wages and quality of teaching performance accounts generally for 60 or 70 per cent of the total number of points. As stated in the 2000 OECD *Economic Survey on Mexico*, it is better designed than *Carrera Magisterial*.

74. OECD estimates based on data provided by national authorities put the ratio at 4.4 and 4.7 per cent of GDP in 1999 and 2000, respectively (see OECD, 2003, *Education at a Glance*).

75. As in many other low-income countries, teachers' salaries are relatively high in terms of GDP per capita (see OECD, *Education at a Glance*, several issues).

76. So far, the amount spent in the form of grants and student loans only represents about 5 per cent of the overall public spending on tertiary education compared with about 20 per cent on average in other OECD countries. One recent example of scholarships for students from low-income background in Mexico is the National Programme of scholarships and financing for superior studies, (*Programa Nacional de Becas y Financiamiento para Estudios de Tipo Superior*, PRONABES) created in 2001 (see *http://sesic.sep.gob.mx/pe/pronabes/inicio.htm*). On returns on education in Mexico see López-Acevedo (2001b) and Psacharopoulos and Patrinos (2002).

77. See Márquez (2001). According to this study, this situation reflects the fact that when operating in a low skill environment, firms do not create skilled vacancies because of the difficulties of filling them (even if filling that vacancy would increase their profits). Workers, in turn, do not have the incentives to obtain training given the lack of skilled vacancies (even if obtaining them would increase their productivity and wages). Comparable result was found on the manufacturing sector by Tan and Lopez-Acevedo (2003).

78. These programmes were renamed in 2002 by the new administration. Evaluations, mainly by the World Bank, have shown mixed success, which are partly due to the use of these schemes as income support in periods of high unemployment (1995-96).

79. CONOCER is in charge of defining norms for and fostering a standardized system of labour competency certification in Mexico. Surveys among firms and students that

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have used programmes certified by CONOCER show that it has a positive effect on productivity.

80. The mean and median job tenures in Mexico are among the lowest in OECD. Moreover, workers' mobility in the informal sector is twice as large as in the formal sector: according to Calderon Madrid (2000), in 1995, one-half of informal employees had been working in the same firm for more than two years compared with 3/4 of formal employees.

81. Sixty per cent of the working population has not accumulated pension rights, but there is a minimum pension equivalent to one minimum wage financed on general budget (see Larre and Bonturi (2001) for details on financial and labour market impacts of the pension reform).

82. See OECD *Implementing the* OECD *Jobs Strategy: assessing performance and policy*, Annex I.A.

83. Some analysts cite the existence of a minimum wage as a source of high wage cost. However, the minimum wage (which in Mexico is lower than the official poverty line) has been less and less binding in the last decade. In real terms, it has decreased more than average wages and the share of the urban employees paid between one and two minimum wages has declined from about 40 to 25 per cent, while the share of those paid more than two minimum wages increased (see Annex Figure 23).

84. The worker used for Mexico in international comparisons earns four times the minimum wage. According to IMSS data 61 per cent of insured workers earn less than that amount. For workers earning the minimum wage Health and Maternity contribution rate can be 10 percentage points higher. The effect on total wedge of this regressive pattern is however limited by the existence of a salary tax credit (see below).

85. This Federal payroll tax of 3 per cent (increased to 4 per cent by the 2003 budget) was introduced to finance the "salary tax credit". This was an optional tax in the sense that the employer could choose from the following two options: either to pay, obtaining in full the salary tax credit as in the past; or, not to pay the tax, but then the government deducted the payroll tax liability from the entitlement to salary credit. See OECD 2002 *Taxing Wages*.

86. However, most firms do not comply with the legislation – see 1997 OECD *Economic Survey on Mexico*, which gives more details on this point of the labour law, confirmed by more recent anecdotal evidence.

87. Severance payments associated with fair dismissals are not much higher than in other OECD countries, but the definition of fair dismissal is the tightest in OECD, with redundancies and poor performance not being normally legal grounds for dismissals. In that case, severance payments are of three months, plus 20 days per year of service.

88. Most allowed temporary jobs in Mexico are seasonal jobs in the agriculture sector.

89. See Maloney (1999).

90. According to Azuara (2003), to receive a pension exceeding the minimum guarantee (equivalent to one minimum wage), rates of return for a worker earning two minimum wages during 40 years have to exceed 8 per cent (5.5 per cent for workers earning three minimum wages). These levels are well above current rates of return.

91. See Samaniego *et al.* (2000).

92. Garro and Melendez (2003) defend the view that the regressive structure of Sickness and Maternity contribution rates reduces formal employment opportunities for low-income workers. However, the main reason for not having a progressive tax system is

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that this would tend to increase the possibility and profitability of tax evasion in the form of underdeclaration of company incomes and payrolls.

93. Increasing contributions on higher revenues would be a step backwards from the 1997 IMSS reform (OECD *Economic Surveys* 1998 and 2000) and it would probably induce tax evasion or avoidance from higher income groups.

94. "Paper unions" (also called "white unions") have also been used extensively by large enterprises to avoid existing regulation. The progressive welcome democratisation of the industrial relation system will however limit this possibility in the future. These practices also have contributed to making the legal and regulatory environment uncertain and inequitable.

95. There is international evidence that part-time work favours women participation to the labour market. In the case of Mexico, one could argue that it can favour formal labour market participation.

96. In Chile, the introduction of such accounts has not been followed by a significant reduction of severance payments and Mexico should be careful to avoid this problem (see OECD *Economic Survey of Chile*, forthcoming).

97. This is a much better option than limiting withdrawals to unfair dismissals. In Brazil, the necessity of being dismissed without a just cause has several drawbacks (see OECD *Economic Survey of Brazil* 2001).

98. Another important issue is the choice of a scheme for dismissed workers whose amount saved on the personal account is insufficient to cover minimum withdrawals. One option is the use of a solidarity fund as in Chile (see Annex Box 8).

99. Spending of 3.5 pesos on the PET generates only 1 peso of income for the workers. See World Bank (2001).

100. This is the case of the proposal to reinforce of the indirect employer relationship (according to which a firm is responsible for employers' obligations towards workers employed by its sub-contractor). This could raise business costs of subcontracting and result in discrimination against small providers.

101. OECD, 2003, *The Sources of Economic Growth in* OECD *Countries*.

102. See Nicoletti *et al.* (2003).

103. Practically, all the final decisions issued by the Judiciary, especially those of the Supreme Court of Justice, regarding the constitutionality of Competition Law have been favourable to the Commission's position.

104. Esquivel *et al.* (2002) show for instance that phone density is playing an increasingly important role in explaining the differences in productivity gains between regions.

105. Article 28 of the Constitution lists strategic areas where functions should be exercised exclusively by the State: postal services, telegraph and radiotelegraphy, petroleum and other hydrocarbons, basic petrochemicals, radioactive minerals, nuclear energy, electric power, and the functions of the central bank in producing coins and paper currency. Changing this list necessitates voting a constitutional amendment, which requires a two-thirds majority of both houses of Congress as well as approval by a majority of the state legislatures. Private sector participation in the distribution of power in particular would require a constitutional change.

106. If the state-owned Federal Electricity Commission (CFE), which generates 80 per cent of Mexican electricity, were responsible for all the investment needed, its net debt

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would be multiplied by 5 and its debt/asset ratio would increase to 104 per cent by the year 2011 (compared with 16 per cent in 2001).

107. TELMEX has a market share of 77.1 per cent in domestic long-distance, 71.3 per cent in outgoing international long-distance, and 41.7 per cent in incoming international long-distance market.

108. This is one of the issues in the WTO dispute proceeding against Mexico.

109. See decisions of 2001, 2002 and 2003.

110. The project law also aims at eliminating barriers to enter the industry by simplifying the requirements to obtain concessions for operating or using public telecommunication networks, as well as the requirements for concessionaires to allow interconnection to their networks through procedures explicitly stipulated in the law.

111. In the 2002 *Economic Survey on Mexico*, it was suggested that all operators should be required to contribute to the universal fund.

112. The CFC has stated that if the two major airlines, Aeromexico and Mexicana, were not sold to independent owners, the resulting entity would constitute an unlawful concentration subject to attack under the Federal Law of Economic Competition.

113. Nicoletti *et al.* (2003).

114. SARE concerns "*low public risk activities*" as defined by INEGI 1999, which represent more than two-thirds of SMEs creation. Formalities for firm creation in general have to be completed within three months. According to the *Consejo Coordinador Empresarial* survey, in 2001, the average number of days to open a business varied from 27 in Durango State to 88 in the Federal District (down from nearly 120 in 2000).

115. See OECD, *Regulatory Policies in* OECD *Countries*, 2002, Box 6.

116. Besides the Federal Registry of Formalities and Services, other important tools include: the system of electronic tax declarations, various one-stop shop initiatives, the use of regulatory impact analysis and the reinforcement criteria for the simplification of formalities See OECD 2003, *From red tape to smart tape*.

117. According to the *Global Competitiveness Report* 2002-2003, Mexico ranks 47th out of 75 countries on the corruption index and 62nd on laws and contracts. Mexico holds the 58th position on the quality of public institutions (based on a combined index) and it ranks poorly in UN statistics concerning crime (in particular robbery and homicide).

118. Efforts are already made at the federal level. For instance, an institute (*Instituto Federal de Especialistas de Concursos Mercantiles*) providing the federal judges with highly skilled professionals to assist them on technical issues was created in 2003.

119. See Chapter I above; and OECD (2003) *The Sources of Economic Growth in* OECD *Countries*.

120. Mexico also lags behind OECD countries in sciences and technology and innovation system. The weaknesses in those areas are emphasised in reports by CONACyT, which also set out policy orientations both in qualitative and quantitative terms.

121. These programmes are discussed in greater detail in OECD (2003) *Environmental Performance Review: Mexico*.

122. For example, the Huelco Bolson aquifer, used for drinking water supplies by El Paso and Ciudad Juarez may be exhausted by 2025 at current pumping rates. There are a further 18 critical bi-national aquifers (EPA, 2003).

123. Indeed, the number of employees in the federal water utility did not fall after the privatisation of large part of its functions. In the Federal District the number of water utility

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employees per 1000 connections averaged 13.4 in 1993, against 4.6 in Lima, Peru and 2.7 in the United States.

124. 63 per cent of the foreigners were US citizens and 45 per cent of the foreigners are between 5 and 19 years old in 2000 (CONAPO, 2001).

125. In the early 1980s, about 40 000 refugees came to Mexico from Guatemala, but 75 per cent of these have now returned to Guatemala, the others having been granted permanent residence or Mexican nationality.

126. Based on Mexico's National Institute of Migration statistics.

127. The Binational Study (1997) – a study launched in 1994 at the initiative of both governments to analyse the nature, causes and consequences of migration for both countries, named the temporary migrants "sojourners" and the more permanent ones "settlers". Permanent flows are estimated from the Mexican and US census. Some uncertainty remains, however, about the definitive character of the change in residence (some Mexicans residing in the United States may decide to come back at one point, either during their working life, or when retiring).

128. Temporary migration flows are based on EMIF data. Temporary outflows for labour motives refer to people stating work or work search as their motive for crossing the border.

129. Source: CONAPO. 80 per cent of the Mexicans who had been to the United States between 1992 and 1997 (and had come back to Mexico) did it without authorisation (ENADID 1997).

130. See STPS, *Encuestas sobre Migración en la Frontera Norte de México* (EMIF), Table 3.2.

131. 62 per cent of migrants crossing the border to the United States were from urban origin in 1998-2003, against 55 percent in the 1993-1997 period. 61 per cent of Mexicans lived in urban areas in 2000, against 57 per cent in 1990. In part, the urbanisation of migration flows is related to the geographical diversification, since migrants from the non-traditional sending states tend to be more urban than those from the traditional sending region. According to the 1997 demographic survey (ENADID), 60 per cent of the migrants came from urban areas in the non-traditional zone of international migration, against 40 per cent in the traditional area (Lozano, 2001).

132. Still, according to the 1990 census, 13 per cent of the Mexican-born were employed in the agricultural sector and since then the share has increased by more than 3 percentage points.

133. See ENADID 1992 data shown in Durand *et al.* (2001).

134. Estimations by the Consejo Estatal de Poblacion (COESPO) of Guanajuato, based on census data.

135. 59 per cent of the migrants proceeding from the south had no previous international migration experience in 1998/99, against 30 per cent in 1993/94 (source EMIF).

136. International migrants most often come from municipalities with a medium degree of marginalisation. The index of marginalisation, used by the Mexican authorities to target their anti-poverty policies, is calculated taking into account the access to primary education, housing conditions, the income level and access to infrastructure and health services.

137. EMIF data of Table 13 show a slightly inferior educational attainment than data from the 2000 census for the 25-49 population (24 per cent in the primary incomplete category, 39 per cent in the primary complete and lower secondary, and 31 per cent above

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that level). Given that the universe considered in the EMIF is the population aged over 12 and 25 as in the census, this probably tends to lower the average somewhat (as youngsters have not been able to reach upper secondary and tertiary levels).

138. According to the 1997 demographic survey, 40 per cent of the migrants who returned to Mexico had a previous experience of internal migration. According to the 2000 Census, the internal migration rate was about 7 per cent.

139. This section is largely based on the Binational Study Migration between Mexico and the United States (1997), Orrenius (2001), US-Mexico Migration Panel (2001) and Weintraub *et al.* (1997).

140. Welfare reform legislation denied undocumented immigrants and many legally residing immigrants access to most public benefits, including Medicaid.

141. CONAPO, population aged 12-64 years old.

142. The *Matricula Consular* contains holograms and embedded designs. Strong opposition to its recognition has been encountered in some states and some Federal agencies, but support is also provided, notably by deputies at the US Congress.

143. This programme has existed since 1974. In 2001, it involved about 11 000 Mexican workers. Workers are assured the minimum wage and have right to healthcare, pension, insurance accident and housing.

144. A network of people fighting human trafficking reports that the US Consulate in Monterrey once discovered a labour recruiter who was charging people $1 500 for the permits, while the processing cost usually falls below $200 [*http://fpmail.friends-partners.org/pipermail/stop-traffic/*2000-N*ovember*/001120.*html*]. More recently, Reuters News (6 February 2003) also reports the apprehensions of four persons (including an employee of the Monterrey US consulate) charged with selling H2A and H2B visas for 1 500 dollars.

145. The offices of state services to migrants co-ordinate the various programmes directed at migrants and their families and now have a national co-ordination. La Coordinación Nacional de Oficinas Estatales de Atención a Migrantes (CONOFAM) was created in 2000, and has 24 participating states. As to improving knowledge, the Consejo Estatal de Población of Guanajuato, for example, is developing a survey at the state level with its federal counterpart (CONAPO) and the Colegio de la Frontera Norte. They have also developed a Guanajuato module for the Northern Border Survey (EMIF).

146. OECD (2003), T*rends in* I*nternational* M*igration*: SOPEMI 2002 Edition.

147. Source: Instituto Nacional de Migración (INM).

148. The P*lan* S*ur* increased the number of border agents, established new blocks and increased military participation. 150 000 persons were deported in 2000, and about 140 000 in 2001, sending them back directly to their country of origin instead of leaving them at the border.

149. Source: EREM survey, conducted in 6 towns of the states of Jalisco, Guanajuato and Zacatecas (Papail, 2002). Data are corrected for the age structure. While the average annual wage in Mexico in 2002 was 5 894 US dollars, the average for Mexican-born in the United States was 28 818 US dollars, and for non-Hispanic Americans, it was 37 514 US dollars (Banamex, May 2003).

150. According to the survey at the northern border (EMIF), 70 per cent of the emigrants to the United States were in search of a job.

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151. Their share is much higher in the Los Angeles region, where they represented already more than 50 per cent in of the workers in 1990. While Mexican *maquiladoras* have specialised in the mass production of standardized garments, the leading garment districts in New York and Los Angeles retained jobs in design, as well as in direct production for the small-batch, high-fashioned garments, the latter relying on low-cost immigrant labour.

152. This is particularly the case for apparel, electronics and computer-related goods.

153. The wage in the formal sector is measured based on the average contribution wage to the IMSS.

154. The returns to education, defined as the incremental revenue from additional years of education relative to the average earnings, are higher in Mexico than in the United States.

155. As noted by Chiquiar and Hanson (2002), the better educated Mexicans may be better able to manage the US immigration process and may be less subject to credit constraints. This finding goes against the hypothesis of a negative selection bias developed by Borjas (1987).

156. For an analysis of the international mobility of the highly-skilled workers, see Guellec and Cervantes (2002).

157. It is interesting to note that US students are also increasingly going to Mexico; in 2000/01, they were 8 360. Mexican students have a low mobility intensity, about half the OECD average (Tremblay, 2002).

158. The study by Adams uses data from the 2000 US Population Census, referring to foreign-born individuals residing in the US, aged 25 and over. The figure for Mexico may somewhat over-estimate the outflow of the highly-educated Mexicans, since a number of Mexican-born in the US may have come as children and received their education in the US. Considering a wider set of data on immigration to OECD countries, international migration takes much more than 10 per cent of those with tertiary education in other countries, as well. For instance it takes more than 30 per cent in Turkey, Morocco and Tunisia.

159. Source: EMIF 1998-1999.

160. Papail (2002). Training was provided mostly to the more educated (over 10 years of education).

161. The returns to US experience are also found to be twice as high as those of getting secondary education, thereby providing persons who do not plan to continue their education beyond the lower secondary a powerful incentive to migrate at a young age.

162. The data used came from a 1 per cent subsample of the 2000 Mexican Census. School attendance is compulsory until age 10, and after 15, children often no longer live with their parents, which makes it impossible to link them to their original family. Potential simultaneity biases between household migration behaviour and schooling outcomes are controlled for by treating household migration behaviour as endogenous, using as instruments the interaction of historical state migration patterns and household characteristics. Controlling for the various factors affecting schooling other than migration, such as household resource constraints, the education level of parents, the location in a rural or urban area, and local labour market conditions, the estimated increase ranges from 0.7 year to 1.6 years of schooling, with a much larger impact for girls than for boys, and for older children than for younger ones, suggesting that schooling outcomes of these groups are more sensitive to fluctuations in household income.

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163. Other changes, which are particularly difficult to measure, include i) the shortfall in the taxable profits of the enterprise in which the migrant was previously working, and ii) the increase in the taxable profits of the banks or other companies operating the transfers of remittances to Mexico.

164. From 1994 to 2001, the annual border-enforcement budget of the INS nearly tripled to more than 2.5 billion dollars, and as a result of heightened security concerns following the September 11 events it may exceed 5 billion dollars in the fiscal year 2003.

165. In 2000, for example, 369 persons died when trying to cross the border, mostly due to harsh climate conditions faced during the long walks in the desert.

166. Coyotes are generally previous migrants. Spener (2001) emphasizes the fact that few of them are part of major smuggling rings, and that they are rarely connected with organised crimes. In Mexico, they are more perceived as "service providers" than as criminals.

167. With only a small part likely to remain in the United States to pay for the costs associated with the conveying of migrants on the US territory.

168. To some extent, remittances might include transfers stemming from money laundering, which would explain part of the discrepancies among statistics.

169. Improved technology and secure transactions, the entry of banks in the remittance industry and the increased number of distribution outlets in Mexico and recollection outlets in the United States have all been factors contributing to increase the share of remittances sent by emigrants through formal channels.

170. The average allowance received through PROGRESA for food and education was around 340 pesos a month in 2000 (SEDESOL data), *i.e.* about US$560 a year.

171. Source: Papail (2002) and own calculation based on Zárate-Hoyos (1999). These figures take into account both the remittances per se and the savings accumulated by migrants in the United States that are brought or sent back on return to Mexico. Data from the Mexican Migration Project used by Zárate-Hoyos (1999) show that the propensity to invest savings in productive activities is 17 per cent, against 3.7 per cent for remittances. Savings represent about 25 per cent of the amounts remitted.

172. Indeed, the increasing integration of the economy is probably reducing the indirect effects of remittances at the local level, just as the integration of Mexico into the North American market may reduce their indirect effect at the national level.

173. See Corona (2000).

174. Source: Goldring (2003).

175. In fact, the cost of sending remittances from the United States to Latin America has remained higher than to the rest of the world (O'Neil, 2003).

176. See "Analiza EU acceso bancario a inmigrantes", Reforma, 12 February 2003.

177. Based on data provided by Orozco (2000) for Western Union.

178. For example, some banks charge monthly maintenance fees. Other charge extra fees for withdrawing money via ATM, when a sender or a recipient contacts a person in the bank for inquiries regarding the remittance, and/or when there is more than one withdrawal during a given period of time, etc. (Orozco, 2003).

179. See Garcia Zamora (2000).

180. BANSEFI would help them to manage their liquidity and would provide accounting services. For details on BANSEFI see Annex III of the OECD *Economic Survey* 2002.

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181. BANSEFI is already distributing social assistance provided under *Oportunidades* (previously PROGRESA); here it would also make the link with housing subsidies provided by various public housing funds (FONAPO, FOVI, INFONAVIT).

182. This would, among others, allow them to benefit through BANSEFI from the new transfer system put in place by the central bank and thus to lower the exchange rates fees.

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Annex I

Background information to Chapter I

I.A. Output growth and productivity: figures and tables
I.B. The transition of the agricultural sector
I.C. Recent economic developments
I.D. Poverty alleviation

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Annex I.A

Output growth and productivity: Figures and tables

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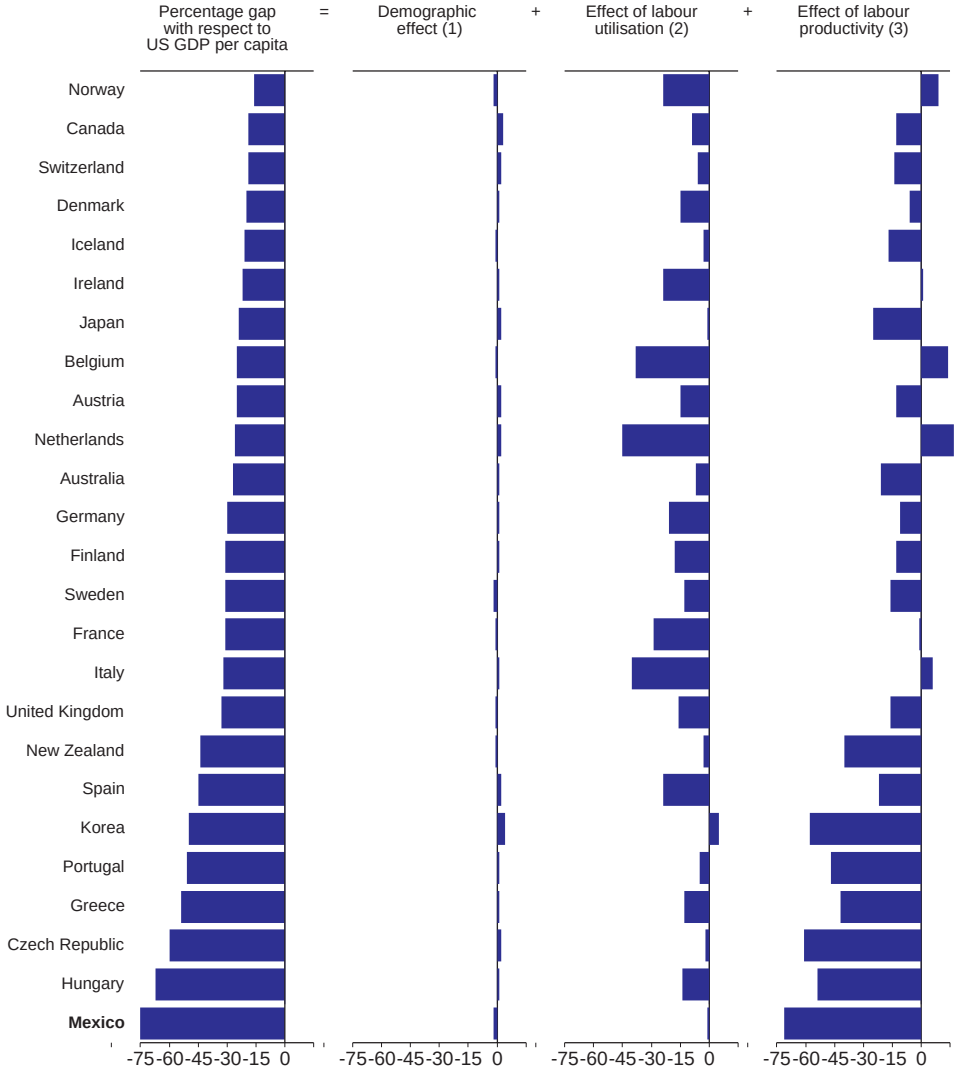
Annex Figure I.A.1. **Per capita GDP in OECD countries, 2000**

Percentage point difference in trend, PPP-based GDP per capita with respect to the United States



Note: GDP growth performance in OECD countries over past decades is shown in Annex I.A. Table 1 below.
1. Based on the ratio of working-age population (15-64 years) to total population.
2. Based on employment rates and average hours worked.
3. GDP per hour worked.

Source: OECD, *The Sources of Economic Growth in OECD Countries* (2003).

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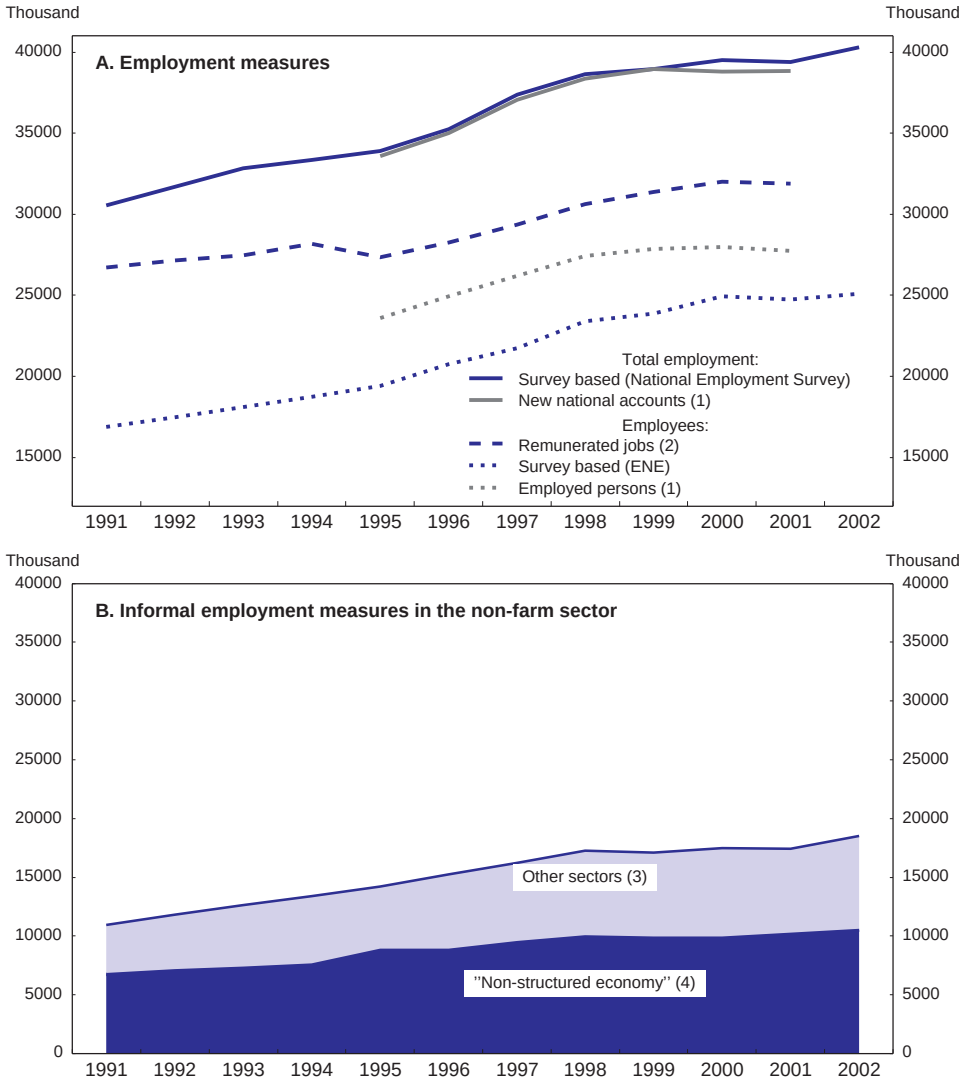
Annex Figure I.A.2. **Employment data**
Persons aged 12 years and over



1. National Accounts (1995-2000).
2. National Accounts (1988-2001).
3. Includes workers in registered establishments and households, but who have no work contract or receive no payment; OECD estimates, see Annex Box 1 in Chapter I.
4. Workers in the "non-structured sectors", estimated by INEGI on ILO definition, see note to next Figure. Data before 1994 are OECD estimates (ILO-based).

Source: OECD; INEGI.

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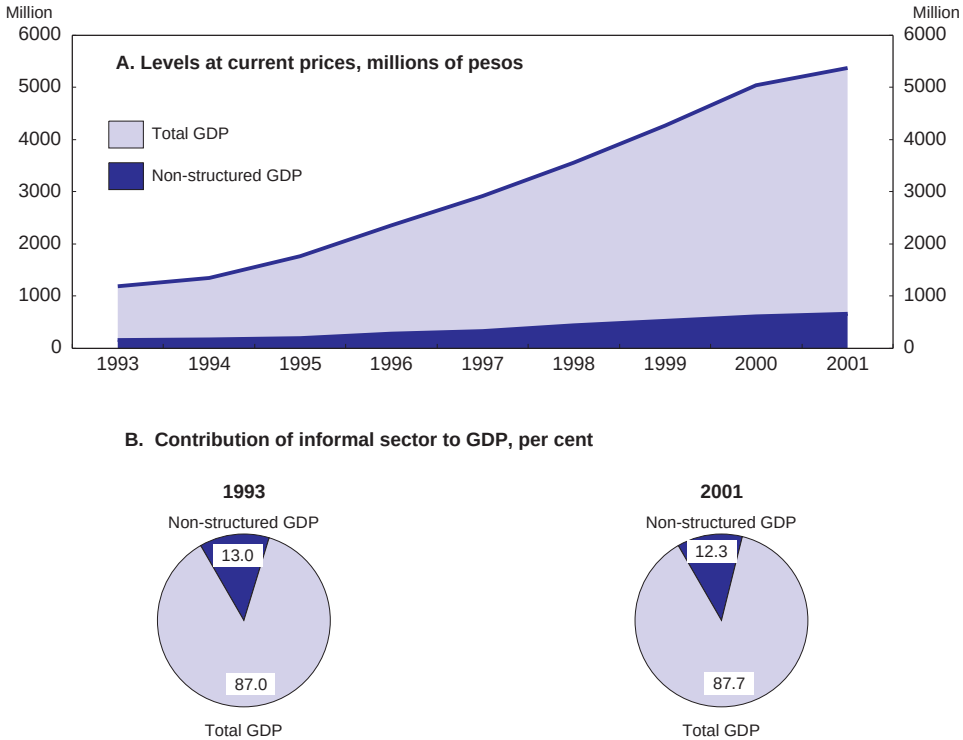
Annex Figure I.A.3. **Total and non-structured GDP**[1]
Per cent



1. According to the ILO definition, the non structured sector is regarded as a group of household enterprises (uncorporated enterprises owned by households) that includes:
   – informal own account enterprises, which may employ contributing family workers and employees on an occasional basis; and
   – enterprises of informal employers, which employ one or more employees on a continuous basis (these enterprises must fulfil one or both of the following criteria: size of unit below a specified level of employment, and non registration of the enterprise or its employees).

Source: INEGI, *Cuenta Satélite del Subsector informal de los Hogares*.

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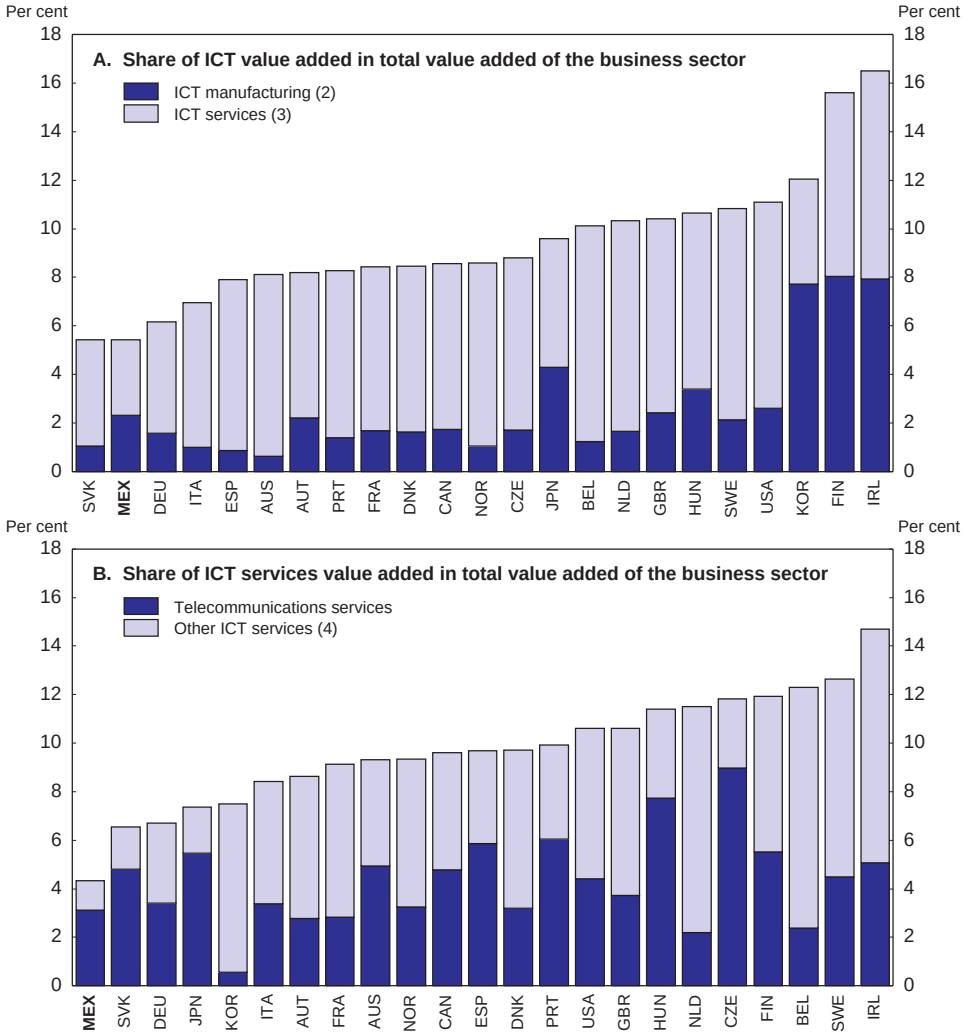
Annex Figure I.A.4. **Size of the ICT industry in selected OECD countries**

2000[1]



1. 1999 for Germany, Hungary, Italy, Korea, Portugal and the Slovak Republic; 1998 for Canada.
2. ICT manufacturing is the sum of ISIC Rev. 3 sectors: 3000, 3130, 3210, 3220, 3230, 3312, 3313.
3. ICT services is the sum (when available) of ISIC Rev. 3 sectors: 5150, 7123, 6420, 72.
4. "Other ICT services" is the sum (when available) of ISIC Rev. 3 sectors : 5150, 7123.

Source: OECD estimates based on national sources; STAN and National Accounts databases, September 2002.

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Annex Table I.A.1. **GDP growth in OECD countries**

Average annual rates of change, 1970-2000

	Actual growth of GDP				Actual growth of GDP per capita				Trend growth of GDP per capita		
	1970-1980	1980-1990	1990[1]-2000	1996-2000	1970-1980	1980-1990	1990[2]-2000	1996-2000	1980-1990	1990[2]-2000	1996-2000
Australia	3.2	3.2	3.5	4.2	1.5	1.7	2.3	3.0	1.6	2.4	2.8
Austria	3.6	2.3	2.3	2.7	3.5	2.1	1.8	2.6	2.1	1.9	2.3
Belgium	3.4	2.1	2.1	3.2	3.2	2.0	1.8	3.0	2.0	1.9	2.3
Canada	4.3	2.8	2.8	4.4	2.8	1.5	1.7	3.5	1.4	1.7	2.6
Czech Republic	..	..	1.5	0.1	..	..	1.6	0.2	..	1.7	1.4
Denmark	2.2	1.9	2.3	2.8	1.8	1.9	2.0	2.4	1.9	1.9	2.3
Finland	3.5	3.1	2.2	5.3	3.1	2.7	1.8	5.0	2.2	2.1	3.9
France	3.3	2.4	1.8	2.9	2.7	1.8	1.4	2.6	1.6	1.5	1.9
Germany[3]	2.7	2.2	1.6	2.0	2.6	2.0	1.3	2.0	1.9	1.2	1.7
Greece	4.6	0.7	2.3	3.7	3.6	0.2	1.9	3.5	0.5	1.8	2.7
Hungary	..	..	2.3	4.7	..	..	3.4	5.1	..	2.3	3.5
Iceland	6.3	2.7	2.6	4.6	5.2	1.6	1.6	3.4	1.7	1.5	2.6
Ireland	4.7	3.6	7.3	10.4	3.3	3.3	6.4	9.2	3.0	6.4	7.9
Italy	3.6	2.2	1.6	2.1	3.1	2.2	1.4	1.9	2.3	1.5	1.9
Japan	4.4	4.1	1.3	0.7	3.3	3.5	1.1	0.5	3.3	1.4	0.9
Korea	7.6	8.9	6.1	4.3	5.8	7.6	5.1	3.3	7.2	5.1	4.2
Luxembourg	2.6	4.5	5.9	7.1	1.9	3.9	4.5	5.7	4.0	4.5	4.6
Mexico	**6.6**	**1.8**	**3.5**	**5.6**	**3.3**	**–0.3**	**1.7**	**4.2**	**0.0**	**1.6**	**2.7**
Netherlands	2.9	2.2	2.9	3.8	2.1	1.6	2.2	3.2	1.6	2.4	2.7
New Zealand	1.6	2.5	2.6	2.6	2.2	0.5	1.9	1.2	1.4	1.2	1.8
Norway[4]	4.4	1.5	2.8	2.6	3.8	1.1	2.2	2.0	1.4	2.0	2.2
Poland	..	..	3.6	4.9	..	..	3.5	4.9	..	4.2	4.8
Portugal	4.7	3.2	2.7	3.6	3.4	3.1	2.5	3.2	3.1	2.8	2.7
Slovak Republic	..	..	4.6	3.6	..	..	4.4	3.5	..	..	..
Spain	3.5	2.9	2.6	4.1	2.5	2.6	2.5	4.0	2.3	2.7	3.2
Sweden	1.9	2.2	1.7	3.3	1.6	1.9	1.4	3.2	1.7	1.5	2.6
Switzerland	1.4	2.1	0.9	2.2	1.2	1.5	0.2	1.8	1.4	0.4	1.1
Turkey	4.1	5.2	3.6	3.1	1.8	2.8	1.8	1.5	2.1	2.1	1.9
United Kingdom	1.9	2.7	2.3	2.9	1.8	2.5	1.9	2.4	2.2	2.1	2.3
United States	3.2	3.2	3.2	4.2	2.1	2.2	2.2	3.3	2.1	2.3	2.8

1. 1991 for Germany and Hungary, 1992 for Czech Republic, 1993 for Slovak Republic.
2. 1991 for Germany, 1992 for Czech Republic and Hungary, 1993 for Slovak Republic.
3. Western Germany before 1991.
4. Mainland only.

Source: OECD, *Economic Outlook*, No. 70.

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Annex Table I.A.2. **Intra-firm trade of the United States, with selected trading partners**

As a per cent of all goods trade with partner country

	Level 1999	Change, 1992-99
Imports		
Japan	73.7	–1.3
Mexico	66.4	3.1
Korea	49.3	22.5
Canada	43.1	–2.9
Eastern Europe	32.1	20.1
Taiwan	20.8	4.9
China	17.6	7.1
Total	46.7	1.7
Exports		
Mexico	44.3	5.6
Canada	42.4	–2.9
Japan	36.3	0.1
Taiwan	16.0	6.0
Eastern Europe	12.3	2.6
China	11.6	5.0
Total	32.1	1.2

Note: Partner countries shown are those with the highest level of intra-firm trade in 1999 or the largest increase over the period 1992-99.

Source: OECD, *Economic Outlook* No. 71 (June 2002); United States Department of Commerce (2001).

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Annex Table I.A.3. **Manufacturing intra-industry trade**

As percentage of total manufacturing trade

	1988-91	1992-95	1996-2000	Change
High and increasing intra-industry trade				
Czech Republic	n.a.	66.3	77.4	11.1
Slovak Republic	n.a.	69.8	76.0	6.2
Mexico	62.5	74.4	73.4	10.9
Hungary	54.9	64.3	72.1	17.2
Germany	67.1	72.0	72.0	5.0
United States	63.5	65.3	68.5	5.0
Poland	56.4	61.7	62.6	6.2
Portugal	52.4	56.3	61.3	8.9
High and stable intra-industry trade				
France	75.9	77.6	77.5	1.6
Canada	73.5	74.7	76.2	2.7
Austria	71.8	74.3	74.2	2.4
United Kingdom	70.1	73.1	73.7	3.6
Switzerland	69.8	71.8	72.0	2.2
Belgium/Luxembourg	77.6	77.7	71.4	–6.2
Spain	68.2	72.1	71.2	3.0
Netherlands	69.2	70.4	68.9	–0.3
Sweden	64.2	64.6	66.6	2.4
Denmark	61.6	63.4	64.8	3.2
Italy	61.6	64.0	64.7	3.1
Ireland	58.6	57.2	54.6	–4.0
Finland	53.8	53.2	53.9	0.1
Low and increasing intra-industry trade				
Korea	41.4	50.6	57.5	16.1
Japan	37.6	40.8	47.6	10.0
Low and stable intra-industry trade				
New Zealand	37.2	38.4	40.6	3.4
Turkey	36.7	36.2	40.0	3.3
Norway	40.0	37.5	37.1	–2.9
Greece	42.8	39.5	36.9	–5.9
Australia	28.6	29.8	29.8	1.2
Iceland	19.0	19.1	20.1	1.1

Note: Countries are classified as having "high" or "low" level of intra-industry trade according to whether intra-industry trade is above or below 50 per cent of total manufacturing trade on average over all periods shown and "increasing" or "stable" according to whether intra-industry trade increases by more than 5 percentage points between the first and last periods, as shown in the final column.

Source: OECD, *Economic Outlook No.* 71 (June 2002), based on OECD International Trade Statistics.

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Annex I.B

The transition of the agricultural sector

To supplement the analysis of Mexico's overall growth performance, this annex reviews the adjustment of the agricultural sector to trade liberalisation as well as the policies to support the sector.

Overview

Between 1980 and 2000, the share of the agricultural sector in GDP dropped from 8.2 to 5.5 per cent, but the sector still accounts for nearly 20 per cent of employment, based on national accounts (17 per cent based on the national employment survey). Relative productivity in the sector is particularly low, suggesting that it is predominantly this sector which absorbs excess labour by providing casual jobs and part-time employment. Ten years after the beginning of NAFTA, Mexico's agricultural exports (including livestock, processed food, beverages and tobacco) have nearly doubled and there has been a significant diversification of export products. Nevertheless, the Mexican agricultural sector continues to struggle and has not yet achieved the benefits that were expected from a free-trade agreement with the United States and Canada. The agricultural sector acted as a cushion during the 1995 recession, but it was largely isolated from the expansion phase that followed, growing at lower growth rates than the rest of the economy. Rural poverty increased during most of the 1990s, decreasing only in 1999-2000.

How has agricultural trade reform affected the sector?[1]

Expected impact and needs for accompanying measures

Agricultural trade reform, in general, is expected to have important short-term costs for producers in the import sector, while benefiting producers of "exportables". In Mexico, the liberalisation of maize production that accompanied the process of trade liberalisation was expected to have a major impact on Mexico's rural economy. As support prices were set to disappear, commercial agriculture in irrigated lands was expected to shift to other crops, while subsistence maize agriculture, unable to compete with imports, was expected to be hurt by liberalisation. Losses of jobs and lower wage income, which were to accompany the process, were expected to lead to increased migration to urban areas and to the United States.

There is a consensus that to minimise short-term costs, in general the transition should include three lines of actions:

i) Direct transfers to cushion the impact;

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ii) Upskilling and training of labour to allow increased productivity both in the sector and in other sectors;

iii) Credit to allow the reconversion and to invest in new technology.[2]

Liberalisation in Mexico started before the entry into force of NAFTA and the trade agreement set gradual steps for the process. This gradual process was established with a view to facilitate the adjustment of the sector. Mexico has been phasing out subsidies (guaranteed prices to producers) and increasing direct cash transfers, which have been provided by PROCAMPO (Programme for direct support to rural areas, see below), with a view to compensate growers affected by the trade liberalisation. This strategy has been appropriate. The direct transfers have cushioned the negative impact of trade liberalisation on commercial producers of importable crops (grains, and in particular maize). It has also benefited subsistence farmers, who were relatively isolated from the markets and were not receiving price support under the previous arrangement.

However, the second and third lines of action for a smooth transition were insufficient or simply missing. Over the 1990s, there has been a clear lack of credit to the sector, affecting mostly the small producers. Three-quarters of national producers use land of five or fewer hectares (half of them producing for families' own consumption). They have also had limited access to modern technology and machinery. In addition, the sector has suffered from the impact of budget consolidation, which has meant a reduction of resources for roads and other transport infrastructure.

The impact on the small farm sector

Producers of exportable products (mostly fruit and vegetables including tropical and semi-tropical products) overall have benefited from the trade liberalisation, with exports of these products growing considerably.[3] But the adjustment of the rest of the sector has not been as fast as expected. Producers of other crops have had problems in shifting their production to more competitive commodities. Eight years after NAFTA and the introduction of PROCAMPO, maize imports had soared, but the domestic supply of maize had not fallen.[4] Because of market failure and heterogeneity across households, overall maize production increased rather than declining as projected.[5] As most households engaged in staples production have other sources of income the impact of staple price reduction was buffered. Only large commercial farms on irrigated lands were induced to increase yields, thus lowering unit costs. Many small farmers, on the other hand, because of high transaction costs (distance to purchase points and uncertainties) did not sell to the government at the guaranteed, high, prices in the period of extensive price support and they were not affected by the phasing out of subsidies. In an environment lacking credit, income transfers were essentially used to stimulate production on these small farms. Emigration from rural areas did not happen on the scale expected, and overall employment in the agricultural sector did not change as much as anticipated.[6]

A microeconomy-wide model (computable general equilibrium) was estimated by Taylor for a village-town economy of a migrant sending state.[7] Under a scenario with high transaction costs, hence no transactions in outside staple markets, producers and consumers are isolated from government policy (*i.e.* from the support price change in this case). The income transfers increase household income, pushing up the price of the staple; in this context the agricultural reform did not discourage production, it might even have had an expansionary impact for the small-farm sector.

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Phasing out the guaranteed price support scheme

With the opening up of the economy to international competition, Mexico has been reducing the intervention of government agencies in national markets for agricultural products. There has been a shift in policy from price support measures for certain grains and oilseeds to direct payments to producers (income support). Until 1989, the main price intervention mechanism came from the National Basic Food Company (CONASUPO), which used to buy main grains, beans and oilseeds from producers at guaranteed prices as well as controlling imports through licensing. By 1991, CONASUPO intervened only in the case of maize and dry edible beans and the implementation of general tortilla subsidy. Simultaneously, the Mexican Government implemented a transitory concerted price scheme for some crops in regions facing marketing difficulties, given the lack of infrastructure and the poor knowledge of market conditions.

A Presidential Decree ordered the closure of CONASUPO in 1999. This has allowed the entry of new agents that did not participate in the market previously. It also has allowed the development of new marketing channels and strengthening of regional markets.

Policies in place for the agricultural and rural sector

Recent experience shows that domestic production of basic crops did not collapse with the trade liberalisation. Nonetheless, as progress continues in the NAFTA agenda, the need to cushion the impact will continue to be felt.[8] A number of programmes directly aimed to the agricultural sector are already in place. The main ones are the following:

The Programme for rural areas: PROCAMPO[9]

This programme, introduced in 1993-4, substituted the previous guaranteed price support scheme by the provision of direct payments to facilitate adjustment after NAFTA for producers of basic crops.[10] PROCAMPO cash transfers are "decoupled" from production (unlike guaranteed prices which distorted production incentives). Its main objectives have been to provide income support to farmers affected by market liberalisation, and to promote a shift in production towards products that have comparative advantages. This programme also benefits subsistence farmers who did not receive any kind of price support previously.[11] To be eligible for this programme, land must have been cultivated with one of the basic crops prior to August 1993. The payments are granted to those who actually cultivate the land, being owners or renters. Farmers can devote land to any crop, livestock, forestry or environmental activity.

The programme provides income support to 2.8 million producers, of which 2.3 million are farmers on *ejidos* (a community-based form of ownership, see Box below).[12] In 2002, it had a planned expenditure of 12.4 billion pesos (1.2 per cent of public programmable spending). The number of crops that can be cultivated has been expanded in the last years to allow for crop switching. The money transfers received through the programme have reached up to 40 per cent of household income for low-income families. Although providing income support to low-income families, the programme has not actually entailed much switching to more productive crops. Payments have often been used to maintain unprofitable subsistence production. In addition, because of difficulty in proving the rental of eligible plots, cultivators have sometimes not been able to receive payments.[13]

The programme for agricultural marketing support

Since 1991, the Marketing Commercialization Support Agency (ASERCA) is the institution in charge of promoting the marketing of some agricultural products; in order to support

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Annex Box I.B.1. The ejido sector

An *ejido* is a Mexican community based form of land tenure that originated from the breakdown of large holdings of land during the post revolutionary land reform after 1917. E*jidatarios*, the people farming this land, were given only usufruct rights over the land and could not alienate it as if it were private property: they could not sell it or rent it, nor use it as a collateral, thus limiting their ability to access credit. E*jidatarios* had to work the land directly, being prohibited from hiring labour or to share crop. If an *ejidatario* could no longer farm his land and had no successors in the family able to do so, or if he was absent from the land for more than two years, he would lose the usufruct rights, and the land would go to the community for redistribution. All these features have prevented expansion of production scale and limited the flow of capital into *ejidos*.

In 1992, agrarian laws were reformed, giving *ejidatarios* increased rights to their lands, permitting commercial associations of farmers, and allowing peasants to put their land as a collateral for loans. Under the new legislation there are two types of *ejido* land:

- Land for common use. This land cannot be transferred to other individuals, but it can be used in productive associations of farmers or rented up to 30 years.
- P*arcelled* plots. This land continues to be property of the *ejido*, but the individual *ejidatario* has the permanent right to use his plot, can rent it for up to 30 years or sell it to other members of the *ejido*. The *ejidatario* can also decide to separate his plot from the *ejido* and convert it into private property. This allows the former *ejido* member to sell or rent the land to other persons outside the *ejido*, obtain a mortgage or use it as if it were any other type of private property.

After more than 10 years of the agricultural reforms, the efficiency of the *ejido* sector has not increased as expected. At present, *ejido* land accounts for about 63 per cent of the country's farm land, and it continues to be characterised by low productivity. The fragmentation of public lands into individual plots has accentuated the problem of small landholdings, discouraging capital investment and made it difficult to achieve economies of scale in production. Farmers also continue to utilize labour intensive technologies and employ obsolete cultivation techniques, which lead to low yields per hectare. The transition towards private property has also been slow. Since 1992, less than 1 per cent of parcelled lands have been transformed into private property ("Mexican agriculture: the causes of its backwardness". BBVA Bancomer, E*conomic* R*eport*, July 2003).

producers located far away from main consumption centers. ASERCA granted supports to buyers for each produced ton of wheat, sorghum, oilseeds and maize (this last one since 1999). In the case of rice, the payments were given directly to the producers for each produced ton. Since 2001, all payments are granted directly to producers.

The main objective of this programme is to strengthen producer's profitability, competitiveness and crops diversification, as well as to achieve a more efficient marketing of those

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products that were facing surplus production problems in certain regions. However, these commercialization subsidies were concentrated in some products (mainly maize, wheat, sorghum and rice). This situation distorted production decisions, providing an incentive to produce the supported products, instead of other more competitive ones which could represent a better income alternative.[14] In order to reduce market distortions generated by the previous price support policy, in 2003, the Ministry of Agriculture, through ASERCA, established a target income programme for a wider range of crops: it compensates producers for the gap between a previously established target price and the market one.

The Alliance for rural areas: *Alianza para el Campo*[15]

This programme was introduced in 1996 to support investment for medium and small farmers with productive potential. It provides matching grants to agricultural producers to boost investments. Producers tend to contribute with 50 per cent, the state and federal government with the other 50 per cent. The main goals are to increase productivity, to raise producers' income, to improve the agriculture trade balance, and to help food production expand.[16] It is decentralised, each state being responsible for its implementation.

In 2002, it provided 8.7 billion pesos to 4.3 million producers for support investment.[17] *Alianza para el Campo* has benefited a large number of medium and large scale farmers.[18] However, for small farms it is of little use. In particular, *ejidatarios* lack resources to provide the counterpart funding that is necessary. Furthermore, transaction and cooperation costs to form groups to participate are high. As a result, only 11 per cent of *ejidatarios* participated in the programme in 2000.[19]

The Programme of Certification of Ejido Rights: PROCEDE

PROCEDE was created in 1993 as part of the land reform to promote the development of land property rights in the *ejidos* (see box above) and to resolve land boundary problems. The purpose was to give farmers property titles that could be used as collateral for loans, so as to facilitate their access to credit.

By the end of the 1990s this programme had issued certificates to more than 3 million households.[20] It has increased household welfare by allowing participation in off-farm labour markets. However, there has not yet been much increase in credit.

Rural finance

Past government intervention, through the specialised development bank *Banrural* and the trust fund FIRA, has prevented the development of a well-functioning rural financial market. Credit programmes, characterised by poor recovery, large-scale debt forgiveness and subsidized rates undermined discipline. Credit channelled to the agricultural sector was still growing in the early 1990s, but it sharply decreased after the 1995 financial crisis, when the public sector scaled back its operations. Commercial banks did not get involved because of a high default rate from benefiting farmers. Total credit granted to agriculture was 21 per cent higher in 1983-90 than in 1996-2000. In addition, the limited supply of credit in the second half of the 1990s has been channelled in large part to big farmers.

On the saving side, as well, there have been structural obstacles limiting the role of financial institutions, including a lack of confidence in financial institutions. Non-bank savings and loans institutions have been more involved, but the regulatory framework for their activities was lacking, until the creation of *Bansefi* (see 2002, OECD *Economic Survey of Mexico*).

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A new decentralised institution aimed at promoting the development of rural financial markets was created in 2002, *Financiera Rural*. Its purpose is to give access to credit to acquire machinery, equipment and technology, so as to increase productivity in the agricultural sector. It will concentrate on developing financial markets that are not adequately covered by private banks and expanding access to credit of low- and middle-income rural producers. It will include programmes of micro credit. The law pertaining to *Financiera Rural* is designed so as to increase transparency and avoid some of the shortcomings that existed in *Banrural*, including "unethical" behaviour. Operating structures will be simplified to make processes faster.

Summing up

The adjustment of the agricultural sector will continue to take place and substitution of competitive crops (fruit and vegetables) for grains should resume, including substitution within grain production (away from white maize, a surplus crop, to yellow maize, an importable). Commercial producers of grains are likely to be negatively affected. Transfers through PROCAMPO should be continued. But eligibility criteria to this programme could be simplified and information should be widely diffused through various channels to promote better awareness of the programme, especially among the indigenous population.[21] A greater effort should be made through information, training and technical support, to help farmers switch to more productive cropping strategies and long-term alternatives. PROCAMPO payments could be made progressive, so that those having less land receive more support per hectare. *Alianza para el Campo* should also continue but efforts are needed to make *ejidatarios* more aware of the programme and understand how they can have access to resources. Eligibility rules could be eased for the poorest communities. Specific measure could make it easier for indigenous farmers to qualify for assistance.

At the same time, policy action is required in a number of other domains, including measures which are more akin to development policies than reforms in higher-income OECD countries. In particular small-scale farmers, if they are to share the benefits of liberalisation, need to have access to markets for their products and adopt new technology. Efforts to diffuse information to these farmers in particular would be welcome. The strategy should also include measures to develop access to credit and investment, and to secure access to markets. Public support is needed to build rural roads and to help the creation of marketing cooperatives. On the other hand, subsidies for water pumping that are granted implicitly through low electrical tariffs should be eliminated. They benefit mainly large consumers in Northern States in particular, and are thus regressive. In 1999, their amount was nearly twice the amount granted through *Alianza para el Campo* for the whole country.

Furthermore, the rural sector is a pool of low-skilled labour. There are high transaction costs in rural areas, because of geographical remoteness, and because of the cyclicality of agricultural activities. Limited education achievements reflect largely supply factors, with low availability and quality of schools, but also demand factors, because of the high opportunity cost for parents to send children to school. To facilitate the adjustment of the workforce and enable some of those engaged in agricultural activities to find jobs in other sectors, training on a large scale will be needed. Programmes to alleviate poverty will need to continue, providing some kind of safety net to the marginal areas and taking into consideration the low mobility of an increasing share of the poor population in rural areas, because of ageing (see Annex I.D on poverty alleviation).

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Notes

1. This section is based on various papers including those presented at the 2002 OECD Global Forum on "Agricultural trade reform, adjustment and poverty" and reports prepared for the World Bank. See OECD, *Agricultural trade and poverty*, 2003; A. Yunez-Naude (2002) "Lessons from NAFTA: the case of Mexico's agricultural sector", mimeo, Dec.; Levy, S and S. Van Wijnbergen (1992) "Transition problems in economic reform : agriculture in the Mexico-US free trade agreement", *World Bank Policy Research Working Paper*, 967; E. Taylor "Microeconomics of globalisation: evidence from China and Mexico", in OECD *Agricultural trade and poverty*, 2003.
2. These general principles are set out in OECD, *Agricultural trade and poverty*, 2003.
3. Already before NAFTA, this "niche" sector was competitive and cultivated areas had increased in the years prior to the entry into force of NAFTA. Some substitution of exportables for importables had already taken place.
4. Mexico produces mainly white maize, the main use of which is for human consumption; imports are concentrated in yellow maize, used to feed livestock.
5. Domestic productivity for crops which are importables increased only in irrigated land, *i.e.* land used by large commercial farmers. Freer market access has induced an increase in land yields, through technological change, to reduce unit costs. Cultivated areas have not changed since the entry into force of NAFTA, perhaps because market access has remained limited in the transition phase. In rain-fed land, by contrast, production has increased since NAFTA, but yields have not. A. Yunez-Naude (2002) (*op. cit.*).
6. From E. Taylor (2003) (*op. cit.*).
7. The new USA Farm Bill seems to be a step backward in the trade liberalisation for agricultural goods which was agreed in NAFTA. It is likely to compound the on-going adjustment problems of Mexico's rural sector.
8. The new USA Farm Bill seems to be a step backward in the trade liberalisation for agricultural goods which was agreed in NAFTA. It is likely to compound the on-going adjustment problems of Mexico's rural sector.
9. See OECD (2003) *Agricultural Policies in* OECD *Countries: Monitoring and Evaluation*.
10. Barley, beans, maize, cotton, rice, sorghum, soy, sunflower and wheat.
11. See OECD (1997) *Review of Agricultural policies in Mexico*.
12. See Segundo Informe de Gobierno 2002. September. *www.informe.presidencia.gob.mx/index.php*
13. World Bank (2001) "Mexico Land Policy-A decade after the Ejido Reform". *World Bank Report* No. 22187 ME, June.

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14. Until 2002, subsidies for commercialization were granted through price support. This scheme generates a change in relative prices providing an incentive to produce the supported crop.
15. See OECD (1997) *op. cit.*
16. See OECD (1997) *op. cit.*
17. See Segundo Informe de Gobierno 2002. *op. cit.*
18. See FAO and SAGAR (Secretaría de Agricultura, Ganadería y Desarrollo Rural) (2000) *Evaluacion de la Alianza para el Campo : Informe Global*. October. *www.rlc.fao.org/prior/desrural/document/alianza.htm*
19. See World Bank (2001) *op. cit.*
20. Source: Registro Agrario Nacional. March 31, 2003. *www.ran.gob.mx/archivos/12.htm*
21. See World Bank (2001) *Mexico: A comprehensive Development Agenda*.

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Annex I.C

Recent macroeconomic developments

This Annex reviews recent macroeconomic developments in some detail, so as to supplement the discussion of the current cycle included in the *Survey*.

A still hesitant recovery

The recent downturn which started at the end of 2000 has been mild compared with previous ones. It is the first recession in Mexico that was not accompanied, or provoked, by a balance-of-payment crisis nor a currency crisis. Furthermore, Mexico appears to be decoupled from other emerging markets and financial crises in the rest of Latin America did not spread to Mexico. The current cycle is also characterized by a close synchronization with activity in the United States. After a sharp decline in the second half of 2001, Mexico's real GDP turned up in the course of 2002, in pace with the United States, and there was modest growth in the second half of the year (2.3 per cent s.a.a.r. from the previous half year). But the Mexican recovery has been hesitant (Annex Table I.C.1). In the first half of 2003, GDP growth was sluggish and manufacturing output fell, despite stronger final demand in the United States. The delay in the Mexican recovery might reflect to some extent the composition of United States growth where industrial production was weak until mid-2003.

Private sector investment has contracted since the start of 2001 (Annex Table I.C.2), suffering from the uncertain international environment, and also uncertainties regarding the progress in structural reform. Gross capital formation (including stockbuilding) dropped to around 20 per cent of GDP in 2002, close to the ratio in 1995, the previous cyclical trough. Both domestic and foreign saving also came down in 2001 and 2002 (Annex Table I.C.3). Private consumption lagged in the slowdown. But after falling in the latter part of 2002, it turned up in early 2003, recording positive growth for two consecutive quarters (changes from the previous quarter). However, despite the improved economic environment, with low inflation, low interest rates and some recovery in real wages, consumer confidence has been affected by job losses, and short-term indicators on household purchases are still hesitant.

Labour market slack, wages and disinflation

The deterioration of the labour market was briefly interrupted near the end of 2002, when employment in the formal sector (measured by those affiliated to the social security institute, IMSS) recorded positive growth. However, the recovery in formal employment was interrupted in the second quarter of 2003, as the manufacturing sector continued to record severe job losses (Annex Table I.C.4). The open unemployment rate turned upward, as did the broader measures of unemployment (Annex Figure I.C.1). More significantly, employment in the "non-structured sector" (micro-businesses most notably) increased again. Wage increases came down in the course of 2002, even though disinflation stalled, so that in real terms only modest gains were recorded. In the first half of 2003, real wage increases

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Annex Table I.C.1. **Demand and output**
Percentage changes, volume, 1993 prices

	1989-94[1]	1995	1996-98[1]	1999	2000	2001	2002	2003 Q1[2] / 2002 Q4	2003 Q2[2] / 2003 Q1
	Percentage changes from previous period								
Demand									
Private consumption	4.9	–9.5	4.7	4.3	8.2	2.7	1.2	3.2	5.7
Government consumption	3.0	–1.3	1.5	4.7	2.0	–1.2	–1.3	–6.6	8.4
Gross fixed capital formation	7.8	–29.0	15.9	7.7	11.4	–5.8	–1.3	–1.6	6.5
Public sector	. .	–31.3	–4.0	10.7	27.2	–13.0	7.3	12.1	82.4
Private sector	. .	–28.2	21.3	7.2	8.7	–4.4	–2.8	–4.4	–6.5
Final domestic demand	**5.1**	**–12.3**	**6.2**	**5.0**	**8.2**	**0.6**	**0.5**	**1.3**	**6.1**
Stockbuilding[3]	0.2	–2.2	1.0	–0.6	0.3	–0.2	0.5	–5.5	–2.0
Total domestic demand	**5.2**	**–14.0**	**7.1**	**4.3**	**8.3**	**0.4**	**1.0**	**–4.1**	**3.9**
Exports of goods and services	7.8	30.2	13.7	12.4	16.4	–3.6	1.4	–2.9	–2.0
Imports of goods and services	15.9	–15.0	20.7	14.1	21.5	–1.5	1.6	–8.0	–4.3
Foreign balance[3]	–1.3	8.5	–1.3	–0.5	–1.8	–0.7	–0.1	2.1	0.9
GDP at market prices	**3.9**	**–6.2**	**5.6**	**3.7**	**6.6**	**–0.3**	**0.9**	**–2.2**	**4.9**
	Percentage changes from same period of previous year							2003 QI / 2002 QI	2003 Q2 / 2002 Q2
Output									
Agriculture, forestry, fishing	1.8	0.9	1.5	3.6	0.6	3.3	–0.4	–0.2	4.9
Mining (including petroleum)	1.6	–2.7	5.1	–2.1	3.8	0.8	–0.3	1.3	3.7
Manufacturing	4.3	–4.9	9.4	4.2	6.9	–3.7	–0.6	0.9	–4.5
Construction	5.7	–23.5	7.8	5.0	4.1	–5.3	1.7	5.9	1.3
Electricity	3.0	2.1	3.9	7.9	1.0	1.3	3.8	3.0	0.8
Commerce	4.8	–15.5	7.0	3.1	12.2	–1.9	–0.4	1.6	–0.3
Transport and communication	5.0	–4.9	8.2	7.8	9.1	3.8	2.2	4.3	1.0
Financial services	4.7	–0.3	3.0	3.6	5.5	4.6	4.4	4.1	3.9
Community services	2.8	–2.3	2.4	2.1	2.9	–0.3	1.3	2.1	0.9

1. Annual average growth rate.
2. Annual rates, based on data seasonally adjusted by the OECD.
3. As a percentage of GDP in the previous period.

Source: INEGI; Ministry of Finance; OECD.

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Annex Table I.C.2. **Indicators of investment activity**
Percentage changes from same period of previous year[1]

	1995	1996	1997	1998	1999	2000	2001	2002	Jan.-June 2003
Gross fixed capital formation total	–29.0	16.4	21.0	10.3	7.7	11.4	–5.8	–1.3	–1.6
Machinery and Equipment	–35.8	21.5	31.8	16.8	9.7	13.8	–6.7	–4.9	–5.3
Domestic Goods	–36.1	20.5	30.7	14.4	0.7	12.0	–7.0	–0.6	–5.1
Imported goods	–35.6	22.4	32.8	18.8	17.1	15.1	–6.5	–7.9	–5.4
Residential building and construction	–23.6	13.0	13.3	4.9	5.9	9.0	–4.9	2.4	2.0

1. Volume at 1993 constant price basis.
Source: INEGI.

Annex Table I.C.3. **Aggregate saving and investment**
Per cent of GDP

	1994	1995	1996	1997	1998	1999	2000	2001	2002
Gross capital formation, total[1]	21.7	19.8	23.1	25.9	24.3	23.5	23.8	20.9	20.3
Gross domestic saving	14.8	19.3	22.6	24.0	20.5	20.6	20.6	18.0	18.1
Private	9.5	14.1	18.8	21.0	18.1	17.9	17.4	15.1	15.3
Public[2]	5.3	5.2	3.8	3.0	2.4	2.6	3.2	2.9	2.8
Foreign saving	6.9	0.5	0.5	1.9	3.8	2.9	3.1	2.9	2.2

1. Including inventories.
2. The public sector comprises federal government and enterprises under budgetary control. The estimate is based on Ministry of Finance's accounts.
Source: Bank of Mexico; INEGI; Ministry of Finance; OECD.

remained limited, in particular in manufacturing, where they have been below productivity growth for the past year and a half (Annex Figure I.C.2).

Disinflation came to a halt in mid-2002, reflecting the downward rigidity in service prices, and the hike in electricity charges, as well as the weakening of the peso – although there has been less pass-through of exchange rate depreciation than in previous episodes. The 4.5 per cent end-year target for headline inflation was missed, although core inflation was below 4 per cent. Headline inflation, as measured by the CPI, remained on an upward trend till April 2003.* Since then, it has come down substantially, mainly reflecting the wage moderation, but also helped by the appreciation of the peso in the second quarter. Inflation expectations have also

* Regarding inflation developments in 2003: *i*) administered prices were set to rise by 2.7 per cent between December 2002 and December 2003 (although some impact from international prices on domestic gas was expected); *ii*) the minimum wage increase for 2003 was set at 4.5 per cent. Private sector inflation expectations for the year end were around 4.2-4.4 per cent at the start of the year and at the time, nominal wage agreements were about 1 percentage point higher than that. Both have come down since April, with inflation expectations in September and October down to 3.8 per cent, while expected increases of contractual wages were at 4.7 per cent.

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Annex Table I.C.4. **Labour market indicators**

Percentage changes from year earlier

	1997 Thousands	1998	1999	2000	2001	2002	2002 Q3	2002 Q4	2003 Q1	2003 Q2
Insured employment										
(private sector)	10 444.4	7.8	5.7	5.9	–0.5	–0.8	–0.3	0.5	0.7	–0.6
Permanent	9 578.4	4.9	3.5	5.0	–0.5	–1.2	–0.6	0.2	0.2	–0.6
of which:										
Manufacturing	3 423.9	6.5	4.3	5.2	–6.2	–6.6	–4.8	–2.8	–3.1	–5.2
Construction	284.9	10.7	3.0	6.4	2.4	2.0	2.1	1.5	4.5	5.2
Wholesale and retail trade	1 803.9	4.8	4.1	6.0	3.9	2.2	2.5	2.3	2.1	1.1
Transport, storage and communication	509.3	3.9	4.7	5.9	3.5	–0.3	–0.7	0.0	1.5	1.5
Finance, insurance	1 896.8	5.5	4.8	4.0	2.0	0.4	0.4	1.3	1.3	1.5
Personal services	1 044.9	2.0	0.7	7.9	4.4	4.0	3.4	3.2	3.3	3.1
Temporary	866.0	40.1	24.6	12.1	–0.6	1.6	1.5	2.2	4.4	–0.3
Employment in manufacturing										
Non-maquiladoras	. .	3.7	0.6	1.0	–4.4	–5.1	–4.3	–3.8	–3.0	–3.7
Maquiladoras	. .	12.2	12.7	12.9	–6.9	–9.6	–6.3	–0.7	1.8	–0.3

Source: IMSS; INEGI, *Encuesta industrial mensual*; *Estadística mensual de la industria maquiladora de exportación*.

been declining steadily. With real wages likely to decelerate further given the current labour market slack and weak activity, and despite the usual seasonal upturn in the later period of the year, it is expected that inflation will be in the +/– 1 per cent variability interval around the 3 per cent target at end 2003.

The external sector

After a slump in 2001, exports recovered in line with markets during 2002, but in early 2003 their recovery remained sluggish. This weakness was accompanied by lower imports of intermediate goods (80 per cent of the total), and especially of investment goods. The terms of trade improved, largely reflecting the surge in oil prices, both in the period preceding the Iraq war and since. In the context of economic weakness, the current account deficit shrank to a record low in the first half of 2003 (US$3.7 billion, half-yearly data, compared with the 2002 total of US$14 billion). Higher oil shipments reduced the trade deficit, while the surplus on services was boosted by transfers from migrants (largely the effect of better recording) and tourism receipts. On current trends, the current account deficit for the year as a whole is likely to be below 2 per cent of GDP (compared with 2.2 per cent of GDP in 2002). Capital inflows have continued to be dominated by FDI, which, although lower than in 2002, exceeded the current account deficit of the first semester (Annex Tables I.C.5 and I.C.6).

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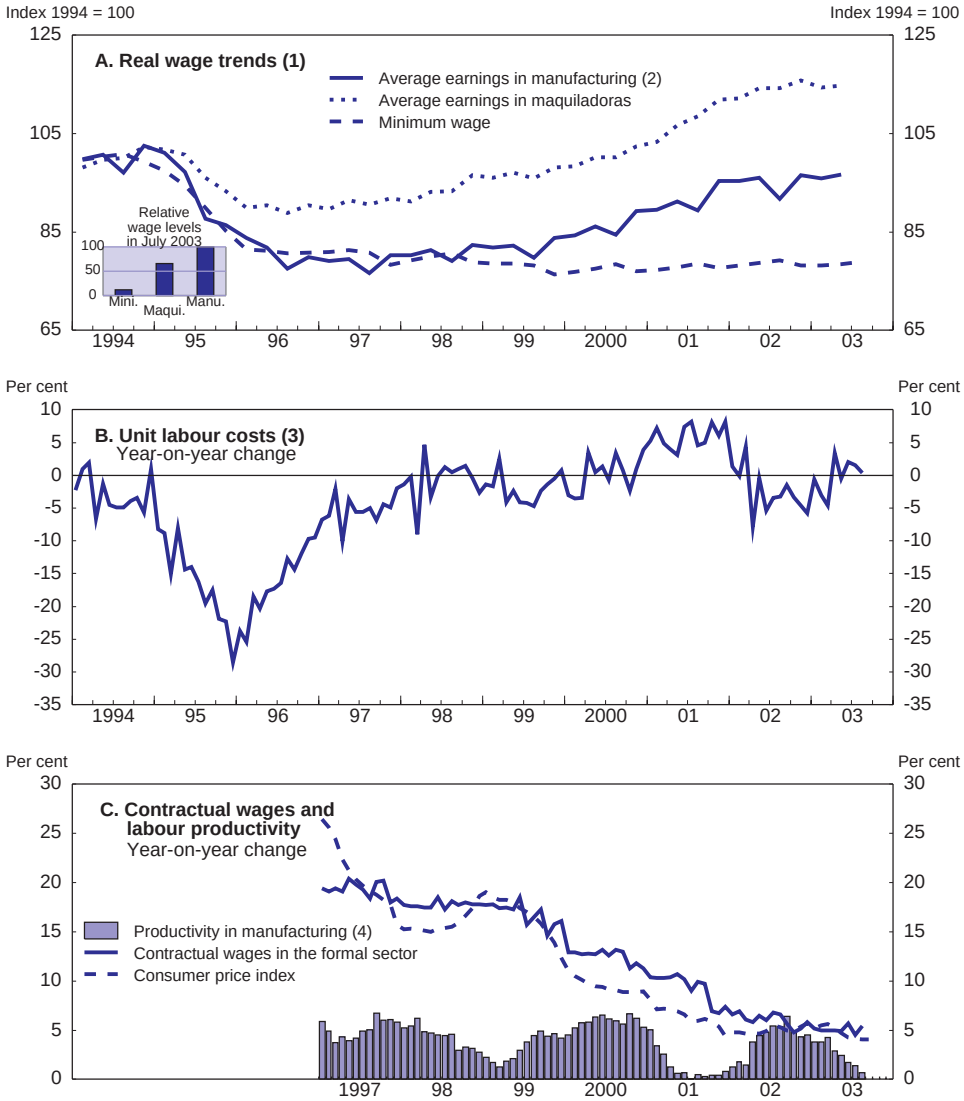
Annex Figure I.C.1. **Wages, labour productivity and unit labour costs**



1. Moving average over three quarters.
2. Manufacturing sector excluding "maquiladoras". Wages, salaries and social benefits based on INEGI monthly industrial survey.
3. Defined as the ratio of real wage to labour productivity, in the manufacturing sector.
4. Real output per persons employed. Index 1994 = 100, moving average over six months.

Source: INEGI; Bank of Mexico; Ministry of Labour and Social Welfare.

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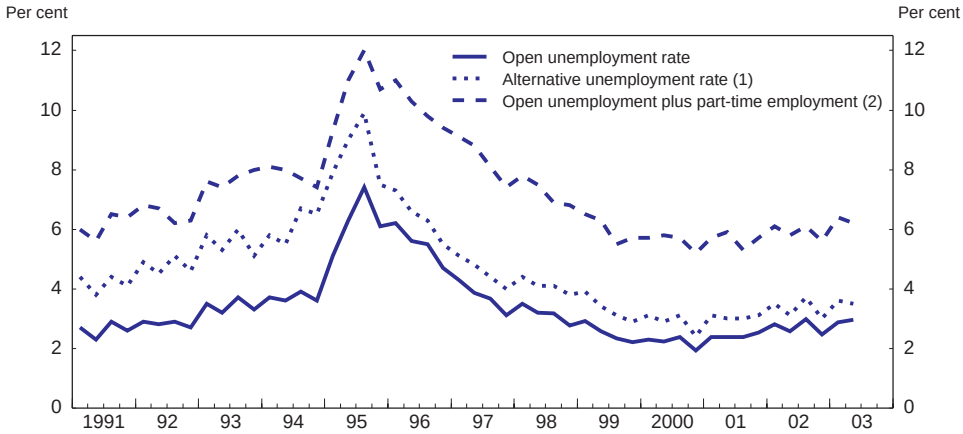
Annex Figure I.C.2. **Unemployment in urban areas**

As a percentage of labour force (12 years and over)



1. Includes those who stopped searching for a job but are still available for work.
2. Part-time refers to employees working less than 15 hours a week.

Source: INEGI.

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Annex Table I.C.5. **Current external account**

US$ billion

	1995	1996	1997	1998	1999	2000	2001	2002	2003 Q1	2003 Q2
Exports, f.o.b.[1]	79.5	96.0	110.4	117.5	136.4	166.5	158.4	160.8	39.2	40.7
Oil	8.4	11.7	11.3	7.1	9.9	16.4	12.8	14.5	4.9	4.3
Non-oil	71.1	84.3	99.1	110.3	126.5	150.1	145.6	145.6	34.3	36.5
Imports, f.o.b.	72.5	89.5	109.8	125.4	142.0	174.5	168.4	168.7	39.6	41.9
Trade balance										
excluding oil exports	–1.3	–5.1	–10.7	–15.0	–15.5	–24.4	–22.8	–22.4	–5.3	–5.5
(Per cent of GDP)	–0.5	–1.5	–2.7	–3.6	–3.2	–4.2	–3.6	–3.5	–3.5	–3.4
Trade balance	**7.1**	**6.5**	**0.6**	**–7.9**	**–5.6**	**–8.0**	**–10.0**	**–7.9**	**–0.4**	**–1.2**
(Per cent of GDP)	2.5	2.0	0.2	–1.9	–1.2	–1.4	–1.6	–1.2	–0.3	–0.8
of which: *Maquiladoras*	4.9	6.4	8.8	10.5	13.4	17.8	19.3	18.8	4.5	4.8
Non-factor services, net	0.7	0.4	–0.7	–0.9	–1.8	–2.3	–3.6	–4.0	–0.6	–1.2
of which: Tourism	3.4	3.6	3.7	3.6	3.6	4.0	3.8	3.9	1.4	1.0
Investment income, net	–13.3	–13.9	–12.8	–13.3	–12.9	–14.8	–13.9	–12.3	–4.1	–2.6
Transfers, net	4.0	4.5	5.2	6.0	6.3	7.0	9.3	10.3	2.9	3.5
Current balance	**–1.6**	**–2.5**	**–7.7**	**–16.1**	**–14.0**	**–18.2**	**–18.1**	**–14.0**	**–2.2**	**–1.5**
(Per cent of GDP)	–0.6	–0.8	–1.9	–3.8	–2.9	–3.1	–2.9	–2.2	–1.5	–0.9

1. Including trade by *Maquiladoras* (in-bond industries).

Source: Bank of Mexico.

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Annex Table I.C.6. **Capital account and the balance of payments**

US$ billion

	1995	1996	1997	1998	1999	2000	2001	2002	2003 Q1	2003 Q2
Capital account	**15.3**	**4.3**	**16.6**	**18.6**	**13.8**	**18.0**	**25.3**	**22.2**	**7.0**	**2.4**
Liabilities	**22.7**	**10.7**	**9.9**	**18.2**	**17.8**	**10.9**	**29.1**	**10.4**	**4.9**	**0.7**
Loans and deposits	22.9	–12.1	–7.9	4.9	–7.3	–4.1	–1.0	–3.5	–0.8	–1.2
Public sector	11.5	–8.9	–6.1	1.5	–4.8	–2.6	–2.6	–2.2	–1.3	–0.7
Development banks	1.0	–1.2	–1.0	0.2	–0.8	0.9	–1.3	–0.5	–0.5	–0.7
Non-financial sector	10.5	–7.7	–5.0	1.3	–4.0	–3.5	–1.3	–1.6	–0.8	0.1
Bank of Mexico	13.3	–3.5	–3.5	–1.1	–3.7	–4.3	0.0	0.0	0.0	0.0
Private sector	–1.9	0.4	1.6	4.5	1.2	2.8	1.6	–1.3	0.5	–0.5
Commercial bank	–5.0	–1.7	–2.0	–0.1	–1.5	–1.8	–2.7	–2.5	0.0	–0.6
Non-financial private sector	3.1	2.1	3.6	4.6	2.7	4.6	4.3	1.2	0.5	0.1
Total foreign investment	–0.2	22.8	17.9	13.3	25.1	14.9	30.1	13.9	5.7	1.9
Direct investment	9.5	9.2	12.8	12.2	13.1	16.1	26.2	14.6	2.6	2.6
Portfolio investment	–9.7	13.6	5.0	1.0	12.0	–1.1	3.9	–0.7	3.2	–0.8
of which:										
Stock market	0.5	2.8	3.2	–0.7	3.8	0.4	0.2	–0.1	–0.3	0.2
Money market	–13.9	0.9	0.6	0.1	–0.9	–0.0	0.9	0.1	0.9	–0.4
Foreign currency securities	3.6	9.9	1.2	1.6	9.2	–1.5	2.8	–0.8	2.5	–0.5
Assets	**–7.4**	**–6.3**	**6.7**	**0.4**	**–4.0**	**7.1**	**–3.8**	**11.8**	**2.1**	**1.7**
In banks abroad	–3.2	–6.1	4.9	0.2	–3.0	3.5	–1.5	10.8	2.1	2.0
Direct investment abroad	0.0	0.0	0.0	0.0	0.0	0.0	–4.4	–1.0	–0.5	–0.4
Credits to non-residents	–0.3	–0.6	–0.1	0.3	0.4	0.4	0.0	0.2	0.0	0.0
External debt guarantees	–0.7	0.5	–0.7	–0.8	–0.8	1.3	3.9	1.1	–0.2	–0.1
Other	–3.3	–0.2	2.7	0.7	–0.6	1.8	–1.8	0.6	0.7	0.1
Memorandum items:										
Current account	–1.6	–2.5	–7.7	–16.1	–14.0	–18.2	–18.1	–14.0	–2.2	–1.5
Capital account	15.3	4.3	16.6	18.6	13.8	18.0	25.3	22.2	7.0	2.4
Errors and omissions	–4.2	0.0	1.5	–0.4	0.8	3.0	0.1	–1.1	–0.8	0.5
Changes in net international reserves (increase = : –)	–9.6	–1.8	–10.5	–2.1	–0.6	–2.8	–7.3	–7.1	–4.0	–1.4

Source: Bank of Mexico.

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Annex I.D

Poverty alleviation

Overview

Poverty alleviation is one of the greatest policy challenges for Mexico. Since 1996, Mexico has made a solid progress in combating poverty, through more sustained income growth and macroeconomic stability, overcoming of the 1994/95 crisis.

According to the World Bank, the share of Mexico's population living in extreme poverty (defined as earning less than US$1 per day) declined from 16.2 per cent in 1989 to 13.2 per cent in 2000. Malnutrition among children under age 5 also decreased from 14.4 per cent in 1988 to 8 per cent in 2000; the under five-years-old mortality rate decreased from 46 per 1 000 in 1990 to 36 in 2000. The reduction in poverty can be attributed to several factors: net job creation in the formal sector, the slight increase in real wages, the increase in migrant remittances, and the expansion of social programmes in the recent years.

Despite this progress, the poverty levels in Mexico are still high. Over 45 million Mexicans live on less than US$2 a day, while 10 million of them "survive" under extreme poverty, having limited access to basic services.[1] In 2000, 27.8 per cent of the population over 15 had not completed primary education, almost 15 per cent lived in homes with dirt floors, and about 12 per cent did not have access to water supply or drainage.[2] Extreme poverty affects mostly the south-eastern part of Mexico (Annex Table I.D.1). An international comparison is presented in Annex Table I.D.2 (below).

Measure of poverty

Poverty is a multidimensional phenomena which is difficult to quantify. Variations in the measurement of poverty make it hard to understand the magnitude of the problem and to evaluate the effectiveness of government policies. As a consequence, in 2002 the Mexican government appointed the Technical Committee for the Measurement of Poverty (*Comité Técnico para la Medición de la Pobreza*) to develop a methodology for poverty measurement that will become the benchmark for government policies. According to the new methodology, there are three definitions of poverty:

- First line: households with an income insufficient to cover basic food necessities, according to the nutritional requirements established by the INEGI-CEPAL food basket (*Línea de Pobreza Alimentaria*).
- Second line: households with an income per capita insufficient to satisfy basic food, health and educational consumption (*Línea de Pobreza Capacidades*).
- Third line: household income is not sufficient to satisfy the basic consumption of food, clothing, shoe, housing, health, public transport, education, and other goods (*Línea de Pobreza Patrimonio*).[3]

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Annex Table I.D.1. **Socioeconomic Indicators by Mexican States, 2000**

Per cent

	Total population	Illiterate population over the age of 15	Population over 15 without primary education	Population living in housing without drainage	Population living in housing without electricity	Population living in housing without water supply	Population living in housing with dirt floor	Housing units where more than 3 persons share a room	Population in communities with less than 5 000 inhabitants	Active population earning up to 2 minimum wages	Marginalisation index	Degree of marginalisation
National	**97 483 412**	**9.46**	**28.46**	**9.90**	**4.79**	**11.23**	**14.79**	**45.94**	**30.97**	**50.99**		
Aguascalientes	944 285	4.84	23.03	3.38	1.78	1.30	3.57	37.82	24.54	42.23	–0.973	Low
Baja California	2 487 367	3.53	19.59	1.95	2.33	6.83	4.59	36.58	11.62	22.22	–1.268	Very low
Baja California Sur	424 041	4.21	20.98	3.71	4.62	6.32	10.42	38.8	25.41	35.82	–0.802	Low
Campeche	690 689	11.81	34.22	17.27	8.79	14.61	14.92	56.63	34.51	64.12	0.702	High
Coahuila de Zaragoza	2 298 070	3.87	18.79	3.42	1.42	2.18	4.55	37.74	13.37	34.68	–1.202	Very low
Colima	542 627	7.16	27.2	2.56	1.96	2.18	12.53	40.90	18.20	48.00	–0.687	Low
Chiapas	3 920 892	22.94	50.31	19.33	12.01	24.99	40.90	65.03	61.21	75.89	2.251	Very high
Chihuahua	3 052 907	4.79	23.3	5.30	6.27	5.88	6.96	36.53	19.64	37.67	–0.780	Low
Distrito Federal	8 605 239	2.91	12.16	0.44	0.17	1.47	1.34	34.82	0.32	42.43	–1.529	Very low
Durango	1 448 661	5.41	28.75	13.67	6.57	7.00	13.73	40.30	42.12	50.12	–0.114	Medium
Guanajuato	4 663 032	11.99	35.75	16.10	3.19	6.86	10.93	47.10	37.39	47.29	0.080	High
Guerrero	3 079 649	21.57	41.92	35.29	11.04	29.54	39.97	59.67	53.44	66.16	2.118	Very high
Hidalgo	2 235 591	14.92	34.09	17.19	7.66	15.25	19.02	49.69	58.52	65.27	0.877	Very high
Jalisco	6 322 002	6.45	26.71	4.93	2.14	6.78	7.36	38.46	19.40	40.93	–0.762	Low
México	13 096 686	6.40	20.84	8.14	1.8	6.23	7.19	47.65	19.38	49.41	–0.605	Low
Michoacán de Ocampo	3 985 667	13.90	40.19	11.40	4.41	10.87	19.90	46.04	43.09	57.29	0.449	High
Morelos	1 555 296	9.25	25.76	7.17	1.40	7.30	14.80	44.26	23.93	54.28	–0.356	Medium
Nayarit	920 185	9.05	31.97	9.52	4.75	9.53	13.25	44.14	43.68	56.25	0.058	High
Nuevo León	3 834 141	3.32	16.49	1.59	1.04	3.62	3.30	36.97	7.57	28.93	–1.393	Very low
Oaxaca	3 438 765	21.49	45.53	18.07	12.54	26.95	41.60	59.45	64.01	71.93	2.079	Very high
Puebla	5 076 686	14.61	35.2	11.89	4.75	16.26	24.09	54.73	41.49	63.90	0.720	High
Querétaro de Arteaga	1 404 306	9.80	26.14	16.37	5.76	6.58	10.06	43.74	42.14	41.72	–0.107	Medium
Quintana Roo	874 963	7.52	25.18	9.23	4.36	5.34	11.37	53.01	21.19	40.37	–0.359	Medium
San Luis Potosí	2 299 360	11.29	34.10	11.43	11.54	20.92	23.70	43.85	44.64	58.82	0.721	High
Sinaloa	2 536 844	7.96	30.06	10.62	3.35	7.22	14.53	47.52	39.17	48.63	–0.100	Medium
Sonora	2 216 969	4.40	22.40	4.19	3.23	3.47	13.18	42.18	21.25	40.95	–0.756	Low
Tabasco	1 891 829	9.73	32.27	8.58	5.85	26.49	13.47	54.52	56.10	62.29	0.655	High
Tamaulipas	2 753 222	5.13	23.35	2.65	4.97	5.01	8.95	42.36	16.89	46.72	–0.691	Low
Tlaxcala	962 646	7.80	23.42	8.43	2.05	2.48	8.98	54.61	36.88	63.38	–0.185	Medium
Veracruz Llave	6 908 975	14.87	39.17	10.21	11.11	29.47	29.29	51.50	48.50	68.64	1.278	Very high

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Annex Table I.D.1. **Socioeconomic Indicators by Mexican States, 2000** (*cont.*)

Per cent

	Total population	Illiterate population over the age of 15	Population over 15 without primary education	Population living in housing without drainage	Population living in housing without electricity	Population living in housing without water supply	Population living in housing with dirt floor	Housing units where more than 3 persons share a room	Population in communities with less than 5 000 inhabitants	Active population earning up to 2 minimum wages	Marginalisation index	Degree of marginalisation
Yucatán	1 658 210	12.30	36.94	24.01	4.12	5.69	5.62	52.52	28.82	67.57	0.381	High
Zacatecas	1 353 610	7.97	37.50	19.68	4.03	11.05	9.12	42.68	55.13	58.91	0.298	High

Note: Estimates, based on INEGI's 2000 Census.
Source: Consejo Nacional de Población (CONAPO).

Annex Table I.D.2. **Poverty indicators, 2000**[1]

	Mexico	Brazil	Turkey	Korea	Poland	Canada	United States
Population living on $1 a day	8	9.9	2	2	2	. .	
Population below minimum level of dietary energy consumption	5	10	. .	. .	. .	. .	. .
Under-5 mortality rate (per 1 000 births)	30	38	43	6	11	7.2	8.7
Access to improved water source	88	87	82	93	. .	100	100
Access to improved sanitation	74	76	90	63	. .	100	100
Adult literacy rate[2]	91.2	87	85	98	99.7	. .	. .
Life expectancy at birth	73	68	70	73	73	79	77

1. Percent of total population.
2. Population above 15 years of age.
Source: World Development Indicators database, April 2002

In 2002, one of every five Mexicans was living below the first line, one out of every four was below the second, and half of the population did not have enough earnings to cover basic consumption as defined in the third line.[4] Under the new methodology, the data indicate a reduction in poverty between 2000 and 2002. However, the reduction in poverty has yet to be confirmed, since some critics have pointed out that the survey used to determine poverty levels in 2002 was not strictly comparable to the one used in the previous years.

Government policies

A wide range of public programmes have been implemented over the years to reduce poverty and improve the well being of the population.[5] Spending on poverty alleviation programmes has increased from 0.7 per cent of GDP to 1.3 per cent in 2002. Spending in 2002 amounted to 77.6 billion pesos (over US$7 billion) of which: 18.4 billion pesos for PROGRESA/O*portunidades*, 3.9 billion pesos for the Temporary Employment Programme, 8.6 billion pesos for the education effort, 5.7 billion pesos for health component and 5.3 billion pesos for nutritional programmes.[6] Through the years, there has been a greater decentralization of programmes, giving states and municipalities greater control over the allocation of resources. There has also been a gradual transition away from general subsidies towards targeted programmes.[7]

An integral approach to poverty alleviation: PROGRESA/O*portunidades*

PROGRESA/O*portunidades* is one of the major programmes of the Mexican government to fight poverty. It began in 1997 under the name PROGRESA (The Education, Health and Nutrition Programme), aiming at developing human capital of poor households by increasing the demand of health and education services.[8] Through monthly monetary transfers it alleviates current poverty, and at the same time it reduces levels of future poverty by building human capital and investing in education and nutrition. In March 2002 PROGRESA was renamed O*portunidades*, its basic structure remaining broadly the same. Cash subsidies are given periodically to eligible women and children in return for increased use of education and health services. The immediate income support represents up to 22 per cent of the beneficiaries' average total income.

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The coverage of the programme was extended in 2002 to a million additional families, including families in semi-urban and urban areas living in extreme poverty.[9] *Oportunidades* also began providing scholarships to upper secondary education. Expenditures of the programme increased from 12.7 billion pesos in 2001 to 16.6 billion pesos in 2002, reaching 4.24 million families (about 21 million Mexicans, three-quarters of them living in rural areas).[10] The programme is organised along three main lines:

- *Nutrition*: Direct payments are granted to families to promote food consumption. In addition, recipients of the payments are invited to attend seminars and courses on nutrition and health. In 2002 about 2.3 million training seminars on nutrition, health and hygiene were taught.
- *Health*: A basic free health service package, including nutritional supplements, is provided to pregnant and nursing women and children under two years of age. In addition, incentives are provided to increase doctor visits. The number of medical appointments provided during 2002 reached more than 27 million.
- *Education*: In 2002 the programme gave more than 4.5 million scholarships (*becas*, which are small cash transfers) to the children of the targeted families. These scholarships are given to ensure school attendance and reduce the incentives for children to work.

There have been some evaluations made on the effectiveness of the programme. For instance, evaluations by National Institute of Public Health conclude that *Oportunidades* has had a positive impact on the welfare of families covered by the programme in terms of income, school attendance, nutritional status and health services. In particular:

- *Malnutrition* of children has decreased by 16 per cent and food consumption has increased by 10.5 per cent.[11]
- Since the programme started in 1997, *illnesses* of children under five have been reduced by 12 per cent and attendance at health clinics has increased by 57 per cent in rural areas that had access to the programme.
- Evaluations by the Ministry of Social Development, SEDESOL, suggest that since the *becas* were introduced, the proportion of children that complete *primary school* increased by 14 per cent and there has been an increase of 23 per cent in the number of students that finished lower secondary school in rural areas. During 2002, when the programme was extended for the first time to students in upper secondary levels of education, the number of students enrolled in this level increased sharply.

Education programmes for the poor

Education has become one of the main instruments in Mexico to fight poverty. In the past 30 years the average number of years in school has more than doubled, from 3.7 years for men (3.1 for women) to 7.8 (7.3). Despite this progress, large disparities remain between socio-economic groups and regions; there exists particularly low educational attainment among the most disadvantaged groups, which include the indigenous population[12] and rural population in marginal zones.

Some government policies to improve access of the poor to education include the Compensatory programmes (*Programas Compensatorios*) directed to children in isolated and marginalised areas[13] and the Communitarian School programme (*Escuelas Comunitarias*) which consists in classes taught by students with a secondary education degree to children in remote areas. Both programmes are administered by CONAFE (*Consejo Nacional del Fomento Educativo*).

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The national programme of *Telesecundaria* (television for lower secondary education) is mainly directed to children and teachers living in small and remote areas. It has also been successful in giving access to education for the poor living in remote areas. This programme began as early as 1968 to meet increasing demand of education in rural areas, where poverty (including acute poverty) is most common.

Another vulnerable group identified by the government as requiring special attention is the children of workers in the agricultural sector who migrate to different states according to the farming season (25 per cent of these children between the age of 4 and 15 has never attended school). For them the government has created specialized programmes like the *Programa Educación Primaria para Niñas y Niños Migrantes* (started in 1997) to provide primary education options that can be adapted to the socio-economic, geographical and cultural conditions of these children.

More details on some of the initiatives above and other education programmes are provided in Annex III.A.

Health programmes

The structure of the health care system in Mexico has left a great proportion of the population, around 55 per cent of the Mexican households, without access to social security.[14] In an attempt to provide some sort of health services to the poorest, the government has implemented the following programmes:

- **IMSS-*Oportunidades*** was created in 1983 as IMSS-Coplamar to improve the well being of rural population in marginalised zones with no access to social security services. The Social Security Institute for the private sector (IMSS) is in charge of it. The medical services are provided at three different levels according to the seriousness of the injury or disease. The services of the first level are general and are provided to anyone, while the second and third levels are more specialized and are located in fewer areas. In 2002, with a budget of 4 billion pesos, it gave medical assistance to small communities in 17 states,[15] helping more than 10 million Mexicans (about 2 million families, half of which are covered by *Oportunidades*) to whom it provided more than 16 million medical and dental consultations.
- **The coverage extension programme** (*Programas de Ampliacion de Cobertura*-PAC) started in 1996, organised by the Ministry of Health (SSA), to extend the access of the rural and indigenous population to medical services. It provides a basic package of free health services and access to surgery and rehabilitation. The services offered in this package include family planning services, prenatal and natal care, infant growth supervision, vaccinations, and anti-parasite treatments. The programme also helps in the maintenance of the existing health centers in poor areas by increasing the supply of doctors, and providing equipment and medication. In addition, doctors and nurses are transported in mobile units to supply health care services to remote areas of the country. This programme is financed by the federal government with contributions from the state governments. Expenditures of PAC in 2002 were 1.2 billion pesos, helping 9 million Mexicans in 19 different states.
- **The popular health insurance programme** (*Seguro Popular de Salud "Salud Para Todos"*) was launched in 2002. Although IMSS-*Oportunidades* and PAC have had a significant positive impact in the population welfare, the SSA reported that the number of Mexicans still not benefiting from health care remained high, and that every year between 2 and 3 million poor families used more than a third of their income paying for health services. To address this problem, the government introduced a programme with the

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objective of providing a "popular" health insurance for the population without coverage. It started at the beginning of 2002 with a budget of 200 million pesos, offering a package of services and medication to those with no access to social security. In its first year of operation it registered 296 000 families of which 60 per cent are headed by single women.

– **Universal coverage programme** is the most recent initiative. To complement the Popular Health Insurance programme, a decree reforming the General Health Law regarding Social Protection (*Ley General de Salud en Materia de Protección Social*) was approved in May 2003 to extend progressively health insurance to the currently non-covered population, with universal coverage targeted by 2010. The purpose of this "Universal Coverage" is to extend social protection by then to around 45 million Mexicans, mainly in the informal sector, who until now have had very limited access to health services. This programme intends to reduce health service costs for the poor, promoting rapid attention to illnesses and reducing the gap between the families covered by social security and those who are not. The resources will be provided by the Federal and State budget, with a contribution by the family, based on its resources (co-payment). It will require a payment of 0 to 35 monthly pesos according to the socioeconomic level of each family.

Nutritional programmes

Three main programmes provide food support to poor families. They benefit mainly the urban poor, except for DICONSA.

– **LICONSA** (*Leche Industrializada Conasupo*) started in 1965 and is the oldest existing Mexican food assistance programme. It provides milk and powdered milk at reduced prices to the poor population, in particular children, pregnant women, and individuals over 60 years of age. The distribution mechanism now is through cards that allow beneficiaries to buy (at roughly half price) a certain amount of milk in authorised stores. LICONSA simultaneously supports small milk producers by buying milk surpluses during certain months at prices higher than the market price. In 2002, LICONSA provided milk daily to almost 2.9 million families.

– **DICONSA** (*Sistema de Distribuidoras Conasupo*) was created in 1972 and distributes basic commodities, including tortilla products at affordable prices to people living in small communities in marginal areas.[16] It has 23 200 stores that benefit more than 29 million Mexicans.

– **DIF** (National System for Integral Family Development) began in 1972, and operates several nutritional sub-programmes, including food support for families with small children, and community kitchens. The largest sub-programme is its school breakfast programme with a budget of 1.8 billion pesos in 2002 which assisted approximately 3.8 million children a day. This sub-programme has been effective in increasing schooling among children between the ages of 8 and 14.[17]

There are several other nutritional programmes including the Tortilla Programme which helps more than 600 000 families get a kilogram of tortillas a day.

Income generating programmes

The main income generating programme is the PET (Temporary Employment Programme). It was first implemented as an emergency programme during the crisis in 1995. Its purpose is to provide jobs in poor rural areas during the low agricultural season, when job opportunities are scarce. This programme offers jobs (with a very low wage), mainly to

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improve basic infrastructure, roads, highways, irrigation and reforesting projects. In 2000, over 1 million temporary jobs were created. Since then, the number has been decreasing to about 900 000 in 2002.[18] The programme, which was created as a federal programme, is now administrated by States.

Apart from PET, the Federal government operates several smaller programmes to help the financing of productive projects in poor areas. Since 2001, there have been 500 000 beneficiaries of micro-credit in Mexico with an average loan of 4 000 pesos that is used towards food stands and small businesses. In 2002, 115 000 micro-credit loans were approved and 1.5 billion pesos supported small businesses through several government programmes.[19]

There are other programmes focusing on providing various forms of employment, credit, and infrastructure for poor communities, including some for the agricultural sector (See Annex I.B). Besides federal programmes, there are also specific programmes administered by the states.

Concluding remarks

Over the last years, the resources channelled to combat extreme poverty in Mexico have increased regularly. Transparency has improved; the operation rules for all these programmes are now published in a Federal Register of Rules. Moreover, anti-poverty programmes in place have been shown to have a positive impact. In particular, as noted above, PROGRESA/*Oportunidades* has had a net positive impact on the welfare of the families covered by it, in terms of improved children's school attendance rates, higher spending on food and children's clothes, better nutritional status of under-five year-olds and increased use of health services.

Despite this progress, much remains to be done and Mexico still faces the challenge of further widening coverage of basic services and improving their quality. For instance, although the situation in the health sector has improved, but there is still a large number of Mexicans that lack access to proper health care. The demand for health services has increased, thanks to programmes such as *Oportunidades*, but the supply has not been keeping up. Because of the lack of medicine and doctors in some areas the quality of some health services has deteriorated. Regarding education, the programmes implemented by the government have been rather successful in increasing enrolment rates for the poor, especially in primary education and reducing failing and repetition rates. But, the quality of education service has not increased at the same pace as the coverage (see Chapter III). Moreover social programmes have not been able to reach many of the small isolated communities, as a result of the high costs involved and the lack of infrastructure. For instance, a programme like *Oportunidades* requires the existence of a health center and a school as a condition of operation, so that some communities might be excluded. Evaluations by UNESCO question the effectiveness of Mexican programmes in helping the most marginalized groups.[20] Finally, during the last decade, policies have been focused towards poverty in rural areas, where extreme poverty is found, but there is a large proportion of population in poverty in urban areas, and efforts will also need to be directed to the urban poor.

Moreover, three important concerns should be taken into consideration in defining the poverty alleviation strategy.

– A better coordination of the social programmes is needed to make sure that some targeted families do not benefit from overlapping programmes, while others are totally excluded from all of them. For instance now that PROGRESA/*oportunidades* targets semi urban and urban population, there might be some overlapping with LICONSA and the

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Tortilla programme and some rationalisation is needed, and discontinuation of LICONSA and Tortilla programmes might even be considered.

– It is important to ensure, via appropriate targeting, that poverty reduction programmes do not create disincentives to work.

– It would also be helpful to undertake regular evaluation of the programmes. When the methodology of poverty measurement undertaken by the newly created "Technical Committee" is fully developed, it should be used to conduct an overall study on the achievements of the poverty reduction strategy conducted in recent years, as well as cost-benefit analysis of some of its components.

Notes

1. World Bank. Country Brief: Mexico (2002). *http://lnweb18.worldbank.org/external/lac/lac.nsf/*
2. SEDESOL 2000, CONAPO and *Censo General de Población* 2000.
3. For the first definition, the poverty line is set at daily incomes lower than 15.4 pesos in rural areas, 20.9 in urban areas based on 2000 data – equivalent to US$1.6 and to US$2.2 respectively. For the second definition it is set at daily earnings lower than 18.9 pesos in rural areas, 24.7 in urban areas – equivalent to US$2 and US$2.6 respectively. And for the third definition it is set at daily earnings lower than 28.1 pesos in rural areas and 41.8 in urban areas – equivalent to US$3 and US$4 respectively.
4. *Plan Nacional de Desarrollo, Informe de Ejecución* 2002. *http://pnd.presidencia.gob.mx*
5. Besides the programmes presented here, there are some specific programmes targeting the indigenous population, which are not described in this annex.
6. The education effort includes the Compensatory programmes and community schools of CONAFE, Indigeneous education programmes, Schools of Quality programme (PEC), and Telesecundaria. Nutritional programmes include DIF (*Sistema Nacional de Desarrollo Integral de la Familia*) breakfasts and Tortilla and Milk programmes. Data on health spending come from *Gasto y principales programas de la superacion de la pobreza, 2ndo Informe Presidencial* 2002.
7. In 1994, targeted programmes received only 39 per cent of overall spending, whereas by the year 2000, 95.5 per cent of all spending was on targeted programmes. Sources: *Subsecretaria de Egresos* 2000. *El propuesto de Egresos de la Federación* 1995 2000, September, and *Poder Ejecutivo Federal* 2000, *Sexto Informe de Gobierno*, September.
8. In 2002, *Oportunidades* provided average transfers of 155 pesos/month (about US$15/month) for food support and scholarships of 315 pesos/month (about US$30/month) on average for children in the first three levels of education. The transfers are currently capped at an amount of 945/month (US$90/month) per family. The programme targets directly the population in extreme poverty living in rural areas. The targeting system is described in the OECD *Economic Survey on Mexico* 2000 Annex II. See also *www.progresa.gob.mx/*
9. The programme had previously been offered only to rural families. Areas are considered semi-urban when there are 2 500 15 000 inhabitants and urban when there are more than 15 000.

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10. Plan Nacional de Desarrollo. Informe de Ejecución 2002. March 2003. *http://pnd.presidencia.gob.mx*
11. See *www.ifpri.org*
12. 50 per cent of the indigenous population over 15 has not completed primary school.
13. This programme benefits mainly the population concentrated in the same areas as the beneficiaries of *Oportunidades*.
14. Ministry of Health, *Encuesta Nacional de Evaluación de Desempeño* 2003. For additional analysis of the Mexican health care sector, see OECD, *Economic Survey on Mexico* (1998).
15. 97 per cent of the communities helped by IMSS-*Oportunidades* have less than 2 500 inhabitants and are zones of difficult access. Source: *Censo de Universo de Trabajo*, INEGI, 2001.
16. The size of the communities it targets is between 500 and 4 000 inhabitants.
17. World Bank. *Mexico: A Comprehensive Development Agenda*, 2001.
18. The number of jobs created by the programme peaked in 1999 and has been decreasing since.
19. FONAES (*Fondo Nacional de Apoyo para las empresas de Solidaridad*), PRONAFIM (*Programa Nacional de Financiamineto a MicroEmpresarios*), FOMMUR (Fondo *de Microfinanciamiento a Mujeres Rurales*) and the *Mujer Campesina Programme*. Source: Segundo *Informe de Gobierno* 2002.
20. Nurit Martinez, *Critican la efectividad the estrategia antipobreza*, *El Universal* 23/12/03.

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Annex II
Background information on fiscal policy

Annex Table II.1. **Federal government budget**
Per cent of GDP

	1998	1999	2000	2001	2002			2003
	Outturn	Outturn	Outturn	Budget[1]	Outturn[2]	Budget[1]	Outturn[2]	Budget[1]
Revenue	**14.2**	**14.7**	**15.8**	**15.5**	**16.1**	**16.6**	**16.1**	**16.2**
Contribution of PEMEX	2.3	2.1	4.0	3.2	3.4	2.3	2.4	2.9
Other non-tax revenue	1.3	1.2	1.2	1.5	1.5	1.3	1.8	1.2
Tax revenue	10.5	11.3	10.6	10.9	11.2	13.0	11.8	12.0
of which:								
Income tax	4.4	4.7	4.7	4.6	4.9	6.0	5.2	5.5
VAT	3.1	3.3	3.5	3.4	3.6	3.6	3.6	3.4
Excise taxes	2.0	2.3	1.5	2.0	1.9	2.9	2.2	2.3
Import duties	0.5	0.6	0.6	0.6	0.5	0.5	0.4	0.4
Expenditure	**15.7**	**16.6**	**17.3**	**16.9**	**17.2**	**18.0**	**18.5**	**17.5**
"Programmable"	10.2	10.2	10.7	10.5	10.8	11.4	12.2	11.5
Current expenditure	8.7	8.9	9.2	9.2	9.3	10.3	10.3	10.3
Capital expenditure	1.7	1.6	1.6	1.6	1.7	1.4	2.1	1.4
Delayed payments	–0.2	–0.2	–0.1	–0.3	–0.2	–0.3	–0.2	–0.3
Non-programmable	5.6	6.4	6.6	6.4	6.3	6.6	6.3	6.0
Revenue sharing	3.0	3.1	3.2	3.2	3.4	3.5	3.5	3.1
Interest payments[3]	2.4	3.1	3.2	3.0	2.9	2.9	2.6	2.4
Other[4]	0.2	0.2	0.2	0.2	0.1	0.2	0.2	0.1
Financial balance	**–1.6**	**–1.9**	**–1.5**	**–1.4**	**–1.1**	**–1.4**	**–2.4**	**–1.3**

1. The projections for 2001, 2002 and 2003 are the approved budgets.
2. Preliminary data.
3. Includes provisions made on account of the support programmes for banks and debtors.
4. Includes accounts payable from previous fiscal period and net expenditure of the Federal Government on behalf of state-owned enterprises.

Source: Ministry of Finance.

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Annex Table II.2. **Public enterprises under budgetary control: financial accounts**[1]

Per cent of GDP

	1998	1999	2000	2001		2002		2003
	Outturn	Outturn	Outturn	Budget[2]	Outturn[3]	Budget[2]	Outturn[3]	Budget[2]
Revenue	**7.9**	**7.9**	**7.5**	**7.4**	**7.5**	**7.9**	**6.3**	**8.2**
PEMEX	2.1	2.2	2.0	2.0	1.8	2.3	1.8	2.3
Other public enterprises	4.1	3.9	3.9	3.7	3.9	3.8	3.0	3.9
Transfers received	1.8	1.7	1.6	1.7	1.8	1.8	1.4	2.0
Expenditure	**7.4**	**7.2**	**7.1**	**6.8**	**7.2**	**7.2**	**5.5**	**7.4**
Current expenditure	5.6	5.5	5.5	5.3	5.8	5.9	4.6	6.2
Capital expenditure	1.5	1.3	1.0	1.1	1.0	0.8	0.7	0.8
Interest payments	0.4	0.4	0.4	0.4	0.4	0.5	0.2	0.4

1. State-owned enterprises under budgetary control, excluding public financial intermediaries (development banks and development trust funds).
2. The projections for 2001, 2002 and 2003 are the approved budgets.
3. Preliminary data.

Source: Ministry of Finance.

Annex Table II.3. **Public expenditure by sector**

	1998	1999	2000	2001	2002
Total,[1] peso billion	600.6	711.2	855.3	937.2	1 078.9
(Percentage changes, in real terms)	– 1.5	3.1	9.1	3.1	10.1
	Per cent of total[2]				
Social development	57.9	60.9	60.7	61.9	59.6
Education	24.7	24.7	24.2	25.1	24.1
Health	14.7	15.5	14.3	14.5	12.3
Social security	10.5	13.4	14.4	13.8	14.8
Labour	0.3	0.3	0.3	0.3	0.2
Social assistance and supply	1.6	1.4	1.3	1.4	1.4
Urban and regionaldevelopment	6.0	5.7	6.2	7.0	6.8
Rural development	5.0	3.7	3.5	3.7	7.9
Environment and fishing	1.0	1.3	1.2	1.1	1.1
Communications and transport	4.0	3.4	2.8	2.1	2.8
Energy	21.5	19.4	19.0	19.2	17.9
Justice and security	5.2	5.4	5.7	5.7	5.0
Administration	5.4	5.8	7.1	6.2	5.3

1. Public expenditure, excluding interest payments and revenue-sharing with state and local governments (*i.e.* "programmable" expenditure).
2. Percentages may not add up because of rounding.

Source: Ministry of Finance.

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Annex Box II.1. Main changes to the tax regime 2002-2003

Corporate income tax

- Tax rate lowered by 1 percentage point a year, starting in 2003 (from 35 per cent in 2002 to 32 per cent in 2005).
- There is no longer a special tax rate for retained earnings.
- As from 2003, immediate deduction of investment carried out outside the three largest metropolitan areas (Mexico City, Monterrey and Guadalajara).[1]
- Starting in 2002, contributions to social security (IMSS), made by firms on behalf of their employees become deductible for income tax purpose (not only those related to workers paid at the minimum wage, as applied in the past).
- Fringe benefits (social security benefits, childcare…) are deductible under certain conditions.[2]

Individual income tax

- The top marginal tax rate reduced from 40 to 35 per cent in 2002, then by a further 1 percentage point in 2003, the reduction continuing in the two following years (to 32 per cent by 2005).
- Starting in 2003, persons are to record income from all sources in their tax return, including real interest – with a view to facilitate tax control; and all income becomes subject to the same progressive tax schedule.
- Starting in 2003, the returns on public bonds are taxed, setting a level-playing field with corporate bonds.
- The scope of tax expenditures is broadened: Starting in 2002, medical insurance premia are deductible from the tax base. Starting in 2003, real interest payments on mortgages are deductible as well (with a maximum value), and voluntary and complementary contributions to retirement savings account or retirement plans (with a limit on the deduction).
- Fringe benefits received by workers (social security benefits, childcare…) are totally or partially exempt from income tax if the employer satisfies certain eligibility requirements (they are fully exempt under certain limitations, except for certain categories of benefits, where no limitations apply).[3]
- Starting in 2003, an individual can set up a trust fund for educational purposes and its income will not be taxable.
- Starting in 2003, the withholding tax rate on interest income becomes 0.5 per cent of the principle amount of accounts in the financial system (it is a provisional payment that can be credited against the annual income tax, and replaces the previous withholding tax which was considered as a final payment).

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Annex Box II.1. **Main changes to the tax regime 2002-2003** (*cont.*)

Special taxes created in 2002 and changed in 2003

- Special tax on telecommunications, on mobile phone services in particular, at a 10 per cent rate (calls with prepaid cards under a threshold for the cost of the call are exempted). Local fixed line calls and long distance calls are not included.
- Special tax of 20 per cent on soft drinks – levied only on those that use anything but sugar as sweetener.
- Sales tax of 5 per cent on luxury goods and services – eliminated in 2003.
- Starting in 2002, payment of a 3 per cent tax on payrolls ("substitute tax") or transferring to enterprises the payment of the "salary credit" that the government was granting wage earners – eliminated in 2003, following a Supreme Court decision.

Changes to special regimes on income tax

- The simplified regime for certain sectors (agriculture, livestock, forestry, fishery and land transportation) is based on cash flow, with a view to reduce tax avoidance.
- The special treatment for publishing (reduction in income tax paid) will be phased out by 2006.
- Changes to the regime for small taxpayers (individuals carrying out commercial activities): the threshold is a yearly gross income of 1.75 million pesos. Rates vary from 0.5 per cent to 2.0 per cent according to gross income levels. Earnings up to 4 minimum wages are exempt from the taxable base. When they have signed an agreement with the Federal government, States can collect the full revenue raised by this tax.

Value added tax

- Starting in 2002, VAT is charged on a cash flow basis (at the time of disbursement), rather than an accrued basis.
- No other significant changes to the VAT regime (multiple rates, wide coverage of zero rating and exemptions remain in place).

Fiscal federalism

- States allowed to charge up to an additional 3 per cent rate as a final sales tax, an option which was not used and was suppressed by Congress in 2003.
- Starting in 2003, taxpayers in the newly-created intermediate regime and those deriving income from the sale of immovable property shall pay to the States 5 per cent of the Federal income tax. This is not an additional burden, since it can be credited against the provisional payment of the Federal income tax of the fiscal year. Taxpayers of the small taxpayer regime shall also make such payments.

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Annex Box II.1. **Main changes to the tax regime 2002-2003** (*cont.*)

- Starting in 2003, States can impose up to an additional 2 per cent rate to individuals under the small taxpayer regime, the proceeds would not be shared with the federal government. By mid-2003, no State had used the option.
- Starting in 2003, States can impose up to an additional 5 per cent rate on net business income of individuals in the newly-created intermediate regime, and to income from the sale of real estate assets.

1. The deduction will also be applicable in three major cities in the case of enterprises which are labour intensive, use non-polluting technology and do not make intensive use of water. By a presidential decree, this benefit was broadened, in particular by raising the percentages of deduction for different assets.
2. These benefits, called "prevision social" (including social security benefits, childcare…), are deductible for union workers if granted on a general basis; for non-union workers, they are deductible under the condition that they are granted to all workers in the enterprise, and provided that the average for every non-union worker is equal or lower than the deductible expenditures of every union worker and as long as the expenditures are not larger than ten times the annual minimum wage. Contributions to savings funds are also deductible when granted in a general form and under a certain ceiling.
3. Fringe benefits are fully exempt when the sum of these and remunerations do not exceed seven minimum wages, otherwise the exemption is one annual minimum wage. The limitation does not apply to pensions, compensations for work risk granted legally, collective contracts (contratos ley), medical, dental, funeral and hospitalization expenses, medical expense insurance, life insurance and savings funds.

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Annex III

Background information to Chapter III

III.A. Education programmes
III.B. Existing support to job search
III.C. The recent development of e-Mexico

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Annex III.A

Education programmes

Several programmes have been implemented by the Mexican government to support education. These programmes are divided in two main categories: broad-based programmes directed to raise general quality of education, and programmes specifically directed to the poor segments of the population to improve access and equity of education.

PEC, *Escuelas de Calidad* (Schools of Quality Programme)[1]

This programme was launched in 2001 to improve the quality of basic education by giving competitive grants for improvement to urban schools with students from low-income households, and to encourage local decision making in the process.[2] The schools participating have to create a project to improve infrastructure and/or quality of education and the local and federal governments will grant the meriting projects 100 000 pesos. About 52 per cent of the expenditures[3] of the programme are used to finance school equipment, books, and teaching materials; 32 per cent go to the construction of new educational centres; 13 per cent are used in school maintenance; and only about 1 per cent goes to teachers' training. In 2002 it helped 2.6 million students, most of which attended primary school or participated in the *telesecundaria* programme.[4]

Programa nacional de lectura[5]

The main purpose of this programme is to strengthen the reading skills and habits of the population as well as to improve teaching practices in schools. It is involved in a constant revision and monitoring of basic education programmes and plans, in the production and actualization of educational material for teachers and students, and in the promotion of reading and writing skills. It creates and improves school and classroom libraries, providing reading material in basic and "normal" education. In 2002 it produced more than 28 million books and distributed them to about 169 000 schools.

Telesecundaria (Lower secondary education with television support)

Launched in 1968, this programme uses the public television network to provide education to distant communities in Mexico. It broadcasts lessons for all three levels of lower secondary school, showing teachers and students on screens as well as making an extensive use of images and video clips. The lessons are transmitted 6 hours per day, Monday through Friday, during the entire school year. The number of students reached by this programme is estimated to be approximately 1.1 million in about 15 800 schools that have implemented the programme. The Ministry of Education has recently started a project to put *telesecundaria*

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in 300 DVD. Shifting to a DVD approach will reduce the costs of the programme, making it possible to reach a greater number of remote communities.

PAREIB[6] (*Programa para Abatir el Rezago en la Educación Inicial y Básica*, 1998-2006)

This is part of the compensatory programmes run by CONAFE (*Consejo Nacional de Fomento Educativo*) to promote equity and give general support to the educational system in marginalized areas.[7] It aims to improve the quality of public education in preschool, primary, and lower secondary school levels, as well as to reduce the dropout and absenteeism rate. It has four basic components:

- **Teaching Materials:** it provides school supplies at the beginning of each school year to students in extreme poverty[8] enrolled in primary education with the purpose of stimulating learning and reducing spending by parents. During the school year 2002-2003, it gave supplies to 4.5 million students and teaching materials to 16 000 primary schools and 4000 *telesecundarias*.
- **Training and Counseling:** directed to teachers and administrative staff in rural areas to improve teaching methods and the quality of education by giving access to technical and economic resources. It also provides economic support to supervisors to better monitor educational projects and activities in schools. In the 2002-2003 school year, it counseled and trained more than 122 000 teachers and school staff in 40 000 schools.
- **Incentives:** teachers and parents associations receive monetary incentives to encourage them to participate more actively in the education of children and facilitate a good learning environment. The incentives consists of two parts, REDES and AGE:
  - *Reconocimiento al Desempeño Docente* (REDES) offers monetary grants to teachers working in marginalized communities with difficult access, as an incentive to stay for longer periods in those areas and to reduce absenteeism among teachers. Teachers who benefit from grants are committed to help students after class and participate in other extracurricular activities.
  - *Apoyo a Gestión Escolar* (AGE) makes funds available to parents associations in preschool and primary schools under extreme poverty (as defined by CONAPO)[9] to buy school material and to finance projects chosen by each school. These grants, which can be from 5 000 to 7 000 pesos a year, require that the parents be involved in a constant evaluation of the performance of teachers and school programmes.
- **Infrastructure:** this component helps in the construction and maintenance of classroom and sanitation facilities to create favorable conditions for learning. During the year 2002, it built and repaired 9 828 educational centres.

Escuelas Comunitarias[10] (Community Schools Programme)

The target of this programme operated by CONAFE is children in rural areas and indigenous population living in small communities (with less than 100 to 500 inhabitants)[11] with limited or no access to drainage, clean water, electricity, and other basic services. These communities include camps for migrant agricultural workers and a few suburban areas. The programme recruits and trains people between the age of 14 and 24 having finished lower or upper secondary education to be the teachers in the community schools for one or two years. The programme covers living expenses of teachers and offers scholarships to continue their education for the next 30 to 60 months after completing service in the community schools.

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Oportunidades

One of the elements of this programme is to improve the education received by children in families under extreme poverty. This programme integrates actions directed at nutrition and health to maximize children's school achievement, so that they be not affected by malnutrition or the need to get a job. It promotes healthier nutritional habits and gives families resources. This programme has been successful in increasing access to education in poor communities (see Annex I.D).

Notes

1. *www.sep.gob.mx/wb2/sep/sep_3414_escuelas_de_calidad*
2. In 2002, the number of participating schools reached 9820 (four times more than in 2001).
3. A total of 1.2 billion pesos in 2002.
4. 78 per cent of the schools benefited are primary schools, 21 per cent are telesecundarias, 0.5 per cent are secondary or technical schools, and 0.5 are preschools.
5. *http://lectura.ilce.edu.mx/documentos/pnl/pnl.html*
6. *www.sep.gob.mx/wb2/sep/sep_3287_programa_para_abatir*
7. There has been a series of compensatory programmes since 1991, each one lasting for about 4 or 5 years and having similar characteristics and structures. Some of these programmes have been: PARE (1991-1996), PRODEI (1993-1997), PAREB (1994-1999), and PIARE (1995-2000).
8. The students receiving school supplies include those participating in *Oportunidades*.
9. *www.conapo.gob.mx/m_en_cifras/principal.html*
10. *www.conafe.edu.mx/menu1.html*
11. The communities in the programme have an average of 50 inhabitants.

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Annex III.B

Existing support to job search in Mexico

There is no unemployment insurance system as such in Mexico. The existence of relatively generous severance payments for workers with indefinite contracts could be considered as a substitute for unemployment benefits.[1] Besides these severance payments, some programmes provide temporary income support (or information services) to jobseekers.

- *Training programmes.* Active labour market policies are presented in Chapter III. However, as mentioned in Chapter III, besides training purposes, programmes such as PROBECAT and CIMO have also been used to provide some income support at a time of emergency (following the 1995-96 recession for instance) in case of job losses.[2] Evaluations made by the World Bank and the Inter American Development Bank at the end of the 1990s have found that they achieved some of their training objectives.[3] However, there is general agreement that that PROBECAT should be more focused on its original objective of improving productivity and human capital.
- *Public Employment Services* (PES). The PES help bring together job offers and demands. In a country with no state unemployment insurance, PES has been rather disconnected from the rest of active labour market policies and has almost exclusively played an information role. There has been significant progress concerning information provision in recent years. The main instruments of the PES are: workshops for the unemployed, where guidance is provided to people with a low education level or difficulties of insertion in the labour market, and job fairs and job finding services. Two examples of such services are: CHAMBATEL which has been operating by telephone since March 2001, and CHAMBANET which has been operating on the internet since March 2002. A persistent problem is the insufficient coordination among states' PES units and between the national and states units.
- *The programa de empleo temporal* (PET). This programme was created after the 1995 crisis. It serves as a rudimentary safety net in remote rural areas by providing temporary employment during the low season in agriculture when work opportunities are scarce. The wage paid for this work is extremely low, so that there is no risk of creating disincentives to work. The programme has now been decentralised and is operated at the local level.
- Some unemployment support comes from the possibility to withdraw up to 10 per cent of the pension savings every five years if non-employed.
- A programme of direct financial support to the jobseekers has been recently launched in 2001 on a pilot basis in the 10 states where employment has contracted the most in the recent slowdown. This *sistema de apoyo econonomico a buscadores de empleo* (SAEBE) provides, under conditions, 1 800 pesos (about US$180) once a year to qualified jobseekers,[4] to enable them to buy a telephone card, finance transportation for job search and

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contribute to family income. This system is mainly designed as a tool to support job search by dismissed workers and complete other public employment services. Its very low cash amount provided means that there is no risk of creating unemployment traps. According to the Ministry of Labour, the employability of beneficiaries of this scheme is 44 per cent higher than the nation average and 57.6 per cent higher than for persons receiving other benefits.

Notes

1. Severance payments associated with fair dismissals are not much higher than in other OECD countries, but the definition of fair dismissal is the tightest in OECD, with redundancies and poor performance not being normally legal grounds for dismissals. This implies that termination of contracts mostly falls under unfair dismissal in which case, severance payments are of three months plus 20 days per year of service.
2. PROBECAT (renamed SICAT in 2002) provides short-term grants and training to the unemployed and CIMO (renamed PAC in 2002) provide the second one for training support in small and medium size enterprises. See OECD Economic Survey 1996 for more details on these programmes.
3. See for instance, Calderon-Madrid A. and B. Trejo (2001) "The Impact of the Mexican Training Programme for Unemployed Workers on Re-employment Dynamics and on Earnings", *mimeo*. World Bank (2001), *Mexico: A Comprehensive Development Agenda for the New Era*.
4. Qualified jobseekers are those who have been contributing to IMSS for at least six months, are 18 to 45 years old and have dependents.

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Annex III.C

The recent development of e-Mexico[1]

E-Mexico is an ambitious project launched in 2001 to eliminate Mexico's digital divide (existing both between various population groups and between Mexico and other OECD countries). It focuses on expanding the country's communication infrastructure and bringing government services to rural and remote communities. The main actions of this project are: *i*) establishing adequate regulatory frameworks for electronic media and e-commerce; *ii*) digitizing government services; *iii*) developing e-learning applications; and *iv*) implementing a nationwide e-health system.

On July 15, 2002, President Fox inaugurated the first phase of the e-Mexico programme, committing to spend 663 million pesos on several subprojects including the creation of 3 200 digital community centres with internet access in 2 445 municipalities by 2003. During the second phase, from 2003 to 2006, the programme is expected to reach 10 000 communities.[2] The ultimate goal of e-Mexico is to install 10 000 digital community centres that serve 25 000 municipalities (95 per cent of the population) by 2025. By then, according to the project, 98 per cent of the population would have access to internet, internet education would be available in all schools, and the legal and regulatory structure for e-commerce would be in place to promote greater consumer confidence. A detailed list of goals has been defined (Annex Box III.C.1).[3] These strategies are incorporated into the four main categories of e-Mexico which are e-Learning, e-Health, e-Economy, and e-Government.

E-Learning

This area is under the responsibility of the Education Ministry jointly with the e-Mexico National System General Coordination Office and the Communication and Transports Ministry (SCT); it aims to modernize education and provide new options using the resources of the internet and new technology. It promotes distance learning through the Human Resource Learning model of the Latin-American Educational Communication Institute (ILCE), offering on-line training, seminars, and courses that cover several subjects, such as business administration, calculus, organisational behaviour, financial math, health, and history of art. Several online seminars for basic and secondary education are also available, including programmes offered by the National Council of Education for Life and Work (CONEVyT). E-Learning provides extensive information about school programmes at all education levels, scholarships, cultural activities, and job listings for graduates. It also makes available school directories and links to their websites.

E-Health

The Health Ministry (*Secretaria de Salud*), the e-Mexico National System General Coordination Office and SCT are in charge of this area with the participation of the Mexican Social

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Annex Box III.C.1. The goals of e-Mexico

1. Give Mexicans opportunities to access knowledge, learning and education by means of an intelligent use of technologies.
2. Articulate the interests of the Federal State and Local Public Administrations, Public Offices, Telecommunications Operators, Chambers and Associations regarding the information and communication technologies, academic and economic agents, and diverse institutions to develop digital services.
3. Promote investment of domestic original equipment manufacturers and suppliers in IT infrastructure and R&D.
4. Develop domestic IT industry.
5. Develop human capital according to the needs of the IT industry.
6. Create a favourable business environment for software development to attract investment.
7. Support the domestic software industry.
8. Increase domestic demand for software and other IT products and services
9. Provide financing to small and medium enterprises (SMEs) to encourage use of IT in their business processes.
10. Promote the use of IT in the administrative systems of SMEs.
11. Develop training programmes for SMEs on the benefits of IT.
12. Integrate public companies into digital supply chains.
13. Finance projects focused on the development of digital supply chains.
14. Develop skills in both managers and staff to administer digital supply chains.
15. Reform legislation to support the use of electronic, optical or any other IT media.
16. Use the Mexican government as a model for the use of IT.
17. Facilitate industry access to key information on how to use IT.
18. Reform regulations to include digitalization of government paperwork and services.
19. Support the transformation of digital processes in the government agencies that offer services to private companies in coordination with the e-Government programme.

Security Institute (IMSS), the Social Security Institute for Government employees (ISSSTE), the Naval Ministry (SEMAR) and the Mexican Oil Company (PEMEX). The initiative of e-Health was set up to improve health services, expand coverage, give better access to information on health service projects and issues, and to help administer the health programmes in a more transparent way. Its main focus is to increase the quality and efficiency of health services in distant communities through the use of modern technologies.

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E-Health informs the community about health service programmes and its requirements, allows residents to make medical appointments through the internet, and uses distance learning to provide continuous education and training to medical staff. In addition, it also promotes portals that offer specialized information to particular groups in the population including women, the elderly, and the handicapped, and it makes available maps and addresses of health centres, pharmacies, and drugstores.

E-Economy

This category is under the responsibility of the Ministry of Economy jointly with the e-Mexico National System General Coordination Office and SCT. It includes a directory of businesses, business opportunities, and regulations of international commerce. It also provides information about international and national commerce, supply and demand of products, credit to enterprises, and training programmes. E-Economy's focus is mainly: to provide services for small and medium sized business; to promote national and international markets; and to enable a better communication among business, customers, and suppliers. Users can get information on how to start and operate a business. There is also virtual assistance on how to register a business, investment opportunities, advertisement programmes, and how to export products, and other questions and business concerns.

A specific project to help small- and medium-sized businesses is being developed with the support and technical assistance of the World Bank[4] and N*afin* (N*acional* F*inanciera*), in partnership with the Ministry of Economy. It will help small companies to get incorporated at the three levels of government (federal, state, and local) in a very quick way using the internet. This should help the government increase the tax base and the company will receive benefits from the incorporation, such as being able to receive credit from financial institutions. The project should also support efforts to integrate e-business into their operations. It will participate in the creation of digital centres on these small businesses and will play an active role in on-the-job training by digital means.

Another purpose of this project is to change the way firms operate in areas such as market research, production, finance and management, and transaction processes via online networks. This should make small businesses more productive, increasing their competitiveness, and helping the creation of better skilled, hence better-paid, jobs.

E-Government

Under the responsibility of the Presidential Office for Government Innovation (O*ficina de la* P*residencia para la* I*nnovación* G*ubernamental*) jointly with the e-Mexico National System General Coordination Office and SCT. It constitutes an information, transaction and interaction bridge between citizens and the Mexican government. Its focus is to multiply government services online. It includes information for foreigners, as well as for Mexican residents, regarding different types of credit, scholarships, entertainment, and government programmes. Users can participate in open forums that allow them to send messages directly to the government authorities. They can also obtain counsel on how to get a job, get credit for a house, or start a small business. It also allows for several transactions to be done through the internet such as paying taxes through e-SAT (Annex Table III.C.1), filling out the paper work to get the voter's card, or querying an electricity bill among others.

Two of the main components of e-Government are *tramitanet* and *compranet*:

– T*ramitanet* was launched at the beginning of 2002. It aims to allow Mexican residents to do some government transactions online such as enquiring for school grades, getting copies of birth certificates, and doing the paper work needed to get a voter's card.

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Table annex III.C.1. **Some e-government applications**[1]

Application/Launch Date	Function	Key Features/Advances
Tramitanet *www.tramitanet.gob.mx* SECODAM January 2002	Interactive transaction portal	Catalogue of 2 064 transactions 20 electronic transactions Electronic signature mechanism
Compranet *www.compranet.gob.mx* SECODAM March 1996	Interactive public procurement portal	37 000 registered providers Detailed information regarding goods/services sought Information regarding sanctioned parties Electronic signature mechanism
e-SAT *www.sat.gob.mx* SAT/SHCP 1997	Interactive tax portal	Online declarations and payments 147 405 electronic declarations (August 2001 2002) More than 180 billion pesos of electronically realized payments It facilitates location of tax evaders, audits and tax litigation Electronic signature mechanism

1. E-government in Mexico: Current Developments and Future Challenges. Robert M. Kossik. March 31, 2003.
Source: OECD.

However, as for now, most of the space in this site offers information about transactions (fees, requirements, government agency in charge), rather than allowing users to perform these transactions. There are currently only 20 different transactions that can be made through the internet, but Mexico is planning to expand services so that every single government transaction can be done on line. This system should help fight corruption since citizens will have fewer direct interactions with public servants.

– Another important component launched some years ago, hence before e-Mexico was created, but now incorporated into the new system is *compranet*, which allows companies to bid on federal contracts over the internet to make acquisitions open to the public, reducing transaction costs and bribery opportunities. But so far the site accounts only for a small percentage of total government purchases.

E-Mexico aims to become an efficient tool to connect different communities to government agencies that can meet their needs. Such is the case of the indigenous community which can use e-Mexico to reach the Presidential Office for the Representation of the Indigenous Population (*Oficina de la Presidencia para la Representacion de los Pueblos Indígenas*, ORPI) and The Indigenous National Institute (INI). There is also a space specially created for the handicapped population which connects them to the Ministry of Health and the Promotion and Social Integration for Handicapped People Representation Office (ORPIS). In addition, a recent project of e-Migrants is being developed to help connect the migrant community with the Mexican Abroad Institute (*Instituto de los Mexicanos en el Exterior*), The Foreign Affairs Ministry (SRE) and 20 other institutions.

Despite all the obstacles in Mexico coming from insufficient infrastructure and human capital, Mexico is developing strong online services and has demonstrated a high e-government capacity. According to a study by the consulting company Accenture in April 2003, Mexico is the country that showed the greatest improvement with respect to last year in digital technology. Several experiences of e-Mexico have been shared with other Latin American countries such

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as Honduras and Colombia in terms of educational contents and Digital Communitarian Centres development

Notable advances have already been made in terms of connectivity. Every municipality in Mexico has at least one connection point, a total of 3200 community centres. However, connectivity alone will not eliminate the digital gap; accomplishment of this goal will require coordination and cooperation among ministries and agencies, organisation of the services around the needs of the citizens, and the sharing of knowledge. There is also the need to teach the Mexican population how to best take advantage of technology and how to use e-Mexico.[5] The goals of e-Mexico will be fully reached only when the indigenous populations living in remote areas and the poor sectors of the population understand how to use it and find its use beneficial.

Notes

1. *www.e-mexico.gob.mx*
2. According to industry analysts, the first two phases will invest US$6.1 billion into Mexico's telecommunications market across the country's 31 states.
3. Source: e-Mexico Project Overview: Telecom Trends Mexico 2003. August 12, 2002.
4. On July 31st the World Bank Board of executive directors approved a US$58.4 million loan to help increase competitiveness and growth of micro and small businesses in low-income urban areas of Mexico.
5. *E-government in Mexico: Current Developments and Future Challenges*. Robert M. Kossik. March 31, 2003.

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Annex IV

Background information to Chapter IV

Information about Mexican international migration is available from a variety of Mexican and foreign sources (mainly from the United States) including: national census data, current surveys, administrative sources, or more specific documentation, such as research projects (including binational studies).

Data collected include information on stock/flows of migrants, legal/undocumented migrants, background characteristics (demographic indicators, labour experience, earnings, education) of migrants.

In Mexico, statistics are mainly produced by the National Institute of Statistics (INEGI), the National Council for Population (*El Consejo Nacional de Población* (CONAPO)), and the National Institute of Migration (INM). In the United States, international migration data are collected by the US Census Bureau, the Bureau of Citizenship and Immigration Services, the US Department of State and teams of researchers. Moreover in both countries surveys conducted by different researchers embodied in national or international teams provide information on migration.

The migration of Mexican to the United States has been widely surveyed. Only those sources used in the special chapter of this Economic Survey are reviewed below.

MEXICAN SOURCES

1. The National Institute of Statistics: INEGI[1]

Population and Housing General Census

– *General objective*: to generate demographic, socio-economic and cartographic information; enhance the historical series of statistical data; allow the construction of sample frameworks.

- *Geographical coverage*: national.
- *Periodicity or time coverage*: decennial, for the years ending with zero. Last census in 2000.
- *Target population*: the usual resident population, households and dwelling units.
- *Summary of thematic coverage*: from the basic questionnaire, information on number and size of households, place of birth, demographic, educational and economic characteristics are gathered. The extended questionnaire contains more specific related migration information such as international migration over the last five

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years, emigration date, country of destination, country of current residency and date of return.

Households Incomes and Expenditures National Survey: ENIGH

- *General objective*: to measure the amount, structure and distribution of households' incomes and expenditures.
- Geographical coverage: national.
- *Periodicity or time coverage*: every two years since 1992. Last survey in 2002.
- *Target population*: households with domestic and foreign members aged 12 and over who reside in private dwelling units within the national territory.
- *Summary of thematic coverage*: contains information on the activity and occupational characteristics of the household; total monetary and non-monetary income and expenditure. The related migration information is the current income derived from the migrants' remittances.

Demographic Dynamics National Survey (ENADID)

- *General objective*: to produce information and determine the basic component of the demographic dynamic: fertility, mortality and migration.
- *Geographical coverage*: national.
- *Periodicity or time coverage*: each five years. Last survey in 1997.
- *Target population*: Usual residents (including resident women aged 15-64 years), international migrant population for the past five years, deceased people over the past five years.
- *Summary of thematic coverage*: demographic, educational and economic characteristics, of which the related migration information: place of residence five years earlier, international migration condition, residence condition, date of emigration trips, number of emigrations, current residence and returning dates; occupation in the United States, number of transfers, date of transfers and return, availability document for work and year when the document was obtained.
- *Remark*: the coverage includes both return migrants (direct information) and individuals who were in the US at the time of the survey (information provided by members of families in Mexico). When provided by third person responses the information may not be accurate and complete families who have emigrated are not identified and not taken into account. Moreover during the reference period (five years) the household might have a different configuration. It implies that the survey may underestimate the magnitude of permanent emigration and may give a wrong image of the economic information. However these data provide valuable information about the migration phenomenon, allowing comparisons between migrants and non-migrants.

2. The National Council of Population: CONAPO[2]

The survey of migration to the Northern Mexican Border (EMIF)

- *General objective*: to study the circular nature of migration by producing direct estimates of the volume of documented and undocumented flows from Mexico to the US, as well as return migration.

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- *Geographical coverage*: identification of "migratory routes" through which circular migration occurs and defined empirical points of observation. In these locations systematic counts and interviews of the migrants are made over specific periods of time.
- *Periodicity or time coverage*: seven one-year phases of this survey have been conducted since the beginning of the Survey in March 1993. The latest was realised from 11 July 2001 to 10 July 2002.
- *Target population*: four types of migrants are under observation:
  1. Illegal and legal migrants preparing to cross the border from Mexico to the United States;
  2. Mexican nationals who are permanent residents of the United States and are returning to Mexico;
  3. Permanent residents of Mexico who have been in the United States legally or illegally and are returning to Mexico; and
  4. Illegal migrants voluntarily deported from the United States.
- *Summary of thematic coverage*: demographic, educational and economic characteristics; place of origin, place of residence; prior labour experience in Mexico and in the United States; prior legal or illegal crossing experience, use of document, use of smugglers; reasons for making the trips (work-related, job-related, other).
- *Remarks*: this survey provides information on the magnitude of the flows (legal and illegal) and on the qualitative and quantitative changes in these flows over time. However the survey focuses only on labour migrants, under-estimating certain groups of migrants, such as women. Moreover, by its nature (observation of flows), the survey is subjected to biases due to multiple countings.

3. Specific surveys

The Mexican Migration Project (MMP)[3]

- *Overview*: the MMP was created in 1982 by an interdisciplinary team of binational researchers, co-directed by Jorge Durand of the University of Guadalajara (Mexico), and Douglas S. Massey of the University of Pennsylvania (United States). It follows the logic of an ethno-survey, with qualitative and quantitative data-gathering techniques.
- *General objective*: to gather and maintain high quality of social and economic information on Mexican-US migration for documented as well for undocumented population.
- *Geographical coverage*: the survey is conducted in Mexico and in the United States. In Mexico, communities in the traditional migrant States were first studied, then the survey was extended to other Mexican States. The communities were chosen to provide a representative sample of communities. This sample is paired with non-random snowball samples, or reputational sample, of settlers in the United States who are out-migrants from the places surveyed in Mexico.
- *Periodicity or time coverage*: several communities surveyed each year for successive years. In Mexico, interviews are conducted during months when migrants return home in large numbers (December/January). In the United States, identical interviews are conducted in the summer following data collection in Mexico.
- *Target population*: (defined above).
- *Summary of thematic coverage*: the information is gathered from the head of household, covering: individual and household demographic, educational and economic charac-

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teristics; migration experience of all household heads; life history of each person of the household from aged 15 and over; basic data on first and latest trips to the United States; the use and cost of smuggling; the use of legal documents.

– *Remarks*: the retrospective questionnaire allows for analysis of migration flows over a long period. The survey captures long-term settlers and temporary migrants; unlike the sample in Mexico, the US sub-sample is not representative and the snowball technique may under-sample people with weak links with their location of origin or living in non-traditional locations in the United States; the way information is gathered from the head of the household may lead to less accurate or incorrect information over past events, and data on migration experience may be biased.

The Zapata Canyon Project[4]

– *Overview*: this project was initiated in September 1987 by *El Colegio de la Frontera Norte* (COLEF), under the direction of Jorge Bustamante. The survey technique consists in a personal, short and standard questionnaire addressed to a randomly selected sample;

– *General objective*: to survey the flows of undocumented migration from Mexico.

– *Geographical coverage*: main crossing sites at the Mexican/US border in the cities of Tijuana, Mexicali, Ciudad Juarez, Nuevo Laredo and Matamoros.

– *Periodicity or time coverage*: since September 1987, interviews are conducted three days per week: Friday to Sunday.

– *Target population*: undocumented emigrants to the US preparing to cross the border.

– *Summary of thematic coverage*: the questionnaire includes information on the demographic characteristics, the labour experiences in Mexico and in the US, the location of border crossing, the cost of the trip, the use of a smuggler, the number of apprehensions by the border patrol, the reason for making the trip.

– *Remarks*: the survey provides extensive time series on illegal border crossing with origin and destination; it is the only data available on illegal crossing apart from the US Border Patrol statistics. However it is not a representative survey, it does not allow a good estimation of total flows of undocumented migrants, focusing only on entry to the US, not on return migration. Lastly, personal information on the migrant is limited by the fact that the migrant is in a hurry to cross clandestinely.

THE BINATIONAL STUDY[5]

1. The Mexico – United States Binational Study on Migration

– *Overview*: it was funded in March 1995 by both the Mexican and the United States government with private sector funding in both countries. It was released in September 1997. The research was conducted by a team of 20 independent researchers, who reviewed existing research, generated new data and analyses, and undertook site visits and consulted with migrants and local residents to gain a joint understanding of the issues raised in this study.

– *General objective*: to contribute to a better understanding and appreciation of the nature, dimensions, and consequences of migration between the two countries.

– *Summary of thematic coverage*: to evaluate the size of the legal and illegal Mexican-born population in the United States, and the size of the migration streams crossing the border; demographic, educational, and income characteristics of Mexican-born

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migrants; factors influencing migration from Mexico; economic and social effects of migration on both Mexico and the United States; and societal responses to migration in both Mexico and the United States (legislation, policy, court decisions, etc.).

– *Remarks*: despite some data limitations, the survey is useful in making a number of specific recommendations for needed research, but there are no mechanisms in place for follow-up.

UNITED STATES SOURCES

1. **The US Census Bureau.** international migration is captured by the information on foreign-born population in the US, which is defined as "people who are not US citizens at birth". It includes immigrants, legal non-immigrants (refugees and persons on student or work visas) and persons illegally residing in the United States. The Census Bureau collects data through the decennial censuses and numerous surveys of the US population, such as the Current Population Survey (CPS).[6] These sources deal with the foreign-born population stock.

The census

– *General objective*: to generate demographic, social, and economic information for the entire US resident population. Two types of questionnaires are used a "short form" and a "long form"; the short form contains questions on demographic and social characteristics, the long form include additionally questions about foreign born residents; the short form was sent in 2000 to approximately five of every six households, while the long form was sent to approximately one of every six households. The data collected are referred to as "sample data" and are used to estimate the characteristics of the population.

– *Geographical coverage*: national.

– *Periodicity or time coverage*: once every 10 years. Last census in 2000.

– *Target population*: US resident population.

– *Summary of thematic coverage*: apart from the demographic, educational and economic characteristics, the related migration information consist of questions about nativity, citizenship status (naturalisation), year of entry and migration history inside the US .

– *Remarks*: in combining the data, the long form provides a detailed description of the foreign born in the census; however it is not designed to collect characteristics which would be particularly relevant to migration: for instance the legal status. This makes it difficult to interpret correctly the shifts in migrant characteristics (for instance a decrease in average earnings may result from an increase in undocumented migrants).

The Current Population Survey

– *General objective*: to collect information on the labour force characteristics of the population; two types of questionnaires are used the basic CPS questionnaire and the monthly supplement questionnaires.

– *Geographical coverage*: national.

– *Periodicity or time coverage*: monthly, with a special annual supplement in March.

– *Target population*: the civilian non-institutional population aged 16 and over.

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– *Summary of thematic coverage*: labour force and demographic information including related migration information such as place of birth, parental nativity, citizen status, year of entry into the US. The March supplement includes questions about geographic mobility in the previous 12 months, including moves from abroad.

– *Remarks*: the survey is a good source of regular information on the foreign born, but its value is limited by its sample size. The population estimates are adjusted for under coverage using the most recent census. Until end 2001, these estimates were adjusted to the 1990 census, hence they are not directly comparable to the foreign-born population from the 2000 census.

2. **The Bureau of Citizenship and Immigration Services (BCIS).** the BCIS collects international migration data about legally admitted immigrants (permanent or temporary), as well as newly naturalized citizens from several administrative records. The data collected give information on flows. It constitutes the major source of information about legal immigration. However these data have two shortcomings: missing relevant information and limited longitudinality on individual characteristics.

3. **The Bureau of Customs and Border Protection (BCBP).** being responsible for control of US borders, the BCBP **gather** data on illegal crossings for aliens and statistics on apprehensions.

Notes

1. For more information see: *www.inegi.gob.mx*
2. For more information see: *www.conapo.gob.mx*
3. For more information see: *www.pop.upenn.edu/mexmig/research*
4. See "Undocumented Migration to the United States: Preliminary Findings of the Zapata Canyon Project", in Frank Bean *et al.* (eds.), *Undocumented Migration to the United States*, three volumes, Mexico City, *Secretaría de Relaciones Exteriores*; Washington, D. C., US Commission on Immigrations Reform, 1998.
5. See *www.utexas.edu/lbj/uscir/binpap-v.html*
6. The other Census Bureau Surveys include: the American Community Survey (ACS), the Survey of Income and Programme Participation (SIPP), the National Health Interview Survey (NHIS), the New York Housing Vacancy Survey, and the American Housing Survey (AHS). For more information see Joseph M. Costanzo *et al.* 2002 “Guide to International Migration Statistics: The Sources, Collection, and Processing of Foreign-Born Population data at the US Census Bureau”, US Census Bureau, Population Division Working Paper No. 68.

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Annex V

Chronology of main economic events

2002

March

A new department in charge of supervision and surveillance of credit unions is created by the National Banking and Security Commission (*Comisión Nacional Bancária y de Valores*, CNBV).

Creation of SARE (*Sistema de Apertura Rápida de Empresas*), a new system to start a business within one day.

April

The Bank of Mexico reduces the *corto* from 360 to 300 million pesos.

Law of Transparency approved. (Effective June 2002.)

Congress approves fiscal adjustment as a response to revenue shortfall.

August

Creation of INEE (*Instituto Nacional de Evaluación de la Educación*) in charge of the evaluation of education institutions.

Electricity reform proposal submitted to Congress.

September

The Bank of Mexico increases the *corto* from 300 to 400 million pesos.

October

The government presents the budget proposal for 2003.

December

The Bank of Mexico increases the *corto* from 400 to 475 million pesos.

The National Commission of Minimum Wages approves a general minimum wage increase of 4.5 per cent for 2003 (the inflation objective is 3 per cent ± 1 per cent by December 2003).

Labour market reform proposal presented to Congress.

Congress approves the budget for 2003.

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Congress approves liquidation of B*anrural* (estimated cost of 48 billion pesos), its last day of operations being 30 June 2003.

2003

January

Part of the tax measures approved at the end of 2001 come into force.

The Bank of Mexico increases the *corto* from 475 to 550 million pesos.

The Bank of Mexico publishes the Monetary Policy Programme for 2003. The Bank of Mexico targets a 3 per cent inflation rate by the end of 2003, with a variability interval of plus or minus 1 per cent. The medium-term objective is to keep the inflation in that range thereafter.

February

The Bank of Mexico increases the *corto* from 550 to 625 million pesos.

March

The Exchange Commission announces a rule-based mechanism to reduce the pace of international reserve accumulation (effective in May).

The Bank of Mexico increases the *corto* from 625 to 700 million pesos.

April

The Senate approved an increase in subsidies for residential electricity prices in some regions "with difficult climatic conditions".

Senate approved creation of a popular health insurance (*seguro popular*) effective in January 2004 to give medical services to more than 40 million inhabitants in the poorest regions.

The Bank of Mexico modifies the way it operates the *corto/largo* mechanism from accumulated balances to daily balances and increases banks' compulsory deposits at the central bank from 25 billion pesos to 30 billion pesos.

Congress approves a reform of the legal framework for secured credit transaction (M*iscelanea de garantías*), including amendments to seven related laws.

May

Presidential decree on health insurance universal coverage is signed.

June

With a last operation, Mexico redeems all remaining Brady Bonds.

August

The newly-created development bank, F*inanciera* R*ural*, begins its operations.

November

The government presents the budget proposal.

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